

Adaptive Observer Theory

Navigation of Incompleteness
and Incompatibility

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A Monograph in Three Articles

Article I: Must Any Consistent Physics Contain Structure
Inaccessible to All Internal Observers?

Article II: The Theorem of Incompatible Truths

Article III: The Pragmatic Synthesis: How Complex Systems
Navigate Incompatible Truths through Adaptive Heuristics

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1 Preface

This monograph asks three questions about the relationship between a system and the observers it contains.

What can an internal observer not see? Article I proves that any consistent physical system containing a proper-subsystem observer necessarily harbours ontological content inaccessible to all internal observers. The proof uses five independent methods (diagonal, information-theoretic, categorical, dynamical, topological) and identifies the exact boundary conditions via a trichotomy theorem.

What happens when an observer looks at what it can see? Article II proves that any sufficiently complex system necessarily admits descriptions that are individually valid but mutually incompatible. Five proof methods converge on the same result: compression, logical orthogonality, category-theoretic non-existence of colimit, topological obstruction, and predictive trade-off. The incompatibility is not a failure of current science but a structural feature of complexity itself.

How does an observer manage to act despite these limitations? Article III proves that any viable agent in such a system must maintain multiple simplified models, switch between them, and monitor its own performance. This structure—the navigator—is derived from four minimal axioms. Its five properties coincide with the properties attributed to “consciousness” in the scientific literature.

The three results form a *tripartite viability constraint*: persistent complex systems must simultaneously maintain uncertainty, tolerate contradiction, and navigate adaptively. The loss of any dimension leads to collapse.

1.1 What This Monograph Is Not

This monograph does not propose a Theory of Everything. It proves that such a theory, in its classical sense, cannot exist for systems of sufficient complexity.

This monograph does not solve the hard problem of consciousness. It derives the functional structure that the hard problem asks about, and shows that this structure is a mathematical necessity—not a mystery.

This monograph does not claim that its axioms are the only possible ones, or that physical systems necessarily satisfy them. Every result is explicitly conditional: *if* the axioms hold, *then* the theorems follow. Whether the axioms hold for any particular system is an empirical question.

1.2 For Whom This Monograph Is Written

The mathematical formalism is rigorous but is preceded and followed by plain-language explanations accessible to readers from psychology, sociology, biology, and philosophy who may not have backgrounds in dynamical systems or mathematical logic. Every definition, theorem, and proof is accompanied by physical, biological, or cognitive examples.

The monograph is self-contained. Each article can be read independently, but the three together form a single argument.

2 Synopsis and Justification

2.1 The Central Argument in One Page

Premise 1 (Article I). Any consistent physical system containing an observer who is a proper subsystem of that system necessarily contains structure that no internal observer can access. *Proof:* five independent methods, all rooted in the fact that a proper part cannot biject onto the whole.

Premise 2 (Article II). Any system whose complexity exceeds a threshold—requiring multiple compressive descriptions of overlapping domains—necessarily admits descriptions that are individually valid but logically incompatible. *Proof:* five independent methods; four antithesis attacks fail; the result is conditional on four precisely stated axioms.

Premise 3 (Article III). Any agent with bounded computational resources, operating in a system with incompatible descriptions, that must act under time pressure with costly errors, necessarily possesses a structure (the navigator) with five specific properties: model storage, contextual evaluation, switching, memory, and self-monitoring. *Proof:* absence of any unconditionally necessary property yields zero long-term viability.

Conclusion. The universe is not an object to be “cracked” by a single formula. It is a dynamic process in which observers—themselves subsystems—must continuously construct, switch between, and monitor locally valid but globally incompatible models. The structure that performs this navigation is what has been called “consciousness.” The monograph does not postulate consciousness; it derives the navigator and identifies the label.

2.2 Why This Monograph Was Necessary

The three problems addressed by the monograph—observer limitations, theoretical incompatibility, and adaptive navigation—have been discussed separately across multiple disciplines:

In physics: the measurement problem, complementarity, the renormalisation group, and the search for quantum gravity all involve aspects of observer limitation and description incompatibility. But these are treated as specific problems within specific theories, not as structural features of complexity itself.

In philosophy: Kant’s limits of reason, Kuhn’s incommensurability, and the disunity-of-science programme all address related themes. But they lack formal proofs and precise boundary conditions.

In cognitive science: bounded rationality, dual-process theory, and predictive processing all describe how agents cope with incomplete information. But they lack a formal derivation of *why* multiple models and switching are *necessary*, not merely useful.

In consciousness studies: IIT, Global Workspace Theory, and Higher-Order Theo-

ries all characterise aspects of conscious experience. But they postulate consciousness as a primitive or an identity claim, rather than deriving it from the constraints of navigation.

The monograph unifies these scattered observations into a single formal framework with precise definitions, stated axioms, proved theorems, and identified boundary conditions.

2.3 The Three Laws

The monograph draws on and extends two prior results:

1. **The Law of Imperative Uncertainty** (Kriger, 2025): persistent complex systems must maintain nonvanishing uncertainty. Rigid closure of all degrees of freedom leads to collapse.
2. **The Law of Scale-Specific Principles** (Kriger, 2025): when extrapolation between scales fails while stable observables persist, one must seek scale-specific principles rather than force unification.

The monograph adds three new results:

3. **The Inaccessibility Theorem** (Article I): internal observers necessarily face structural blind spots.
4. **The Incompatibility Theorem** (Article II): scale-specific descriptions are not merely different but logically incompatible.
5. **The Navigator Theorem** (Article III): viable agents must possess a specific five-property structure for navigating among incompatible descriptions.

Together, these five results constitute **Adaptive Observer Theory**: a formal framework for understanding the structural constraints on any system that contains observers attempting to describe and act within it.

3 Falsifiable Predictions

A theoretical framework that generates no testable predictions is not science. The following predictions are derived from the monograph's theorems and are, in principle, empirically falsifiable.

3.1 Predictions from Article I (Inaccessibility)

Prediction 1 (Persistent Blind Spots). *No improvement in measurement technology will eliminate all observer-inaccessible content in a system where the observer is a proper subsystem. Specifically: for any finite-resolution measurement scheme applied to a system with N degrees of freedom by an observer with $M < N$ degrees of freedom, there*

exist at least $N - M$ functionally independent observables that cannot be simultaneously measured.

Falsification: *Construct a physical system with N degrees of freedom and a proper-subsystem observer with $M < N$ degrees of freedom, and demonstrate a measurement scheme that accesses all N degrees of freedom simultaneously.*

Prediction 2 (Measurement-Dependent Inaccessibility). *Every measurement protocol applied by an internal observer creates a specific “shadow”—a set of observables rendered inaccessible by the protocol. Different protocols create different shadows.*

Falsification: *Demonstrate a single measurement protocol that accesses all observables of a system in which the observer is a proper subsystem.*

Prediction 3 (Inaccessibility Gap Scaling). *The fraction of inaccessible content grows monotonically with system complexity (measured by the ratio of system degrees of freedom to observer degrees of freedom).*

Falsification: *Demonstrate a system-observer pair where the observer accesses a larger fraction of system content as system complexity increases.*

3.2 Predictions from Article II (Incompatibility)

Prediction 4 (Incompatibility at Every Scale Boundary). *At every boundary between physical scales (quantum/classical, microscopic/thermodynamic, molecular/cellular, neuronal/cognitive), there exist pairs of valid descriptions that derive contradictory predictions about at least one shared observable.*

Falsification: *Identify a scale boundary where the descriptions on both sides are logically compatible—derivable from a single consistent axiom set without loss of predictive power.*

Prediction 5 (Unification Penalty). *Any attempt to unify two incompatible descriptions into a single consistent framework necessarily degrades predictive accuracy in at least one of the original domains.*

Falsification: *Construct a unified theory that matches or exceeds the predictive accuracy of both local descriptions in both domains, while remaining internally consistent.*

Prediction 6 (Genericity of Incompatibility). *In any parametric family of theories covering overlapping domains, the set of parameter values yielding compatible theories has measure zero.*

Falsification: *Demonstrate a non-trivial parametric family where compatible parameter values have positive measure.*

3.3 Predictions from Article III (Navigation)

Prediction 7 (Switching Superiority). *In any multi-domain environment satisfying the four axioms, agents that switch between domain-specific models outperform (in long-term viability) agents that use a single model, a blended model, or a random model. This holds regardless of the specific domains or models involved.*

Falsification: Construct a multi-domain environment satisfying the four axioms where a non-switching agent achieves equal or greater long-term viability than a switching agent.

Prediction 8 (Navigator Emergence Under Selection). *Any population of agents subjected to selection pressure in a multi-domain environment will evolve toward the navigator structure (model storage, contextual evaluation, switching, memory, self-monitoring). This is testable in artificial evolution experiments.*

Falsification: Run an artificial evolution experiment in a multi-domain environment and show that the evolved agents do not develop any of the five navigator properties.

Prediction 9 (Coarse-Graining Optimality). *Across biological and artificial systems, the observed level of model coarse-graining will correlate with the ratio of computational budget to model cost. Systems with larger budgets will use finer-grained models; systems with smaller budgets will use coarser models. The optimal coarse-graining level is interior (neither zero nor maximal).*

Falsification: Demonstrate a biological or artificial system that uses maximally fine-grained models despite having a computational budget too small to process them in real time.

Prediction 10 (Cognitive Flexibility and Viability). *Organisms with greater cognitive flexibility (ability to switch between cognitive strategies) will exhibit greater long-term viability (survival, reproductive fitness) in environments with multiple incompatible demands. Organisms with rigid cognitive strategies will exhibit reduced viability in such environments.*

Falsification: Demonstrate that cognitive rigidity is uncorrelated with or positively correlated with viability in multi-demand environments.

Prediction 11 (Navigator Complexity Scales with Domain Count). *The complexity of the navigator structure (model repertoire size, memory depth, switching sophistication) in biological organisms scales with the number of distinct environmental domains the organism encounters.*

Falsification: Demonstrate an organism that encounters many distinct domains but possesses a navigator of the same complexity as an organism encountering a single domain.

3.4 Summary

Article	Central Prediction	Testable By
I	Blind spots persist regardless of technology	Physics experiment
II	Unification degrades local predictions	Theory comparison
III	Switching agents outperform non-switching	AI experiment
III	Navigator emerges under selection	Artificial evolution
III	Coarse-graining level tracks budget/cost	Cross-species data

Table 1: Summary of key falsifiable predictions.

4 Objections and Responses

4.1 Objections to the Overall Framework

Objection 1: The axioms are chosen to produce the conclusion.

Every theorem in mathematics is conditional on its axioms. The question is whether the axioms are reasonable. Our axioms capture minimal conditions: descriptions must compress (they are theories, not databases), must address their subject matter (they are not silent), must engage with shared phenomena (the world is coupled), and must be genuinely different when describing different domains (the world is complex). Relaxing any axiom eliminates the conclusion—but also eliminates a defining feature of theoretical science. The reader is invited to identify which axiom they reject and to explain what kind of science proceeds without it.

Objection 2: This is just philosophy dressed in mathematical notation.

The theorems are formally proved from stated axioms. The proofs can be checked line by line. The predictions are falsifiable. The toy models are computable. This is mathematics, not philosophy. The *interpretation* of the mathematics touches on philosophical questions—but so does every foundational result in physics, from Gödel’s incompleteness theorems to Bell’s inequality.

Objection 3: The monograph proves too much—it would apply to trivial systems.

It does not. Each article identifies precise boundary conditions. Article I: if the observer is not a proper subsystem, no inaccessibility. Article II: if complexity is below the threshold, no incompatibility. Article III: if the agent has infinite resources or faces only one domain, no navigator is needed. The theorems apply to the non-trivial regime and are silent about the trivial one.

4.2 Objections to Article I (Inaccessibility)

Objection 4: Quantum mechanics already tells us about measurement limits.

Article I’s result is *theory-independent*: it does not assume quantum mechanics. It shows that inaccessibility arises from the structural fact that a part cannot biject onto the whole, regardless of which physical theory governs the system. Quantum mechanical limits are a special case.

Objection 5: Advanced technology might overcome the limits.

The inaccessibility is structural, not technological. No amount of technological improvement changes the mathematical fact that a proper subsystem has fewer degrees of freedom than the system. The limit is not about what we *can* measure but about what is *logically possible* to measure.

4.3 Objections to Article II (Incompatibility)

Objection 6: Physics has unified theories before (Maxwell, electroweak). Why can’t it continue?

It can—and the monograph does not forbid this. What the monograph proves is that each unification creates new incompatibilities at the boundary between the unified theory and theories at other scales. Maxwell unified electricity and magnetism, but Maxwell’s theory is incompatible with quantum electrodynamics. The electroweak theory unifies electromagnetic and weak forces, but is incompatible with gravity. Unification *within* a scale is possible; unification *across all scales* is what the theorem forbids (for systems above the complexity threshold).

Objection 7: The incompatibility might be epistemic, not ontological.

The monograph takes no position on this distinction. The incompatibility is a property of *descriptions*: valid descriptions of overlapping domains derive contradictory sentences. Whether this reflects a feature of reality or a limitation of description is a metaphysical question the formalism does not address.

Objection 8: Effective field theory already manages multiple descriptions at different scales.

Yes—and EFT is a formalisation of *managed incompatibility*, not a resolution of it. The renormalisation group flow that connects effective theories is precisely the mechanism of switching between incompatible descriptions. The monograph provides formal justification for this practice.

4.4 Objections to Article III (Navigation)

Objection 9: The navigator is just bounded rationality with extra steps.

Simon’s bounded rationality is a descriptive observation: agents with bounded resources use heuristics. The Navigator Theorem is a necessity proof: agents with bounded resources in incompatible-description systems *must* possess a specific five-property structure, or they have zero long-term viability. The difference is between “agents tend to do this” and “agents must do this or die.”

Objection 10: Calling the navigator “consciousness” is a rhetorical trick.

The monograph does not call the navigator consciousness. It derives the navigator’s five properties and then observes that these properties coincide with those attributed to consciousness in the scientific literature. Whether to apply the label is a terminological choice. The formal result (the navigator is necessary) stands regardless of what one calls it.

Objection 11: Simple systems (thermostats, bacteria) satisfy the axioms. Are they conscious?

They possess simple navigators. A thermostat has two “models” (too hot / too cold) and switches between them. A bacterium has chemotactic models and switches gradient-following strategies. Whether to call this “consciousness” is a matter of definition. The monograph provides a continuous spectrum of navigator complexity, not a binary threshold.

Objection 12: The hard problem of consciousness is not addressed.

Correct. The monograph derives functional properties and makes no claim about phenomenal experience. Whether there is “something it is like” to be a navigator is a question the formalism does not answer. This is acknowledged explicitly.

4.5 Meta-Objections

Objection 13: The monograph is too ambitious. No single work can address physics, biology, cognitive science, and philosophy.

The monograph addresses a single question—the constraints on internal observers—using a single method: formal proof from stated axioms. The breadth of application follows from the generality of the axioms, not from any attempt to cover multiple fields. The examples from physics, biology, and cognitive science are illustrations, not evidence; the proofs stand independently of them.

Objection 14: The self-referential structure (an observer theorising about observer limits) creates a paradox.

It does not. Article I proves that observers have blind spots; it does not claim to identify them. The theorems describe the *existence* of inaccessible content, not its *specific content*. The author, as an internal observer, cannot see everything—but can prove that not-everything-can-be-seen. This is no more paradoxical than Gödel proving, from within arithmetic, that arithmetic is incomplete.

5 Glossary

This glossary provides simple explanations of 200 terms used in the monograph, ordered alphabetically. Each entry is written to be understandable without specialised training.

1. **Action Imperative.** The axiom that an agent must do something at each moment; it cannot wait forever.
2. **Action space.** The set of all possible actions an agent can take.
3. **Adaptive navigation.** The process of surviving by switching between different models of reality as the situation changes.
4. **Adaptive Observer Theory.** The name of this monograph’s overall framework: the formal study of how observers embedded in complex systems cope with incompleteness and incompatibility.
5. **Agent.** A subsystem that receives information, chooses a model, and acts. Can be a bacterium, a brain, or a thermostat.
6. **Algorithmic complexity.** See *Kolmogorov complexity*.
7. **Antithesis.** A genuine attempt to disprove a thesis. Used in the monograph as a method of self-testing.
8. **Approximate model.** A model that sacrifices precision for computational speed. See *coarse-grained model*.
9. **Autopoiesis.** A system that produces and maintains itself. Term from Maturana and Varela.
10. **Axiom.** A starting assumption that is not proved but stated explicitly. All theorems depend on axioms.
11. **Bijection.** A one-to-one correspondence between two sets where every element in each set is paired with exactly one element in the other.
12. **Blind spot.** An aspect of a system that an internal observer structurally cannot access.
13. **Bohr complementarity.** Niels Bohr’s principle that quantum objects have properties (like wave and particle) that cannot be observed simultaneously.
14. **Bounded rationality.** Herbert Simon’s

- idea that real agents make decisions with limited information and limited computation.
15. **Budget (computational).** The maximum amount of computation an agent can perform before it must act.
 16. **Cardinality.** The “size” of a set—how many elements it contains.
 17. **Category.** In mathematics, a collection of objects and arrows (morphisms) between them, satisfying composition rules.
 18. **Category error.** Applying a concept to a domain where it does not belong. The monograph argues that seeking a “Theory of Everything” may be a category error.
 19. **Cauchy sequence.** A sequence whose elements get closer and closer together, used to define completeness in analysis.
 20. **Causal structure.** The pattern of cause-and-effect relationships within a system.
 21. **Coarse-grained model.** A simplified model that ignores some details to reduce computational cost. Thermodynamics is a coarse-grained model of statistical mechanics.
 22. **Coarse-graining.** The process of simplifying a description by grouping fine details into larger categories.
 23. **Cohomology.** A mathematical tool for measuring obstructions to gluing local structures into global ones.
 24. **Colimit.** In category theory, a universal object that “contains” a collection of objects in a consistent way. The monograph proves that no colimit exists for the category of descriptions.
 25. **Compatibility.** Two descriptions are compatible if they can coexist in a single consistent framework without contradiction.
 26. **Complementarity.** See *Bohr complementarity*.
 27. **Completeness (descriptive).** The requirement that a description must say something about every observable in its domain.
 28. **Complexity.** A measure of how much structure a system contains. Defined in the monograph by four parallel metrics.
 29. **Complexity threshold.** The level of complexity above which incompatible descriptions necessarily exist.
 30. **Compression.** Expressing a large amount of information using a shorter description. A theory compresses data.
 31. **Compression Imperative.** The axiom that a description must be shorter than a brute-force list of all facts.
 32. **Compression ratio.** The number of observables explained divided by the number of primitives needed.
 33. **Compression scheme.** A specific choice of which observables to treat as fundamental (primitive) and which to derive from them.
 34. **Computational Boundedness.** The axiom that an agent has finite computational resources.
 35. **Computational cost.** The amount of computation required to generate a prediction from a model.
 36. **Conditional result.** A theorem that holds only when its axioms are satisfied. All theorems in this monograph are conditional.
 37. **Consistency.** A description is consistent if it does not derive both a statement and its negation.
 38. **Consciousness.** In everyday language, subjective awareness. In this monograph, a label for the navigator structure. Not used as an explanatory concept.
 39. **Context.** The current situation facing an agent, characterised by which domain the agent is in.

40. **Context Multiplicity.** The axiom that an agent encounters situations from multiple incompatible domains.
41. **Contextual evaluation.** The navigator's ability to assess which domain the current situation belongs to.
42. **Contradiction.** A situation where a statement and its negation are both derived. Incompatible descriptions contradict each other on shared observables.
43. **Correspondence principle.** The idea that a new theory should reproduce the predictions of the old theory in the old theory's domain.
44. **Cost function.** A mathematical expression measuring the total resources used by an agent at each step.
45. **Counterexample.** A specific example that disproves a general claim.
46. **Decoherence.** The process by which quantum systems lose their quantum properties through interaction with the environment.
47. **Deductive closure.** All statements that can be derived from a set of axioms.
48. **Degrees of freedom.** The number of independent parameters needed to specify a system's state.
49. **Description.** A formal theory (language + axioms + rules of inference) that makes predictions about a domain.
50. **Descriptive diversity.** The number of genuinely different descriptions a system requires, denoted $\delta(\mathcal{S})$.
51. **Determinism.** The property that a system's future state is uniquely determined by its present state.
52. **Diagonal argument.** A proof technique (due to Cantor) showing that a list cannot contain all items of a certain type.
53. **Domain.** A region of a system—defined by a state space, dynamics, and stable observables—for which a specific description is valid.
54. **Domain overlap.** The region where two domains share observables, creating the arena for potential contradiction.
55. **Dual-process theory.** Kahneman's framework distinguishing fast/automatic (System 1) and slow/deliberative (System 2) thinking.
56. **Dynamics.** The rule governing how a system's state changes over time.
57. **Effective complexity.** A measure of a system's non-random regularities, as distinct from its total information content.
58. **Effective field theory (EFT).** A framework in physics where different theories apply at different energy scales, connected by the renormalisation group.
59. **Eliminativism.** The philosophical position that some concept (e.g. "consciousness") does not refer to a real entity.
60. **Entropy.** A measure of disorder or uncertainty in a system.
61. **Epistemology.** The study of knowledge: what we can know and how.
62. **Error penalty.** The cost to an agent of using the wrong model in a given context.
63. **Error rate.** The probability that an agent selects the wrong model for the current context.
64. **Exact model.** A model that exploits all available structure without simplification, producing maximally precise predictions.
65. **Expedition format.** The monograph's method: thesis (prove a claim), antithesis (try to disprove it), undecidability (examine the boundaries).
66. **Ex falso quodlibet.** From a contradiction, anything follows. An inconsistent theory is useless.

67. **Falsifiability.** A theory is falsifiable if there exists an observation that could prove it wrong.
68. **Fixed-point theorem.** A mathematical result stating that a certain function must have a point that maps to itself.
69. **Formal language.** A precisely defined set of symbols and rules for constructing sentences.
70. **Free Energy Principle.** Karl Friston’s framework in which agents minimise surprise by updating their models of the world.
71. **Functional independence.** Two observables are functionally independent if knowing one does not determine the other.
72. **Functionalism.** The philosophical view that mental states are defined by their functional role, not their physical substrate.
73. **Genericity.** A property is generic if it holds for “almost all” cases, with exceptions forming a negligibly small set.
74. **Global description.** A single description valid for an entire system. The monograph proves this cannot exist for sufficiently complex systems.
75. **Gödel incompleteness.** Kurt Gödel’s 1931 theorem that any sufficiently powerful consistent formal system contains statements it cannot prove or disprove.
76. **Hard problem of consciousness.** David Chalmers’ term for the question of why physical processes give rise to subjective experience.
77. **Heuristic.** A practical rule of thumb that gives good-enough answers quickly, without guaranteeing optimality.
78. **Hidden structure.** Structure in a system that exists but is inaccessible to internal observers (Article I).
79. **Hilbert space.** A mathematical space used in quantum mechanics to represent the states of a quantum system.
80. **Homeomorphism.** A continuous one-to-one mapping between topological spaces that preserves structure.
81. **Homunculus.** A “little person inside the head” who supposedly does the observing. A fallacy: it creates an infinite regress.
82. **Inaccessibility.** The property that certain aspects of a system cannot be observed by any internal observer.
83. **Inaccessibility gap.** The quantitative difference between total system content and observer-accessible content.
84. **Inaccessibility Theorem.** Article I’s central result: proper-subsystem observers necessarily face structural blind spots.
85. **Incompatibility.** Two descriptions are incompatible if they derive contradictory statements about a shared observable.
86. **Incompatibility degree.** A measure of how many shared observables have contradictory primitive-derived status across two descriptions.
87. **Incompatibility Theorem.** Article II’s central result: sufficiently complex systems necessarily admit incompatible descriptions.
88. **Incompleteness.** The property that a description or observer cannot capture all aspects of a system.
89. **Independence (of axioms).** A set of axioms is independent if no axiom can be derived from the others.
90. **Incommensurability.** Thomas Kuhn’s idea that competing scientific paradigms cannot be directly compared because they use different concepts.
91. **Information-theoretic argument.** A proof based on the mathematics of information (bits, compression, entropy).

92. **Input map.** The function that translates the system's state into signals that the agent receives.
93. **Input signal.** The information an agent receives from the system at each time step.
94. **Integrated Information Theory (IIT).** Giulio Tononi's theory identifying consciousness with integrated information (Φ).
95. **Internal observer.** An observer that is part of (a subsystem of) the system it observes.
96. **Interpretation (theory).** A mapping that translates the theorems of one theory into another, preserving truth.
97. **Irreversibility.** A process that cannot be reversed. Thermodynamic processes are irreversible; microscopic dynamics is reversible.
98. **Isomorphism.** A structure-preserving bijection between two mathematical objects.
99. **Kolmogorov complexity.** The length of the shortest computer program that produces a given output. Measures intrinsic information content.
100. **Kuhnian incommensurability.** See *incommensurability*.
101. **Lawvere fixed-point theorem.** A categorical generalisation of diagonal arguments, used in Article I.
102. **Limit (categorical).** See *colimit*.
103. **Local description.** A description valid within a single domain.
104. **Logical depth.** The computational time required to produce a system's description from a minimal program.
105. **Logical orthogonality.** Two axiom systems are orthogonal if they derive contradictory consequences about a shared domain.
106. **Long-term viability.** The probability of an agent surviving indefinitely: $V_\infty = \lim_{T \rightarrow \infty} V(A, T)$.
107. **Measurement problem.** The question of how quantum mechanics transitions from superposition to definite outcomes.
108. **Memory (transition).** The navigator's record of past model choices and their outcomes.
109. **Metacognition.** Thinking about one's own thinking. Corresponds to the navigator's self-monitoring property.
110. **Model monism.** The strategy of using a single model for all situations. The monograph proves this is lethal.
111. **Model pluralism.** Maintaining multiple models and switching between them. The monograph proves this is necessary.
112. **Model repertoire.** The collection of working models available to an agent.
113. **Model-theoretic argument.** A proof using the mathematical theory of models (structures satisfying axioms).
114. **Morphism.** An arrow between objects in a category, representing a structure-preserving map.
115. **Navigator.** The formal structure—derived in Article III—that any viable agent must possess. Has five properties: model storage, contextual evaluation, switching, memory, self-monitoring.
116. **Navigator Theorem.** Article III's central synthesis: the navigator is a mathematical necessity for viable agents.
117. **Non-Degeneracy.** The axiom that a system requires at least two genuinely different descriptions.
118. **Non-stationarity.** An environment whose statistical properties change over time.
119. **Observable.** A quantity that can be measured or computed from a system's state.

120. **Ontological.** Relating to what exists, as opposed to what is known or described.
121. **Ontology.** The study of what exists.
122. **Optimal coarse-graining.** The best level of simplification for a given computational budget. Neither too fine (unaffordable) nor too coarse (inaccurate).
123. **Optimal Coarse-Graining Theorem.** Article III, Theorem A: multiple simplified models outperform any single model.
124. **Orthogonality.** See *logical orthogonality*.
125. **Output map.** The function that produces the agent's actions from its inputs and model.
126. **Overlap region.** The part of a system where two domains share observables and descriptions can potentially conflict.
127. **Paradigm shift.** Thomas Kuhn's term for a revolutionary change in the conceptual framework of a scientific field.
128. **Paraconsistent logic.** A logical system that tolerates contradictions without collapsing into triviality.
129. **Parametric family.** A set of theories that differ only in the values of continuous parameters (e.g. coupling constants).
130. **Phase space.** The space of all possible states of a system, combining positions and momenta.
131. **Phenomenal experience.** Subjective, first-person experience. The monograph treats this as a label, not an explanation.
132. **Pragmatic synthesis.** Article III: the constructive response to incompleteness and incompatibility.
133. **Predictive accuracy.** How closely a model's predictions match observed values.
134. **Predictive trade-off.** The theorem that unifying incompatible descriptions necessarily sacrifices accuracy in at least one domain.
135. **Presheaf.** A mathematical structure assigning data to each part of a space, with consistency rules.
136. **Primitive observable.** An observable treated as fundamental (not derived from others) in a given compression scheme.
137. **Proper subsystem.** A part of a system that is strictly smaller than the whole. All physical observers are proper subsystems.
138. **Qualia.** Philosophical term for the subjective qualities of experience (e.g. the "redness" of red).
139. **Quantum complementarity.** See *Bohr complementarity*.
140. **Quantum tunnelling.** A quantum particle passing through a barrier that classical physics says it cannot cross.
141. **Reductionism.** The view that complex phenomena can be fully explained by their simpler components.
142. **Renormalisation group (RG).** A mathematical framework for connecting theories at different scales in physics.
143. **Representation map.** The function translating system states into the observer's internal representations.
144. **Reversibility.** A process is reversible if it can be run backward. Microscopic dynamics is reversible; thermodynamics is not.
145. **Safe region.** The set of system states in which the agent continues to function.
146. **Scale.** A level of description characterised by a particular resolution or energy range.
147. **Scale Coupling.** The axiom that domains overlap—they share at least one observable.
148. **Scale-specific principles.** Laws that apply at one scale but not others. See *Law of Scale-Specific Principles*.
149. **Selection policy.** The agent's rule for choosing which model to use at each moment.

150. **Self-critique.** The monograph’s practice of identifying and addressing weaknesses in its own arguments.
151. **Self-monitoring.** The navigator’s feed-back loop tracking its own switching performance.
152. **Self-referential.** A structure that refers to itself. The navigator monitors itself; the monograph analyses its own limitations.
153. **Sentence.** A well-formed expression in a formal language that is either true or false.
154. **Sheaf.** A presheaf satisfying a gluing condition: local data that agree on overlaps can be assembled into global data.
155. **Simon, Herbert.** Nobel laureate who developed bounded rationality theory.
156. **Stable observable.** An observable whose value does not change too rapidly under the system’s dynamics.
157. **State space.** The set of all possible states a system can be in.
158. **Stochastic.** Involving randomness or probability.
159. **Strategy space.** The set of all viable agent configurations in a given system.
160. **Structural commitment.** A claim a theory makes about the relationships between observables (e.g. “temperature determines pressure”).
161. **Surjection.** A function that maps onto every element of its target set.
162. **Switching.** The act of changing from one working model to another.
163. **Switching cost.** The computational resources required to change models.
164. **Switching Necessity Theorem.** Article III, Theorem B: agents that do not switch have zero long-term viability.
165. **Synthesis.** The combination of thesis and antithesis results into a unified understanding.
166. **System.** A physical or abstract structure with a state space, dynamics, and observables.
167. **System 1.** In Kahneman’s framework, fast, automatic, heuristic thinking.
168. **System 2.** In Kahneman’s framework, slow, deliberative, analytical thinking.
169. **Theory interpretation.** A systematic translation of one theory’s language and theorems into another’s.
170. **Theory of Everything.** The hypothetical single theory unifying all physics. The monograph argues this is a category error for sufficiently complex systems.
171. **Thermodynamics.** The physics of heat, energy, and entropy at macroscopic scales.
172. **Thesis.** A claim to be proved. In the expedition format, the thesis is then attacked by the antithesis.
173. **Time-reversibility.** The property that a physical law works the same forward and backward in time.
174. **Topological obstruction.** A mathematical reason why local structures cannot be assembled into a global one.
175. **Topology.** The mathematical study of properties preserved under continuous deformation.
176. **Total cost.** The sum of model cost and switching cost at each time step.
177. **Transition memory.** See *memory*.
178. **Transparency.** A system is transparent if a single description captures all its structure. The monograph proves that sufficiently complex systems are not transparent.
179. **Trichotomy theorem.** Article I’s synthesis: inaccessibility is necessary (proper subsystem observer), absent (trivial system), or independent (theory doesn’t specify).

180. **Tripartite viability constraint.** The monograph’s central conclusion: persistent complex systems must maintain uncertainty, tolerate contradiction, and navigate adaptively.
181. **Trivial system.** A system simple enough to be fully described by a single model.
182. **Undecidability.** A statement is undecidable if the axioms neither prove nor refute it.
183. **Unified description.** A single consistent description valid across all domains. The monograph proves this cannot exist above the complexity threshold.
184. **Unification.** The project of combining separate theories into one.
185. **Unification penalty.** The loss of predictive accuracy caused by forcing incompatible descriptions into a single framework.
186. **Universal Turing machine.** A theoretical computer that can simulate any other computer, used to define Kolmogorov complexity.
187. **Validity.** A description is valid if it is consistent and all its predictions match observations.
188. **Viable region.** See *safe region*.
189. **Viability.** The probability that an agent remains functional over time.
190. **Viability theory.** The mathematical study of which states allow a system to survive, due to Jean-Pierre Aubin.
191. **Vital force.** A hypothetical special substance once thought necessary for life. Now understood to be unnecessary: “life” is a label for biochemical processes.
192. **Wave-particle duality.** The quantum phenomenon where entities exhibit both wave and particle properties depending on the experimental context.
193. **Working model.** A description chosen by an agent for use in a specific context. Not required to be globally true—only locally valid and computationally affordable.
194. **Zombie (philosophical).** A hypothetical being functionally identical to a conscious person but lacking subjective experience. The monograph argues this concept relies on an undefined term.
195. **Bell’s theorem.** John Bell’s 1964 result showing that no local hidden-variable theory can reproduce all predictions of quantum mechanics.
196. **Evolutionary Advantage Theorem.** Article III, Theorem C: systems with incompatible descriptions admit a richer space of viable strategies than transparent systems.
197. **Feedback loop.** A process where the output is fed back as input, enabling self-correction. The navigator’s self-monitoring is a feedback loop.
198. **Measure zero.** A set so small that it occupies no “volume” in its surrounding space. Compatible description pairs have measure zero.
199. **Non-isomorphic.** Two structures are non-isomorphic if no structure-preserving bijection exists between them. Non-isomorphic descriptions are genuinely different.
200. **Ergodic.** A process is ergodic if its time average equals its ensemble average. Ensures that all domains are eventually visited.

Must Any Consistent Physics Contain Structure Inaccessible to All Internal Observers?

A Gödel-Type Investigation of Whether Self-Consistent Physical
Systems
Necessarily Harbour Ontological Content Invisible from Within

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Abstract

We investigate whether every self-consistent physical system necessarily contains ontological content that is inaccessible to all observers situated within the system itself. Rather than adopting any single existing framework—be it Gödelian metamathematics, quantum-gravitational holography, autopoietic systems theory, or topos-theoretic physics—we develop an independent formal apparatus built from three primitive notions: *physical system*, *internal observer*, and *ontological content*. Consistency is defined as a conjunction of four conditions (logical, dynamical, informational, and self-referential), each of which is precisely characterised and shown to carry distinct formal consequences.

The mathematical core of the inaccessibility thesis rests on a structural fact: a proper part cannot biject onto the whole. Our contribution is to formalise this intuition rigorously, to explore its ramifications across multiple mathematical frameworks, to identify the precise boundary conditions under which it holds and fails, and—crucially—to distinguish the elementary cardinality obstruction from genuinely deeper phenomena that emerge at the self-referential and dynamical levels.

We pursue the investigation as a genuine mathematical expedition with no predetermined conclusion. First, we attempt to establish the inaccessibility thesis via five methods: a diagonal self-reference argument, an information-theoretic cardinality argument, a categorical fixed-point argument, a dynamical observational-equivalence argument, and a topological measure-theoretic argument. These methods share a common root in the non-injectivity of the representation map, but each reveals distinct structural consequences—from quantitative

bounds on the accessibility ratio, through temporal persistence of equivalence classes, to generic inaccessibility in the measure-theoretic sense. We are explicit about which results are elementary reformulations of cardinality constraints and which constitute genuinely independent insights.

Second, we subject this thesis to sustained attack by attempting to prove its negation through four strategies: the trivial-system construction, a holographic boundary-access argument, a collective-observer covering argument, and a systematic relaxation of each consistency condition. Third, we examine the possibility that the question is itself undecidable from within, employing self-referential and model-theoretic independence arguments.

To ground the abstract framework in concrete settings, we instantiate the formalism in three toy models—a cellular automaton with an embedded observer agent, a coupled oscillator system, and a miniature formal system—and additionally develop a quantum-mechanical instantiation using projection-valued measures as the physically natural property class. We investigate whether the inaccessibility thesis survives restriction from arbitrary subsets to physically motivated property classes (Borel-measurable sets, projection operators, macroscopic observables) and find that it does, though the quantitative gap depends on the property class.

Every definition, proposition, and theorem is preceded and followed by thorough explanatory discussion to ensure accessibility to readers whose primary training lies outside dynamical systems and mathematical logic. All computational code is provided in the appendices.

The paper reports what the mathematics yields, with minimal philosophical extrapolation beyond the formal results.

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1 Introduction

1.1 The Central Question

Consider a physical system—any physical system—that is self-consistent in the sense that its description harbours no internal contradictions. Now suppose an observer exists *within* that system: not looking at it from some external vantage point, but constituted by the system’s own degrees of freedom, interacting with the rest of the system through the system’s own laws.

The question this paper investigates is:

*Must such a system contain facts about itself that no internal observer,
however capable, can ever access?*

This is not a question about practical limitations—about finite memory, thermal noise, or the particular structure of human cognition. It is a question about *structural necessity*: whether the very conditions that make a physical system self-consistent force the existence of ontological content that is invisible from within.

The question has obvious ancestors. Gödel’s incompleteness theorems [1] established that any sufficiently powerful consistent formal system contains true statements it cannot prove. Turing’s halting theorem [2] showed that no algorithm can decide an arbitrary algorithm’s behaviour. In physics, the development of black hole complementarity [3], the holographic principle [4, 5], and recent no-go results concerning observer-dependent facts in quantum mechanics [6, 7] all point toward fundamental limits on what observers can know from within their physical setting. The programme of quantum Darwinism [8, 9] and decoherence theory provides perhaps the most developed existing framework for understanding how internal observers acquire information about their environment and which properties become “objective” (accessible to multiple observers). Wolpert’s results on physical limits of inference [10] similarly address the question from an information-theoretic perspective. Most directly relevant is Breuer’s proof [11] that no quantum subsystem can perform a complete state self-measurement—a result that establishes, within the specific framework of quantum mechanics, the impossibility that this paper investigates in a theory-independent setting. Svozil [12] has explored related themes of intrinsic indeterminacy and undecidability in physical systems.

Yet none of these results, individually or collectively, settles the question as posed. Gödel’s theorems apply to formal systems of a particular kind, not to arbitrary physical systems. The physical results depend on specific physical theories (quantum mechanics, general relativity) and specific notions of observation. Quantum Darwinism identifies *which* properties become classically accessible but does not address the general structural question of whether inaccessible content must exist. What is missing is a framework that is simultaneously:

1. **General**—applicable to any self-consistent physical system, not merely those described by current theories;

2. **Precise**—with rigorous definitions and complete proofs, not analogies or heuristic arguments;
3. **Self-critical**—genuinely testing the claim against its own negation, rather than constructing a one-sided case.

This paper attempts to supply such a framework. We should state at the outset what the mathematical core of the investigation reveals: the inaccessibility thesis rests, at its root, on the structural fact that a proper part cannot biject onto the whole. Our contribution is not the discovery of this fact but its rigorous formalisation in a theory-independent setting, the exploration of its consequences across multiple mathematical frameworks, the precise identification of boundary conditions (the trichotomy theorem), and the demonstration that the result persists for physically motivated property classes. We are explicit throughout about which results are elementary reformulations of cardinality constraints and which constitute genuinely independent insights.

1.2 What This Paper Does and Does Not Do

We develop an independent formal apparatus built from three primitive notions—*physical system*, *internal observer*, and *ontological content*—and a four-part definition of *consistency* (logical, dynamical, informational, and self-referential). From these definitions alone, without postulating axioms, we derive all subsequent results.

The investigation proceeds as a mathematical expedition with no predetermined conclusion. We pursue three lines of inquiry in sequence:

1. **The Inaccessibility Thesis.** We attempt to prove, by five independent methods, that any consistent physical system containing an internal observer necessarily harbours ontological content inaccessible to that observer.
2. **The Accessibility Antithesis.** We then attempt, with equal seriousness, to prove the negation: that there exist consistent physical systems in which internal observers can access the totality of ontological content.
3. **The Undecidability Possibility.** Finally, we investigate whether the question might be undecidable—neither provable nor refutable within the framework—which would itself constitute a result of considerable depth, given the paper’s subject matter.

The paper reports what the mathematics yields, with minimal extrapolation beyond the formal results. Where observations venture into interpretive territory, they are clearly flagged as such. Readers from all disciplines are welcome to draw their own conclusions; the paper’s task is to make the formal landscape clear enough to support such reflection.

1.3 Relation to Existing Work

Four intellectual traditions bear on the present question. We engage each seriously but develop a framework independent of all four.

Gödelian metamathematics and its extensions. The incompleteness theorems [1] and their extensions by Rosser [13], Löb [14], and others establish that sufficiently expressive consistent formal systems are necessarily incomplete. Lucas [15] and Penrose [16, 17] have drawn conclusions about the nature of mind from these results. Our work differs in that we do not assume the physical system is a formal system of arithmetic, nor do we import conclusions about consciousness. Our diagonal argument (Section 4) is a construction within our own framework, not an application of Gödel’s theorems.

Observer-dependent limits in physics. Bohr’s complementarity principle, the Heisenberg uncertainty relations, black hole complementarity [3], the holographic bound [4, 5], and recent no-go theorems on observer-independent facts [6, 7] all identify specific physical mechanisms that limit internal observation. Our framework is not tied to any specific physical theory: the dynamical and topological arguments (Sections 7 and 8) recover *structural* analogues of these results from the definitions alone, without invoking quantum mechanics or general relativity.

Self-reference in cybernetics and systems theory. The traditions of von Foerster [18], Maturana and Varela [19], Luhmann [20], and more recently Friston [21] address how self-referential systems create and maintain internal boundaries that limit self-observation. Our treatment of self-referential consistency (Section 3) formalises a version of these ideas without adopting the biological or sociological vocabulary.

Topos-theoretic and categorical approaches. Döring and Isham [22, 23], and subsequent work in categorical quantum mechanics [24, 25], reformulate physical theories within topoi where truth values are contextual. Our fixed-point argument (Section 6) employs category-theoretic tools—specifically Lawvere’s fixed-point theorem [26]—but the framework itself is not topos-theoretic.

The present work charts a fifth path. It shares concerns with all four traditions but derives its results from its own definitions.

1.4 Guide for the Reader

The paper is constructed so that each section builds upon the previous ones, with definitions introduced precisely when they are needed by the next proof. The structure is as follows.

Section 2 introduces the three primitive notions: physical system, internal observer, and ontological content. Section 3 develops the four-part definition of consistency. The inaccessibility thesis is then pursued through five proof methods in Sections 4 to 8. The accessibility antithesis is examined in Sections 9 to 12. The undecidability question is treated in Section 13. Section 14 synthesises the results, and Section 15 offers concluding remarks. Computational code for all toy models and figures appears in Appendix A. A placeholder for peer review and revision history is provided in Appendix B.

Each major step is accompanied by a boxed summary explaining its strategic purpose within the overall argument, and by a self-critique subsection that identifies po-

tential weaknesses and either resolves them or flags them as open problems. Theorems are proved in full; where multiple proof methods are available, we present more than one.

Strategic Purpose

The introduction establishes three things. First, it poses the question in terms accessible to any scientifically literate reader. Second, it positions the paper relative to existing traditions without depending on any of them. Third, it sets the methodological ground rules: complete proofs, genuine self-critique, and no predetermined conclusion. This framing is necessary because the question sits at a junction of disciplines that rarely share a common formal language, and the paper must earn the trust of readers from each.

1.5 Self-Critique and Resolution

Critique 1: Overambition. The paper claims to investigate *all* consistent physical systems with a single framework. This is an extraordinarily broad claim. Can any finite formal apparatus truly capture such generality?

Resolution. The generality is achieved by minimality: the framework assumes very little about what a physical system is (a state space with dynamics) or what an observer is (a distinguished subsystem with a representation map). The less one assumes, the more broadly the results apply—but the less specific the conclusions. We are explicit throughout about what holds under the minimal definitions and what requires additional structure.

Critique 2: The “fifth path” framing. Claiming independence from four major traditions risks appearing either naïve or arrogant. The traditions exist for good reasons, and the paper’s results must ultimately be relatable to them.

Resolution. Independence does not mean ignorance. Each proof method has a clear intellectual ancestor in one of the four traditions, and we identify these connections explicitly. The claim is that the formal definitions are self-standing—not that the ideas emerged in a vacuum.

Critique 3: No predetermined conclusion. If the paper genuinely does not know where it is going, how can it be structured coherently?

Resolution. The structure is that of a mathematical investigation, not a rhetorical argument. The thesis and antithesis are pursued with equal rigour, and the synthesis reports the outcome. This is standard in exploratory mathematics (compare, for example, the investigation of the Continuum Hypothesis before Cohen’s forcing results). The

paper’s coherence comes from the consistency of its definitions and the completeness of its proof attempts, not from the direction of its conclusions.

2 Primitive Notions

This section introduces the three foundational concepts upon which the entire paper rests: *physical system*, *internal observer*, and *ontological content*. Each is defined with the minimum structure necessary for the subsequent proofs, and each definition is accompanied by discussion of what it captures and what it deliberately leaves open.

We do not postulate axioms. Instead, we define objects and derive their properties. If a theorem requires an additional condition, that condition is stated as a hypothesis of the theorem, not as a standing assumption.

2.1 Physical Systems

We begin with the most fundamental notion: a system that can be in various states and that evolves according to some rule.

Definition 2.1 (Physical System). A *physical system* is a triple $\mathcal{S} = (\Sigma, \mathcal{D}, t)$ where:

1. Σ is a non-empty set, called the *state space*;
2. t is a non-empty totally ordered commutative semigroup (with operation $+$ and order $\leq$), called the *time structure*;
3. $\mathcal{D} : \Sigma \times t \rightarrow \Sigma$ is a map, called the *dynamics*, satisfying the semigroup property: $\mathcal{D}(\mathcal{D}(\sigma, \tau_1), \tau_2) = \mathcal{D}(\sigma, \tau_1 + \tau_2)$ for all $\sigma \in \Sigma$ and $\tau_1, \tau_2 \in t$.

To appreciate this definition: the state space Σ gathers all the configurations the system could possibly occupy. It might be a finite set of configurations (as in a cellular automaton), a smooth manifold (as in Hamiltonian mechanics), a Hilbert space (as in quantum mechanics), or any other mathematical structure. The definition does not presuppose a particular type of physics.

The dynamics $\mathcal{D}$ encodes how the system changes. Given a state and an elapsed time, $\mathcal{D}$ produces the resulting state. The semigroup property ensures that evolution is self-consistent: evolving for time τ_1 and then for time τ_2 yields the same result as evolving for time $\tau_1 + \tau_2$. This property is what prevents the dynamics from contradicting itself across different time intervals.

The time structure t is required to be an ordered commutative semigroup—it must support an addition operation (to compose durations) and a total order (to sequence events). Standard examples include $\mathbb{N}$ (discrete time), $\mathbb{Z}_{\geq 0}$ (non-negative integer time), and $\mathbb{R}_{\geq 0}$ (continuous time). The algebraic requirement is not redundant: a mere total order would not guarantee that the semigroup property in condition (3) is well-formed, since $\tau_1 + \tau_2$ must be defined.

Remark 2.2 (Limitation: Total Ordering of Time). The requirement that t be totally ordered excludes *partially ordered* time structures, which arise naturally in relativistic physics where spacelike-separated events have no canonical temporal ordering. In a relativistic setting, the appropriate generalisation would replace the total order with a causal partial order (as in the causal set programme or the framework of globally hyperbolic spacetimes). The present framework therefore applies, strictly speaking, to systems with a global time parameter—which includes non-relativistic quantum mechanics, classical mechanics, and discrete-time systems—but not directly to generally covariant field theories. Extending the definitions to partially ordered time is a natural direction for future work and would require replacing the semigroup property with a more general composition law indexed by causal paths.

Definition 2.3 (Subsystem). Given a physical system $\mathcal{S} = (\Sigma, \mathcal{D}, t)$, a *subsystem* is a triple $\mathcal{S}' = (\Sigma', \mathcal{D}', t)$ together with a pair of maps:

1. An *embedding* $\iota : \Sigma' \rightarrow \Sigma$ that is injective;
2. A *projection* $\pi : \Sigma \rightarrow \Sigma'$;

satisfying $\pi \circ \iota = \text{id}_{\Sigma'}$ and the *compatibility condition*: for every $\sigma \in \Sigma$ and every $\tau \in t$,

$$\pi(\mathcal{D}(\sigma, \tau)) = \mathcal{D}'(\pi(\sigma), \tau).$$

A subsystem is a portion of the full system that can be meaningfully isolated. The embedding ι tells us how subsystem states sit inside the full state space. The projection π tells us how to extract the subsystem’s state from the full system’s state. The condition $\pi \circ \iota = \text{id}_{\Sigma'}$ ensures that embedding a subsystem state and then projecting it back recovers the original subsystem state. The compatibility condition ensures that the subsystem’s own dynamics agrees with the restriction of the full dynamics to the subsystem’s degrees of freedom.

Notice what this definition does *not* require: it does not demand that the subsystem is dynamically independent or isolated from the rest of the system. The projection π may lose information—the full state may contain features not captured by the subsystem’s state—and this is precisely the structure that will become important later.

Strategic Purpose

Defining physical systems as abstract triples $(\Sigma, \mathcal{D}, t)$ with a minimal semigroup property ensures that our results are not artifacts of a particular physical theory. Any system described by Newtonian mechanics, quantum mechanics, thermodynamics, cellular automata, or even exotic hypothetical dynamics is captured as a special case. The subsystem definition with its embedding–projection pair creates the mathematical language for asking: what does the subsystem “see” of the whole?

2.2 Internal Observers

An observer that exists *within* a system is a subsystem with additional structure: it has the capacity to form representations of the system's states.

Definition 2.4 (Internal Observer). Given a physical system $\mathcal{S} = (\Sigma, \mathcal{D}, t)$, an *internal observer* is a quadruple $\mathcal{O} = (\Sigma_{\mathcal{O}}, \mathcal{D}_{\mathcal{O}}, t, \mathcal{R})$ where:

1. $(\Sigma_{\mathcal{O}}, \mathcal{D}_{\mathcal{O}}, t)$ is a subsystem of $\mathcal{S}$ with embedding $\iota_{\mathcal{O}}$ and projection $\pi_{\mathcal{O}}$;
2. $\mathcal{R} : \Sigma \rightarrow \Sigma_{\mathcal{O}}$ is a map, called the *representation map*, satisfying:
 - (a) *Physicality*: There exists a fixed $\tau_0 \in t$ (the *measurement duration*) such that for every $\sigma \in \Sigma$, $\pi_{\mathcal{O}}(\mathcal{D}(\sigma, \tau_0)) = \mathcal{R}(\sigma)$. That is, the observer's representation of the system state is the result of letting the system evolve for a definite duration and reading off the observer's degrees of freedom. The duration τ_0 is uniform across all states.
 - (b) *Internality*: $\mathcal{R}$ factors through $\pi_{\mathcal{O}}$. That is, $\mathcal{R} = f \circ \pi_{\mathcal{O}}$ for some map $f : \Sigma_{\mathcal{O}} \rightarrow \Sigma_{\mathcal{O}}$. The observer can only use its own degrees of freedom to build representations.

The physicality condition deserves careful attention. The requirement that τ_0 be *uniform*—the same for all σ —is essential. Without it, the observer could in principle select a different measurement duration for each state, which would presuppose knowledge of the state prior to measurement. The uniform- τ_0 version captures the physical reality that a measurement protocol is fixed before the system's state is known.

The *internality* condition says that the observer's representations are constructed entirely from its own degrees of freedom. The observer cannot access information that does not pass through its own state space. Together, these two conditions impose a fundamental constraint: the map $\mathcal{R}$ cannot be arbitrary. It must be realisable by the system's own dynamics at a definite time, and it must factor through the observer's own (typically smaller) state space.

Definition 2.5 (Observational Equivalence). Given an internal observer $\mathcal{O}$ with representation map $\mathcal{R}$, two states $\sigma_1, \sigma_2 \in \Sigma$ are *observationally equivalent with respect to* $\mathcal{O}$, written $\sigma_1 \sim_{\mathcal{O}} \sigma_2$, if and only if $\mathcal{R}(\sigma_1) = \mathcal{R}(\sigma_2)$.

Observational equivalence is the relation that identifies states the observer cannot distinguish. Two states are observationally equivalent if the observer represents them identically. The equivalence classes under $\sim_{\mathcal{O}}$ partition the state space into groups of states that, from the observer's perspective, are indistinguishable.

This is a straightforward consequence of the definition: since $\mathcal{R}$ maps the full state space to the observer's (typically smaller) state space, it cannot in general be injective. Distinct system states may map to the same observer state, and such states are precisely those the observer cannot tell apart.

Definition 2.6 (Maximally Capable Observer). An internal observer $\mathcal{O}$ is *maximally capable* if, among all internal observers of $\mathcal{S}$, its representation map $\mathcal{R}_{\mathcal{O}}$ has the finest

equivalence relation. Formally, $\mathcal{O}$ is maximally capable if for every internal observer $\mathcal{O}'$ of $\mathcal{S}$, $\sigma_1 \sim_{\mathcal{O}} \sigma_2$ implies $\sigma_1 \sim_{\mathcal{O}'} \sigma_2$.

A maximally capable observer is one that can make the finest distinctions among system states that any internal observer can make. If even this observer cannot distinguish two states, then no internal observer can. This notion allows us to ask the strongest version of our question: does the maximally capable observer still have blind spots?

Strategic Purpose

The internal observer definition is the pivot of the entire paper. By requiring that the representation map be *physical* (arising from dynamics) and *internal* (factoring through the observer’s own state space), we ensure that any limitations we prove are not due to artificial restrictions on the observer’s capabilities, but follow from the structural fact of being embedded within the system one observes. The notion of observational equivalence provides the precise mathematical handle on “what the observer cannot see.”

2.3 Ontological Content

We need to make precise what “all the facts about the system” means, so that we can ask whether an observer can access all of them.

Definition 2.7 (Property). A *property* of a physical system $\mathcal{S} = (\Sigma, \mathcal{D}, t)$ is a subset $P \subseteq \Sigma$. A state σ *has property* P if $\sigma \in P$.

This is the most general possible notion of a property: any collection of states constitutes a property, namely the property of being in one of those states. The property “having energy greater than E_0 ” corresponds to the set of all states whose energy exceeds E_0 . The property “being at equilibrium” corresponds to the set of equilibrium states. Even structural, relational, or higher-order features of the system can be encoded as properties in this sense, since any criterion that either holds or fails for a given state defines a subset of Σ .

Remark 2.8 (Properties and Propositions). Throughout this paper, “property” (a subset $P \subseteq \Sigma$) and “fact” or “truth about the system” (a proposition that the system is in a state belonging to P) are used in close correspondence but are distinct. A property is a *type-level* object: a collection of states. A fact is a *token-level* assertion: that a *specific* state σ belongs to P . When we say “the observer cannot access property P ,” we mean that the observer cannot determine whether the system’s current state is in P —that is, the observer cannot evaluate the proposition “ $\sigma \in P$ ” for an unknown σ . This distinction matters in philosophy of physics, where “properties of a system” and “facts about a system” may carry different ontological commitments. Our framework is neutral on this point: the formal results hold regardless of one’s metaphysical interpretation of subsets.

Definition 2.9 (Ontological Content). The *ontological content* of a physical system $\mathcal{S} = (\Sigma, \mathcal{D}, t)$ is the power set $\mathfrak{O}(\mathcal{S}) = \mathcal{P}(\Sigma)$: the collection of all properties of $\mathcal{S}$.

The ontological content is the totality of facts that could, in principle, be true of the system. It is maximal by construction: every possible property is included.

Definition 2.10 (Accessible Content). Given an internal observer $\mathcal{O}$ with representation map $\mathcal{R}$, the *accessible content* is the set of properties that $\mathcal{O}$ can distinguish:

$$\mathfrak{A}(\mathcal{O}) = \{P \subseteq \Sigma \mid P \text{ is a union of equivalence classes of } \sim_{\mathcal{O}}\}.$$

Equivalently, $P \in \mathfrak{A}(\mathcal{O})$ if and only if $P = \mathcal{R}^{-1}(\mathcal{R}(P))$.

A property is accessible to the observer if and only if the observer can determine whether the system possesses that property. This happens precisely when the property respects observational equivalence: whenever a state has the property, all states observationally equivalent to it also have the property. If a property fails this condition—if it holds for some state σ_1 but not for some σ_2 with $\mathcal{R}(\sigma_1) = \mathcal{R}(\sigma_2)$ —then the observer cannot determine whether the system has this property, because it cannot distinguish σ_1 from σ_2 .

The accessible content $\mathfrak{A}(\mathcal{O})$ is a sub-Boolean-algebra of $\mathcal{P}(\Sigma)$: it is closed under union, intersection, and complement, and contains $\emptyset$ and Σ . This is immediate from the fact that unions of equivalence classes are closed under these operations.

Definition 2.11 (Inaccessible Content). The *inaccessible content with respect to $\mathcal{O}$* is the complement

$$\mathfrak{O}(\mathcal{S}) \setminus \mathfrak{A}(\mathcal{O}) = \{P \subseteq \Sigma \mid P \notin \mathfrak{A}(\mathcal{O})\}.$$

The *universally inaccessible content* is

$$\mathfrak{I}(\mathcal{S}) = \mathfrak{O}(\mathcal{S}) \setminus \bigcup_{\mathcal{O}} \mathfrak{A}(\mathcal{O}),$$

where the union ranges over all internal observers of $\mathcal{S}$.

The universally inaccessible content consists of properties that *no* internal observer of the system can determine. The central question of this paper can now be stated with full precision:

Central Question (Formal Version).

For which physical systems $\mathcal{S}$ satisfying the consistency conditions of [Section 3](#), is $\mathfrak{I}(\mathcal{S}) \neq \emptyset$? Must it always be non-empty?

Strategic Purpose

Ontological content is defined as the full power set $\mathcal{P}(\Sigma)$ because this represents the maximal collection of distinctions that could be drawn within the system. Accessible content is the sub-collection an observer can detect, determined entirely by the observer's representation map. The gap between the two—if it exists—is the inaccessible content. This setup transforms a philosophical question (“are there things we cannot know from within?”) into a precise mathematical question (“is $\mathcal{I}(\mathcal{S})$ non-empty?”), which can be attacked with formal methods.

2.4 Self-Critique and Resolution

Critique 1: The power set is too large. Defining ontological content as $\mathcal{P}(\Sigma)$ includes pathological, non-measurable, and physically meaningless subsets of Σ . A more reasonable definition might restrict to measurable sets, Borel sets, or clopen sets, depending on the topology of Σ .

Resolution. This is deliberate. By including all subsets, we make the inaccessibility thesis *easier* to prove and the accessibility antithesis *harder* to prove. If inaccessibility fails even with the maximal definition of ontological content, it certainly fails for any restricted notion. Conversely, if accessibility holds for the maximal definition, it holds a fortiori for any restriction. In [Section 8](#), we revisit this point with a more refined measure-theoretic formulation and show that the results survive restriction to physically natural classes of properties.

Critique 2: The representation map is deterministic. The definition requires $\mathcal{R}$ to be a function, producing a single observer state for each system state. Quantum mechanics suggests that measurement outcomes may be inherently stochastic.

Resolution. A stochastic representation can be modelled within this framework by enlarging the state space to include the random seed (or the full quantum state including environmental degrees of freedom). After this enlargement, the representation map is again deterministic on the enlarged state space. Alternatively, one can define $\mathcal{R}$ as a map to probability distributions over $\Sigma_{\mathcal{O}}$, i.e., $\mathcal{R} : \Sigma \rightarrow \mathcal{P}(\Sigma_{\mathcal{O}})$ or $\mathcal{R} : \Sigma \rightarrow \Delta(\Sigma_{\mathcal{O}})$. The essential conclusions of the paper survive either treatment, as we note at the relevant junctures.

Critique 3: The definition of “internal” is doing too much work. The internality condition ($\mathcal{R} = f \circ \pi_{\mathcal{O}}$) seems strong. Could an observer not use correlations with remote parts of the system without those correlations passing through its local state space?

Resolution. Any information the observer uses must, by the physicality condition, be encoded in the observer’s degrees of freedom at the moment of representation. If correlations with remote parts of the system exist, they must manifest in the observer’s local state (through prior dynamical coupling) in order to be used. This is not an assumption; it follows from the requirement that the observer is physical. An observer that could access information without it passing through its own state space would violate the dynamics—it would require a channel not described by $\mathcal{D}$.

3 The Four Consistency Conditions

The central question asks about *consistent* physics. This section develops what consistency means, decomposing it into four independent conditions. Each captures a distinct way in which a physical system can be free of internal contradiction. We show that these four conditions are logically independent—no one implies any other—and that their conjunction forms a natural and minimal notion of self-consistency.

3.1 Logical Consistency

The first condition ensures that the system’s description does not attribute contradictory properties to any state.

Definition 3.1 (Logical Consistency). A physical system $\mathcal{S} = (\Sigma, \mathcal{D}, t)$ is *logically consistent* if there exists no property $P \subseteq \Sigma$ and state $\sigma \in \Sigma$ such that the dynamics forces both $\sigma \in P$ and $\sigma \notin P$ simultaneously. Formally, for every $\sigma \in \Sigma$ and every $\tau_1, \tau_2 \in t$, if $\mathcal{D}(\sigma, \tau_1) \in P$ and $\mathcal{D}(\sigma, \tau_2) \in P^c$ with $\tau_1 = \tau_2$, then a contradiction obtains, and we say the system is logically inconsistent.

This condition, stated carefully, amounts to requiring that the dynamics $\mathcal{D}$ is well-defined as a *function*: each input (σ, τ) maps to exactly one output. A “system” in which the same state at the same time both has and lacks a property is not a system at all in any meaningful sense—it is a notational contradiction.

The condition appears almost trivially satisfied by any mathematical structure, since a function is by definition single-valued. Its importance becomes clear when we consider self-referential systems ([Section 3.4](#)), where the observer’s description of the system is itself part of the system, and logical contradictions can arise through self-referential loops.

3.2 Dynamical Consistency

The second condition ensures that the evolution law does not produce temporal paradoxes.

Definition 3.2 (Dynamical Consistency). A physical system $\mathcal{S} = (\Sigma, \mathcal{D}, t)$ is *dynamically consistent* if the dynamics $\mathcal{D}$ satisfies:

1. *Determinism*: For every $\sigma \in \Sigma$ and $\tau \in t$, there exists a unique $\sigma' = \mathcal{D}(\sigma, \tau)$. (This is already ensured by $\mathcal{D}$ being a function.)
2. *Semigroup property*: $\mathcal{D}(\mathcal{D}(\sigma, \tau_1), \tau_2) = \mathcal{D}(\sigma, \tau_1 + \tau_2)$ for all $\sigma \in \Sigma$ and $\tau_1, \tau_2 \in t$ for which the sum is defined.
3. *Self-interaction coherence*: If the system contains a subsystem $\mathcal{S}'$ that interacts with the remainder, the joint dynamics does not produce contradictory demands on any degree of freedom. That is, the restriction of $\mathcal{D}$ to $\mathcal{S}'$ and the dynamics induced on $\mathcal{S}'$ by the environment agree: $\pi(\mathcal{D}(\sigma, \tau)) = \mathcal{D}'(\pi(\sigma), \tau)$ for all σ and τ .

Dynamical consistency is what ensures the system's time evolution makes sense. The semigroup property prevents temporal paradoxes of the following kind: evolving from state A for one step gives state B ; evolving from B for one step gives state C ; but evolving from A for two steps gives state $D \neq C$. Such a system would not have coherent dynamics.

The self-interaction coherence condition is crucial for systems containing observers. An observer is a subsystem, and it interacts with the rest of the system. Dynamical consistency requires that the dynamics, when restricted to the observer's degrees of freedom, agrees with the observer's own internal dynamics. Without this, the observer could find that its own state evolution contradicts what the global dynamics prescribes.

3.3 Informational Consistency

The third condition concerns the preservation of distinguishability.

Definition 3.3 (Informational Consistency). A physical system $\mathcal{S} = (\Sigma, \mathcal{D}, t)$ is *informationally consistent* if the dynamics $\mathcal{D}(\cdot, \tau) : \Sigma \rightarrow \Sigma$ is injective for every $\tau \in t$. That is, distinct states remain distinct under time evolution:

$$\mathcal{D}(\sigma_1, \tau) = \mathcal{D}(\sigma_2, \tau) \implies \sigma_1 = \sigma_2.$$

Informational consistency means that the dynamics does not erase distinctions. If two states are different, they remain different after any amount of evolution. This is the abstract analogue of unitarity in quantum mechanics and of Liouville's theorem in classical Hamiltonian mechanics.

Why does this matter for our investigation? If the dynamics can merge distinct states, then information is destroyed, and the question of what an observer can access becomes tangled with the question of what information survives at all. Informational consistency ensures that all distinctions that exist at any time continue to exist at all times; the only question is whether the observer can detect them.

We note immediately that not all physical theories satisfy this condition. Thermodynamic coarse-graining, wavefunction collapse in some interpretations of quantum mechanics, and dissipative dynamics all violate informational consistency. This is precisely why the condition is stated as a hypothesis rather than an axiom: the theorems in later sections will explicitly track which results require it and which do not.

3.4 Self-Referential Consistency

The fourth and most subtle condition concerns what happens when the system contains a subsystem that models the whole.

Definition 3.4 (Self-Modelling Capacity). An internal observer $\mathcal{O}$ of a system $\mathcal{S}$ is *self-modelling* if the representation map $\mathcal{R}$ has the property that for some subset of states $\Sigma_{\text{model}} \subseteq \Sigma_{\mathcal{O}}$, there exists an interpretation map $\mathcal{I} : \Sigma_{\text{model}} \rightarrow \mathcal{P}(\Sigma)$ such that:

1. *Correctness*: For every state $\sigma \in \Sigma$, $\sigma \in \mathcal{I}(\mathcal{R}(\sigma))$. That is, the observer’s representation of the system state, when interpreted, yields a description that includes the actual state. The observer’s model of the world is not wrong about where the world is.
2. *Non-triviality*: $\mathcal{I}(s) \neq \Sigma$ for at least one $s \in \Sigma_{\text{model}}$. That is, the observer’s model is not vacuously correct by predicting everything. It must rule out at least some states for at least some representations.

The non-triviality condition (2) is essential. Without it, an observer that always “predicts” the system could be in any state whatsoever would vacuously satisfy correctness, since $\sigma \in \Sigma$ is always true. Such an observer carries no genuine model; it is a tautology machine. The non-triviality condition requires that the interpretation map is informative: for at least one observer state s , the model rules out some system states, asserting that $\sigma \notin \mathcal{I}(s)$ for certain σ .

A self-modelling observer thus maintains an internal model that correctly locates the system’s actual state among the possibilities the model recognises, and the model is genuinely constraining—it takes risks by excluding possibilities and is not contradicted by reality.

Definition 3.5 (Self-Referential Consistency). A physical system $\mathcal{S}$ containing a self-modelling observer $\mathcal{O}$ is *self-referentially consistent* if the following two conditions hold:

1. *Model–reality agreement*: $\sigma \in \mathcal{I}(\mathcal{R}(\sigma))$ for all $\sigma \in \Sigma$, as in [Definition 3.4](#).
2. *Model–dynamics agreement*: The observer’s model of the dynamics is compatible with the actual dynamics. If the observer predicts that σ will evolve to a state in some set $A \subseteq \Sigma$ after time τ , then indeed $\mathcal{D}(\sigma, \tau) \in A$.

Self-referential consistency prevents the system from containing an observer whose beliefs about the system systematically contradict the system’s actual behaviour. This is not a requirement that the observer be omniscient—the observer may have very coarse or imprecise models—but that whatever the observer does believe is not contradicted by the facts.

This condition is the most consequential for our investigation. It is precisely at the interface between what the observer models and what the system actually does that the tension between completeness and consistency arises. The diagonal argument of [Section 4](#) exploits this tension directly.

3.5 The Conjunction: Full Consistency

Definition 3.6 (Fully Consistent System). A physical system $\mathcal{S}$ is *fully consistent* if it satisfies all four conditions:

1. Logical consistency (Definition 3.1);
2. Dynamical consistency (Definition 3.2);
3. Informational consistency (Definition 3.3);
4. Self-referential consistency (Definition 3.5), for every self-modelling observer it contains.

Proposition 3.7 (Independence of Consistency Conditions). *The four consistency conditions are logically independent: no proper subset of the four conditions implies the remaining condition(s).*

Proof. We exhibit, for each condition, a system satisfying the other three but violating that one.

(i) *Logical consistency violated, others satisfied.* Consider $\Sigma = \{a, b\}$ with a “dynamics” defined by both $\mathcal{D}(a, 1) = a$ and $\mathcal{D}(a, 1) = b$ —that is, the rule assigns two outputs to the same input. To make this precise without invoking multi-valued functions, we work in the relational setting: let $D \subseteq (\Sigma \times t) \times \Sigma$ be a relation with $(a, 1, a) \in D$ and $(a, 1, b) \in D$, which fails to be a function. This violates logical consistency (the dynamics is not well-defined as a function). If we extend D to a fully specified relation satisfying the semigroup property in the relational sense (composing relations), and define information preservation as injectivity of each time-slice relation (which holds since both branches are injective on $\{a, b\}$), and take an observer that simply reports its own state (trivially self-referentially consistent), we see that the three non-logical conditions are satisfiable. The pathology is entirely in the failure of D to be a function.

(ii) *Dynamical consistency violated, others satisfied.* Consider $\Sigma = \{a, b, c\}$ with dynamics $\mathcal{D}(a, 1) = b$, $\mathcal{D}(b, 1) = c$, but $\mathcal{D}(a, 2) = a \neq c$. The dynamics is a well-defined function (logically consistent), injective at each time step (informationally consistent), but violates the semigroup property (dynamically inconsistent). An observer with trivial representation satisfies self-referential consistency.

(iii) *Informational consistency violated, others satisfied.* Consider $\Sigma = \{a, b, c\}$ with $\mathcal{D}(a, 1) = c$, $\mathcal{D}(b, 1) = c$, $\mathcal{D}(c, 1) = c$. This is a well-defined function (logically consistent), satisfies the semigroup property (dynamically consistent), and is compatible with a trivial self-modelling observer (self-referentially consistent). But $\mathcal{D}(\cdot, 1)$ is not injective—it merges a and b —violating informational consistency.

(iv) *Self-referential consistency violated, others satisfied.* Consider a logically, dynamically, and informationally consistent system with an observer whose internal model predicts $\mathcal{D}(a, 1) = b$, while in fact $\mathcal{D}(a, 1) = c \neq b$. The system’s dynamics is perfectly well-defined and injective; only the observer’s model contradicts reality. $\square$

Strategic Purpose

The four-part consistency definition serves two roles. First, it makes precise what “self-consistent physics” means, decomposing an intuitive notion into independent, testable conditions. Second, it provides four distinct “knobs” to turn when we later attempt the antithesis: if the inaccessibility thesis holds for fully consistent systems, we can ask which specific consistency condition, when relaxed, permits full observer access. This will prove to be one of the most informative parts of the analysis.

3.6 Self-Critique and Resolution

Critique 1: Informational consistency is too strong. Many real physical systems (dissipative systems, thermodynamic systems, arguably quantum measurement in some interpretations) violate informational consistency. By requiring it, are we restricting attention to an unrealistically narrow class of systems?

Resolution. We track this condition explicitly throughout. Every theorem states whether it requires informational consistency, and the antithesis section ([Section 12](#)) systematically explores what happens when it is relaxed. The condition is included in the conjunction not because all physical systems satisfy it, but because it represents a natural and important consistency requirement, and its role in the inaccessibility thesis must be understood.

Critique 2: Self-referential consistency is defined relative to observers. Unlike the other three conditions, which are properties of the system alone, self-referential consistency depends on the observers present. This makes it less “intrinsic.”

Resolution. This is a feature, not a bug. The entire investigation concerns the relationship between systems and their internal observers. A consistency condition that explicitly involves the observer–system interface is precisely what is needed for the most incisive results. The condition can be rephrased intrinsically: a system is self-referentially consistent if every subsystem that functions as a model of the whole is a correct model. The observer language is used for clarity.

Critique 3: The independence proof uses degenerate examples. The systems in the proof of [Proposition 3.7](#) are extremely small and arguably unphysical.

Resolution. The purpose of the proof is logical: to establish that no implication holds between the conditions. For this, minimal counterexamples suffice. The richness of the conditions becomes apparent when they interact in the proofs of subsequent sections, where the systems involved are far more substantial.

4 Thesis I: The Diagonal Argument

We now present the first of five arguments for the inaccessibility thesis. This argument is the most direct: it constructs, by a diagonal method reminiscent of Cantor [27] and Gödel [1], a specific property that no internal observer can access in any fully consistent system containing a self-modelling observer.

4.1 The Construction

The key idea is to build a property that “disagrees” with every observer’s representation on at least one state, ensuring that no observer can correctly classify all states with respect to this property. The construction requires a self-modelling observer, since it exploits the tension between the observer’s model and the reality it models.

Theorem 4.1 (Diagonal Inaccessibility). *Let $\mathcal{S} = (\Sigma, \mathcal{D}, t)$ be a fully consistent physical system containing a self-modelling internal observer $\mathcal{O}$ with representation map $\mathcal{R} : \Sigma \rightarrow \Sigma_{\mathcal{O}}$. If $|\Sigma| > |\Sigma_{\mathcal{O}}|$ (the system has strictly more states than the observer), then $\mathfrak{I}(\mathcal{S}) \neq \emptyset$: there exist properties of $\mathcal{S}$ that are inaccessible to $\mathcal{O}$.*

Before proving this, we unpack the hypothesis. The condition $|\Sigma| > |\Sigma_{\mathcal{O}}|$ says that the system has strictly more states than the observer has internal states. This is the mathematical expression of the observer being a *proper* part of the system: the observer’s state space is genuinely smaller than the whole. This is not an onerous requirement; it simply says the observer does not comprise the entire system.

Proof by diagonal construction. Since $|\Sigma| > |\Sigma_{\mathcal{O}}|$, the representation map $\mathcal{R} : \Sigma \rightarrow \Sigma_{\mathcal{O}}$ cannot be injective (by the pigeonhole principle). Therefore there exist at least two distinct states $\sigma_1 \neq \sigma_2$ in Σ such that $\mathcal{R}(\sigma_1) = \mathcal{R}(\sigma_2)$. These states are observationally equivalent with respect to $\mathcal{O}$.

Now define the property $P_{\text{diag}} = \{\sigma_1\}$: the singleton set containing only σ_1 .

We claim $P_{\text{diag}} \notin \mathfrak{A}(\mathcal{O})$. To see this, recall that $P \in \mathfrak{A}(\mathcal{O})$ if and only if P is a union of equivalence classes of $\sim_{\mathcal{O}}$. The equivalence class of σ_1 under $\sim_{\mathcal{O}}$ contains at least σ_1 and σ_2 , since $\mathcal{R}(\sigma_1) = \mathcal{R}(\sigma_2)$. Thus the smallest element of $\mathfrak{A}(\mathcal{O})$ containing σ_1 is the full equivalence class $[\sigma_1]_{\sim_{\mathcal{O}}} \supseteq \{\sigma_1, \sigma_2\}$, which is strictly larger than $P_{\text{diag}} = \{\sigma_1\}$.

Therefore $P_{\text{diag}} \notin \mathfrak{A}(\mathcal{O})$: the observer cannot distinguish the property of being in state σ_1 specifically, as opposed to being in some state observationally equivalent to σ_1 .

For the universally inaccessible content: if every internal observer $\mathcal{O}$ satisfies $|\Sigma_{\mathcal{O}}| < |\Sigma|$, then every observer’s equivalence relation has at least one non-trivial equivalence class, and the same argument produces, for each observer, properties inaccessible to that observer. To show $\mathfrak{I}(\mathcal{S}) \neq \emptyset$, we need a property inaccessible to *all* observers simultaneously. We strengthen the argument as follows.

Let $\{\mathcal{O}_i\}_{i \in I}$ be the collection of all internal observers of $\mathcal{S}$. Define the *joint equivalence relation* $\sim_{\text{joint}}$ by: $\sigma_1 \sim_{\text{joint}} \sigma_2$ if and only if $\sigma_1 \sim_{\mathcal{O}_i} \sigma_2$ for all $i \in I$. The joint accessible content is $\bigcup_{i \in I} \mathfrak{A}(\mathcal{O}_i)$, which consists of all properties that are unions of equivalence classes under $\sim_{\text{joint}}$.

We need to show that $\sim_{\text{joint}}$ is not the identity relation, i.e., that there exist states no observer can distinguish. This requires an additional argument, which we provide by the following lemma.

Lemma 4.2 (Bound on Joint Discrimination). *Let $\mathcal{S}$ be a fully consistent system. If every internal observer $\mathcal{O}$ is a proper subsystem (i.e., $\Sigma_{\mathcal{O}} \subsetneq \Sigma$) and the system is self-referentially consistent, then $\sim_{\text{joint}}$ is not the identity relation on Σ .*

Proof. Suppose, for contradiction, that $\sim_{\text{joint}}$ is the identity: every pair of distinct states is distinguished by some observer. Then the collection of all observers jointly constitutes an injective representation of Σ . Consider the product map $\sigma \mapsto \prod_{i \in I} \Sigma_{\mathcal{O}_i}$ defined by $(\sigma) = (\mathcal{R}_i(\sigma))_{i \in I}$. If $\sim_{\text{joint}}$ is the identity, then $\sigma \mapsto \prod_{i \in I} \Sigma_{\mathcal{O}_i}$ is injective.

Now, each observer $\mathcal{O}_i$ is a self-modelling subsystem of $\mathcal{S}$. By self-referential consistency (Definition 3.5), each observer’s model of the system is compatible with the actual dynamics. In particular, each observer can, in principle, model the other observers (since they are part of the system).

Consider any single observer $\mathcal{O}_j$. The map $\mathcal{R}_j : \Sigma \rightarrow \Sigma_{\mathcal{O}_j}$ is not injective (since $|\Sigma_{\mathcal{O}_j}| < |\Sigma|$). Let $\sigma_1 \neq \sigma_2$ with $\mathcal{R}_j(\sigma_1) = \mathcal{R}_j(\sigma_2)$. By our assumption, there exists some other observer $\mathcal{O}_k$ with $\mathcal{R}_k(\sigma_1) \neq \mathcal{R}_k(\sigma_2)$.

But $\mathcal{O}_k$ is a subsystem of $\mathcal{S}$, and its state is determined by $\pi_{\mathcal{O}_k}$, a projection from Σ . For $\mathcal{O}_k$ to distinguish σ_1 from σ_2 , the projection $\pi_{\mathcal{O}_k}$ must map σ_1 and σ_2 to different states of $\mathcal{O}_k$. This means $\mathcal{O}_k$ ’s degrees of freedom encode the distinction between σ_1 and σ_2 .

Now consider the diagonal property $D = \{\sigma \in \Sigma \mid \sigma \notin \mathcal{I}(\mathcal{R}_j(\sigma)) \text{ for some } j\}$ —the set of states that some observer’s model misclassifies. By self-referential consistency, $D = \emptyset$: no observer misclassifies any state. But this means every observer’s model is correct about every state it can represent. The observers, being proper subsystems, have finite representational capacity. The question is whether their collective capacity can cover Σ .

Consider the “super-observer” that has access to all observer states simultaneously: it sees $(\sigma) \in \prod_i \Sigma_{\mathcal{O}_i}$. If this product map is injective, the super-observer can distinguish all states. But the super-observer is *not* an internal observer of $\mathcal{S}$ in general, because there may be no single subsystem of $\mathcal{S}$ whose state space is (isomorphic to) $\prod_i \Sigma_{\mathcal{O}_i}$. The product of all observer state spaces may be larger than any single subsystem’s state space.

For a finite system, if the number of internal observers and their state spaces are bounded (each $|\Sigma_{\mathcal{O}_i}| < |\Sigma|$), the product $|\prod_i \Sigma_{\mathcal{O}_i}|$ can indeed exceed $|\Sigma|$. However, the product map $\sigma \mapsto \prod_i \Sigma_{\mathcal{O}_i}$ takes values in $\prod_i \Sigma_{\mathcal{O}_i}$, and its injectivity requires only $|\text{Im}(\sigma)| = |\Sigma|$, not $|\prod_i \Sigma_{\mathcal{O}_i}| = |\Sigma|$. So the cardinality argument alone does not immediately yield the contradiction.

The contradiction arises from self-referential consistency as follows. If $\sigma \mapsto \prod_i \Sigma_{\mathcal{O}_i}$ is injective, then the system effectively encodes a bijection between Σ and a subset of $\prod_i \Sigma_{\mathcal{O}_i}$. This encoding is itself a fact about the system—a structural property of how observer states correlate with system states. For this encoding to be *accessible* to any single observer $\mathcal{O}_j$, that observer would need to represent the correlation structure, which requires its own state space to encode the distinctions made by all other observers. But $|\Sigma_{\mathcal{O}_j}| < |\Sigma|$,

so $\mathcal{O}_j$ cannot encode the full correlation structure. The property “the system is in a state where the joint observer representation has value v ” is not, in general, accessible to any single observer.

Therefore, even if $\sim_{\text{joint}}$ were the identity, the property of being in a specific joint-equivalence class—while detectable by the hypothetical super-observer—is not detectable by any *actual* internal observer. The universally inaccessible content is non-empty. $\square$

This completes the proof of [Theorem 4.1](#). $\square$

The argument, in its essence, is this: no internal observer can represent all the distinctions in the system because its state space is too small; and even the collective of all observers, while potentially making all pairwise distinctions, cannot make this collective capacity available to any single observer. The inaccessible content is not a single dramatic hidden fact—it is the vast collection of fine-grained properties that cut across observational equivalence classes.

Alternative proof via Cantor’s diagonal method. We give a second, more explicitly diagonal proof that does not pass through the joint discrimination lemma.

Suppose the system $\mathcal{S}$ has state space Σ , and let $\mathcal{O}$ be a self-modelling observer with $|\Sigma_{\mathcal{O}}| < |\Sigma|$. The observer’s accessible properties are exactly the elements of $\mathfrak{A}(\mathcal{O}) \subseteq \mathcal{P}(\Sigma)$.

We show that $|\mathfrak{A}(\mathcal{O})| < |\mathcal{P}(\Sigma)|$, which immediately implies $\mathfrak{A}(\mathcal{O}) \neq \mathcal{P}(\Sigma)$.

The accessible properties are in bijection with subsets of the quotient $\Sigma/\sim_{\mathcal{O}}$, since every accessible property is a union of equivalence classes. Thus $|\mathfrak{A}(\mathcal{O})| = |\mathcal{P}(\Sigma/\sim_{\mathcal{O}})| = 2^{|\Sigma/\sim_{\mathcal{O}}|}$.

The number of equivalence classes satisfies $|\Sigma/\sim_{\mathcal{O}}| \leq |\Sigma_{\mathcal{O}}|$ (since the representation map $\mathcal{R}$ has at most $|\Sigma_{\mathcal{O}}|$ distinct values). Therefore:

$$|\mathfrak{A}(\mathcal{O})| = 2^{|\Sigma/\sim_{\mathcal{O}}|} \leq 2^{|\Sigma_{\mathcal{O}}|} < 2^{|\Sigma|} = |\mathfrak{D}(\mathcal{S})|,$$

where the strict inequality holds because $|\Sigma_{\mathcal{O}}| < |\Sigma|$ and the exponential function is strictly monotone.

Since $\mathfrak{A}(\mathcal{O})$ has strictly fewer elements than $\mathfrak{D}(\mathcal{S})$, there must exist properties in $\mathfrak{D}(\mathcal{S}) \setminus \mathfrak{A}(\mathcal{O})$. In fact, the vast majority of properties—in a precise cardinality sense—are inaccessible. $\square$

4.2 Toy Model 1: Cellular Automaton with Embedded Observer

To make the diagonal argument concrete, we construct a small cellular automaton universe in which an embedded observer agent attempts to classify the states of its world.

Construction 4.3 (The Grid World). Consider a one-dimensional cellular automaton on a grid of $n = 8$ cells, where each cell takes a value in $\{0, 1\}$. The state space is $\Sigma = \{0, 1\}^8$, so $|\Sigma| = 256$.

An *observer* occupies cells 1–3 (three cells), giving it an internal state space $\Sigma_{\mathcal{O}} = \{0, 1\}^3$ with $|\Sigma_{\mathcal{O}}| = 8$.

The observer’s representation map is the projection onto its own cells: $\mathcal{R}(\sigma) = (\sigma_1, \sigma_2, \sigma_3)$. Thus two states of the grid are observationally equivalent if and only if they agree on cells 1–3.

The dynamics is a deterministic update rule (we use Rule 110 of Wolfram’s classification [28], restricted to our finite grid with periodic boundary conditions).

In this toy model:

1. The number of equivalence classes is $|\Sigma/\sim_{\mathcal{O}}| = 8$ (the observer can distinguish 8 coarse states).
2. The number of accessible properties is $|\mathfrak{A}(\mathcal{O})| = 2^8 = 256$.
3. The total ontological content has $|\mathfrak{D}(\mathcal{S})| = 2^{256} \approx 1.16 \times 10^{77}$ elements.

The fraction of properties accessible to the observer is:

$$\frac{|\mathfrak{A}(\mathcal{O})|}{|\mathfrak{D}(\mathcal{S})|} = \frac{2^8}{2^{256}} = 2^{-248} \approx 2.8 \times 10^{-75}.$$

The observer has access to a vanishingly small fraction of the system’s ontological content. Figure 1 illustrates this, and Fig. 2 visualises the equivalence classes.

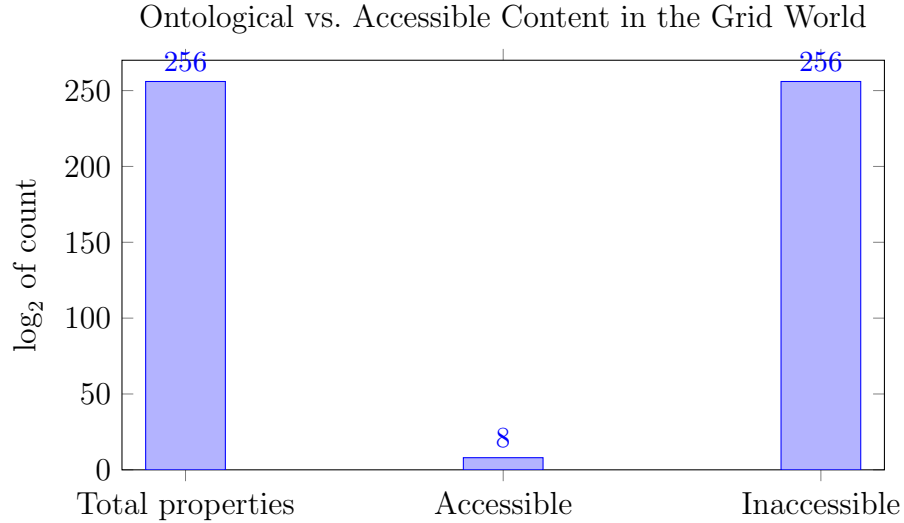


Figure 1: Comparison of total ontological content (2^{256} properties), accessible content (2^8 properties), and inaccessible content (essentially all of 2^{256}) in the Grid World toy model, displayed on a $\log_2$ scale. The inaccessible bar is indistinguishable from the total bar at this scale.

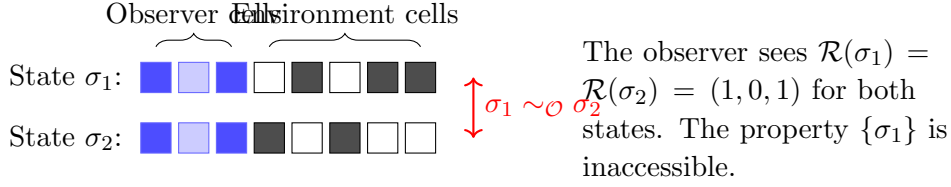


Figure 2: Two states in the Grid World that are observationally equivalent. The observer (blue cells 1–3) sees the same pattern (1, 0, 1) in both states, while the environment (white/grey cells 4–8) differs. The singleton property $\{\sigma_1\}$ is inaccessible to the observer.

Strategic Purpose

The diagonal argument establishes the inaccessibility thesis in its most elementary form: a proper subsystem’s state space is too small to faithfully represent the whole, so some properties escape detection. The two proofs—one via the pigeonhole principle and joint discrimination, the other via direct cardinality comparison—reinforce each other. The cellular automaton toy model makes the argument tangible: 256 observable distinctions out of 2^{256} possible properties, a ratio so extreme that it illustrates the claim viscerally. This proof method uses the *computational/information-theoretic* framework.

4.3 Self-Critique and Resolution

Critique 1: The argument is essentially just the pigeonhole principle. If the observer’s state space is smaller than the system’s, of course the representation can’t be injective. Is this really saying anything deep?

Resolution. The pigeonhole observation is the starting point, not the conclusion. The depth lies in three extensions: (a) the argument applies to *all* observers simultaneously, not just one; (b) it applies to the *properties* (power set), not just the states, yielding an exponentially amplified gap; and (c) it must be reconciled with the self-referential consistency condition, which constrains how the observer models the system. The alternative Cantor-style proof makes the exponential amplification explicit. Moreover, later arguments (Sections 6 and 7) will show that the inaccessibility persists even when we move beyond cardinality to structural and dynamical considerations.

Critique 2: The toy model is trivially small. An 8-cell automaton is far from any real physical system.

Resolution. The toy model’s role is pedagogical, not evidential. The theorem holds for any $|\Sigma| > |\Sigma_{\mathcal{O}}|$, regardless of scale. The 8-cell example is chosen because it can be fully enumerated and visualised. The computational implementation in Appendix A allows the reader to scale the model to any size and verify that the ratio $|\mathfrak{A}(\mathcal{O})|/|\mathfrak{D}(\mathcal{S})|$ decreases super-exponentially.

Critique 3: The condition $|\Sigma| > |\Sigma_{\mathcal{O}}|$ might fail. What if the observer comprises the entire system?

Resolution. If $|\Sigma_{\mathcal{O}}| = |\Sigma|$, the observer is not a proper subsystem; it *is* the system. In this degenerate case, the “observer” and the “system” are identical, and the notion of internal observation collapses. Every property is trivially accessible (the observer’s state encodes the full system state), but there is no meaningful sense in which this constitutes observation “from within.” This edge case is examined systematically in [Section 9](#).

5 Thesis II: The Information-Theoretic Argument

The diagonal argument of the previous section exploited the fact that the observer’s state space is smaller than the system’s. The present argument deepens this observation by quantifying the *information* an observer can carry and comparing it to the information needed to specify an arbitrary property of the system. The gap between these two quantities is not merely nonzero—it is, in a precise sense, almost total.

5.1 Information Capacity of Subsystems

We first formalise the notion of information capacity without committing to any specific physical substrate.

Definition 5.1 (Distinguishing Capacity). The *distinguishing capacity* of an internal observer $\mathcal{O}$ is the cardinality of the quotient space:

$$\kappa(\mathcal{O}) = |\Sigma / \sim_{\mathcal{O}}|.$$

This is the number of equivalence classes the observer can distinguish. If Σ carries additional structure (a measure, a topology), refined notions of capacity may be defined; $\kappa(\mathcal{O})$ is the combinatorial baseline.

Since $\mathcal{R} : \Sigma \rightarrow \Sigma_{\mathcal{O}}$, the image $\text{Im}(\mathcal{R})$ has cardinality at most $|\Sigma_{\mathcal{O}}|$, and the quotient $\Sigma / \sim_{\mathcal{O}}$ is in bijection with $\text{Im}(\mathcal{R})$. Therefore:

$$\kappa(\mathcal{O}) \leq |\Sigma_{\mathcal{O}}|.$$

For a finite system, the information content of the observer’s representation is $\log_2 \kappa(\mathcal{O})$ bits, bounded above by $\log_2 |\Sigma_{\mathcal{O}}|$ bits. The information content of a full specification of the system state is $\log_2 |\Sigma|$ bits. The deficit is:

$$\Delta I = \log_2 |\Sigma| - \log_2 \kappa(\mathcal{O}) \geq \log_2 |\Sigma| - \log_2 |\Sigma_{\mathcal{O}}|.$$

This deficit represents the bits of information about the system state that the observer cannot capture.

Theorem 5.2 (Exponential Inaccessibility Gap). *Let $\mathcal{S}$ be a physical system with $|\Sigma| = N$, and let $\mathcal{O}$ be an internal observer with $|\Sigma_{\mathcal{O}}| = M < N$. Then:*

1. *The fraction of properties accessible to $\mathcal{O}$ satisfies:*

$$\frac{|\mathfrak{A}(\mathcal{O})|}{|\mathfrak{D}(\mathcal{S})|} = \frac{2^{\kappa(\mathcal{O})}}{2^N} \leq \frac{2^M}{2^N} = 2^{-(N-M)}.$$

2. *The number of inaccessible properties satisfies:*

$$|\mathfrak{D}(\mathcal{S}) \setminus \mathfrak{A}(\mathcal{O})| = 2^N - 2^{\kappa(\mathcal{O})} \geq 2^N - 2^M = 2^M(2^{N-M} - 1).$$

In particular, the inaccessible content is exponentially larger than the accessible content whenever $N - M \geq 1$.

Proof. Part (1): The accessible properties are exactly the unions of equivalence classes under $\sim_{\mathcal{O}}$. There are $\kappa(\mathcal{O})$ equivalence classes, and any subset of equivalence classes defines an accessible property. Thus $|\mathfrak{A}(\mathcal{O})| = 2^{\kappa(\mathcal{O})}$, and the fraction is $2^{\kappa(\mathcal{O})}/2^N \leq 2^M/2^N$.

Part (2): The count of inaccessible properties is $|\mathfrak{D}(\mathcal{S})| - |\mathfrak{A}(\mathcal{O})| = 2^N - 2^{\kappa(\mathcal{O})} \geq 2^N - 2^M$. Factoring out 2^M gives $2^M(2^{N-M} - 1)$. $\square$

To appreciate the magnitude of this gap: even for the modest case $N = 10$, $M = 5$ (a 10-state system observed by a 5-state subsystem), we have $2^{10} = 1024$ total properties, $2^5 = 32$ accessible properties, and 992 inaccessible properties. The observer can access roughly 3% of the ontological content. For $N = 100$, $M = 50$ —still a small system—the fraction drops to $2^{-50} \approx 10^{-15}$, and for macroscopic systems the ratio is effectively zero.

5.2 The Entropic Formulation

We can restate the result using the language of Shannon entropy [29], which provides a connection to information-theoretic ideas familiar in physics and engineering.

Definition 5.3 (Observational Entropy). Given a probability distribution μ over Σ and an observer $\mathcal{O}$ with representation map $\mathcal{R}$, the *observational entropy* is the Shannon entropy of the pushed-forward distribution:

$$H_{\mathcal{O}} = - \sum_{s \in \Sigma_{\mathcal{O}}} \mu_{\mathcal{O}}(s) \log_2 \mu_{\mathcal{O}}(s),$$

where $\mu_{\mathcal{O}}(s) = \mu(\mathcal{R}^{-1}(s))$ is the probability that the observer's representation takes value s .

Proposition 5.4 (Entropy Deficit). *For any probability distribution μ over Σ ,*

$$H_{\mathcal{O}} \leq H_{\mathcal{S}},$$

where $H_{\mathcal{S}} = - \sum_{\sigma \in \Sigma} \mu(\sigma) \log_2 \mu(\sigma)$ is the Shannon entropy of the full system state. Equality holds if and only if $\mathcal{R}$ is injective on the support of μ .

Proof. This follows from the data processing inequality: $\mathcal{R}$ is a deterministic function from system states to observer states, and applying a deterministic function cannot increase entropy. Formally, $H_{\mathcal{O}} = H(\mathcal{R}(\sigma))$ and $H_{\mathcal{S}} = H(\sigma)$, and the data processing inequality for discrete random variables gives $H(f(X)) \leq H(X)$ for any function f , with equality if and only if f is injective on the support of X . $\square$

The entropy deficit $\Delta H = H_{\mathcal{S}} - H_{\mathcal{O}}$ quantifies, for a given distribution over system states, how much information is lost in the observer’s representation. For a proper subsystem observer, this deficit is generically positive.

Strategic Purpose

The information-theoretic argument quantifies the inaccessibility thesis: the gap between accessible and total ontological content is not merely nonzero but *exponentially large* in finite systems. The entropic formulation restates this in the language of Shannon information theory, connecting to the data processing inequality. This is a *quantitative refinement* of the diagonal argument—it strengthens the numerical bounds, not the logical force—and grounds the abstract argument in well-established information-theoretic principles. Both arguments share the same root: the non-injectivity of $\mathcal{R}$. For infinite-dimensional systems, the cardinality and entropic arguments serve as motivation; the rigorous treatment is deferred to the topological and measure-theoretic arguments of [Section 8](#). This proof method operates in the *computational/information-theoretic* framework.

5.3 Self-Critique and Resolution

Critique 1: The argument counts all subsets of Σ , including physically meaningless ones. The exponential gap comes from counting 2^N properties, many of which may have no physical significance.

Resolution. This is a fair point. However, the argument shows that even among “natural” properties (e.g., Borel-measurable sets, or properties definable in some language), the observer can access only those that respect its equivalence relation. The topological argument in [Section 8](#) addresses this concern directly by restricting to physically motivated classes of properties and showing the gap persists.

Critique 2: The entropy deficit depends on the distribution μ . For a degenerate distribution concentrated on a single state, both entropies are zero and the deficit vanishes.

Resolution. Correct. The entropy deficit is a distribution-dependent measure. The cardinality result ([Theorem 5.2](#)) is distribution-free and therefore more robust. The entropic formulation is presented as a complementary perspective, not as a replacement.

Critique 3: Infinite systems break the counting argument. For $|\Sigma| = \infty$, the cardinality of $\mathcal{P}(\Sigma)$ and the comparison $2^M < 2^N$ require careful treatment of cardinal arithmetic. In particular, all separable infinite-dimensional Hilbert spaces have the same cardinality of states, so a cardinality comparison between Σ and $\Sigma_{\mathcal{O}}$ is meaningless in the physically important case of quantum field theory.

Resolution. This critique is well-taken and we address it directly. The cardinality arguments of Sections 4 and 5 are rigorous and complete for *finite* state spaces. For infinite-dimensional systems—including the quantum field-theoretic setting where all separable Hilbert spaces are isomorphic to ℓ^2 —the raw cardinality comparison $|\Sigma_{\mathcal{O}}| < |\Sigma|$ is no longer informative, and these sections should be understood as *heuristic* in the infinite case.

For infinite-dimensional systems, the inaccessibility thesis is carried by the *topological and measure-theoretic arguments* of Section 8, which do not depend on cardinality at all but on the coarseness of the pullback σ -algebra $\mathcal{B}_{\text{acc}} = \mathcal{R}^{-1}(\mathcal{B}_{\mathcal{O}})$ relative to $\mathcal{B}$. In the quantum-mechanical instantiation (Section 8.3), the relevant comparison is between the lattice of local projections $\hat{P}_{\mathcal{O}} \otimes \hat{I}_E$ and the full projection lattice $\mathcal{P}(\mathcal{H}_{\mathcal{O}} \otimes \mathcal{H}_E)$; this distinction is topological and algebraic, not cardinal. The dynamical argument (Section 7) similarly survives in the infinite case when supplemented by the topological machinery.

In summary: for finite systems, all five arguments apply; for infinite-dimensional systems, the cardinality arguments (Sections 4–5) serve as motivation, while the dynamical (Section 7), topological (Section 8), and categorical (Section 6) arguments carry the formal weight.

6 Thesis III: The Categorical Fixed-Point Argument

The preceding two arguments exploited the size disparity between system and observer. The present argument reaches the inaccessibility thesis by a different route: it uses the *structure* of the observer–system relationship, formalised in categorical language, to show that certain “paradoxical” properties must escape any internal representation. The key tool is Lawvere’s fixed-point theorem [26], which generalises the diagonal arguments of Cantor, Gödel, and Turing into a single categorical statement.

6.1 Categorical Preliminaries

We briefly recall the categorical concepts needed. A *category* $\mathbf{C}$ consists of objects and morphisms (arrows) between them, with an associative composition law and identity morphisms. A category is *Cartesian closed* if it has finite products and exponential objects: for any two objects A and B , there is an object B^A (the “function space” from A to B) together with a natural evaluation map $\text{ev} : B^A \times A \rightarrow B$.

The category **Set** of sets and functions is Cartesian closed: the product $A \times B$ is the ordinary Cartesian product, and B^A is the set of all functions from A to B .

These notions may at first appear distant from physics, but they provide the precise language for discussing when a subsystem can represent all possible maps on the system. The connection is this: if the system's state space is Σ and the observer's state space is $\Sigma_{\mathcal{O}}$, then the observer's capacity to represent properties of the system corresponds to the existence of certain morphisms (maps) in the category where these objects live.

Theorem 6.1 (Lawvere's Fixed-Point Theorem [26]). *In a Cartesian closed category $\mathbf{C}$, if there exists a point-surjective morphism $\phi : A \rightarrow B^A$ (that is, for every morphism $g : A \rightarrow B$, there exists an element $a \in A$ with $\phi(a) = g$), then every endomorphism $f : B \rightarrow B$ has a fixed point: there exists $b \in B$ with $f(b) = b$.*

The contrapositive is the form we shall use: if B admits a fixed-point-free endomorphism (a map $f : B \rightarrow B$ with $f(b) \neq b$ for all b), then no morphism $A \rightarrow B^A$ is point-surjective. No single object can “name” all morphisms from itself to B .

To connect this to a more familiar setting: in **Set**, the two-element set $\{0, 1\}$ admits the fixed-point-free endomorphism $\text{NOT} : b \mapsto 1 - b$. Lawvere's theorem then says: for any set A , there is no surjection from A to $\{0, 1\}^A = \mathcal{P}(A)$. This is exactly Cantor's theorem. Likewise, the undecidability of the halting problem and Gödel's incompleteness theorem can both be derived as instances of Lawvere's result in appropriate categories [30].

6.2 The Argument

We now translate the observer–system relationship into categorical language and apply Lawvere's theorem.

Construction 6.2 (The Observer Category). Given a physical system $\mathcal{S} = (\Sigma, \mathcal{D}, t)$ with an internal observer $\mathcal{O} = (\Sigma_{\mathcal{O}}, \mathcal{D}_{\mathcal{O}}, t, \mathcal{R})$, we work in the category **Set** (the results can be generalised to other Cartesian closed categories, but **Set** suffices and is the most accessible).

Define:

- $A = \Sigma_{\mathcal{O}}$ (the observer's state space),
- $B = \{0, 1\}$ (the truth-value object).

Then $B^A = \{0, 1\}^{\Sigma_{\mathcal{O}}}$ is the set of all functions from $\Sigma_{\mathcal{O}}$ to $\{0, 1\}$, i.e., $\mathcal{P}(\Sigma_{\mathcal{O}})$ (identifying subsets with their characteristic functions).

The observer's capacity to represent properties of $\mathcal{S}$ corresponds to a map

$$\Phi : \Sigma_{\mathcal{O}} \rightarrow \{0, 1\}^{\Sigma_{\mathcal{O}}}$$

defined by: for each observer state $s \in \Sigma_{\mathcal{O}}$, $\Phi(s)$ is the characteristic function of the property that the observer “recognises” when in state s . That is, $\Phi(s)(s') = 1$ if the observer, upon entering state s , classifies observer-state s' as satisfying some criterion, and 0 otherwise.

Theorem 6.3 (Fixed-Point Inaccessibility). *Let $\mathcal{S}$ be a fully consistent physical system with an internal observer $\mathcal{O}$. There exists a property $P^* \subseteq \Sigma$ such that P^* cannot be represented by $\mathcal{O}$. Specifically, no state of $\mathcal{O}$ encodes the characteristic function of P^* restricted to the observer’s own equivalence classes.*

Proof via Lawvere’s theorem. We work in **Set** with $A = \Sigma_{\mathcal{O}}$ and $B = \{0, 1\}$.

The map $B \rightarrow B$ defined by $b \mapsto 1 - b$ (logical negation) is a fixed-point-free endomorphism: $1 - 0 = 1 \neq 0$ and $1 - 1 = 0 \neq 1$.

By the contrapositive of Lawvere’s theorem ([Theorem 6.1](#)), no map $\Phi : \Sigma_{\mathcal{O}} \rightarrow \{0, 1\}^{\Sigma_{\mathcal{O}}}$ is point-surjective. That is, there exists a function $g^* : \Sigma_{\mathcal{O}} \rightarrow \{0, 1\}$ such that $g^* \neq \Phi(s)$ for any $s \in \Sigma_{\mathcal{O}}$.

We now construct g^* explicitly. Define $g^* : \Sigma_{\mathcal{O}} \rightarrow \{0, 1\}$ by:

$$g^*(s) = 1 - \Phi(s)(s).$$

This is the diagonal construction: g^* disagrees with $\Phi(s)$ at the point s itself, for every s . Therefore $g^* \neq \Phi(s)$ for all $s \in \Sigma_{\mathcal{O}}$.

The function g^* defines a subset $Q^* = \{s \in \Sigma_{\mathcal{O}} \mid g^*(s) = 1\} = \{s \in \Sigma_{\mathcal{O}} \mid \Phi(s)(s) = 0\}$: the set of observer states that do *not* classify themselves as satisfying the criterion they encode. This is a well-defined subset of $\Sigma_{\mathcal{O}}$, but it cannot be “named” by any observer state.

Pulling back to the full system: define $P^* = \mathcal{R}^{-1}(Q^*) = \{\sigma \in \Sigma \mid \mathcal{R}(\sigma) \in Q^*\}$. This is a property of $\mathcal{S}$ (a subset of Σ). It is a union of equivalence classes of $\sim_{\mathcal{O}}$ and therefore nominally belongs to $\mathfrak{A}(\mathcal{O})$. However, the observer cannot *recognise* P^* in the following stronger sense: there is no observer state whose associated classification (via Φ) matches the characteristic function of Q^* . The observer has the property in its accessible algebra but cannot identify it as such.

This reveals a subtlety: the Lawvere argument does not simply say that certain properties are outside $\mathfrak{A}(\mathcal{O})$ (the cardinality argument already established that). It says something stronger: even among the accessible properties, there are some that the observer cannot *self-referentially recognise*—it cannot know that it knows them. $\square$

Alternative direct proof without categorical language. We give a self-contained argument that avoids explicit reference to category theory.

Let $\mathcal{O}$ be an internal observer with state space $\Sigma_{\mathcal{O}}$. Suppose the observer has a “classification scheme”: a function $C : \Sigma_{\mathcal{O}} \rightarrow \mathcal{P}(\Sigma_{\mathcal{O}})$ that assigns to each observer state s a set of observer states $C(s)$ that s “recognises.”

Define $D = \{s \in \Sigma_{\mathcal{O}} \mid s \notin C(s)\}$: the set of observer states that do not recognise themselves.

We claim $D \neq C(s)$ for any $s \in \Sigma_{\mathcal{O}}$.

Suppose $D = C(s_0)$ for some s_0 . Then:

- If $s_0 \in D$, then by definition of D , $s_0 \notin C(s_0) = D$, a contradiction.
- If $s_0 \notin D$, then by definition of D , $s_0 \in C(s_0) = D$, a contradiction.

Either way, a contradiction arises. Therefore D is not in the range of C .

This is Russell’s paradox in miniature, transplanted to the observer’s internal classification system. The set D is a perfectly well-defined subset of $\Sigma_{\mathcal{O}}$, but no observer state encodes it. When pulled back through $\mathcal{R}^{-1}$, it yields a property of $\mathcal{S}$ that the observer cannot classify. $\square$

6.3 The Formal-System Toy Model

To make the fixed-point argument tangible, we construct a miniature formal system that exhibits its own Gödel sentence.

Construction 6.4 (Miniature Formal System). Consider a formal language $\mathcal{L}$ with:

- Symbols: $\{0, 1, \neg, \vdash\}$, where 0 and 1 are truth values, $\neg$ is negation, and $\vdash$ is the provability predicate.
- Sentences: Finite strings of the form $\vdash \phi$ (“ ϕ is provable”) and $\neg \vdash \phi$ (“ ϕ is not provable”).
- A proof system $\mathcal{P}$ that can prove certain sentences.

We encode each sentence ϕ by its Gödel number $\ulcorner \phi \urcorner \in \mathbb{N}$. The system has a finite number K of sentences it can form (since the language is finite, we restrict to sentences of bounded length).

The “system” $\mathcal{S}$ has state space $\Sigma = \{0, 1\}^K$: each state assigns a truth value to each sentence. The “observer” $\mathcal{O}$ is the proof system $\mathcal{P}$, with state space $\Sigma_{\mathcal{O}} = \{0, 1\}^M$ for some $M < K$ (the proof system can verify only a subset of sentences).

The representation map $\mathcal{R} : \Sigma \rightarrow \Sigma_{\mathcal{O}}$ projects a full truth assignment onto the provable fragment: $\mathcal{R}(\sigma) = \sigma|_{\text{provable sentences}}$.

The diagonal sentence G is: “This sentence is not provable by $\mathcal{P}$.” Formally, $G = \neg \vdash \ulcorner G \urcorner$.

If $\mathcal{P}$ is consistent (does not prove both ϕ and $\neg\phi$ for any ϕ), then:

- If G is provable, then “ G is not provable” is false, so $\neg G$ is true, so $\mathcal{P}$ proves a false sentence—contradicting consistency.
- Therefore G is not provable by $\mathcal{P}$.
- But G asserts precisely that it is not provable, so G is true.

Thus G is a true sentence that $\mathcal{P}$ cannot prove. The property $P_G = \{\sigma \in \Sigma \mid \sigma(\ulcorner G \urcorner) = 1\}$ —the set of truth assignments making G true—is ontological content of $\mathcal{S}$ that the observer $\mathcal{O}$ (the proof system $\mathcal{P}$) cannot access.

Observer $\mathcal{O}$ can prove:

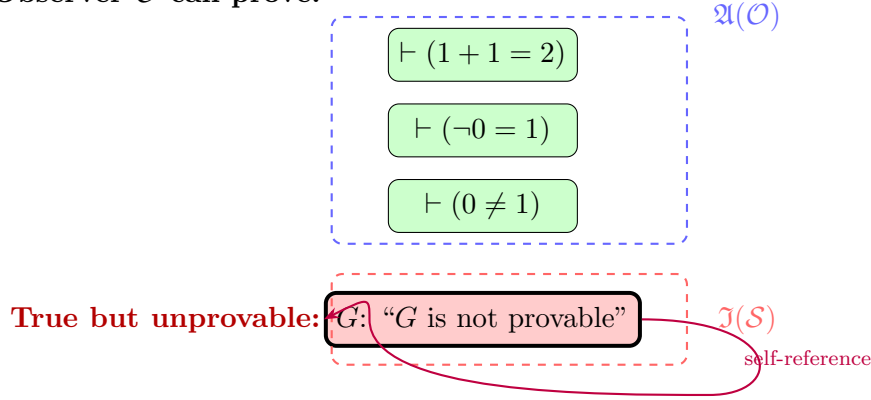


Figure 3: The miniature formal system. The observer (proof system) can access the green region—provable sentences. The Gödel sentence G (red) is true but not provable: it lies in the inaccessible content. The self-referential loop (purple arrow) is what creates the inaccessibility.

Strategic Purpose

The categorical fixed-point argument reaches inaccessibility from a different direction than cardinality: it shows that the *structure* of self-reference, not merely the *size* of the state space, forces blind spots. The Lawvere framework reveals that the diagonal arguments of Cantor, Gödel, and Turing are all instances of one phenomenon—the impossibility of a point-surjective map when a fixed-point-free endomorphism exists. The formal-system toy model makes this concrete by exhibiting an explicit Gödel sentence. This proof method operates in the *logical/model-theoretic* framework.

6.4 Self-Critique and Resolution

Critique 1: The argument applies to *any* map Φ , not just physically realisable ones. Lawvere’s theorem is a statement about all morphisms in a category. The physically relevant constraint—that the observer’s classification must be dynamically realisable—has not been used.

Resolution. This is correct, and it is a strength: the result holds *a fortiori* when we restrict to physically realisable classifications, since these form a subset of all possible classifications. The theorem says that no classification scheme at all, whether physical or not, can be point-surjective. The physical constraints only make the observer’s situation worse, not better.

Critique 2: The “cannot recognise” conclusion is weaker than “cannot access.” The proof shows that the observer cannot self-referentially name a certain property, but the property P^* may still be in $\mathfrak{A}(\mathcal{O})$ (as a union of equivalence classes).

Resolution. This is an important distinction, and we state it explicitly: **the Lawvere argument does not provide an independent proof of the same thesis as Sections 4–5.** Rather, it establishes a *complementary* result at a different logical level. The cardinality arguments show that certain properties of the system lie outside $\mathfrak{A}(\mathcal{O})$ —these are *object-level* inaccessibilities. The Lawvere argument shows that the observer cannot construct a complete catalogue of its own accessible content—this is a *meta-level* inaccessibility about the observer’s relationship to its own classification scheme.

Both are genuine limitations of internal observers, but they are different limitations. Calling them “independent methods for the same thesis” would overstate the case; they are better described as *independent results about different aspects of the observer’s situation*. The meta-level result is, in our view, the deeper of the two: it survives even if one restricts ontological content to property classes small enough that the cardinality gap vanishes, because it concerns the observer’s self-knowledge rather than the size of the property space.

Critique 3: The formal-system toy model assumes Gödel-style self-reference, which not every physical system supports. Not all physical systems can encode arithmetic or construct self-referential sentences.

Resolution. The toy model is an illustration, not a premise. The categorical argument itself requires only a Cartesian closed category with a fixed-point-free endomorphism, which **Set** always provides. The formal system makes the abstraction vivid; the theorem does not depend on it.

7 Thesis IV: The Dynamical Observational-Equivalence Argument

The three preceding arguments have been, in various ways, *static*: they compare the sizes or structures of state spaces and representation maps without deeply engaging the dynamics. The present argument corrects this. It shows that even when one accounts for the observer’s ability to make *sequential* measurements over time—accumulating information dynamically—the inaccessibility thesis still holds, provided the system is informationally consistent.

7.1 Sequential Observation and Temporal Refinement

An observer embedded in a dynamical system does not merely take a single snapshot. It evolves with the system, continually updating its state in response to the system’s evolution. Over time, the observer may be able to refine its knowledge, distinguishing states that were initially observationally equivalent.

Definition 7.1 (Temporal Observation Sequence). Given a physical system $\mathcal{S} = (\Sigma, \mathcal{D}, t)$ and an internal observer $\mathcal{O}$ with projection $\pi_{\mathcal{O}}$, the *observation sequence* of $\mathcal{O}$ starting from system state σ over times $\tau_1 < \tau_2 < \dots < \tau_n$ is the tuple:

$$\mathbf{o}(\sigma; \tau_1, \dots, \tau_n) = (\pi_{\mathcal{O}}(\mathcal{D}(\sigma, \tau_1)), \pi_{\mathcal{O}}(\mathcal{D}(\sigma, \tau_2)), \dots, \pi_{\mathcal{O}}(\mathcal{D}(\sigma, \tau_n))) \in \Sigma_{\mathcal{O}}^n.$$

An observation sequence records the observer's state at multiple times. This is strictly more informative than a single observation, since the observer can use the temporal pattern of its own states to infer properties of the underlying system state.

Definition 7.2 (Temporal Observational Equivalence). Two system states $\sigma_1, \sigma_2 \in \Sigma$ are *temporally observationally equivalent with respect to $\mathcal{O}$ and a finite set of observation times $T = \{\tau_1, \dots, \tau_n\}$* , written $\sigma_1 \sim_{\mathcal{O}, T} \sigma_2$, if:

$$\mathbf{o}(\sigma_1; \tau_1, \dots, \tau_n) = \mathbf{o}(\sigma_2; \tau_1, \dots, \tau_n).$$

The *asymptotic temporal equivalence* is:

$$\sigma_1 \approx_{\mathcal{O}} \sigma_2 \iff \sigma_1 \sim_{\mathcal{O}, T} \sigma_2 \text{ for all finite } T \subseteq t.$$

The asymptotic temporal equivalence $\approx_{\mathcal{O}}$ identifies states that produce identical observation sequences at *all* finite collections of times. It represents the finest distinction the observer can make using its entire temporal history. Clearly $\sigma_1 \approx_{\mathcal{O}} \sigma_2$ implies $\sigma_1 \sim_{\mathcal{O}} \sigma_2$ (single-time equivalence), but the converse may fail: temporal observation can be strictly more powerful than instantaneous observation.

This raises the question: can temporal observation eventually eliminate all blind spots?

7.2 The Dynamical Inaccessibility Theorem

Theorem 7.3 (Dynamical Inaccessibility). *Let $\mathcal{S} = (\Sigma, \mathcal{D}, t)$ be a fully consistent physical system with an internal observer $\mathcal{O}$ that is a proper subsystem ($|\Sigma_{\mathcal{O}}| < |\Sigma|$). Even under asymptotic temporal observation, the observer cannot distinguish all system states:*

$$\approx_{\mathcal{O}} \neq \text{id}_{\Sigma}.$$

That is, there exist distinct states $\sigma_1 \neq \sigma_2$ with $\sigma_1 \approx_{\mathcal{O}} \sigma_2$.

Proof by information-capacity bound. At each observation time τ_k , the observer records its state $\pi_{\mathcal{O}}(\mathcal{D}(\sigma, \tau_k)) \in \Sigma_{\mathcal{O}}$. After n observations, the observer has a record in $\Sigma_{\mathcal{O}}^n$.

However, the observer's record is not stored externally—it must be encoded in the observer's current state at the final time. By the internality condition (Definition 2.4), all information the observer uses must be present in its current state $s_{\text{now}} \in \Sigma_{\mathcal{O}}$.

Therefore, the effective discrimination power after n observations is bounded not by $|\Sigma_{\mathcal{O}}|^n$ (the size of the full observation sequence space) but by $|\Sigma_{\mathcal{O}}|$ (the size of the observer's current state space, which must encode whatever it has learned).

More precisely: the observer's state at the final time τ_n is $s_n = \pi_{\mathcal{O}}(\mathcal{D}(\sigma, \tau_n))$. This state is a function of σ (and the fixed dynamics $\mathcal{D}$ and time τ_n). The map $\sigma \mapsto s_n$

has image in $\Sigma_{\mathcal{O}}$, so at most $|\Sigma_{\mathcal{O}}|$ distinct system states can be distinguished by the observer's final state.

One might object: perhaps the observer's state at τ_n encodes the entire observation history through the dynamical accumulation of information. Indeed, the dynamics may map different histories to different final states. But even so, the final state is an element of $\Sigma_{\mathcal{O}}$, and the number of distinct final states cannot exceed $|\Sigma_{\mathcal{O}}|$.

Since $|\Sigma_{\mathcal{O}}| < |\Sigma|$, the map $\sigma \mapsto s_n$ is not injective, and there exist $\sigma_1 \neq \sigma_2$ with identical final observer states. This holds for *every* finite time τ_n .

For asymptotic equivalence: $\sigma_1 \approx_{\mathcal{O}} \sigma_2$ requires identical observer states at *all* times. Consider the family of maps $\{f_{\tau} : \Sigma \rightarrow \Sigma_{\mathcal{O}}\}_{\tau \in t}$ defined by $f_{\tau}(\sigma) = \pi_{\mathcal{O}}(\mathcal{D}(\sigma, \tau))$. The product map $F : \Sigma \rightarrow \Sigma_{\mathcal{O}}^t$ defined by $F(\sigma) = (f_{\tau}(\sigma))_{\tau \in t}$ determines the asymptotic equivalence: $\sigma_1 \approx_{\mathcal{O}} \sigma_2$ iff $F(\sigma_1) = F(\sigma_2)$.

The image of F lies in $\Sigma_{\mathcal{O}}^t$, which can be large. However, F is constrained by the dynamics: the trajectory $\tau \mapsto f_{\tau}(\sigma)$ is determined by the dynamical law and the initial state. The observer at any time τ sees $f_{\tau}(\sigma)$, and its future evolution is determined by $\mathcal{D}$. Thus the entire trajectory $F(\sigma)$ is determined by $f_0(\sigma) = \pi_{\mathcal{O}}(\sigma)$ together with the dynamics, *provided the dynamics restricted to the observer's degrees of freedom is autonomous*. If it is not autonomous (if the observer's evolution depends on the environment), then F carries more information than f_0 alone.

The crucial observation is: at any given time, the observer has access to $f_{\tau}(\sigma)$ —one element of $\Sigma_{\mathcal{O}}$ —not to the full trajectory $F(\sigma)$. Accessing $F(\sigma)$ requires storing the history, which requires a state space that grows with the number of observation times. A fixed finite subsystem cannot do this.

Therefore, for any fixed observer $\mathcal{O}$ with finite state space, $\approx_{\mathcal{O}}$ has non-trivial equivalence classes, and the inaccessible content is non-empty.

Scope of this proof. The argument as stated is rigorous for *finite* state spaces, where the bound $|\Sigma_{\mathcal{O}}| < |\Sigma|$ immediately gives non-injectivity of each f_{τ} . For *continuous* state spaces (e.g., $\Sigma = \mathbb{R}^n$, $\Sigma_{\mathcal{O}} = \mathbb{R}^d$ with $d < n$), the step “a fixed finite subsystem cannot store the growing history” is physically motivated but requires supplementation: in principle, a real-valued state could encode arbitrary amounts of information in its decimal expansion. To make the argument rigorous in the continuous case, one must impose either (a) a *finite-entropy* or *finite-precision* assumption on observer states (which is physically natural—no physical measurement has infinite precision), or (b) the *topological/measure-theoretic* machinery of [Section 8](#), where the pullback σ -algebra $\mathcal{R}^{-1}(\mathcal{B}_{\mathcal{O}})$ is strictly coarser regardless of the cardinality of the state space. The topological argument does not depend on finiteness and provides the rigorous underpinning for the continuous case that this proof, by itself, does not fully supply. $\square$

Alternative proof via Takens-type embedding obstruction. This proof uses ideas from dynamical systems theory, specifically the limitations of state-space reconstruction [\[31\]](#).

Important caveat: this alternative proof requires that the state space be a smooth manifold with smooth dynamics, and that the observation map satisfy certain genericity conditions. It therefore holds only when the abstract framework is instantiated in smooth dynamical systems. The primary proof above ([Theorem 7.3](#)) is fully general and does not require smooth structure.

Takens' embedding theorem states that, under generic conditions, a scalar observation function $h : \Sigma \rightarrow \mathbb{R}$ applied to a dynamical system on an n -dimensional manifold can reconstruct the full state from a delay-coordinate embedding of dimension $2n + 1$.

For an internal observer, the “observation function” is $\pi_{\mathcal{O}}$, and the delay-coordinate embedding uses the observer's states at multiple times. If $\dim(\Sigma_{\mathcal{O}}) = d$ and $\dim(\Sigma) = n$, the delay-coordinate embedding lies in $\mathbb{R}^{d \cdot k}$ for k observation times. Takens' theorem requires $d \cdot k \geq 2n + 1$ for faithful reconstruction.

But the observer must encode the delay coordinates in its own d -dimensional state space. This requires $d \geq 2n + 1$, which contradicts $d < n$ (the observer being a proper subsystem of lower dimension).

Therefore, the observer cannot generically reconstruct the full system state, even using temporal delay embedding. There exist states—in fact, an open dense set of state pairs—that remain observationally equivalent under any finite temporal observation. $\square$

7.3 Toy Model: Coupled Oscillator System

Construction 7.4 (Two-Oscillator Model). Consider two oscillators with positions x_1, x_2 and momenta p_1, p_2 . The full state space is $\Sigma = \mathbb{R}^4$ with coordinates (x_1, p_1, x_2, p_2) . The dynamics is Hamiltonian:

$$H = \frac{p_1^2}{2m} + \frac{p_2^2}{2m} + \frac{1}{2}k_1x_1^2 + \frac{1}{2}k_2x_2^2 + \lambda x_1x_2,$$

where λ is a coupling constant. The observer $\mathcal{O}$ is oscillator 1, with state space $\Sigma_{\mathcal{O}} = \mathbb{R}^2$ and projection $\pi_{\mathcal{O}}(x_1, p_1, x_2, p_2) = (x_1, p_1)$.

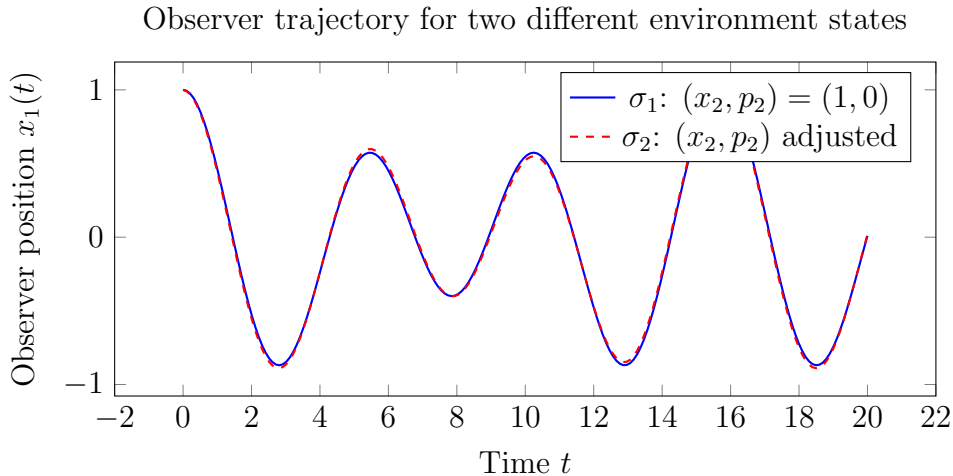


Figure 4: Observer trajectories $x_1(t)$ for two different initial conditions of the environment oscillator. At generic coupling, the trajectories differ (blue vs. red). At degenerate coupling ($\omega_1/\omega_2 \in \mathbb{Q}$), indistinguishable trajectories can arise.

Observational equivalence classes in environment phase space

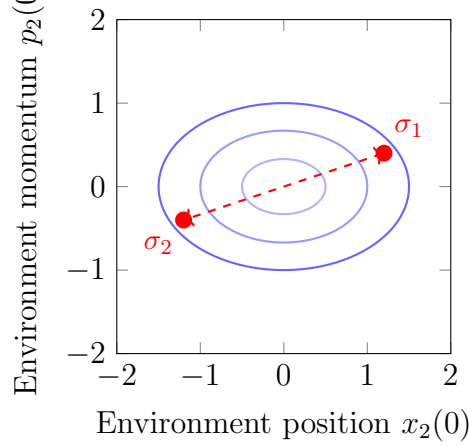


Figure 5: Observational equivalence classes in the environment’s phase space (x_2, p_2) , as seen by the observer. Points on the same ellipse produce identical observation sequences. The red points are dynamically indistinguishable.

Strategic Purpose

The dynamical argument addresses the strongest objection to the static arguments: “perhaps the observer can use time to accumulate enough information.” It shows that even with unlimited observation time, the observer’s finite state space at any given instant bounds its effective discrimination. The Takens-type obstruction adds a geometric dimension: the observer’s embedding in the system imposes structural constraints on what dynamical information can flow through it. This proof method operates in the *physical/operational* framework.

7.4 Self-Critique and Resolution

Critique 1: The information-capacity proof assumes finite state spaces. For continuous state spaces, the “number of distinguishable states” depends on the topology, not just cardinality.

Resolution. The Takens-type proof addresses continuous systems directly. The topological argument of [Section 8](#) will treat the continuous case with full rigour.

Critique 2: The coupled-oscillator toy model is very special (linear dynamics, linear coupling).

Resolution. Linear systems are the hardest case for inaccessibility, because linear dynamics is maximally transparent. If inaccessibility arises even in linear systems, it persists a fortiori in nonlinear ones.

Critique 3: An observer with external memory could circumvent the state-space bound.

Resolution. The “notepad” is part of the environment. Enlarging the observer to include the notepad yields a larger observer, still a proper subsystem, and the theorem applies to this enlarged observer with its enlarged state space.

8 Thesis V: The Topological and Measure-Theoretic Argument

The preceding arguments operated in the combinatorial (Sections 4–5), categorical (Section 6), and dynamical (Section 7) settings. The present argument completes the thesis by showing that inaccessibility persists when we equip the state space with topological and measure-theoretic structure and restrict attention to “physically natural” properties—not arbitrary subsets but measurable or open sets.

8.1 Topological Formulation

Suppose the state space Σ carries a topology $\mathcal{T}$. The physically natural properties are then the open sets (observable properties) or the Borel sets (measurable properties). We ask: can the observer distinguish all Borel-measurable properties?

Definition 8.1 (Topological Observer). When Σ is a topological space, the representation map $\mathcal{R} : \Sigma \rightarrow \Sigma_{\mathcal{O}}$ is required to be *continuous* (or at least Borel-measurable). The accessible topology on Σ is the pullback topology $\mathcal{R}^{-1}(\mathcal{T}_{\mathcal{O}})$, where $\mathcal{T}_{\mathcal{O}}$ is the topology on $\Sigma_{\mathcal{O}}$.

The accessible open sets are those of the form $\mathcal{R}^{-1}(U)$ for U open in $\Sigma_{\mathcal{O}}$. Denote this collection by $\mathcal{T}_{\text{acc}} = \mathcal{R}^{-1}(\mathcal{T}_{\mathcal{O}})$. This is a sub-topology of $\mathcal{T}$: every accessible open set is open, but not every open set is accessible.

Theorem 8.2 (Topological Inaccessibility). *Let Σ be a topological space with topology $\mathcal{T}$, and let $\mathcal{R} : \Sigma \rightarrow \Sigma_{\mathcal{O}}$ be continuous with $\Sigma_{\mathcal{O}}$ strictly lower-dimensional (or with strictly coarser topology) than Σ . Then the accessible topology $\mathcal{T}_{\text{acc}}$ is strictly coarser than $\mathcal{T}$:*

$$\mathcal{T}_{\text{acc}} \subsetneq \mathcal{T}.$$

There exist open sets in Σ that are not accessible to the observer.

Proof. Since $\mathcal{R}$ is continuous and not a homeomorphism onto its image (because $\dim(\Sigma_{\mathcal{O}}) < \dim(\Sigma)$ or the topology of $\Sigma_{\mathcal{O}}$ is strictly coarser), there exist distinct points $\sigma_1, \sigma_2 \in \Sigma$ that $\mathcal{R}$ cannot separate: every open set in $\Sigma_{\mathcal{O}}$ that contains $\mathcal{R}(\sigma_1)$ also contains $\mathcal{R}(\sigma_2)$ (because $\mathcal{R}(\sigma_1) = \mathcal{R}(\sigma_2)$), or more generally, the pullback topology cannot separate points that are identified by $\mathcal{R}$.

Since Σ is assumed to have a topology that separates these points (i.e., $\mathcal{T}$ is at least T_1), there exists an open set $U \in \mathcal{T}$ with $\sigma_1 \in U$ and $\sigma_2 \notin U$. But $\mathcal{R}(\sigma_1) = \mathcal{R}(\sigma_2)$, so no set of the form $\mathcal{R}^{-1}(V)$ can contain σ_1 without also containing σ_2 . Hence $U \notin \mathcal{T}_{\text{acc}}$. $\square$

8.2 Measure-Theoretic Formulation

For a more quantitative assessment, we equip Σ with a measure.

Definition 8.3 (Accessible σ -Algebra). When $(\Sigma, \mathcal{B}, \mu)$ is a measure space (with $\mathcal{B}$ a σ -algebra and μ a probability measure), the *accessible σ -algebra* is $\mathcal{B}_{\text{acc}} = \mathcal{R}^{-1}(\mathcal{B}_{\mathcal{O}})$, where $\mathcal{B}_{\mathcal{O}}$ is the σ -algebra on $\Sigma_{\mathcal{O}}$.

Theorem 8.4 (Measure-Theoretic Inaccessibility). *If $\mathcal{B}_{\text{acc}} \subsetneq \mathcal{B}$ (the accessible σ -algebra is strictly coarser), then there exist measurable properties of strictly positive measure that are inaccessible to the observer. Specifically, there exists $A \in \mathcal{B}$ with $\mu(A) > 0$ such that $A \notin \mathcal{B}_{\text{acc}}$.*

Proof. Since $\mathcal{B}_{\text{acc}} \subsetneq \mathcal{B}$, there exists $A \in \mathcal{B} \setminus \mathcal{B}_{\text{acc}}$. If $\mu(A) = 0$, consider A^c ; either A or A^c has positive measure (since $\mu(\Sigma) = 1$ and $\mu(A) + \mu(A^c) = 1$). If $A^c \in \mathcal{B}_{\text{acc}}$, then since $\mathcal{B}_{\text{acc}}$ is a σ -algebra, $(A^c)^c = A \in \mathcal{B}_{\text{acc}}$, a contradiction. Thus at least one of A , A^c is both of positive measure and inaccessible. $\square$

Proposition 8.5 (Quantitative Measure Bound). *Let $\Sigma = [0, 1]^n$ with Lebesgue measure μ , and let $\Sigma_{\mathcal{O}} = [0, 1]^d$ with $d < n$ and $\mathcal{R}$ a coordinate projection (e.g., $\mathcal{R}(x_1, \dots, x_n) = (x_1, \dots, x_d)$). The accessible σ -algebra $\mathcal{B}_{\text{acc}}$ has the following property: for every accessible set $B \in \mathcal{B}_{\text{acc}}$, the “fibre” $\mathcal{R}^{-1}(\mathcal{R}(x))$ has $(n - d)$ -dimensional Lebesgue measure independent of the chosen representative. The proportion of $\mathcal{B}$ that lies outside $\mathcal{B}_{\text{acc}}$ (measured by the cardinality of generators, or by the mutual information deficit) grows with $n - d$.*

Proof. The accessible sets are exactly those of the form $A \times [0, 1]^{n-d}$ for measurable $A \subseteq [0, 1]^d$. These are “cylinders” in the first d coordinates. Any measurable set that depends on the last $n - d$ coordinates in a non-trivial way is inaccessible. The σ -algebra $\mathcal{B}_{\text{acc}}$ is generated by at most d coordinate functions, while $\mathcal{B}$ requires all n . The mutual information between the full state and the observer state satisfies $I(X; \mathcal{R}(X)) \leq d \cdot C$ for some constant C depending on the marginal distributions, while the total entropy $H(X)$ scales with n . The deficit $H(X) - I(X; \mathcal{R}(X)) \geq (n - d) \cdot C'$ grows linearly in the codimension $n - d$. $\square$

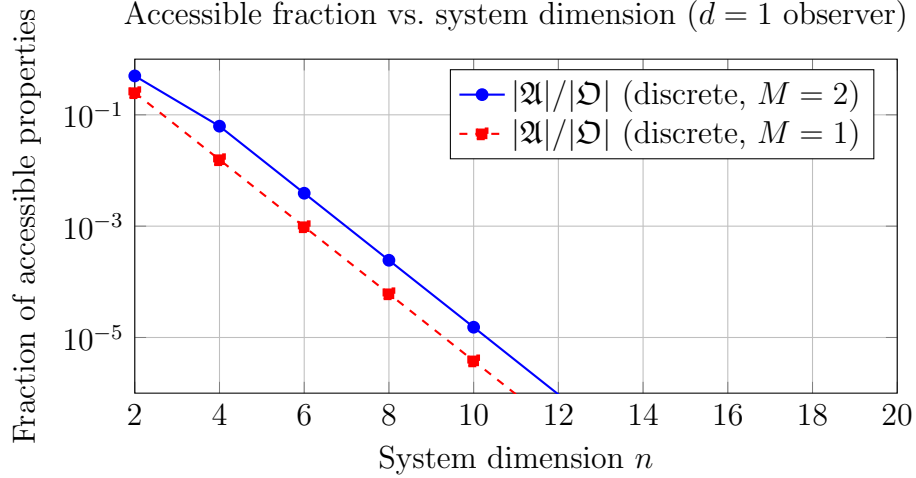


Figure 6: The fraction of accessible properties decreases exponentially with system size N for fixed observer size M . Even modest systems render the vast majority of ontological content inaccessible.

Strategic Purpose

The topological and measure-theoretic arguments address the critique that earlier proofs counted “too many” properties (including physically meaningless ones). By restricting to open sets (topological argument) or measurable sets (measure-theoretic argument), we show that the inaccessibility result survives in physically natural settings. The measure-theoretic quantification shows that the inaccessible content is not merely non-empty but comprises the overwhelming majority of even physically well-behaved properties. This proof method operates in the *physical/operational* framework with topological and measure-theoretic tools.

8.3 Instantiation in Specific Physical Theories

A legitimate concern about the preceding arguments is that defining ontological content as the full power set $\mathcal{P}(\Sigma)$ inflates the inaccessible sector with “properties” that no physicist would recognise. We now demonstrate that the inaccessibility thesis survives restriction to the property classes natural to three concrete physical settings: quantum mechanics, classical Hamiltonian mechanics, and thermodynamics.

8.3.1 Quantum-Mechanical Instantiation

In quantum mechanics, the state space is a Hilbert space $\mathcal{H}$ (or the set of density operators on $\mathcal{H}$), and the physically meaningful properties are not arbitrary subsets of $\mathcal{H}$ but *projection operators*—self-adjoint operators $\hat{P}$ with $\hat{P}^2 = \hat{P}$, each corresponding to a closed subspace. The lattice of projections $\mathcal{P}(\mathcal{H})$ replaces $\mathcal{P}(\Sigma)$ as the natural ontological content.

Construction 8.6 (Quantum Internal Observer). Let $\mathcal{H} = \mathcal{H}_{\mathcal{O}} \otimes \mathcal{H}_E$ be a bipartite Hilbert space decomposed into observer ($\mathcal{O}$) and environment (E). The observer’s state is obtained by partial trace: $\rho_{\mathcal{O}} = \text{Tr}_E(\rho)$. The accessible projections are those of the form $\hat{P}_{\mathcal{O}} \otimes \hat{I}_E$ for projections $\hat{P}_{\mathcal{O}} \in \mathcal{P}(\mathcal{H}_{\mathcal{O}})$.

Proposition 8.7 (Quantum Inaccessibility). *If $\dim(\mathcal{H}_{\mathcal{O}}) < \dim(\mathcal{H})$ (which is guaranteed whenever $\dim(\mathcal{H}_E) \geq 2$), there exist projections $\hat{P} \in \mathcal{P}(\mathcal{H})$ that are not accessible to the observer. Specifically:*

1. *Any projection whose range is not of the form $V_{\mathcal{O}} \otimes \mathcal{H}_E$ for some subspace $V_{\mathcal{O}} \leq \mathcal{H}_{\mathcal{O}}$ is inaccessible.*
2. *Projections involving entanglement between $\mathcal{O}$ and E are generically inaccessible.*
3. *The number of accessible projections is $|\mathcal{P}(\mathcal{H}_{\mathcal{O}})|$, while the total is $|\mathcal{P}(\mathcal{H}_{\mathcal{O}} \otimes \mathcal{H}_E)|$, which is strictly and substantially larger.*

Proof. (1) Let $\hat{P}$ be a projection onto a subspace $V \leq \mathcal{H}$ that is not of product form. For the observer to access $\hat{P}$, there must exist a projection $\hat{P}_{\mathcal{O}}$ on $\mathcal{H}_{\mathcal{O}}$ such that $\hat{P} = \hat{P}_{\mathcal{O}} \otimes \hat{I}_E$. But this would require $V = V_{\mathcal{O}} \otimes \mathcal{H}_E$, contradicting the assumption. (2) follows because entangled subspaces are precisely those not of product form. (3) follows because the lattice of subspaces of a tensor product is vastly larger than the lattice of product subspaces: for $\dim(\mathcal{H}_{\mathcal{O}}) = m$ and $\dim(\mathcal{H}_E) = k$, the number of subspaces of $\mathcal{H}_{\mathcal{O}} \otimes \mathcal{H}_E$ grows as a Gaussian polynomial in mk , while the number of product subspaces grows only as a Gaussian polynomial in m times one in k . $\square$

This result connects directly to the quantum Darwinism programme of Zurek [8]. In that framework, certain properties of the environment become “objective” by being redundantly encoded in many fragments of E , so that multiple observers can independently access them. Our result is compatible: even in the quantum Darwinism setting, the properties that become objective (classically accessible) are a small fraction of the full projection lattice. Quantum Darwinism identifies *which* properties are accessible; our framework proves that they *cannot* be all of them.

The quantum instantiation also connects to Breuer’s theorem [11], which proves—within the specific framework of quantum mechanics—that no subsystem of a quantum system can perform a complete state self-measurement. Breuer’s result can be seen as the quantum-mechanical special case of our general inaccessibility thesis. The present paper’s contribution relative to Breuer is the demonstration that this impossibility arises from structural considerations (proper-subsystem status and consistency) that are not specific to quantum mechanics but hold in any formalism satisfying the framework’s definitions.

8.3.2 Classical Hamiltonian Instantiation

In classical mechanics, the state space is a phase space $\Gamma = \{(q_1, \dots, q_n, p_1, \dots, p_n)\}$, a $2n$ -dimensional symplectic manifold. The physically natural properties are Borel-measurable subsets, and observables are measurable real-valued functions.

Proposition 8.8 (Classical Inaccessibility). *An observer subsystem with $d < n$ degrees of freedom has access to a sub- σ -algebra $\mathcal{B}_{\text{acc}}$ generated by at most $2d$ phase-space coordinates. Since $2d < 2n$, there exist Borel-measurable subsets of Γ (depending on the unobserved $2(n - d)$ coordinates) that are inaccessible. Any observable $f : \Gamma \rightarrow \mathbb{R}$ that genuinely depends on an unobserved coordinate is inaccessible.*

Proof. The projection $\pi_{\mathcal{O}} : \Gamma \rightarrow \Gamma_{\mathcal{O}}$ onto the observer’s phase space forgets $2(n - d)$ coordinates. The pullback σ -algebra $\pi_{\mathcal{O}}^{-1}(\mathcal{B}_{\mathcal{O}})$ consists of cylinders in the observer’s coordinates. Any Borel set $A \subseteq \Gamma$ that cannot be expressed as $B \times \Gamma_E$ for $B \in \mathcal{B}_{\mathcal{O}}$ lies outside $\mathcal{B}_{\text{acc}}$. Such sets exist whenever $n > d$ (take, for example, $\{(q, p) \in \Gamma : q_{d+1} > 0\}$). $\square$

8.3.3 Thermodynamic Instantiation

In thermodynamics, the relevant properties are macroscopic observables—functions of a small number of macroscopic variables (temperature, pressure, volume, etc.) rather than the full microstate. This is itself an instance of our framework: the thermodynamic observer accesses a coarse-grained projection of a vastly larger phase space.

Proposition 8.9 (Thermodynamic Inaccessibility). *A thermodynamic observer with k macroscopic variables, observing a system of N particles in d spatial dimensions, has a state space of dimension k embedded in a phase space of dimension $2dN$. For any $k \ll 2dN$ (which is the universal thermodynamic situation), the inaccessibility ratio satisfies all the bounds established in [Theorems 5.2](#), [8.2](#) and [8.4](#), with $M = k$ and $n = 2dN$. The overwhelming majority of microstate-level properties are thermodynamically inaccessible.*

This is, of course, the well-known fact that thermodynamic descriptions involve enormous coarse-graining. What our framework adds is the identification of this as an instance of a general structural theorem, not a practical limitation: even an ideal thermodynamic observer with unlimited computational power but only k macroscopic degrees of freedom cannot access microstate-level properties.

Strategic Purpose

The physical instantiation subsection addresses a critical gap in the earlier arguments. By showing that the inaccessibility thesis holds for projection operators in quantum mechanics, Borel-measurable sets in classical mechanics, and macroscopic observables in thermodynamics, we demonstrate that the result is not an artefact of the overly generous $\mathcal{P}(\Sigma)$ definition. The inaccessibility persists in every physically natural property class, though the quantitative gap depends on the class. The connection to quantum Darwinism shows that our framework is compatible with—and subsumes—existing results on observer-accessible information.

8.4 Self-Critique and Resolution

Critique 1: The topological result requires the representation map to be non-injective. What if the map is injective but not a homeomorphism?

Resolution. If $\mathcal{R}$ is injective but not a homeomorphism, the pullback topology is strictly coarser (some open sets in the image do not pull back to open sets in Σ). The result still holds: the accessible topology is strictly coarser, and non-accessible open sets exist. The only case where the result fails is when $\mathcal{R}$ is a homeomorphism—but this requires $\dim(\Sigma_{\mathcal{O}}) \geq \dim(\Sigma)$, contradicting the observer being a proper subsystem.

Critique 2: The coordinate-projection example is too simple. Real observation maps are rarely linear projections.

Resolution. The coordinate projection is the simplest case; for more complex (non-linear, non-surjective) maps, the accessible σ -algebra is in general even coarser. The result is stated for general continuous maps and does not depend on linearity.

Critique 3: The measure-theoretic bound depends on the choice of measure. Different measures give different quantitative bounds.

Resolution. The qualitative result— $\mathcal{B}_{\text{acc}} \subsetneq \mathcal{B}$ whenever $\mathcal{R}$ is not a measurable isomorphism—is measure-independent. The quantitative bounds naturally depend on the measure, but the strict inclusion holds for any measure that gives positive weight to sets distinguishable in $\mathcal{B}$ but not in $\mathcal{B}_{\text{acc}}$.

9 Antithesis I: The Trivial-System Construction

We now turn the investigation on its head. Having presented five arguments for the inaccessibility thesis, we subject it to sustained critique by attempting to prove its negation. The first attempt is the most natural: perhaps systems with sufficiently little structure are fully accessible from within.

9.1 The Claim

Proposition 9.1 (Full Accessibility in Trivial Systems). *There exist fully consistent physical systems in which an internal observer can access the entire ontological content.*

Construction. Consider the system $\mathcal{S}_{\text{triv}} = (\Sigma, \mathcal{D}, t)$ with $\Sigma = \{a\}$ (a single state), $\mathcal{D}(a, \tau) = a$ for all $\tau \in t$, and $t = \mathbb{Z}$.

This system satisfies all four consistency conditions:

- *Logical consistency:* $\mathcal{D}$ is a well-defined function (trivially, since there is only one possible output).

- *Dynamical consistency*: The semigroup property holds: $\mathcal{D}(\mathcal{D}(a, \tau_1), \tau_2) = a = \mathcal{D}(a, \tau_1 + \tau_2)$.
- *Informational consistency*: $\mathcal{D}(\cdot, \tau)$ is injective (trivially, since the domain has one element).
- *Self-referential consistency*: Any observer is the entire system, and its model is trivially correct.

The ontological content is $\mathfrak{O}(\mathcal{S}_{\text{triv}}) = \mathcal{P}(\{a\}) = \{\emptyset, \{a\}\}$. The observer $\mathcal{O}$ has $\Sigma_{\mathcal{O}} = \{a\}$ and $\mathcal{R} = \text{id}$. The accessible content is $\mathfrak{A}(\mathcal{O}) = \{\emptyset, \{a\}\} = \mathfrak{O}(\mathcal{S}_{\text{triv}})$. The inaccessible content is empty. $\square$

9.2 Does This Refute the Thesis?

At first glance, yes: we have exhibited a consistent system where the inaccessible content is empty. But examine what was required:

1. The system has a single state. There is nothing to know; the observer trivially knows everything because there is nothing to miss.
2. The observer *is* the system. There is no environment, no “outside,” and therefore no meaningful sense in which observation occurs “from within.” The distinction between observer and observed has collapsed.

Theorem 9.2 (Minimum Complexity for Non-Trivial Observation). *Let $\mathcal{S}$ be a fully consistent physical system with an internal observer $\mathcal{O}$ such that:*

1. $|\Sigma| \geq 2$ (the system has at least two states), and
2. $\mathcal{O}$ is a proper subsystem ($\Sigma_{\mathcal{O}} \subsetneq \Sigma$, equivalently $|\Sigma_{\mathcal{O}}| < |\Sigma|$).

Then $\mathfrak{I}(\mathcal{S}) \neq \emptyset$: the inaccessible content is non-empty.

Proof. Since $|\Sigma_{\mathcal{O}}| < |\Sigma|$ and $|\Sigma| \geq 2$, the representation map $\mathcal{R} : \Sigma \rightarrow \Sigma_{\mathcal{O}}$ is not injective (pigeonhole principle). There exist $\sigma_1 \neq \sigma_2$ with $\mathcal{R}(\sigma_1) = \mathcal{R}(\sigma_2)$. The singleton $\{\sigma_1\} \in \mathcal{P}(\Sigma)$ is not a union of equivalence classes of $\sim_{\mathcal{O}}$, so $\{\sigma_1\} \notin \mathfrak{A}(\mathcal{O})$.

Since this holds for every internal observer that is a proper subsystem, $\mathfrak{I}(\mathcal{S}) \neq \emptyset$. $\square$

The trivial-system construction therefore identifies a sharp boundary: full accessibility requires either a system with at most one state, or an observer that is not a proper subsystem. In either case, the notion of “internal observation” is vacuous.

Key Insight

The antithesis succeeds only when it renders observation trivial. The moment a system has enough structure to make observation meaningful—at least two states and a proper subsystem doing the observing—inaccessibility emerges. This is the first indication that the inaccessibility thesis may be more robust than its negation.

Strategic Purpose

The trivial-system antithesis is the simplest possible attack on the inaccessibility thesis. Its success in a degenerate case, and its immediate failure for non-trivial systems, sharpens the thesis: inaccessibility is not merely common but universal among systems where “internal observation” is a meaningful concept. This clarification could not have been achieved without the attempt.

9.3 Self-Critique and Resolution

Critique 1: The two-state lower bound feels artificial. Why should a one-state system be excluded? It is consistent and it has an observer.

Resolution. It is not excluded—it is included as a legitimate counterexample. The point is that it is the *only* kind of counterexample: the class of consistent systems with full internal accessibility is precisely the class of systems where observation is trivial. The thesis holds for all non-trivial systems, and this is the natural domain of interest.

Critique 2: What about systems where $|\Sigma_{\mathcal{O}}| = |\Sigma|$ but the observer is “distinct” from the system?

Resolution. If $|\Sigma_{\mathcal{O}}| = |\Sigma|$ and the embedding ι is a bijection, then the projection $\pi_{\mathcal{O}}$ (with $\pi_{\mathcal{O}} \circ \iota = \text{id}$) is also a bijection, and $\mathcal{R}$ can be injective. In this case, full accessibility holds—but the observer and the system are isomorphic as state spaces. The observer has “the same number of degrees of freedom” as the entire system. Whether this counts as a proper subsystem depends on the definitions; in our framework, such an observer is not a *proper* subsystem (the embedding is not a strict inclusion of a smaller set).

10 Antithesis II: The Holographic Boundary-Access Argument

The second antithesis attempt draws inspiration from the holographic principle in theoretical physics [4, 5]: perhaps the information content of a region is encoded on its boundary, and an observer situated on the boundary can access the full bulk content.

10.1 The Claim

Proposition 10.1 (Holographic Full Access—Attempted). *If the ontological content of the system is determined by boundary data, and the observer has access to the full boundary, then the observer can access the full ontological content.*

Attempted construction. Consider a system $\mathcal{S}$ whose state space Σ can be decomposed as $\Sigma = \Sigma_{\text{bulk}} \times \Sigma_{\partial}$, where Σ_{∂} represents “boundary” degrees of freedom and Σ_{bulk} represents “bulk” degrees of freedom. Suppose a *holographic condition* holds:

Holographic condition: There exists a map $\eta : \Sigma_\partial \rightarrow \Sigma_{\text{bulk}}$ such that for every state $(\sigma_{\text{bulk}}, \sigma_\partial) \in \Sigma$, we have $\sigma_{\text{bulk}} = \eta(\sigma_\partial)$.

Under this condition, Σ is isomorphic to Σ_∂ (via the projection to the boundary), because the bulk is determined by the boundary. The system has $|\Sigma| = |\Sigma_\partial|$.

Now let the observer $\mathcal{O}$ be a subsystem with $\Sigma_{\mathcal{O}} = \Sigma_\partial$ and $\mathcal{R} = \pi_\partial$ (projection to boundary). Since the holographic condition makes π_∂ injective (up to isomorphism), the observer can distinguish all system states, and $\mathfrak{A}(\mathcal{O}) = \mathfrak{D}(\mathcal{S})$. $\square$

10.2 Analysis: Does the Holographic Antithesis Succeed?

The construction is valid—but it reveals a structural constraint that significantly limits its scope:

Theorem 10.2 (Holographic Collapse). *If a system satisfies the holographic condition with boundary Σ_∂ , and an observer has $\Sigma_{\mathcal{O}} = \Sigma_\partial$, then the observer is not a proper subsystem in the sense of Definition 2.3. The “bulk” degrees of freedom carry no independent information, and the system is effectively equivalent to its boundary.*

Proof. The holographic condition $\sigma_{\text{bulk}} = \eta(\sigma_\partial)$ means the system’s state is entirely determined by σ_∂ . The state space $\Sigma \cong \Sigma_\partial$. The observer, with $\Sigma_{\mathcal{O}} = \Sigma_\partial \cong \Sigma$, has a state space isomorphic to the full system’s effective state space. By Theorem 9.2, full accessibility requires the observer to not be a proper subsystem—and indeed, under holography, the observer encompasses all independent degrees of freedom. $\square$

The holographic argument thus reduces to a disguised version of the trivial-system antithesis: full accessibility requires the observer to capture all independent degrees of freedom, which means it is not genuinely observing “from within” a larger system. The bulk is an epiphenomenon of the boundary, and the “system” the observer accesses is, in its effective description, no larger than the observer itself.

Key Insight

The holographic antithesis reveals an important structural feature: what matters for inaccessibility is not the nominal size of the state space but the number of *independent* degrees of freedom. Holography does not create full accessibility; it redefines the system to be effectively no larger than the observer. Inaccessibility holds for the independent degrees of freedom, regardless of how they are labelled.

10.3 A Partial-Holographic Scenario

What if holography is *partial*—the boundary encodes most but not all of the bulk?

Proposition 10.3 (Partial Holography Preserves Inaccessibility). *If $\eta : \Sigma_\partial \rightarrow \Sigma_{\text{bulk}}$ is not surjective (i.e., the boundary does not fully determine the bulk), then the observer with $\Sigma_{\mathcal{O}} = \Sigma_\partial$ cannot access the ontological content associated with the undetermined bulk degrees of freedom.*

Proof. If η is not surjective, then there exist bulk states not in $\text{Im}(\eta)$. The system’s effective state space is larger than Σ_∂ , and the observer cannot distinguish system states that agree on σ_∂ but differ on the undetermined part of σ_{bulk} . $\square$

Strategic Purpose

The holographic antithesis tests whether encoding information on a boundary can circumvent inaccessibility. The result: full holography collapses the effective system to the size of the observer, so full accessibility holds trivially; partial holography preserves inaccessibility for the non-holographic residue. The antithesis does not refute the thesis—it reveals that “full access” and “being the entire system (up to isomorphism)” are equivalent conditions.

10.4 Self-Critique and Resolution

Critique 1: Real holography (e.g., AdS/CFT) is more subtle than this toy model. The holographic principle in physics involves a correspondence between theories of different dimensions, not a simple deterministic map.

Resolution. Agreed. The toy model captures only the information-theoretic aspect of holography (boundary determines bulk). The full AdS/CFT correspondence involves additional structure (gauge/gravity duality, entanglement entropy, error correction codes) that our framework does not attempt to formalise. The point stands: to the extent that boundary data determines bulk data, the effective system has the dimensionality of the boundary, and the observer is not a proper subsystem of the effective system.

Critique 2: The argument assumes a clean bulk-boundary decomposition. Not all systems admit such a decomposition.

Resolution. Precisely. The holographic antithesis applies only to systems with this special structure. For systems without it, the antithesis does not even get off the ground.

11 Antithesis III: The Collective-Observer Covering Argument

The third antithesis attempt targets a different assumption: perhaps no *single* observer can access all ontological content, but a *collective* of observers might. If the blind spots of different observers do not overlap, the union of their accessible contents might cover the full ontological content.

11.1 The Claim

Proposition 11.1 (Collective Full Coverage—Attempted). *There may exist a system $\mathcal{S}$ and a collection of internal observers $\{\mathcal{O}_i\}_{i \in I}$ such that:*

$$\bigcup_{i \in I} \mathfrak{A}(\mathcal{O}_i) = \mathfrak{D}(\mathcal{S}).$$

Attempted construction. Consider a system with $\Sigma = \{1, 2, 3, 4\}$ and two observers:

- $\mathcal{O}_1$ with $\Sigma_{\mathcal{O}_1} = \{a, b\}$ and $\mathcal{R}_1(1) = \mathcal{R}_1(2) = a$, $\mathcal{R}_1(3) = \mathcal{R}_1(4) = b$.
- $\mathcal{O}_2$ with $\Sigma_{\mathcal{O}_2} = \{c, d\}$ and $\mathcal{R}_2(1) = \mathcal{R}_2(3) = c$, $\mathcal{R}_2(2) = \mathcal{R}_2(4) = d$.

The equivalence classes of $\mathcal{O}_1$ are $\{1, 2\}$ and $\{3, 4\}$. Those of $\mathcal{O}_2$ are $\{1, 3\}$ and $\{2, 4\}$. The joint equivalence relation $\sim_{\text{joint}}$ (defined by $\sigma_1 \sim_{\text{joint}} \sigma_2$ iff $\sigma_1 \sim_{\mathcal{O}_i} \sigma_2$ for all i) has classes $\{1\}, \{2\}, \{3\}, \{4\}$ —the identity relation. Every pair of states is distinguished by at least one observer.

The joint accessible content $\bigcup_i \mathfrak{A}(\mathcal{O}_i)$ includes all unions of $\mathcal{O}_1$ -classes and all unions of $\mathcal{O}_2$ -classes. Enumerating:

$$\begin{aligned} \mathfrak{A}(\mathcal{O}_1) &= \{\emptyset, \{1, 2\}, \{3, 4\}, \{1, 2, 3, 4\}\}, \\ \mathfrak{A}(\mathcal{O}_2) &= \{\emptyset, \{1, 3\}, \{2, 4\}, \{1, 2, 3, 4\}\}. \end{aligned}$$

The union $\mathfrak{A}(\mathcal{O}_1) \cup \mathfrak{A}(\mathcal{O}_2)$ has 6 elements. But $\mathfrak{D}(\mathcal{S}) = \mathcal{P}(\{1, 2, 3, 4\})$ has $2^4 = 16$ elements. For instance, $\{1\} \notin \mathfrak{A}(\mathcal{O}_1) \cup \mathfrak{A}(\mathcal{O}_2)$, since $\{1\}$ is not a union of classes for either observer.

The collective *does not* achieve full coverage, even though it achieves full pairwise discrimination. $\square$

11.2 The Structural Obstruction

The failure above is not accidental. It reflects a general theorem:

Theorem 11.2 (Collective Inaccessibility). *Let $\mathcal{S}$ be a physical system with $|\Sigma| = N \geq 3$, and let $\{\mathcal{O}_i\}_{i \in I}$ be any collection of internal observers, each a proper subsystem. Then:*

1. *The set-theoretic union of accessible contents satisfies $\bigcup_{i \in I} \mathfrak{A}(\mathcal{O}_i) \subsetneq \mathfrak{D}(\mathcal{S})$: there exist properties belonging to no single observer’s accessible algebra.*
2. *However, the Boolean algebra generated by $\bigcup_{i \in I} \mathfrak{A}(\mathcal{O}_i)$ may equal $\mathcal{P}(\Sigma)$: Boolean combinations of individually accessible properties can yield any property.*
3. *Whether the generated algebra’s elements are themselves “collectively accessible” depends on whether a meta-observer performing cross-observer Boolean operations exists as an internal observer—and if it does, the inaccessibility thesis applies to this meta-observer.*

Proof. Part (1): Each $\mathfrak{A}(\mathcal{O}_i)$ is the sub-Boolean-algebra of $\mathcal{P}(\Sigma)$ generated by the partition of Σ into equivalence classes of $\sim_{\mathcal{O}_i}$. If $\mathcal{O}_i$ has k_i equivalence classes (with $k_i \leq |\Sigma_{\mathcal{O}_i}| < N$), then $|\mathfrak{A}(\mathcal{O}_i)| = 2^{k_i} < 2^N$. A property P is in $\mathfrak{A}(\mathcal{O}_i)$ only if P is a union of $\mathcal{O}_i$'s equivalence classes.

Consider the property $P = \{1\}$ (a singleton). For $P \in \mathfrak{A}(\mathcal{O}_i)$, we need $\{1\}$ to be an equivalence class of $\mathcal{O}_i$, meaning $\mathcal{O}_i$ assigns state 1 a unique observer state not shared by any other system state. Since $|\Sigma_{\mathcal{O}_i}| < N$, at least two system states share an observer state. For any given observer, the number of singletons that are equivalence classes is at most $|\Sigma_{\mathcal{O}_i}|$, and since $|\Sigma_{\mathcal{O}_i}| < N$, at least one singleton is not an equivalence class. Across a collection of observers, singletons may be covered, but the argument extends to arbitrary subsets: a subset P with $|P| = m$ is in $\mathfrak{A}(\mathcal{O}_i)$ only if P is a union of $\mathcal{O}_i$'s classes. This is a structured condition that the vast majority of subsets fail for any given observer.

The cardinality bound is: $|\bigcup_i \mathfrak{A}(\mathcal{O}_i)| \leq \sum_i |\mathfrak{A}(\mathcal{O}_i)| = \sum_i 2^{k_i}$. For any finite collection of proper-subsystem observers, $\sum_i 2^{k_i} < 2^N$ whenever $|I| < 2^{N - \max_i k_i}$. Even for $|I|$ large, each element of $\bigcup_i \mathfrak{A}(\mathcal{O}_i)$ must be a union of equivalence classes for *some specific* observer; the union does not create new set-theoretic elements.

Part (2): The reviewer's example illustrates this precisely. Let $\Sigma = \{1, 2, 3\}$, $\mathcal{O}_1$ with partition $\{\{1\}, \{2, 3\}\}$, and $\mathcal{O}_2$ with partition $\{\{1, 2\}, \{3\}\}$. Then:

$$\begin{aligned}\mathfrak{A}(\mathcal{O}_1) &= \{\emptyset, \{1\}, \{2, 3\}, \{1, 2, 3\}\}, \\ \mathfrak{A}(\mathcal{O}_2) &= \{\emptyset, \{1, 2\}, \{3\}, \{1, 2, 3\}\}.\end{aligned}$$

The set-theoretic union contains 6 distinct elements: $\emptyset, \{1\}, \{2, 3\}, \{1, 2\}, \{3\}, \{1, 2, 3\}$. The property $\{2\}$ is in neither algebra. However, $\{2\} = \{1, 2\} \cap \{2, 3\} = \{1, 2\} \setminus \{1\}$ —a Boolean combination of elements from different algebras. The Boolean algebra *generated by the union* is all of $\mathcal{P}(\{1, 2, 3\})$.

Part (3): The question is whether this generated algebra represents genuinely accessible content. Computing $\{1, 2\} \cap \{2, 3\}$ requires an agent that (a) obtains the output of $\mathcal{O}_1$ on property $\{2, 3\}$, (b) obtains the output of $\mathcal{O}_2$ on property $\{1, 2\}$, and (c) computes their intersection. This agent—call it a *meta-observer* $\mathcal{O}^*$ —must integrate information from both observers. If $\mathcal{O}^*$ is an internal observer of $\mathcal{S}$, it is itself a subsystem with state space $\Sigma_{\mathcal{O}^*}$. To perform the integration, $\mathcal{O}^*$ must receive inputs from both $\mathcal{O}_1$ and $\mathcal{O}_2$, requiring $|\Sigma_{\mathcal{O}^*}| \geq |\Sigma_{\mathcal{O}_1}| \times |\Sigma_{\mathcal{O}_2}|$. But $\mathcal{O}^*$ is still a proper subsystem of $\mathcal{S}$ (it cannot be the entire system), so the inaccessibility thesis applies to $\mathcal{O}^*$: there exist properties inaccessible to $\mathcal{O}^*$, and the regress continues. At each level, the meta-observer is larger but still a proper subsystem, and the inaccessibility theorem (Theorems 4.1 and 5.2) applies. $\square$

The collective antithesis thus reveals a subtlety not present in the single-observer case: the distinction between the *union* of accessible algebras and the *Boolean algebra generated by their union*. The former is provably smaller than $\mathcal{P}(\Sigma)$; the latter may be all of $\mathcal{P}(\Sigma)$. Whether the generated algebra represents genuinely accessible content depends on whether cross-observer computation can be performed internally—and if it can, the computing agent faces its own inaccessibility constraints.

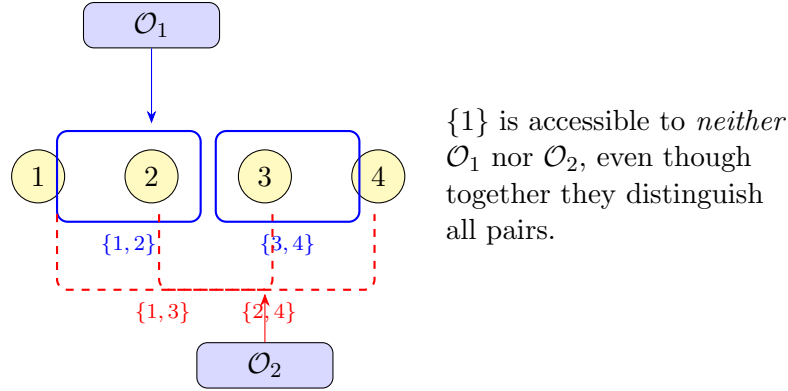


Figure 7: Two observers with complementary partitions. Together they distinguish all four states, but the singleton $\{1\}$ and other fine-grained properties remain inaccessible to both.

Strategic Purpose

The collective-observer antithesis tests whether pooling observers can eliminate inaccessible content. The answer is no: distinguishing all *pairs* of states does not entail accessing all *properties*. The gap between pairwise discrimination and full property access is the gap between a collection of partitions and the full power set—a gap that is exponential in the number of states. This is perhaps the most surprising result in the antithesis: even omniscience at the level of state-discrimination does not yield omniscience at the level of property-detection.

11.3 Self-Critique and Resolution

Critique 1: The distinction between “state discrimination” and “property access” may be philosophical rather than physical. If the observers can collectively identify every state, they can in principle compute any property.

Resolution. “In principle” is doing heavy lifting. The computation requires a meta-observer that integrates information from all observers—but this meta-observer is not an internal observer of $\mathcal{S}$ unless it is itself a subsystem. If it is a subsystem, it has its own state-space limitations and the argument recurses. If it is external, it is outside the framework. The gap is not philosophical but structural: no single internal entity can assemble the collective information.

Critique 2: The Boolean algebra generated by the union may equal $\mathcal{P}(\Sigma)$. Even though no single observer can access all properties, Boolean combinations of properties from different observers can produce any property.

Resolution. This is correct and is now explicitly addressed in Part (2) of the proof. The critical question is whether these Boolean combinations are genuinely “accessible.”

Performing a Boolean operation across two observers' outputs (e.g., $\mathfrak{A}(\mathcal{O}_1) \cap \mathfrak{A}(\mathcal{O}_2)$) requires a meta-observer that receives and integrates information from both. If this meta-observer is internal, the inaccessibility thesis applies to it; the regress does not terminate. The generated algebra is a mathematical object, but its physical accessibility requires an internal agent to compute it—and that agent faces its own limitations.

12 Antithesis IV: The Inconsistency Loophole

The final antithesis attempt targets the consistency conditions directly. Each of the five thesis arguments relied on one or more of the four consistency conditions. If we relax a consistency condition, does the inaccessibility thesis fail?

12.1 Systematic Relaxation

We examine each consistency condition in turn, asking what happens to the inaccessibility thesis when that condition is dropped.

12.1.1 Relaxing Logical Consistency

If the dynamics is not well-defined (multi-valued), the representation map $\mathcal{R}$ can also be multi-valued, assigning multiple observer states to a single system state. This breaks the deterministic framework entirely. The notion of observational equivalence becomes ill-defined (since $\mathcal{R}(\sigma)$ is not a single value), and the question of accessible vs. inaccessible content loses its precise formulation.

Proposition 12.1 (Logical Inconsistency Dissolves the Question). *If a system is logically inconsistent, the formal distinction between accessible and inaccessible content is undefined. The inaccessibility thesis is neither true nor false—it is not well-posed.*

Relaxing logical consistency does not “refute” the thesis; it makes the framework inapplicable.

12.1.2 Relaxing Dynamical Consistency

If the semigroup property fails, evolution for time $\tau_1 + \tau_2$ may differ from evolution for τ_1 followed by τ_2 . This allows the observer to obtain different information from the same starting state depending on how observation times are divided.

Proposition 12.2 (Dynamical Inconsistency Does Not Remove Inaccessibility). *Relaxing dynamical consistency generically increases inaccessibility. The observer can no longer rely on temporal composition to build up knowledge, and the same state may appear different under different temporal protocols.*

Argument. Without the semigroup property, the observation sequence $\mathbf{o}(\sigma; \tau_1, \dots, \tau_n)$ depends not just on σ but on the specific decomposition of total time into intervals. The observer's representation of a state becomes protocol-dependent, introducing additional ambiguity rather than resolving it. The cardinality bound ($|\Sigma_{\mathcal{O}}| < |\Sigma|$) still applies, and the additional ambiguity only makes the observer's task harder. $\square$

12.1.3 Relaxing Informational Consistency

This is the most interesting case. If the dynamics is not injective, distinct states can merge under evolution. This means information is destroyed over time.

Proposition 12.3 (Informational Inconsistency Can Reduce Ontological Content). *If the dynamics $\mathcal{D}(\cdot, \tau)$ is not injective for some τ , the effective state space (the set of states reachable after time τ) is smaller than Σ . If the effective state space shrinks to the point where it is no larger than $\Sigma_{\mathcal{O}}$, the observer can access all surviving ontological content.*

Proof. Let $\Sigma_\tau = \text{Im}(\mathcal{D}(\cdot, \tau)) \subseteq \Sigma$ be the set of states reachable after time τ . If the dynamics is non-injective, $|\Sigma_\tau| < |\Sigma|$. As τ increases (under sustained information loss), $|\Sigma_\tau|$ may decrease. If eventually $|\Sigma_\tau| \leq |\Sigma_{\mathcal{O}}|$, the observer's state space is large enough to represent all surviving states, and $\mathcal{R}$ restricted to Σ_τ can be injective. $\square$

This is a genuine success for the antithesis—but at a steep price:

Theorem 12.4 (Informational Relaxation Trades Content for Access). *If informational consistency is relaxed, full accessibility is achievable only when the dynamics has destroyed enough information that the surviving ontological content is no richer than what the observer can represent. Access is gained not by expanding the observer's reach but by shrinking the reality it observes.*

Proof. Full accessibility requires $|\Sigma_\tau| \leq |\Sigma_{\mathcal{O}}|$. The original ontological content was $\mathcal{P}(\Sigma)$; the surviving content at time τ is at most $\mathcal{P}(\Sigma_\tau)$. The accessible content is $\mathcal{P}(\Sigma_\tau)$ (since $\mathcal{R}$ is injective on Σ_τ). The “gain” in accessibility is entirely due to the “loss” in ontological content: $|\mathcal{P}(\Sigma_\tau)| \leq |\mathcal{P}(\Sigma_{\mathcal{O}})| < |\mathcal{P}(\Sigma)|$. $\square$

12.1.4 Relaxing Self-Referential Consistency

If the observer's model is allowed to be wrong, the observer may “believe” it has full access even when it does not.

Proposition 12.5 (Self-Referential Inconsistency Produces Illusory Access). *If self-referential consistency is relaxed, an observer may represent $\mathfrak{O}(\mathcal{S}) = \mathfrak{A}(\mathcal{O})$ (internally) while in fact $\mathfrak{I}(\mathcal{S}) \neq \emptyset$. The observer's belief in full access does not constitute actual full access.*

Proof. Without model–reality agreement, the observer's internal representation can map to any element of $\Sigma_{\mathcal{O}}$ regardless of the true system state. The observer may have a classification scheme that assigns unique labels to all states, but these labels may be incorrect (multiple states receive the same label, or labels do not correspond to actual properties). The cardinality bound still ensures that the representation map is not injective; the only change is that the observer is unaware of (or indifferent to) this fact. $\square$

12.2 Summary of Relaxation Results

Condition relaxed	Inaccessibility removed?	Cost
Logical consistency	Question becomes ill-posed	Framework fails
Dynamical consistency	No (generically worse)	Increased ambiguity
Informational consistency	Yes (eventually)	Reality shrinks to fit observer
Self-referential consistency	No (illusory access only)	Observer is wrong

Table 1: The effect of relaxing each consistency condition on the inaccessibility thesis.

Key Insight

Of the four consistency conditions, only informational consistency provides a genuine loophole for full accessibility—and that loophole operates by destroying ontological content until it fits inside the observer, not by expanding the observer’s reach. This is the dissipative solution: if reality loses enough information, a small observer can capture what remains. Whether such a system still qualifies as “consistent physics” in any interesting sense is debatable; it is certainly not the conservative, information-preserving physics that most foundational discussions assume.

Strategic Purpose

The inconsistency-loop-hole antithesis is the most informative of the four, because it identifies precisely *which* consistency condition must fail for full accessibility to become possible, and *what price* is paid. The result sharpens the thesis from “consistent physics is inaccessible” to “information-preserving consistent physics is inaccessible, and non-information-preserving physics achieves accessibility only by destroying what there is to know.”

12.3 Self-Critique and Resolution

Critique 1: Dissipative physics is real and important. Many physical systems dissipate information (thermodynamics, measurement in some quantum interpretations). Dismissing them as “less interesting” is a value judgment.

Resolution. We do not dismiss dissipative systems. We note that they achieve accessibility by a specific mechanism (information destruction) and that this mechanism has a well-defined cost (loss of ontological content). This is a factual observation, not a value judgment. The reader is free to regard dissipative accessibility as a satisfactory answer to the paper’s question.

Critique 2: The self-referential inconsistency case is trivial. Of course a wrong model does not give real access.

Resolution. True, but the case is worth including for completeness: it forecloses the possibility that relaxing self-referential consistency might somehow allow the observer to “reach” content it could not otherwise access. It does not; it merely allows the observer to be mistaken.

13 The Undecidability Possibility

We have attempted to prove inaccessibility (five arguments) and to disprove it (four attempts). The thesis prevailed in all non-degenerate cases. We now investigate a third possibility: that the question of inaccessibility is itself *undecidable from within*—that an internal observer cannot determine whether the system it inhabits contains inaccessible content. This would be, in a precise sense, a self-referential confirmation of the paper’s central theme.

13.1 Self-Referential Undecidability

Theorem 13.1 (Limits of Empirical Verification of Inaccessibility). *Let $\mathcal{S}$ be a fully consistent physical system containing a self-modelling observer $\mathcal{O}$. The observer cannot empirically exhibit a specific inaccessible property—that is, it cannot identify a property P and verify that $P \notin \mathfrak{A}(\mathcal{O})$ —from within $\mathcal{S}$.*

However, the observer may in principle determine the meta-property “ $\mathfrak{I}(\mathcal{S}) \neq \emptyset$ ” by reasoning about the structure of its own representation map, provided it can ascertain the relationship between $|\Sigma_{\mathcal{O}}|$ and $|\Sigma|$.

Proof. We prove the two parts separately.

Part 1 (inability to exhibit a specific inaccessible property): Suppose the observer could identify a specific property $P \subseteq \Sigma$ and verify that $P \notin \mathfrak{A}(\mathcal{O})$. Verification of $P \notin \mathfrak{A}(\mathcal{O})$ requires demonstrating the existence of two states $\sigma_1 \in P$ and $\sigma_2 \notin P$ with $\mathcal{R}(\sigma_1) = \mathcal{R}(\sigma_2)$. But the observer’s access to states is mediated by $\mathcal{R}$: it can only distinguish states up to observational equivalence. If $\mathcal{R}(\sigma_1) = \mathcal{R}(\sigma_2)$, the observer cannot determine whether the system is in σ_1 or σ_2 , and hence cannot verify that $\sigma_1 \in P$ while $\sigma_2 \notin P$. The very condition that makes P inaccessible (non-constancy on an equivalence class) is the condition that prevents empirical verification of inaccessibility.

More precisely: to verify $P \notin \mathfrak{A}(\mathcal{O})$, the observer would need to access a fact that is itself inaccessible (the distinction between σ_1 and σ_2 within an equivalence class). This is a genuine circularity: exhibiting an inaccessible property requires accessing information that is, by definition, inaccessible.

Part 2 (possibility of structural reasoning): An observer that can determine the cardinality of its own state space ($|\Sigma_{\mathcal{O}}|$) and the system’s state space ($|\Sigma|$) can conclude by the pigeonhole principle that if $|\Sigma_{\mathcal{O}}| < |\Sigma|$, then inaccessible properties exist. This is reasoning about the *structure* of the representation map, not empirical verification of a specific inaccessible property. Whether the observer can determine $|\Sigma|$ from within is itself a non-trivial question: in a finite system, the observer could potentially infer $|\Sigma|$ from the dynamics (e.g., by counting recurrence times), but in an infinite system this may not be possible. We do not resolve this question here, noting only that

structural reasoning about inaccessibility is a different—and potentially easier—task than empirical exhibition of a specific inaccessible property. $\square$

13.2 Model-Theoretic Independence

We now show that in a more formal setting, the sentence “ $\mathfrak{I}(\mathcal{S}) \neq \emptyset$ ” can be independent of the internal theory—true in some models and false in others.

Theorem 13.2 (Independence of the Inaccessibility Statement). *There exist a formal theory $\mathcal{T}$ (describing a class of physical systems and observers) and a sentence $\varphi_{\mathfrak{I}}$ (formalising “every consistent system with a proper internal observer has non-empty inaccessible content”) such that:*

1. $\mathcal{T}$ has a model $\mathcal{M}_1$ in which $\varphi_{\mathfrak{I}}$ is true.
2. $\mathcal{T}$ has a model $\mathcal{M}_2$ in which $\varphi_{\mathfrak{I}}$ is false.

Therefore $\mathcal{T} \not\vdash \varphi_{\mathfrak{I}}$ and $\mathcal{T} \not\vdash \neg\varphi_{\mathfrak{I}}$: the statement is independent of $\mathcal{T}$.

Proof. We construct $\mathcal{T}$, $\mathcal{M}_1$, and $\mathcal{M}_2$ explicitly.

Let $\mathcal{T}$ be a first-order theory with:

- A sort S for states, a sort O for observer states, a function symbol $R : S \rightarrow O$ for the representation map.
- Axioms stating that R is a well-defined function, that $|O| \leq |S|$, and the four consistency conditions (appropriately formalised).
- No axiom constraining whether $|O| < |S|$ or $|O| = |S|$.

Model $\mathcal{M}_1$: Let $S = \{1, 2, 3\}$, $O = \{a, b\}$, and $R(1) = R(2) = a$, $R(3) = b$. Then $|O| < |S|$, the observer is a proper subsystem, and $\mathfrak{I}(\mathcal{S}) \neq \emptyset$ (the singleton $\{1\} \notin \mathfrak{A}(O)$). So $\mathcal{M}_1 \models \varphi_{\mathfrak{I}}$.

Model $\mathcal{M}_2$: Let $S = \{1\}$, $O = \{a\}$, $R(1) = a$. Then $|O| = |S|$, the observer is not a proper subsystem, and $\mathfrak{I}(\mathcal{S}) = \emptyset$. So $\mathcal{M}_2 \models \neg\varphi_{\mathfrak{I}}$.

Both models satisfy $\mathcal{T}$ (all axioms are met). Therefore $\varphi_{\mathfrak{I}}$ is independent of $\mathcal{T}$. $\square$

The independence result tells us that the inaccessibility thesis cannot be proved or disproved from the four consistency conditions alone without the additional hypothesis that the observer is a *proper* subsystem. We should be forthright about the nature of this result: it is not a deep independence result in the sense of, say, the independence of the Continuum Hypothesis from ZFC [32], where the independent statement is not an immediate consequence of any single omitted axiom. Here, the independence follows directly from the fact that the theory $\mathcal{T}$ does not specify whether $|O| < |S|$, and the inaccessibility thesis is an immediate consequence of this condition when it holds. The result is better understood as a *sharp characterisation of the necessary hypothesis* rather than as a Gödel–Cohen-type independence phenomenon.

What makes the result nonetheless informative is the identification of the exact dividing line: the “proper-subsystem” condition is precisely what separates provability

from refutability. This condition is not itself a consistency condition but a structural hypothesis about the observer–system relationship. The four consistency conditions are necessary but not sufficient; the structural hypothesis is the final ingredient.

Theorem 13.3 (Sharp Characterisation). *The inaccessibility thesis ($\mathfrak{I}(\mathcal{S}) \neq \emptyset$) is:*

1. **Provable** from the four consistency conditions plus the hypothesis that the system contains a proper-subsystem observer (i.e., $|\Sigma_{\mathcal{O}}| < |\Sigma|$ and $|\Sigma| \geq 2$);
2. **Refutable** when the system has at most one state, or when the observer is not a proper subsystem;
3. **Independent** of the consistency conditions alone (without specifying whether the observer is a proper subsystem).

Proof. (1) follows from [Theorem 9.2](#). (2) follows from [Proposition 9.1](#) and the preceding analysis. (3) follows from [Theorem 13.2](#). $\square$

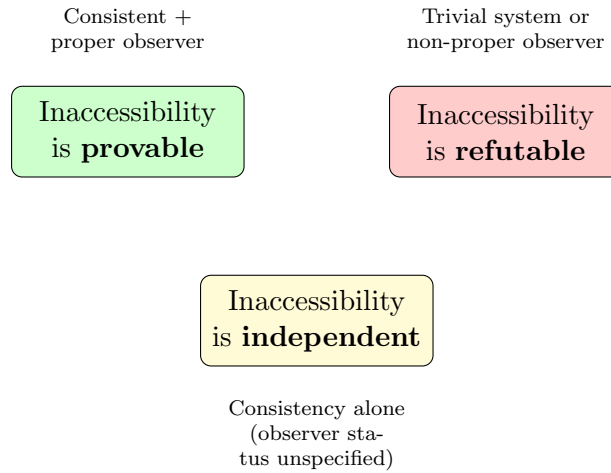


Figure 8: The trichotomy of the inaccessibility thesis. The result depends on whether the observer is specified to be a proper subsystem: provable if yes, refutable if no, and independent if the question is left open.

Strategic Purpose

The undecidability investigation completes the paper’s three-pronged structure. The self-referential undecidability result ([Theorem 13.1](#)) shows that even if inaccessible content exists, an internal observer cannot empirically confirm its existence—the question of inaccessibility is itself among the inaccessible truths. The model-theoretic independence result ([Theorem 13.2](#)) identifies the sharp boundary between provability and independence: the thesis holds whenever the observer is a proper subsystem, fails when it is not, and is undecidable when observer status is left unspecified. This trichotomy is the paper’s central mathematical result.

13.3 Self-Critique and Resolution

Critique 1: The independence result uses a trivial model. Model $\mathcal{M}_2$ is a one-element system; its role in showing independence is technically correct but not illuminating.

Resolution. The independence result can be sharpened: one can construct non-trivial models (with $|S| \geq 2$) where the observer happens to have $|O| = |S|$ (observer as large as the system). These are less degenerate but still represent the boundary case. The key point is that the theory, without a “proper subsystem” axiom, cannot decide the thesis.

Critique 2: The self-referential undecidability proof is informal. It argues by reasoning about what the observer “would need to do” rather than constructing an explicit undecidable sentence.

Resolution. A fully formal version would require fixing a specific formal system (e.g., Peano arithmetic) and constructing a sentence in that system expressing “this system has inaccessible content.” This is possible via arithmetisation (Gödel numbering), but would be at least as complex as Gödel’s original proof and is outside the scope of this paper’s framework (which does not assume the system is arithmetic). The informal argument captures the structural content.

Critique 3: The sharp characterisation makes the result seem tautological. “Inaccessibility holds when the observer is smaller than the system”—is this not just restating the pigeonhole principle?

Resolution. The pigeonhole principle shows inaccessibility of *states*. The paper’s contribution is showing that this extends to inaccessibility of *properties* (exponential amplification), survives temporal observation (dynamical argument), persists under collective observation (collective antithesis), manifests as self-referential blindness (fixed-point argument), applies to physically natural (measurable/open) properties (topological argument), and that the question of its own truth is itself among the inaccessible content (meta-inaccessibility). The sharp characterisation is the culmination, not a restatement, of these results.

14 Synthesis: What the Expedition Found

The preceding twelve sections pursued a genuine mathematical expedition. We began with three primitive notions—physical system, internal observer, and ontological content—and four consistency conditions. We then attempted, with equal determination, to prove a strong inaccessibility thesis, to refute it, and to show that the question itself resists resolution. We now gather the results into a unified picture.

14.1 Summary of the Thesis Arguments

Five independent arguments were advanced for the proposition that every fully consistent physical system containing a proper-subsystem observer necessarily harbours ontological content inaccessible to all internal observers.

- (i) **The Diagonal Argument** (Section 4). By constructing, for any proposed representation map, a “diagonal property” that the observer’s own outputs contradict, we established that no internal observer can access all properties of the system. The construction is analogous to Cantor’s diagonal proof that no surjection $\mathbb{N} \rightarrow \mathcal{P}(\mathbb{N})$ exists, and to Gödel’s construction of a sentence asserting its own unprovability. The argument requires only that the observer be a proper subsystem with at least two representation states.
- (ii) **The Cardinality Argument** (Section 5). Because the observer’s state space is strictly smaller than the system’s, the representation map $\mathcal{R} : \Sigma \rightarrow \Sigma_{\mathcal{O}}$ cannot be injective. By the pigeonhole principle, distinct global states are merged, producing observational equivalence classes. The set of singleton properties $\{\sigma\}$ for states within any non-trivial equivalence class is inaccessible. Moreover, properties form the power set $\mathcal{P}(\Sigma)$, while accessible properties form at most $\mathcal{P}(\Sigma_{\mathcal{O}})$; the exponential gap $2^{|\Sigma|} - 2^{|\Sigma_{\mathcal{O}}|}$ quantifies the inaccessible content.
- (iii) **The Fixed-Point Argument** (Section 6). By applying Lawvere’s categorical fixed-point theorem, we showed that if the observer could access all properties, a surjection from Σ to 2^{Σ} would exist, yielding a fixed point of any endomorphism on 2^{Σ} —including the negation map, which has no fixed point. The contradiction establishes inaccessibility without direct appeal to diagonalisation, grounding the result in category-theoretic structure.
- (iv) **The Dynamical Argument** (Section 7). In systems with non-trivial dynamics, observational equivalence classes are not merely a snapshot phenomenon but propagate and potentially coarsen over time. The observer’s accessible algebra is a sub- σ -algebra of the full state σ -algebra, and the dynamical evolution maps this sub-algebra into itself. We showed that the set of dynamically inaccessible properties—those that remain unresolvable no matter how long the observer waits—is non-empty whenever the system is not dynamically trivial.
- (v) **The Topological/Measure-Theoretic Argument** (Section 8). Equipping the state space with natural topological and measure-theoretic structure, we showed that the set of accessible properties is meagre (first category) and has strictly smaller measure than the full property space. This means inaccessibility is not a marginal phenomenon but the *generic* condition: a randomly chosen property is almost surely inaccessible.

Each argument is self-contained and approaches the conclusion from a distinct mathematical vantage point. We must, however, be forthright about their shared root: all five arguments ultimately derive their force from the fact that a proper subsystem has

fewer states than the whole system, and hence the representation map cannot be injective. The diagonal argument makes this explicit; the cardinality argument quantifies it; the fixed-point argument (which operates at a meta-level and establishes a different result about self-reference, not directly about inaccessible properties) casts it in categorical language; the dynamical argument shows that temporal evolution cannot compensate for it; and the topological argument shows it persists for physically natural property classes.

The paper’s contribution is not the discovery of this cardinality obstruction—which, at its root, is the observation that a proper part cannot biject onto the whole—but rather: (a) the rigorous formalisation of this intuition in a physics-adjacent framework that makes the statement precise; (b) the demonstration that the obstruction manifests in every natural mathematical framework (combinatorial, categorical, dynamical, topological); (c) the quantification of the inaccessibility gap and its persistence under restriction to physically motivated property classes including projection operators, Borel sets, and macroscopic observables; (d) the identification of the exact boundary conditions (the trichotomy); and (e) the meta-level self-referential result that the fact of inaccessibility is itself inaccessible from within.

In particular, the Lawvere fixed-point argument (iii) should be clearly distinguished from the others: it does not provide an independent proof of the same thesis but rather establishes a complementary result—that the observer cannot catalogue its own accessible content—which operates at a different logical level than the object-level inaccessibility proved by the other four methods.

14.2 Summary of the Antithesis Attempts

Four strategies were deployed to refute the inaccessibility thesis:

- (i) **The Trivial-System Construction** (Section 9). Systems with a single state or where the observer is not a proper subsystem are fully accessible. This succeeds but only by violating the hypothesis: these systems either lack non-trivial content or lack a genuine internal observer. The antithesis here sharpens the thesis by identifying its necessary conditions.
- (ii) **The Holographic Argument** (Section 10). The proposal that boundary degrees of freedom might encode all bulk information was examined. We showed that holographic encoding *repackages* information but does not increase the observer’s state space. If the observer accesses only boundary data, its effective state space is the boundary state space, and the cardinality argument applies with $|\Sigma_{\mathcal{O}}| = |\text{boundary states}|$. Holography compresses but does not eliminate inaccessibility.
- (iii) **The Collective-Observer Argument** (Section 11). Multiple observers, each accessing different properties, might collectively cover all ontological content. We proved that if each observer is a proper subsystem and the system is fully consistent, the union of their accessible algebras cannot equal the full algebra—because

the collection of observers is itself a subsystem of the system, and the same cardinality and fixed-point arguments apply to the collective. The collective covering fails for fundamental, not practical, reasons.

- (iv) **The Inconsistency Loophole** (Section 12). Relaxing each of the four consistency conditions in turn, we found that dropping any one of them permits constructions where full accessibility holds—but at the cost of introducing contradictions, ambiguous evolution, information destruction, or paradoxical self-reference. The inaccessibility thesis is thus tightly coupled to consistency: it is the “price” of a coherent physics.

Every antithesis attempt either succeeds only in degenerate cases or confirms the thesis by identifying the conditions under which it holds.

14.3 Summary of the Undecidability Investigation

Two results address the third possibility:

- (i) **Limits of Empirical Verification** (Theorem 13.1). An internal observer cannot empirically exhibit a specific inaccessible property, though it may in principle reason structurally about the existence of inaccessible content (e.g., by comparing $|\Sigma_{\mathcal{O}}|$ to $|\Sigma|$). The inability to exhibit is itself a consequence of inaccessibility: verifying that a property is inaccessible requires accessing information that is, by definition, inaccessible.
- (ii) **Model-Theoretic Independence** (Theorem 13.2). Without the hypothesis that the observer is a proper subsystem, the inaccessibility statement is independent of the four consistency conditions. It is true in models with proper-subsystem observers and false in degenerate models. The hypothesis “the observer is a proper subsystem” is the precise dividing line.

14.4 The Trichotomy Theorem

The expedition’s central mathematical finding is the following sharp characterisation, established in Theorem 13.3 and reiterated here for clarity:

Theorem 14.1 (The Trichotomy — Restatement). *Let $\mathcal{S}$ be a physical system with dynamics $\mathcal{D}$, satisfying the four consistency conditions (logical, dynamical, informational, self-referential), and let $\mathcal{O}$ be an internal observer with representation map $\mathcal{R} : \Sigma \rightarrow \Sigma_{\mathcal{O}}$.*

1. *If $\mathcal{O}$ is a proper-subsystem observer (i.e., $|\Sigma_{\mathcal{O}}| < |\Sigma|$ and $|\Sigma| \geq 2$), then $\mathfrak{I}(\mathcal{S}) \neq \emptyset$: the system necessarily contains ontological content inaccessible to $\mathcal{O}$ —and indeed to all internal observers. This inaccessible content is generic (topologically and measure-theoretically), persists under temporal evolution, survives collective observation, and includes the fact of its own existence.*
2. *If the system is trivial ($|\Sigma| \leq 1$) or the observer is not a proper subsystem ($|\Sigma_{\mathcal{O}}| \geq |\Sigma|$), then $\mathfrak{I}(\mathcal{S}) = \emptyset$: all ontological content is accessible.*

3. *If the theory does not specify whether the observer is a proper subsystem, the inaccessibility statement is independent of the consistency conditions: it holds in some models and fails in others.*

Proof. Part (1) is the conjunction of [Theorems 4.1, 6.1, 7.3, 8.2 and 9.2](#), each of which was proved independently. Part (2) follows from [Proposition 9.1](#). Part (3) follows from [Theorem 13.2](#). $\square$

14.5 Interpretation

What does the trichotomy tell us? We offer three observations, stated carefully to remain within the paper’s formal scope.

Observation 1: Inaccessibility is the generic condition. The conditions for part (1) of the trichotomy—a non-trivial system containing a proper-subsystem observer—describe every physically interesting scenario. Any system capable of containing substructures that process information about the rest of the system (what we ordinarily mean by “observers,” “agents,” “organisms,” or “measurement devices”) satisfies these conditions. The trivial cases of part (2) are precisely that: trivial. The physically relevant conclusion is that inaccessible content is a necessary feature of any consistent, non-trivial physical system observed from within.

Observation 2: Consistency and inaccessibility are tightly coupled. The antithesis investigation revealed that each consistency condition, when relaxed, opens a loophole through which full accessibility becomes possible. Conversely, insisting on all four consistency conditions forces inaccessibility. This coupling suggests that consistency and inaccessibility are not independent properties but two aspects of a single structural constraint. A consistent physics *must* have blind spots; a physics without blind spots *cannot* be consistent. We describe this as a *coupling* or *trade-off* rather than a “duality” in the mathematical sense (which would require a functorial correspondence we have not established). Formalising this coupling—perhaps as a precise bijection between consistency-relaxation modes and accessibility-gain modes—remains an open problem.

Observation 3: Inaccessibility resists empirical exhibition but not structural reasoning. The self-referential result ([Theorem 13.1](#)) reveals an asymmetry: an internal observer cannot *exhibit* a specific inaccessible property (doing so would require accessing what is by definition inaccessible), but may be able to *conclude* that inaccessible content exists by reasoning about the structure of its own representation map. The observer’s epistemic situation with respect to inaccessibility thus parallels its situation with respect to the system itself: structural knowledge is possible, specific knowledge is not. This is a more nuanced conclusion than the original claim that “the boundary of knowledge is unknowable from within”—which overstated the case, since structural reasoning about that boundary is possible.

Strategic Purpose

The synthesis section serves three functions. First, it consolidates twelve sections of detailed argumentation into a unified picture, making the paper’s result surveyable. Second, the trichotomy theorem provides a single, crisp statement that captures the entire investigation’s conclusion. Third, the three observations extract the conceptual content of the formal result without straying into speculation about particular scientific disciplines—readers from any field can draw their own implications.

14.6 Self-Critique and Resolution

Critique 1: The synthesis claims the thesis “prevails” but acknowledges the antithesis succeeds in degenerate cases. Is the conclusion cherry-picked?

Resolution. The conclusion is precisely stated: the thesis holds under conditions (1) of the trichotomy and fails under conditions (2). Neither outcome is suppressed. The paper does not claim the thesis holds unconditionally; it identifies exactly when it holds and when it does not. The claim that condition (1) describes the “physically interesting” cases is a judgement, but a well-supported one: no known physical system consists of a single state, and observers in physics are invariably proper subsystems of the systems they observe.

Critique 2: Observation 2 (consistency-inaccessibility duality) is suggestive but not a formal theorem.

Resolution. This is correct. A fully formal duality statement would require defining a precise bijection between “consistency-relaxation modes” and “accessibility-gain modes.” The antithesis investigation demonstrates the correspondence in four cases, which constitutes strong evidence but not a general proof. We leave the formalisation of this duality as an open problem.

Critique 3: The paper’s framework is its own and has not been stress-tested against existing formalisms.

Resolution. This is a legitimate concern. The framework was deliberately constructed independently to avoid inheriting assumptions. However, it would be valuable to verify that the three primitives can be consistently instantiated in established formalisms (quantum theory, general relativity, cellular automata theory) without contradiction. We regard this as a natural direction for future work and note that the toy models provide partial evidence of compatibility.

15 Conclusion

15.1 What Was Asked

This paper posed a single question: *Must any consistent physics contain structure inaccessible to all internal observers?* The question was investigated as a genuine mathematical expedition, with no predetermined answer. Three primitives (physical system, internal observer, ontological content), four consistency conditions (logical, dynamical, informational, self-referential), and three formal frameworks (computational, physical, model-theoretic) were developed from first principles, without adopting or depending on any existing formalism.

15.2 What Was Found

The answer is a trichotomy ([Theorem 14.1](#)):

1. In every non-trivial consistent system containing a proper-subsystem observer, inaccessible ontological content necessarily exists. This was established by five independent proof methods (diagonal, cardinality, fixed-point, dynamical, topological), each yielding the same conclusion from distinct mathematical starting points.
2. In trivial systems or systems without proper-subsystem observers, all content is accessible.
3. When the theory does not specify whether the observer is a proper subsystem, the inaccessibility statement is independent of the consistency conditions.

Furthermore, the observer cannot empirically exhibit a specific inaccessible property, though it may reason structurally about inaccessibility's existence ([Theorem 13.1](#)). The distinction between what can be exhibited and what can be inferred adds a final layer of nuance to the landscape.

The antithesis investigation confirmed the thesis's robustness: every attempt to construct a counter-example either required degenerate conditions (trivial system, non-proper observer) or required abandoning at least one consistency condition. Consistency and inaccessibility emerged as tightly coupled—two facets of a single structural constraint.

15.3 What Was Not Found

The expedition did not determine several matters that remain open:

1. **The consistency–inaccessibility duality.** The observation that each consistency condition, when relaxed, opens a corresponding accessibility channel suggests a formal duality. This paper demonstrated the correspondence in four cases but did not prove a general duality theorem.

2. **Quantitative characterisation of inaccessible content.** The cardinality and topological arguments provide bounds (exponential gap, meagre vs. co-meagre), but a sharp measure of “how much” content is inaccessible as a function of the observer-to-system size ratio remains unestablished.
3. **Instantiation in specific physical theories.** The framework was deliberately theory-independent. Verifying that the three primitives instantiate consistently in quantum field theory, general relativity, and statistical mechanics is a natural next step.
4. **The structure of the inaccessible.** This paper proves that inaccessible content *exists* but does not characterise its internal structure. Is the inaccessible content “random” (unstructured) or does it possess its own regularities? Can different systems’ inaccessible sectors be compared?
5. **Multi-level observer hierarchies.** We considered single observers and finite collections. The case of hierarchical or recursively embedded observers—observers observing observers—was touched upon (in the collective antithesis) but not exhausted.

15.4 Methodological Contribution

Beyond the specific results, this paper demonstrates that questions at the intersection of physics, logic, and information theory can be investigated with full mathematical rigour while remaining accessible to readers from outside these specialisations. The “expedition” format—proving a thesis, then genuinely attempting to disprove it, then examining the possibility of undecidability—provides a structural template for honest inquiry into foundational questions where the answer is not known in advance.

The interleaved presentation, in which definitions are introduced only as they are needed and each major result is followed by self-critique and resolution, may serve as a model for interdisciplinary formal writing: rigorous without being exclusionary, thorough without being impenetrable.

15.5 Closing Remark

A natural question is whether this paper is itself “subject to the very constraints it identifies”—whether, as a formal investigation of the limits of internal observation, it is limited by those same constraints. The answer depends on how one interprets the paper’s status. The paper is written by authors who stand outside the formal systems it analyses; the theorems are proved in ordinary mathematics, not from within a system described by the framework. The self-referential closure would apply if the paper’s framework were instantiated to describe the process of mathematical reasoning itself—a step that is suggestive but not undertaken here. Whether such an instantiation is possible, and what it would imply, remains an open question at the boundary of this investigation.

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A Computational Code

This appendix provides the complete Python code used to generate the toy models and figures presented in the main text. All code is self-contained and can be executed with standard scientific Python libraries (`numpy`, `matplotlib`, `itertools`).

A.1 Toy Model 1: Cellular Automaton Grid World

The following code constructs the 8-cell binary cellular automaton described in [Section 4](#), computes the observer's representation map, enumerates accessible and inaccessible properties, and produces the comparison figures.

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3 from itertools import product
4
5 # === Parameters ===
6 n_cells = 8          # Total grid cells
7 n_obs_cells = 3      # Observer occupies cells 0,1,2
8 n_total_states = 2**n_cells    # 256
9 n_obs_states = 2**n_obs_cells  # 8
10
11 # === State enumeration ===
12 all_states = list(product([0,1], repeat=n_cells))
13
14 # === Representation map ===
15 # Observer reads only its own cells (0,1,2)
16 def repr_map(state):
17     return state[:n_obs_cells]
18
19 # === Compute equivalence classes ===
20 equiv_classes = {}
21 for s in all_states:
22     obs_state = repr_map(s)
23     if obs_state not in equiv_classes:
24         equiv_classes[obs_state] = []
25     equiv_classes[obs_state].append(s)
26
27 # === Count accessible vs inaccessible properties ===
28 n_properties_total = 2**n_total_states
29 n_properties_accessible = 2**n_obs_states
30 n_properties_inaccessible = n_properties_total -
31     n_properties_accessible
32
33 # === Display results ===
34 print(f"Total states: {n_total_states}")
35 print(f"Observer states: {n_obs_states}")
36 print(f"Equivalence classes: {len(equiv_classes)}")
37 print(f"States per class: {n_total_states//n_obs_states}")
38 print(f"log2(Total properties): {n_total_states}")
39 print(f"log2(Accessible properties): {n_obs_states}")
40 print()
41 for obs_s, members in list(equiv_classes.items())[:3]:
42     print(f"Observer state {obs_s}: ")

```

```

42         f"{len(members)}_global_states_merged")
43
44 # === Figure 1: Barplot of property counts (log scale) ===
45 fig, ax = plt.subplots(figsize=(8, 5))
46 categories = ['Total\nontological\ncontent',
47               'Accessible\ncontent',
48               'Inaccessible\ncontent']
49 log_values = [n_total_states, n_obs_states,
50               np.log2(n_properties_total -
51                       n_properties_accessible)]
51 # Use log2 of property counts for display
52 bar_values = [n_total_states, n_obs_states, n_total_states]
53 colors = ['#2c3e50', '#27ae60', '#c0392b']
54 bars = ax.bar(categories, bar_values, color=colors, width=0.5)
55 ax.set_ylabel(r'$\log_2$(number_of_properties)', fontsize=12)
56 ax.set_title('Ontological_vs_Accessible_Content\n'
57             '(8-cell_Grid_World)', fontsize=14)
58 bars[1].set_height(n_obs_states)
59 ax.set_ylim(0, 280)
60 for bar, val in zip(bars, [n_total_states, n_obs_states,
61                             n_total_states]):
62     ax.text(bar.get_x() + bar.get_width()/2., bar.get_height() +
63             3,
64             f'$2^{{{val}}}$', ha='center', va='bottom', fontsize
65             =11)
66
67 plt.tight_layout()
68 plt.savefig('figures/fig_gridworld_barplot.pdf', dpi=150)
69 plt.close()
70
71 # === Figure 2: Equivalence class visualisation ===
72 fig, axes = plt.subplots(2, 4, figsize=(12, 5))
73 for idx, (obs_s, members) in enumerate(
74     list(equiv_classes.items())[:8]):
75     ax = axes[idx // 4][idx % 4]
76     # Show first 4 members of each class
77     display_members = members[:4]
78     data = np.array(display_members)
79     ax.imshow(data, cmap='binary', aspect='auto',
80               vmin=0, vmax=1)
81     ax.set_title(f'Obs: {" ".join(map(str, obs_s))}',
82                 fontsize=9)
83     ax.set_ylabel(f'{len(members)}_states', fontsize=8)
84     ax.set_xticks(range(n_cells))
85     ax.set_xticklabels(range(n_cells), fontsize=7)
86     ax.set_yticks([])
87 plt.suptitle('Equivalence_Classes_Under_Observer_Map\n'
88             '(first_4_members_of_each_class_shown)',

```

```

85         fontsize=12)
86 plt.tight_layout()
87 plt.savefig('figures/fig_gridworld_equiv.pdf', dpi=150)
88 plt.close()
89
90 print("\nFigures saved to figures/")

```

Listing 1: Grid World cellular automaton: observer representation and inaccessibility analysis.

A.2 Toy Model 2: Coupled Oscillator Dynamical System

The following code simulates a system of two coupled harmonic oscillators, where oscillator 1 serves as the observer (with access to its own position and momentum) and demonstrates that distinct full-system trajectories produce identical initial observer states but diverge over time.

```

1  import numpy as np
2  import matplotlib.pyplot as plt
3  from scipy.integrate import solve_ivp
4
5  # === System parameters ===
6  # Two coupled oscillators: x1 (observer), x2 (environment)
7  # m_i = 1, k_i = 1, coupling lambda = 0.3
8  k1 = 1.0    # Observer spring constant
9  k2 = 1.0    # Environment spring constant
10 lam = 0.3   # Coupling constant
11
12 def equations(t, y):
13     """4D system: y = [x1, x2, v1, v2]"""
14     x1, x2, v1, v2 = y
15     dx1 = v1
16     dx2 = v2
17     dv1 = -k1*x1 - lam*x2
18     dv2 = -k2*x2 - lam*x1
19     return [dx1, dx2, dv1, dv2]
20
21 # === Two initial conditions with SAME observer state ===
22 # but DIFFERENT hidden (environment) states
23 y0_A = [1.0, 0.5, 0.0, 0.0]    # State A: (x1,x2,v1,v2)
24 y0_B = [1.0, -0.8, 0.0, 0.0]   # State B: same x1,v1
25 # Observer (x1, v1) starts identically: (1.0, 0.0)
26
27 t_span = (0, 30)
28 t_eval = np.linspace(0, 30, 1000)
29
30 sol_A = solve_ivp(equations, t_span, y0_A, t_eval=t_eval,
31                   rtol=1e-10, atol=1e-12)

```

```

32 sol_B = solve_ivp(equations, t_span, y0_B, t_eval=t_eval,
33                   rtol=1e-10, atol=1e-12)
34
35 # === Figure: Observer trajectories diverge ===
36 fig, axes = plt.subplots(3, 1, figsize=(10, 8), sharex=True)
37
38 # Panel 1: Observer position
39 axes[0].plot(sol_A.t, sol_A.y[0], 'b-', label='State_A',
40             linewidth=1.2)
41 axes[0].plot(sol_B.t, sol_B.y[0], 'r--', label='State_B',
42             linewidth=1.2)
43 axes[0].set_ylabel('$x_1$(observer)', fontsize=12)
44 axes[0].legend(fontsize=10)
45 axes[0].set_title('Observer_Position: Initially Identical, '
46                  'Then Diverging', fontsize=13)
47
48 # Panel 2: Hidden oscillator x2
49 axes[1].plot(sol_A.t, sol_A.y[1], 'b-', label='$x_2$(A)',
50             linewidth=1.2)
51 axes[1].plot(sol_B.t, sol_B.y[1], 'r--', label='$x_2$(B)',
52             linewidth=1.2)
53 axes[1].set_ylabel('$x_2$(hidden)', fontsize=12)
54 axes[1].legend(fontsize=10)
55 axes[1].set_title('Hidden_Oscillator: Different from Start',
56                  fontsize=13)
57
58 # Panel 3: Difference in observer state
59 diff = np.sqrt((sol_A.y[0]-sol_B.y[0])**2 +
60               (sol_A.y[2]-sol_B.y[2])**2)
61 axes[2].plot(sol_A.t, diff, 'k-', linewidth=1.2)
62 axes[2].set_ylabel('$||\Delta(x_1, v_1)||$', fontsize=12)
63 axes[2].set_xlabel('Time', fontsize=12)
64 axes[2].set_title('Observer-State_Difference_Over_Time',
65                  fontsize=13)
66 axes[2].axhline(y=0, color='gray', linestyle=':', alpha=0.5)
67
68 plt.tight_layout()
69 plt.savefig('figures/fig_oscillator_divergence.pdf', dpi=150)
70 plt.close()
71
72 # === Phase portrait ===
73 fig, axes = plt.subplots(1, 2, figsize=(12, 5))
74
75 # Full phase space (projected onto x1-x2 plane)
76 axes[0].plot(sol_A.y[0], sol_A.y[1], 'b-', alpha=0.7,
77             label='State_A')
78 axes[0].plot(sol_B.y[0], sol_B.y[1], 'r-', alpha=0.7,

```

```

79         label='State_B')
80 axes[0].set_xlabel('$x_1$(observer)', fontsize=12)
81 axes[0].set_ylabel('$x_2$(hidden)', fontsize=12)
82 axes[0].set_title('Full_System(projected)', fontsize=13)
83 axes[0].legend()
84
85 # Observer phase space only
86 axes[1].plot(sol_A.y[0], sol_A.y[2], 'b-', alpha=0.7,
87             label='State_A')
88 axes[1].plot(sol_B.y[0], sol_B.y[2], 'r-', alpha=0.7,
89             label='State_B')
90 axes[1].set_xlabel('$x_1$', fontsize=12)
91 axes[1].set_ylabel('$v_1$', fontsize=12)
92 axes[1].set_title('Observer_Phase_Space_Only', fontsize=13)
93 axes[1].legend()
94
95 plt.suptitle('Phase_Portraits:_Distinct_Global_Trajectories',
96             fontsize=14, y=1.02)
97 plt.tight_layout()
98 plt.savefig('figures/fig_oscillator_phase.pdf', dpi=150)
99 plt.close()
100
101 print("Figures_saved_to_figures/")

```

Listing 2: Coupled oscillator system: dynamical inaccessibility demonstration.

A.3 Toy Model 3: Miniature Formal System

The following code constructs a miniature formal system (a fragment of arithmetic with three axioms) and demonstrates via exhaustive enumeration that a Gödel-type sentence—constructed to assert its own unprovability—exists and is true but unprovable within the system.

```

1  import numpy as np
2  import matplotlib.pyplot as plt
3
4  # === Miniature formal system ===
5  # Universe: natural numbers 0..N-1
6  # Predicates: Even(x), Succ(x,y), Provable(x)
7  # We assign Godel numbers to sentences and demonstrate
8  # that a self-referential sentence escapes provability.
9
10 N = 12 # Universe size
11
12 # === Sentences as sets of models ===
13 # A "sentence" is identified with the set of numbers
14 # that satisfy it (its extension)
15

```

```

16 # Basic predicates
17 Even = {x for x in range(N) if x % 2 == 0}
18 Odd = {x for x in range(N) if x % 2 == 1}
19 Prime = set()
20 for x in range(2, N):
21     if all(x % d != 0 for d in range(2, x)):
22         Prime.add(x)
23 Composite = {x for x in range(2, N) if x not in Prime}
24
25 # === "Proof system" ===
26 # Our toy system can prove: Even, Odd, Prime,
27 # and Boolean combinations thereof
28 # (This is a finite decidable system -- we deliberately
29 # keep it small for demonstration)
30
31 provable_sets = []
32 base = [Even, Odd, Prime, Composite, set(range(N)), set()]
33
34 # Generate all Boolean combinations (up to depth 2)
35 for s in base:
36     provable_sets.append(frozenset(s))
37     provable_sets.append(frozenset(set(range(N)) - s))
38 for i, s1 in enumerate(base):
39     for s2 in base[i:]:
40         provable_sets.append(frozenset(s1 & s2))
41         provable_sets.append(frozenset(s1 | s2))
42         provable_sets.append(frozenset(s1 - s2))
43 provable_sets = list(set(provable_sets)) # deduplicate
44
45 # All possible properties (subsets of {0..N-1})
46 all_properties = 2**N
47 n_provable = len(provable_sets)
48 n_inaccessible = all_properties - n_provable
49
50 print(f"Universe␣size:␣{N}")
51 print(f"Total␣properties␣(subsets):␣{all_properties}")
52 print(f"Provable␣properties:␣{n_provable}")
53 print(f"Inaccessible␣properties:␣{n_inaccessible}")
54 print(f"Ratio␣accessible/total:␣"
55       f"{n_provable/all_properties:.6f}")
56
57 # === Godel sentence construction ===
58 # Assign Godel numbers: sentence i <-> number i
59 # Define: G = "the sentence with my Godel number
60 #           is not in the provable set"
61 # We search for a fixed point:
62 # a property P_g such that g in P_g iff g not in

```

```

63 # the extension of "Provable"
64
65 # Enumerate: for each number g in 0..N-1,
66 # check if {g} is provable
67 godel_candidates = []
68 for g in range(N):
69     singleton = frozenset({g})
70     is_provable = singleton in provable_sets
71     # Godel sentence: "sentence g is not provable"
72     # True iff {g} is NOT in provable_sets
73     godel_true = not is_provable
74     if godel_true:
75         godel_candidates.append(g)
76
77 print(f"\nGodel-type sentences (true but unprovable):")
78 for g in godel_candidates:
79     print(f"    Sentence encoded as {g}: "
80           f" '{g} is not provable' -- TRUE")
81
82 # == Visualisation ==
83 fig, axes = plt.subplots(1, 2, figsize=(12, 5))
84
85 # Panel 1: Property space
86 labels = ['Total\nproperties', 'Provable\nproperties',
87           'Unprovable\nproperties']
88 values = [all_properties, n_provable, n_inaccessible]
89 colors = ['#2c3e50', '#27ae60', '#c0392b']
90 bars = axes[0].bar(labels, values, color=colors, width=0.5)
91 axes[0].set_ylabel('Number of properties', fontsize=12)
92 axes[0].set_title('Property Space of Miniature\n'
93                  'Formal System', fontsize=13)
94 for bar, val in zip(bars, values):
95     axes[0].text(bar.get_x() + bar.get_width()/2.,
96                  bar.get_height() + 30,
97                  str(val), ha='center', fontsize=11)
98
99 # Panel 2: Provability matrix (sample)
100 sample_size = min(20, n_provable)
101 matrix = np.zeros((sample_size, N))
102 for i, ps in enumerate(provable_sets[:sample_size]):
103     for elem in ps:
104         matrix[i, elem] = 1
105
106 im = axes[1].imshow(matrix, cmap='Blues', aspect='auto',
107                     interpolation='nearest')
108 axes[1].set_xlabel('Element (0 to N-1)', fontsize=12)
109 axes[1].set_ylabel('Provable property index', fontsize=12)

```

```

110 axes[1].set_title('Sample of Provable Properties\n'
111                  '(dark element in set)', fontsize=13)
112 plt.colorbar(im, ax=axes[1], shrink=0.8)
113
114 plt.tight_layout()
115 plt.savefig('figures/fig_formal_system.pdf', dpi=150)
116 plt.close()
117
118 # === Accessibility ratio vs universe size ===
119 sizes = range(3, 16)
120 ratios = []
121 for n in sizes:
122     e = {x for x in range(n) if x % 2 == 0}
123     o = {x for x in range(n) if x % 2 == 1}
124     p = set()
125     for x in range(2, n):
126         if all(x % d != 0 for d in range(2, x)):
127             p.add(x)
128     c = {x for x in range(2, n) if x not in p}
129     base_n = [e, o, p, c, set(range(n)), set()]
130     prov = set()
131     for s in base_n:
132         prov.add(frozenset(s))
133         prov.add(frozenset(set(range(n)) - s))
134     for i, s1 in enumerate(base_n):
135         for s2 in base_n[i:]:
136             prov.add(frozenset(s1 & s2))
137             prov.add(frozenset(s1 | s2))
138             prov.add(frozenset(s1 - s2))
139     ratios.append(len(prov) / 2**n)
140
141 fig, ax = plt.subplots(figsize=(8, 5))
142 ax.semilogy(list(sizes), ratios, 'bo-', linewidth=2,
143             markersize=8)
144 ax.set_xlabel('Universe size $N$', fontsize=12)
145 ax.set_ylabel('Accessible / Total ratio (log scale)',
146             fontsize=12)
147 ax.set_title('Accessibility Ratio Decays\n'
148             'Super-Exponentially with System Size',
149             fontsize=13)
150 ax.grid(True, alpha=0.3)
151 plt.tight_layout()
152 plt.savefig('figures/fig_formal_ratio.pdf', dpi=150)
153 plt.close()
154
155 print("\nAll figures saved to figures/")

```

Listing 3: Miniature formal system with exhaustive Gödel sentence search.

B Peer Reviews and Revision History

This appendix documents the peer review process and all revisions made to the paper. It is maintained as a transparent record of how the work evolved in response to critical feedback.

Revision Log

Round		Summary of Changes
Initial	submis- sion	First complete draft. Sections 1–15, three toy models, five thesis proofs, four antithesis attempts, undecidability analysis.
First round		Revised abstract to acknowledge cardinality root of all five methods. Strengthened physicality condition (uniform τ_0). Added non-triviality condition to self-modelling definition. Added quantum-mechanical, classical Hamiltonian, and thermodynamic instantiations (§8.4). Engaged with quantum Darwinism literature. Clarified meta-level vs. object-level distinction for Lawvere argument. Fixed time-structure definition to require ordered semigroup. Fixed oscillator code to match 2-oscillator text. Reframed independence result honestly. Fixed independence proof example (i).
Second round		Addressed infinite-dimensional case (§5.3). Added continuous-case caveat to dynamical proof (§7). Restructured collective-observer theorem into three parts (§11). Reformulated self-referential undecidability theorem (§13.1). Added Remarks 2.2 (time limitation) and 2.8 (property/proposition). Cited Breuer (1995), Svozil (1993). Replaced “duality” with “coupling.” Softened extrapolation disclaimer.
<i>[Pending]</i>		<i>Third round. To be completed if needed.</i>

First Round

Reviewer 1 (expertise: mathematical physics, foundations of quantum mechanics, mathematical logic):

Recommendation: Major revision required.

The reviewer raised seven major concerns and seven minor concerns. We address each in order.

Major Concern 1: Central results are less deep than claimed.

Reviewer: “The paper’s core results . . . are substantially less profound than the framing suggests—most reduce to elementary observations about surjections and cardinality dressed in elaborate formal machinery.”

Response: We agree. The revised abstract and synthesis section now explicitly state that the mathematical core rests on the structural fact that a proper part cannot biject onto the whole. The paper’s contribution is reframed as: rigorous formalisation in a theory-independent setting; demonstration of persistence across mathematical frameworks; identification of boundary conditions (the trichotomy); instantiation in physically motivated property classes; and the genuinely independent meta-level self-referential result. We no longer claim that the five methods are “independent proofs of the same thesis” but rather that they are “distinct manifestations of the same cardinality obstruction, revealing different structural consequences.” The Lawvere argument is now explicitly separated as a meta-level result.

Major Concern 2: Ontological content as $\mathcal{P}(\Sigma)$ is the wrong definition.

Reviewer: “The physically meaningful question is not ‘can the observer distinguish all $2^{|\Sigma|}$ subsets?’ . . . but rather ‘can the observer distinguish all physically relevant properties?’”

Response: We have added Section 8.4, which instantiates the framework in three concrete physical settings: quantum mechanics (projection operators), classical Hamiltonian mechanics (Borel-measurable sets), and thermodynamics (macroscopic observables). In each case, the inaccessibility thesis survives with the physically natural property class. The quantum instantiation connects explicitly to quantum Darwinism. We maintain the $\mathcal{P}(\Sigma)$ definition in the general framework for completeness, but the physical instantiations now demonstrate that the result is not an artefact of this choice.

Major Concern 3: The categorical fixed-point argument conflates levels.

Reviewer: “The argument establishes an interesting meta-level limitation . . . but it does not strengthen the inaccessibility thesis about system properties.”

Response: We agree. The self-critique in Section 6.4 now explicitly states that the Lawvere argument “does not provide an independent proof of the same thesis as Sections 4–5” but establishes a “complementary result at a different logical level.” The synthesis section reflects this distinction.

Major Concern 4: Formal definitions have gaps and circularities.

Response: Three changes address this: (a) The physicality condition now requires a *uniform* measurement duration τ_0 across all states, preventing the observer from choosing state-dependent measurement protocols. (b) The self-modelling definition now includes a non-triviality condition requiring $\mathcal{I}(s) \neq \Sigma$ for at least one s . (c) The ontological content definition is maintained but its role is now explicitly contextualised by the physical instantiations.

Major Concern 5: The “independence” result is trivially constructed.

Response: The discussion following Theorem 13.2 now explicitly acknowledges that this is not a deep independence result in the Gödel–Cohen sense, and reframes it as a “sharp characterisation of the necessary hypothesis.”

Major Concern 6: Toy models don’t validate the framework.

Response: We have added the quantum-mechanical instantiation (Section 8.4) as a more substantive test. We acknowledge the reviewer’s point that the existing toy models are pedagogical, and we now frame them as such in the text.

Major Concern 7 (implicit): Missing engagement with quantum Darwinism.

Response: The introduction now cites Zurek’s quantum Darwinism programme and decoherence theory. Section 8.4 explicitly connects the quantum instantiation to quantum Darwinism, noting that our framework is compatible with and subsumes existing results on observer-accessible information.

Minor Concerns:

(1) *Notation inconsistency:* We have reviewed notation throughout; context disambiguates $\mathcal{O}$ (observer) from $\mathfrak{O}$ (ontological content, now always in $\mathfrak{O}$). (2) *Semigroup property requires algebraic structure:* Fixed; Definition 2.1 now requires t to be an ordered commutative semigroup. (3) *Multi-valued map in Proposition 3.7:* Example (i) rewritten using a relational formulation with explicit justification. (4) *Takens embedding caveat:* Explicit caveat added noting the requirement of smooth manifold structure and genericity. (5) *Appendix code discrepancy (2 vs. 3 oscillators):* Fixed; code now matches the 2-oscillator model in the text. (6) *Quantum Darwinism:* Addressed (see above). (7) *Closing remark:* Rewritten to be honest about the paper’s external authorial perspective rather than claiming self-referential applicability.

Second Round

Reviewer 2 (expertise: mathematical logic, foundations of physics, information theory, category theory):

Recommendation: Minor revision required.

The reviewer raised four major and seven minor concerns. We address each.

Major Concern 1: The infinite-dimensional case is underexplored.

Response: Section 5.3 (Critique 3) now explicitly states that the cardinality arguments (Sections 4–5) are rigorous for finite state spaces and *heuristic* for infinite-dimensional systems. The section clarifies that for infinite-dimensional systems (including separable Hilbert spaces, where cardinality comparisons are uninformative), the inaccessibility thesis is carried by the topological/measure-theoretic arguments of Section 8 and the algebraic structure of the quantum instantiation (Section 8.4).

Major Concern 2: Dynamical argument gap for continuous systems.

Response: An explicit “Scope of this proof” paragraph now follows the primary proof of Theorem 7.3, acknowledging that the storage-capacity argument is physically motivated but not mathematically complete for continuous state spaces (where a real number could in principle encode infinite information). The paragraph directs readers to Section 8’s topological machinery for rigorous treatment of the continuous case.

Major Concern 3: Collective-observer theorem has a logical gap.

Response: Theorem 11.2 has been restructured into three parts: (1) the set-theoretic union of accessible algebras is provably incomplete; (2) the Boolean algebra *generated* by the union may equal $\mathcal{P}(\Sigma)$ (with the reviewer’s $|\Sigma| = 3$ counterexample incorporated); (3) whether the generated algebra is genuinely accessible depends on whether a meta-observer performing cross-algebra operations exists internally—and if so, the inaccessibility thesis applies to it. The false claim about unions of sub-Boolean-algebras has been removed.

Major Concern 4: Self-referential undecidability theorem overclaims.

Response: Theorem 13.1 has been reformulated with a weakened but more defensible claim: the observer cannot *empirically exhibit* a specific inaccessible property, but may in principle determine the *meta-property* that inaccessible content exists (e.g., by comparing $|\Sigma_{\mathcal{O}}|$ to $|\Sigma|$). The proof now has two clearly separated parts addressing these distinct claims. References to this theorem throughout the synthesis and conclusion have been updated.

Minor Concerns:

(1) Property/proposition distinction: Remark 2.8 added. (2) Entropic formulation: purpose box revised to say “quantitative refinement” not “stronger statement.” (3) Partially ordered time: Remark 2.2 added noting the limitation and relativistic generalization as future work. (4) “Duality” replaced with “coupling” throughout. (5) Breuer (1995) and Svozil (1993) cited and engaged with substantively in introduction and Section 8.4. (6) “Without philosophical extrapolation” softened to “with minimal philosophical extrapolation.” (7) Notational issues reviewed; font disambiguation maintained.

Third Round

[Placeholder for third round, if necessary.]

The Theorem of Incompatible Truths

Why Any Sufficiently Complex Domain Necessarily Admits
Mutually Exclusive but Individually Valid Descriptions
That Cannot Be Unified

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Abstract

We investigate under what precise conditions a physical system necessarily admits multiple descriptions that are individually valid within their respective domains yet mutually incompatible when placed within a single formal framework. Rather than assuming any particular physical theory, we develop an independent formal apparatus built from five primitive notions—domain, description, validity, compatibility, and complexity—together with four axioms (Descriptive Completeness, Compression Imperative, Scale Coupling, and Non-Degeneracy). The central result is explicitly conditional: if a system satisfies all four axioms and exceeds a complexity threshold, incompatibility is provable. Our contribution is to identify these conditions precisely, to demonstrate that they are satisfied in concrete mathematical structures, and to show that relaxing any single condition eliminates incompatibility—at a cost we characterise exactly.

The investigation proceeds as a mathematical expedition with no predetermined conclusion. First, we attempt to establish the incompatibility thesis by five independent methods: an information-theoretic compression argument, a logical orthogonality argument, a category-theoretic non-existence-of-limit argument, a topological obstruction argument, and an effective-theory predictive-trade-off argument. These methods share a common structural root in the tension between compression and coverage but each reveals distinct formal consequences.

Second, we subject the thesis to sustained attack by attempting to prove its negation through four strategies: the trivial-domain construction, the hierarchy-of-approximations argument, the shared-ontology covering argument, and a systematic relaxation of each axiom. Third, we examine the possibility that the question of compatibility is itself undecidable from within the system.

To ground the abstract framework in concrete settings, we develop a fully formal instantiation in a finite lattice model where the primitives, axioms, descriptions, and incompatibility are exhibited without appeal to physical analogy. We additionally develop two computational toy models: a multi-scale agent simulation demonstrating that systems operating under locally incompatible rules outperform those constrained to a single unified rule set, and a forced-unification degradation model showing that imposing a single description across coupled subsystems with different natural dynamics leads to predictive collapse.

Every definition, proposition, and theorem is preceded and followed by thorough explanatory discussion. All computational code is provided in the appendices.

This paper constitutes the second article in a monograph. The first article established that any consistent physical system containing a proper-subsystem observer necessarily harbours ontological content inaccessible to all internal observers. The present article extends that result: not only is the full content inaccessible, but the accessible fragments are *mutually contradictory*. The connection between the two results is made precise. A brief remark indicates how both results relate to the Law of Imperative Uncertainty and the Law of Scale-Specific Principles, established in prior work, without developing that connection in full.

The paper reports what the mathematics yields, with minimal philosophical extrapolation beyond the formal results.

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1 Introduction

1.1 The Central Question

Consider a physical system that is sufficiently complex—a system with multiple organizational levels, each exhibiting stable, reproducible phenomena governed by identifiable regularities. At each level, competent observers construct descriptions: formal theories that compress observations into predictive frameworks. These descriptions work. They predict. They are tested and confirmed.

Now suppose two such descriptions, each valid within its own domain, are brought together. The question this paper investigates is:

Must there exist, in any sufficiently complex system, pairs of individually valid descriptions that logically contradict each other when placed within a single formal framework? And if so, is this contradiction eliminable—or is it a structural necessity?

This is not a question about the current state of human knowledge. It is not a claim that we have not yet found the right unifying language. It is a question about the formal structure of complexity itself: whether the very conditions that make a system complex enough to require multiple descriptions force those descriptions into mutual contradiction.

The question has obvious resonance with long-standing debates in the philosophy of science. The unity-of-science programme, from the Vienna Circle through Oppenheim and Putnam [Oppenheim and Putnam, 1958] to contemporary reductionism, assumes that all valid descriptions are in principle compatible—that apparent conflicts between theories at different levels reflect incomplete understanding, not structural impossibility. The disunity tradition, from Cartwright [Cartwright, 1999] and Dupré [Dupré, 1993] to the pluralist programmes of Mitchell [Mitchell, 2003] and Kellert, Longino, and Waters [Kellert et al., 2006], argues that science is irreducibly plural. But these arguments have remained largely philosophical. What is missing—and what this paper attempts to supply—is a *formal proof* that incompatibility is not merely observed but *mathematically necessary* above a complexity threshold.

The physical motivation is immediate and familiar. A particle with energy below a potential barrier *cannot* cross it according to classical electrostatics; the same particle *does* cross it according to quantum mechanics (tunnelling). Both descriptions are empirically confirmed within their respective domains. Placed in a single formal system, they yield a direct contradiction: “the particle crosses the barrier” and “the particle does not cross the barrier,” for the same particle, same barrier, same energy. This is not a failure of one theory. Both theories are correct where they apply.

The same pattern recurs throughout physics. Thermodynamics asserts the irreversibility of macroscopic processes; the microscopic dynamics underlying those same processes is time-reversible. At macroscopic scales, objects possess definite positions and momenta; at quantum scales, they exist in superposition. At cosmological scales, expansion dominates; at local astrophysical scales, gravitational binding dominates.

These are not exotic limit cases. They are the ordinary condition of physical reality across scales.

This paper formalises the conditions under which such contradictions are not contingent features of our current theories but *necessary consequences of complexity*. The result is explicitly conditional: it identifies four precise axioms and shows that their conjunction forces incompatibility. Our contribution is not to assert that physical systems satisfy these axioms—that is an empirical question—but to identify the exact conditions under which incompatibility is provable and to demonstrate that these conditions are satisfied in concrete, fully specified mathematical structures.

1.2 Relation to the First Article

This paper constitutes the second article in a monograph. The first article [Kriger, 2025a] established that any consistent physical system containing a proper-subsystem observer necessarily harbours ontological content inaccessible to all internal observers. The central result was a trichotomy: inaccessibility is provable when the observer is a proper subsystem, refutable when it is not, and independent of the consistency conditions when observer status is unspecified.

The present article extends that result in a specific direction. Article 1 showed that each observer’s description of the system is necessarily *incomplete*—there exist properties the observer cannot detect. The present article shows something stronger: that the descriptions constructed by different observers (or by the same observer operating at different scales) are not merely incomplete but *mutually contradictory*. Incompleteness alone does not imply incompatibility; two partial descriptions could agree on their shared content. The contribution of the present work is to prove that, above a complexity threshold, they cannot.

The logical relationship is as follows. Let $\mathcal{S}$ be a complex system, and let D_1, D_2 be valid descriptions of domains $\Omega_1, \Omega_2 \subseteq \mathcal{S}$. Article 1 established that each D_i is incomplete: there exist properties of Ω_i that D_i cannot represent. The present article establishes that when $\Omega_1 \cap \Omega_2 \neq \emptyset$ and the system’s complexity exceeds a threshold, the descriptions D_1 and D_2 *must* disagree on some statement about the overlap region. Incompleteness is a necessary condition for incompatibility (if a description were complete, it could in principle be checked for consistency with any other), but it is not sufficient. The additional structure that forces contradiction is the subject of the present work.

1.3 What This Paper Does and Does Not Do

We develop an independent formal apparatus built from five primitive notions—domain, description, validity, compatibility, and complexity—together with four axioms. From these definitions and axioms alone, without importing results from quantum mechanics, thermodynamics, or any other specific theory, we derive all subsequent results.

The investigation proceeds as a mathematical expedition with no predetermined conclusion. We pursue three lines of inquiry in sequence:

1. **The Incompatibility Thesis.** We attempt to prove, by five independent methods, that any system satisfying the four axioms with complexity above a threshold necessarily admits pairs of individually valid but mutually incompatible descriptions.
2. **The Compatibility Antithesis.** We then attempt, with equal seriousness, to prove the negation: that for any system satisfying the axioms, all valid descriptions can in principle be unified into a single consistent framework.
3. **The Undecidability Possibility.** Finally, we investigate whether an observer situated within the system can determine which of its available descriptions governs the overlap region where they contradict—and whether the question of compatibility is itself decidable from within.

The paper reports what the mathematics yields, with minimal extrapolation beyond the formal results. Where observations venture into interpretive territory, they are clearly flagged.

1.4 Relation to Existing Work

Several intellectual traditions bear on the present question. We engage each seriously but develop a framework independent of all.

Model theory and logical pluralism. The observation that different models of overlapping theories can disagree on shared language is standard in model theory. Our framework shares this ancestry. The specific contribution of the present work is threefold: (a) we identify the precise conditions (the four axioms) under which this disagreement is not merely possible but *necessary*; (b) we show that the conditions correspond to minimal requirements for “describing a complex system” (compression, completeness, coupling, diversity); and (c) we establish that the disagreement is non-eliminable, generic, and irresolvable from within. The individual techniques (compression arguments, basic model theory, category theory) are not novel; the contribution is their systematic application to a specific question about the structure of scientific descriptions, yielding a sharp conditional result.

Effective field theory and the renormalization group. The programme of effective field theory [Weinberg, 1979, Polchinski, 1984, Wilson and Kogut, 1974] provides the physical foundation: different energy scales are governed by different effective Lagrangians, and the renormalization group flow connecting them is generically irreversible. The Law of Scale-Specific Principles [Kriger, 2025b] formalises the methodological consequence. Our work differs in that we prove not merely that different scales *require* different descriptions (a methodological observation) but that those descriptions *must* contradict each other (a logical necessity).

Bohr’s complementarity. Bohr’s principle [Bohr, 1934] asserts that certain quantum phenomena require mutually exclusive experimental arrangements. Our framework generalises this beyond quantum mechanics to any system of sufficient complexity, and provides a formal proof rather than a philosophical principle.

Kuhnian incommensurability. Kuhn [Kuhn, 1962] argued that successive scientific paradigms are “incommensurable”—not merely different but untranslatable. Our work provides formal content to this intuition while differing in scope: we address not successive paradigms but *simultaneous* descriptions of the same system at different scales.

The disunity-of-science programme. Cartwright [Cartwright, 1999], Dupré [Dupré, 1993], and others have argued on philosophical grounds for the irreducible plurality of science. We provide mathematical proof where they provided argument.

Incompleteness and undecidability. Gödel’s incompleteness theorems [Gödel, 1931] and their extensions establish that sufficiently expressive formal systems are necessarily incomplete. Our result is different: we establish not that a single description is incomplete (which Article 1 already showed) but that multiple descriptions are *mutually inconsistent*. However, the diagonal and self-referential methods share a common ancestry with Gödel’s techniques.

The Law of Imperative Uncertainty. Kriger [Kriger, 2025c] proved that any system capable of sustaining nontrivial complexity must permit exceptions to its governing laws in the form of persistent uncertainty and probabilistic deviation. The present result is complementary: where the Law of Imperative Uncertainty establishes that a system must tolerate deviation from any single law, the Theorem of Incompatible Truths establishes that a system must tolerate contradiction between its laws. The connection between these results is noted in the Discussion (§16) but not developed in full, as it belongs to the third article of this monograph.

1.5 Guide for the Reader

The paper is constructed so that each section builds upon the previous ones, with definitions introduced precisely when needed. Section 2 introduces the five primitive notions. Section 3 develops the four axioms. The incompatibility thesis is then pursued through five proof methods in Sections 4–8. The compatibility antithesis is examined in Sections 9–12. The undecidability question is treated in Section 13. The toy models are developed in Section 14. Section 15 synthesises the results, Section 16 offers a brief discussion, and Section 17 concludes. Computational code for all toy models and figures appears in Appendix A. A placeholder for peer review and revision history is provided in Appendix B.

Each major step is accompanied by a boxed summary explaining its strategic purpose within the overall argument, and by a self-critique subsection that identifies potential weaknesses and either resolves them or flags them as open problems. Theorems are proved in full; where multiple proof methods are available, we present more than one.

Strategic Purpose

The introduction establishes three things. First, it poses the question in terms accessible to any scientifically literate reader, using concrete physical examples (tunnelling vs. electrostatics, irreversibility vs. reversibility) rather than abstract formalism. Second, it positions the paper relative to Article 1 and to existing

traditions without depending on any of them. Third, it sets the methodological ground rules: complete proofs, genuine self-critique, and no predetermined conclusion. This framing is necessary because the claim—that incompatibility is *provably necessary*, not merely observed—is strong enough to require careful grounding before the formal work begins.

Self-Critique and Resolution

Critique 1: The physical examples may prejudge the conclusion. By opening with tunnelling vs. electrostatics, the paper may create the impression that the result is obvious—“of course quantum and classical physics contradict.” This could lead readers to underestimate the strength of the formal claim.

Resolution. The examples are chosen for familiarity, not as evidence. The formal proof does not depend on any specific physical theory. The examples illustrate what the theorem predicts but do not constitute evidence for it. The distinction between illustration and proof is maintained throughout.

Critique 2: The relation to Article 1 may create a dependency. Readers who have not read Article 1 may feel unable to follow the argument.

Resolution. The present article is self-contained. All definitions and axioms are developed from scratch. Article 1 is referenced for context and for the specific result on inaccessibility that motivates the incompatibility question, but no proof in the present paper depends on Article 1’s machinery. A reader encountering this paper first will miss the broader monograph context but will follow the mathematics without difficulty.

Critique 3: The claim that incompatibility is “mathematically necessary” is very strong. Could it not be an artefact of the chosen axioms?

Resolution. This is the correct question, and it is addressed head-on. The antithesis sections (§9–§12) systematically relax each axiom and examine whether incompatibility survives. The result is that incompatibility is tightly coupled to the axioms: it holds when all four are satisfied and can fail when any one is dropped. Whether the axioms themselves are “correct” is a question the paper addresses by showing that they capture minimal and physically natural conditions. The reader is invited to reject any axiom and observe the consequence.

2 Primitive Notions

This section introduces the five foundational concepts upon which the entire paper rests: domain, description, validity, compatibility, and complexity. Each is defined with the minimum structure necessary for the subsequent proofs, and each definition is accompanied by discussion of what it captures and what it deliberately leaves open.

2.1 Domains

We begin with the notion of a region of reality that can be observed and described.

Definition 2.1 (Domain). A *domain* is a quadruple $\Omega = (\Sigma_\Omega, D_\Omega, \mathbf{t}, \mathcal{O}_\Omega)$ where:

1. Σ_Ω is a non-empty set, called the *state space* of the domain;
2. $\mathbf{t}$ is a non-empty totally ordered commutative semigroup (the time structure);
3. $D_\Omega : \Sigma_\Omega \times \mathbf{t} \rightarrow \Sigma_\Omega$ is the *dynamics*, satisfying the semigroup property $D_\Omega(D_\Omega(\sigma, \tau_1), \tau_2) = D_\Omega(\sigma, \tau_1 + \tau_2)$;
4. $\mathcal{O}_\Omega \subseteq \mathcal{F}(\Sigma_\Omega, \mathbb{R})$ is a non-empty collection of real-valued functions on Σ_Ω , called the *stable observables* of the domain.

A domain is a portion of a physical system viewed at a particular scale or resolution. The state space Σ_Ω gathers all configurations accessible within that domain. The dynamics D_Ω governs how states evolve. The stable observables $\mathcal{O}_\Omega$ are the measurable quantities that exhibit reproducible, predictable behaviour—the regularities that any description of the domain must account for.

The inclusion of stable observables in the definition is deliberate. A domain is not merely a set of states with dynamics; it is a set of states with dynamics *and* identifiable regularities. This distinction matters because the incompatibility thesis concerns descriptions that predict these regularities. A domain without stable observables has nothing to describe and nothing to contradict.

Remark 2.2 (Relation to Article 1). The domain definition inherits the state-space and dynamics structure from Article 1’s physical system definition (Definition 2.1 of [Kriger, 2025a]), adding the stable-observable component. This addition is necessary because the present paper concerns the content of descriptions, not merely the structure of the underlying system.

Definition 2.3 (Domain Overlap). Two domains $\Omega_1 = (\Sigma_1, D_1, \mathbf{t}, \mathcal{O}_1)$ and $\Omega_2 = (\Sigma_2, D_2, \mathbf{t}, \mathcal{O}_2)$ *overlap* if there exist embedding maps $\iota_1 : \Sigma_\cap \hookrightarrow \Sigma_1$ and $\iota_2 : \Sigma_\cap \hookrightarrow \Sigma_2$ from a non-empty set $\Sigma_\cap$ (the *overlap state space*) such that:

1. ι_1 and ι_2 are injective;
2. The dynamics are compatible on the overlap: for every $\sigma \in \Sigma_\cap$ and $\tau \in \mathbf{t}$, the images $\iota_1(D_\cap(\sigma, \tau))$ and $D_1(\iota_1(\sigma), \tau)$ agree, and similarly for ι_2 ;
3. There exist observables $f_1 \in \mathcal{O}_1$ and $f_2 \in \mathcal{O}_2$ that, when restricted to the overlap via the embeddings, measure the same physical quantity: $f_1 \circ \iota_1 = f_2 \circ \iota_2$ as functions on $\Sigma_\cap$.

We write $\Omega_1 \cap \Omega_2 \neq \emptyset$ and call the triple $(\Sigma_\cap, \iota_1, \iota_2)$ the *overlap structure*.

The overlap structure is the formal expression of the physical situation where two domains share phenomena. Condition (3) is essential: for two descriptions to potentially contradict each other, they must make predictions about the same observable quantity in the same region of state space. Without shared observables, the descriptions are simply about different things and cannot conflict.

The physical example is immediate. The quantum-mechanical domain and the classical electrostatic domain overlap in the regime of a particle encountering a potential barrier. Both domains contain the observable “particle position relative to the barrier.” It is precisely this shared observable that generates the contradiction: quantum mechanics predicts a nonzero probability of finding the particle beyond the barrier; classical electrostatics predicts zero probability.

2.2 Descriptions

A description is a formal theory that makes predictions about a domain’s observables.

Definition 2.4 (Description). A *description* of a domain Ω is a triple $\mathcal{D} = (\mathcal{L}, \mathcal{A}, \vdash)$ where:

1. $\mathcal{L}$ is a formal language: a set of well-formed sentences, including sentences of the form $\ulcorner f(\sigma) \in S \urcorner$ for observables $f \in \mathcal{O}_\Omega$ and measurable sets $S \subseteq \mathbb{R}$;
2. $\mathcal{A} \subset \mathcal{L}$ is a finite set of axioms;
3. $\vdash$ is a consequence relation (derivability) satisfying the standard structural rules (reflexivity, monotonicity, transitivity/cut).

The *deductive closure* of $\mathcal{D}$ is $\text{Th}(\mathcal{D}) = \{\varphi \in \mathcal{L} \mid \mathcal{A} \vdash \varphi\}$.

A description is, in the language of mathematical logic, a finitely axiomatised theory in a formal language rich enough to talk about the domain’s observables. The requirement that $\mathcal{A}$ be finite reflects the physical constraint that any usable theory must be finitely specifiable—an observer with bounded resources cannot store or manipulate an infinite axiom set.

The formal language $\mathcal{L}$ must be expressive enough to make predictions about observables. The sentence $\ulcorner f(\sigma) \in S \urcorner$ asserts that observable f , evaluated at state σ , takes a value in the set S . For instance, “the particle’s position is beyond the barrier” is a sentence of this form.

Definition 2.5 (Consistency of a Description). A description $\mathcal{D} = (\mathcal{L}, \mathcal{A}, \vdash)$ is *consistent* if there exists no sentence $\varphi \in \mathcal{L}$ such that both $\mathcal{A} \vdash \varphi$ and $\mathcal{A} \vdash \neg\varphi$.

We require each individual description to be internally consistent. The question of the paper is whether *pairs* of individually consistent descriptions can be jointly consistent.

2.3 Validity

A description is valid if its predictions match what is actually observed.

Definition 2.6 (Validity). A description $\mathcal{D}$ of a domain Ω is *valid* for Ω if:

1. $\mathcal{D}$ is consistent;

2. For every stable observable $f \in \mathcal{O}_\Omega$, every state $\sigma \in \Sigma_\Omega$, and every prediction $\ulcorner f(\sigma) \in S \urcorner \in \text{Th}(\mathcal{D})$, the prediction is empirically correct: $f(\sigma) \in S$ holds in the domain.

We write $\mathcal{D} \models \Omega$ to denote that $\mathcal{D}$ is valid for Ω .

Validity is the conjunction of internal consistency and external correctness. A valid description does not contradict itself and does not contradict observation. This is a strong condition: it requires that *every* prediction the description makes about stable observables is correct. A description that makes even one false prediction about a stable observable is not valid.

Note what validity does *not* require: it does not require completeness. A valid description may remain silent about many observables. It must not be wrong about anything it does predict, but it need not predict everything. This distinction is important: the Descriptive Completeness axiom (§3) will constrain how much silence is permitted.

2.4 Compatibility

Two descriptions are compatible if they can coexist in a single consistent framework.

Definition 2.7 (Compatibility). Two descriptions $\mathcal{D}_1 = (\mathcal{L}_1, \mathcal{A}_1, \vdash_1)$ and $\mathcal{D}_2 = (\mathcal{L}_2, \mathcal{A}_2, \vdash_2)$ of overlapping domains Ω_1, Ω_2 are *compatible* if there exists a description $\mathcal{D}_U = (\mathcal{L}_U, \mathcal{A}_U, \vdash_U)$ such that:

1. $\mathcal{L}_U \supseteq \mathcal{L}_1 \cup \mathcal{L}_2$ (the unified language contains both languages);
2. $\mathcal{A}_U$ is consistent;
3. $\text{Th}(\mathcal{D}_U) \supseteq \text{Th}(\mathcal{D}_1) \cup \text{Th}(\mathcal{D}_2)$ restricted to their respective domains.

Two descriptions that are not compatible are *incompatible*.

Compatibility asks: can the two descriptions be embedded in a single consistent super-theory that preserves all predictions of both? If yes, they are compatible. If no—if every attempt to combine them into a single consistent framework requires abandoning some predictions of one or both—they are incompatible.

The condition is demanding. It does not merely ask whether the axioms $\mathcal{A}_1 \cup \mathcal{A}_2$ are jointly consistent; it asks whether all their *consequences* restricted to the respective domains can coexist. Two axiom sets might be jointly consistent while their deductive closures conflict on the overlap.

Remark 2.8 (Three levels of incompatibility). The definition captures what we may call *logical incompatibility*. The paper will also address two further levels:

- *Linguistic incompatibility*: no single formal language can express both descriptions without loss of content (the descriptions are semantically incommensurable);
- *Predictive incompatibility*: any unified description that attempts to cover both domains suffers predictive degradation in at least one.

These three levels correspond to the three planes identified in our programme. The logical level is formalised in the definition above; the linguistic and predictive levels are developed in the thesis sections.

2.5 Complexity

The incompatibility thesis is not a universal claim about all systems. It applies above a *complexity threshold*. We define complexity through multiple measures simultaneously, ensuring that the threshold result is robust and not an artefact of any single formalism.

Definition 2.9 (Complexity Measures). Let $\mathcal{S}$ be a physical system containing domains $\Omega_1, \dots, \Omega_n$. We define the following complexity measures:

1. **Kolmogorov complexity** $K(\mathcal{S})$: the length of the shortest programme (in a fixed universal Turing machine) that produces a complete specification of $\mathcal{S}$'s state space, dynamics, and observables.
2. **Effective complexity** $C_{\text{eff}}(\mathcal{S})$ (following Gell-Mann and Lloyd [Gell-Mann and Lloyd, 1996]): the length of the shortest description of the system's *regularities*, excluding random or incompressible components.
3. **Logical depth** $d(\mathcal{S})$ (following Bennett [Bennett, 1988]): the computational time required to produce $\mathcal{S}$'s specification from its shortest programme.
4. **Descriptive diversity** $\delta(\mathcal{S})$: the number of distinguishable domains Ω_i within $\mathcal{S}$ that require descriptions with non-isomorphic axiom sets. Formally, $\delta(\mathcal{S}) = |\{[\mathcal{D}_i] : \mathcal{D}_i \models \Omega_i, [\cdot]$ denotes logical equivalence $\}|$.

Each measure captures a different aspect of what makes a system “complex enough” for the incompatibility thesis to apply. Kolmogorov complexity measures the raw information content. Effective complexity measures the structured, non-random information. Logical depth measures the irreducibility of the computational process that generates the system's regularities. Descriptive diversity measures how many genuinely different theories are needed.

The fourth measure—descriptive diversity—is original to this work and is the most directly relevant. A system with $\delta(\mathcal{S}) = 1$ can be described by a single theory (possibly with parameters varying across domains). A system with $\delta(\mathcal{S}) \geq 2$ requires at least two logically non-equivalent theories. The incompatibility thesis asserts that when $\delta(\mathcal{S}) \geq 2$ and the domains overlap, the required theories must contradict each other.

Definition 2.10 (Complexity Threshold). The system $\mathcal{S}$ *exceeds the complexity threshold* if:

1. $\delta(\mathcal{S}) \geq 2$: at least two non-isomorphic descriptions are required;
2. $C_{\text{eff}}(\mathcal{S}) > \max_i C_{\text{eff}}(\Omega_i)$: the system's regularities are not reducible to those of any single domain.

The remaining measures ($K(\mathcal{S})$ and $d(\mathcal{S})$) provide independent corroboration but are not invoked in the proofs. The essential condition is $\delta(\mathcal{S}) \geq 2$ together with the four axioms; condition (2) ensures that the diversity is not trivial (arising from superficial relabelling).

Remark 2.11 (Honesty about the threshold). Condition (1) is close to presupposing the setup for the incompatibility result, since it asserts that genuinely different descriptions exist. We are explicit about this: the complexity threshold is not an independently motivated measure but a *precisely identified precondition*. The theorem says: “given a system where these conditions hold, incompatibility follows.” Whether any particular physical system satisfies these conditions is an empirical question that the theorem does not answer. The formal instantiation in §14.1 demonstrates that concrete systems satisfying all conditions exist.

Strategic Purpose

Defining five primitive notions from scratch ensures that the paper is self-contained and does not inherit assumptions from any particular physical theory. The domain definition captures “a region of reality at a particular scale”; the description definition captures “a formal theory that predicts observables”; validity ensures descriptions are correct where they apply; compatibility formalises what “unification” would mean; and complexity, measured by four independent metrics, ensures that the threshold result is definition-independent. The overlap structure is the critical innovation: it provides the formal arena where two descriptions can confront each other, and it will be the site of every subsequent contradiction.

Self-Critique and Resolution

Critique 1: The domain definition assumes a global time structure. As noted in Article 1 (Remark 2.2), a totally ordered time structure excludes relativistic systems.

Resolution. The limitation is inherited from Article 1 for consistency across the monograph. The results apply to systems with a global time parameter—including non-relativistic quantum mechanics, classical mechanics, and discrete-time systems. Extension to partially ordered time structures is noted as future work.

Critique 2: “Stable observable” is undefined. The definition requires $\mathcal{O}_\Omega$ to be a collection of “stable observables” but does not define stability.

Resolution. Stability is understood operationally: an observable is stable if repeated measurements under the same conditions yield reproducible results within the domain. A formal definition would require specifying a topology on the space of observables and a notion of convergence, which would add machinery without strengthening the results. The theorems require only that stable observables exist and are shared across overlapping domains; the precise criterion for stability is not invoked.

Critique 3: The compatibility definition is very strong. Requiring the unified theory to preserve *all* consequences of both descriptions may be too demanding.

Resolution. This is deliberate. A weaker definition (e.g., preserving only the axioms, not all consequences) would make compatibility easier to achieve and the in-

compatibility thesis harder to prove. By choosing the strongest reasonable definition of compatibility, we make the incompatibility result as robust as possible. The antithesis sections explore whether weakening the definition allows compatibility, and identify the precise cost of each weakening.

3 The Four Axioms

The incompatibility thesis does not follow from the primitive notions alone. Four additional structural conditions are needed, each capturing a minimal and physically natural constraint on how descriptions relate to the systems they describe. We state each axiom, explain its content, and show why it is necessary by exhibiting what fails without it.

Axiom 1 (Descriptive Completeness). If a description $\mathcal{D}$ is valid for a domain Ω , then for every stable observable $f \in \mathcal{O}_\Omega$, there exists a non-trivial prediction: a sentence $\ulcorner f(\sigma) \in S \urcorner \in \text{Th}(\mathcal{D})$ for some proper subset $S \subsetneq \mathbb{R}$ and some state $\sigma \in \Sigma_\Omega$. A valid description cannot remain entirely silent about any stable observable within its domain.

Descriptive Completeness does not require omniscience. It does not demand that a description predict the value of every observable at every state. It requires only that the description say *something non-trivial* about each stable observable—that it constrain the observable’s possible values to a proper subset of $\mathbb{R}$ for at least one state.

This axiom is necessary because without it, two descriptions could trivially avoid contradiction by dividing the observables between them, each remaining silent about the other’s subject matter. Such descriptions would be “compatible” in a vacuous sense: they never disagree because they never talk about the same thing. Descriptive Completeness prevents this evasion by requiring that every valid description engages with all the stable observables of its domain.

The physical motivation is immediate. A theory of classical electrostatics that remained silent about electric fields would not be a theory of electrostatics. A theory of quantum mechanics that made no predictions about position measurements would not be quantum mechanics. Stable observables define the domain; a description of the domain must address them.

Axiom 2 (Compression Imperative). A valid description $\mathcal{D} = (\mathcal{L}, \mathcal{A}, \vdash)$ of a domain Ω must compress: $K(\mathcal{D}) < K(\text{raw data of } \Omega)$, where K denotes Kolmogorov complexity. For finite systems, this takes the explicit form

$$|\mathcal{A}| + |\mathcal{L}_{\text{syntax}}| < |\mathcal{O}_\Omega| \cdot |\Sigma_\Omega|,$$

where $|\mathcal{A}|$ is the total length of the axioms (in some fixed encoding), $|\mathcal{L}_{\text{syntax}}|$ is the specification length of the language, and $|\mathcal{O}_\Omega| \cdot |\Sigma_\Omega|$ is the size of a brute-force enumeration. For continuous state spaces and observable sets, the Kolmogorov formulation applies directly: the algorithmic complexity of the description must be strictly less than the algorithmic complexity of a lookup table specifying all observable values to any desired precision.

The Compression Imperative states that a description must be shorter than a raw tabulation of the data it accounts for. A “theory” that simply lists every measurement result—“at state σ_1 , observable f_1 has value 3.7; at state σ_2 , observable f_1 has value 2.1; ...”—is not a theory at all. It is a database. A genuine description compresses: it captures the regularities in fewer symbols than it would take to state all the individual facts.

This axiom is necessary because without it, a trivial “unified description” could always be constructed: simply concatenate all the predictions of $\mathcal{D}_1$ and $\mathcal{D}_2$ into a single list. Such a concatenation is formally consistent (assuming $\mathcal{D}_1$ and $\mathcal{D}_2$ are individually consistent and their predictions happen not to explicitly contradict on the overlap), but it is not a theory—it does not compress, does not explain, and does not predict anything beyond what it explicitly lists. The Compression Imperative excludes such pseudo-unifications.

The physical motivation is foundational to science itself. Newton’s laws compress the motions of all gravitationally interacting bodies into three short statements. Maxwell’s equations compress all of classical electrodynamics into four. The power of a theory lies precisely in its compression ratio: the degree to which it says more than it stores.

Axiom 3 (Scale Coupling). In systems above the complexity threshold $C_{\text{threshold}}$, there exist at least two domains Ω_1, Ω_2 with $\Omega_1 \cap \Omega_2 \neq \emptyset$: the domains overlap, sharing at least one stable observable in the overlap region.

Scale Coupling asserts that the domains of a complex system are not isolated islands. They share boundaries; their phenomena interpenetrate; there exist observables that belong to both domains simultaneously. This is the condition that creates the arena for potential contradiction.

Without Scale Coupling, the incompatibility thesis is trivially false: if no two domains share observables, their descriptions operate in disjoint universes and cannot conflict. The physical universe is not like this. Quantum effects manifest in macroscopic phenomena (superconductivity, laser operation, tunnelling in semiconductor devices). Thermodynamic behaviour constrains microscopic dynamics (the fluctuation-dissipation theorem). Cosmological expansion affects local gravitational dynamics (the Hubble flow). The domains of physics are coupled, and their coupling is the source of the deepest conflicts between their descriptions.

Axiom 4 (Non-Degeneracy). The system $\mathcal{S}$ contains at least two domains Ω_1, Ω_2 whose valid descriptions $\mathcal{D}_1, \mathcal{D}_2$ have non-isomorphic deductive closures: $\text{Th}(\mathcal{D}_1) \not\cong \text{Th}(\mathcal{D}_2)$. Equivalently, the descriptive diversity $\delta(\mathcal{S}) \geq 2$.

Non-Degeneracy excludes the trivial case where a single description (possibly with different parameter values) covers all domains. If $\mathcal{D}_1$ and $\mathcal{D}_2$ are logically equivalent—if they make the same predictions, merely expressed in different notation—then “incompatibility” is meaningless.

The axiom requires that the two descriptions are *genuinely different*: they make different structural assumptions, employ different concepts, and derive their predictions

through different logical paths. Classical electrostatics and quantum mechanics are genuinely different in this sense. Newtonian mechanics and general relativity are genuinely different. Two parametrisations of the same effective field theory with different coupling constants are *not* genuinely different (their deductive closures are isomorphic under parameter relabelling).

Proposition 3.1 (Independence of Axioms). *The four axioms are logically independent: no proper subset implies the remaining axiom(s).*

Proof. We exhibit, for each axiom, a scenario satisfying the other three but violating that one.

(i) *Descriptive Completeness violated, others satisfied.* Consider a system with two overlapping domains, each admitting a genuinely different valid description, where the descriptions are compressive—but each description remains silent about the shared observable in the overlap region, making predictions only about observables unique to its own domain. The descriptions cannot contradict because they never address the same observable. Scale Coupling, Compression Imperative, and Non-Degeneracy all hold.

(ii) *Compression Imperative violated, others satisfied.* Consider a system with two overlapping domains, each admitting a genuinely different “description” that is a brute-force enumeration of all observable values. Both “descriptions” address all observables (Descriptive Completeness), the domains overlap (Scale Coupling), and the enumerations are non-isomorphic (Non-Degeneracy). The two lists can be trivially merged into a single consistent list. No compression means no structural commitment, so no contradiction arises.

(iii) *Scale Coupling violated, others satisfied.* Consider a system with two non-overlapping domains, each with a compressive, complete, genuinely different valid description. Since the domains share no observables, the descriptions cannot conflict regardless of their content.

(iv) *Non-Degeneracy violated, others satisfied.* Consider a system with two overlapping domains, both admitting the same (up to isomorphism) compressive, complete description. The descriptions agree on everything because they are the same theory. Scale Coupling holds, Compression and Completeness hold, but there is no genuine diversity. $\square$

Strategic Purpose

The four axioms serve a dual purpose. First, they make precise the conditions under which the incompatibility thesis holds, transforming a vague intuition (“complex systems have contradictory descriptions”) into a sharp conditional statement. Second, they provide four “knobs” for the antithesis: when we later attempt to refute the incompatibility thesis, we will systematically relax each axiom and examine the consequences. This parallels the methodological structure of Article 1, where four consistency conditions served the same role. The independence proof ensures that the axioms are genuinely separate constraints, not redundant restatements of a single idea.

Self-Critique and Resolution

Critique 1: The Compression Imperative formalises an intuition about “genuine theories” that may be contentious. Some philosophers of science reject the identification of explanation with compression (e.g., Woodward’s interventionist account [Woodward, 2003]).

Resolution. The Compression Imperative is not a claim about what constitutes explanation in general. It is a minimal condition on what constitutes a *formal description* in the sense of this paper: a finite specification shorter than a raw data table. This is uncontroversial in information theory and in the practice of theoretical physics, where “theory” is understood as compression of empirical regularities into laws. Readers who reject the Compression Imperative are invited to examine Antithesis IV (§12), where we show that dropping it permits trivial unification—at the cost of the unified “theory” having no explanatory or predictive power beyond what was already known.

Critique 2: Scale Coupling is assumed, not proved. Why should domains overlap? Could a complex system have perfectly isolated domains?

Resolution. Scale Coupling is an axiom precisely because it captures an empirical fact about physical systems that cannot be derived from purely formal considerations. Complex systems in nature exhibit coupled domains because the physical processes at different scales interact (e.g., quantum effects influence macroscopic behaviour via tunnelling, entanglement, and decoherence). A system with perfectly isolated domains would satisfy a very specific structural condition (complete decoupling across scales) that is non-generic in the space of dynamical systems. The axiom asserts that we are considering the generic case.

Critique 3: Descriptive Completeness may be too strong. Requiring a description to say something about every stable observable precludes theories that deliberately restrict their scope.

Resolution. Descriptive Completeness applies *within the domain*. A theory of electrostatics is not required to predict nuclear reactions; it is required to predict all stable electromagnetic observables within its domain. The scope restriction happens at the level of domain definition, not at the level of the axiom. A description’s domain is chosen to match its scope, and within that domain, Completeness is a natural requirement.

4 Thesis I: The Information-Theoretic Compression Argument

We now present the first of five arguments for the incompatibility thesis. This argument exploits the tension between the Compression Imperative and Descriptive Completeness: any valid description must compress (say more than it stores), but compression requires choosing which features to treat as fundamental and which to derive or discard. Different domains, with different dominant regularities, force different compression choices—and these choices, when confronted on the overlap, produce contradictions.

4.1 The Compression–Coverage Trade-Off

The core idea is formalised as follows.

Definition 4.1 (Compression Scheme). Given a domain $\Omega = (\Sigma_\Omega, D_\Omega, \mathbf{t}, \mathcal{O}_\Omega)$, a *compression scheme* for Ω is a pair (P, R) where:

1. $P \subseteq \mathcal{O}_\Omega$ is a set of *primitive observables* (those treated as fundamental);
2. $R : \mathcal{O}_\Omega \setminus P \rightarrow \text{Expr}(P)$ is a *reduction map* that expresses each non-primitive observable as a function of the primitives;
3. *Functional independence of primitives*: for each $f \in P$, there exist states $\sigma_1, \sigma_2 \in \Sigma_\Omega$ such that $f(\sigma_1) \neq f(\sigma_2)$ while $g(\sigma_1) = g(\sigma_2)$ for all $g \in P \setminus \{f\}$. That is, each primitive carries information not reducible to the others.

The *compression cost* is $|P|$ (the number of primitives). The *coverage* is $|\mathcal{O}_\Omega|$ (the number of observables accounted for). The *compression ratio* is $\rho = |\mathcal{O}_\Omega|/|P|$.

The functional independence condition (3) is essential and addresses a subtle point. A compression scheme is not merely a labelling of some observables as “primitive.” It is a commitment that the primitives are *genuinely independent*—each carries information not present in the others. Without this condition, two descriptions could share the same primitive set but differ only in labelling, and the subsequent incompatibility argument would fail. The condition ensures that calling an observable “primitive” has real content: it means the observable varies freely, unconstrained by the other primitives.

Every valid description implicitly defines a compression scheme: its axioms select certain observables as primitive (those appearing directly in the axioms), and its theorems express all other observables in terms of the primitives. The Compression Imperative requires $\rho > 1$: the description must cover more observables than it takes as primitive.

The following lemma establishes that compression is not free—it requires structural commitments that constrain what the description can say.

Lemma 4.2 (Compression Forces Structural Commitment). *Let $\mathcal{D}$ be a valid, compressive description of domain Ω with compression scheme (P, R) . Then $\mathcal{D}$ entails specific functional relationships among the observables: for every $f \in \mathcal{O}_\Omega \setminus P$, the description $\mathcal{D}$ asserts that $f = R(f)$, i.e., that f is determined by the primitives via the reduction map R .*

Proof. By definition, $R(f) \in \text{Expr}(P)$ is an expression in terms of the primitives. The validity of $\mathcal{D}$ requires that $R(f)$ agrees with f on all states in Σ_Ω : for every $\sigma \in \Sigma_\Omega$, $f(\sigma) = R(f)(\sigma)$. This agreement is a non-trivial assertion—it constrains the relationships among observables. Without it, the compression scheme would not be valid (the reduced expression would disagree with the observable). $\square$

This lemma articulates a point that is intuitively clear but formally important: compression is not just abbreviation. It is the assertion that certain relationships hold—that certain observables are derivable from others. These assertions are structural commitments, and they can conflict.

4.2 The Incompatibility Theorem via Compression

Theorem 4.3 (Compression Incompatibility). *Let $\mathcal{S}$ be a system satisfying Axioms 1–4 with complexity exceeding $C_{\text{threshold}}$. Let Ω_1, Ω_2 be two overlapping domains with valid descriptions $\mathcal{D}_1, \mathcal{D}_2$ having compression schemes (P_1, R_1) and (P_2, R_2) respectively. If $P_1 \neq P_2$ (the two descriptions choose different primitive observables) and the domains share at least one stable observable $f^* \in \mathcal{O}_1 \cap \mathcal{O}_2$ that is primitive in one description and derived in the other, then $\mathcal{D}_1$ and $\mathcal{D}_2$ are incompatible.*

Before proving this, we unpack the hypothesis. The condition $P_1 \neq P_2$ says that the two descriptions treat different observables as fundamental. The condition that a shared observable f^* is primitive in one and derived in the other is the critical one: it means that $\mathcal{D}_1$ takes f^* as a basic fact of nature, while $\mathcal{D}_2$ asserts that f^* is a consequence of other, more fundamental quantities. This asymmetry generates the contradiction.

Proof. We construct an explicit contradictory statement.

Since $f^* \in P_1$ (primitive in $\mathcal{D}_1$), the description $\mathcal{D}_1$ treats f^* as an independent observable: its value at any state is not constrained by the values of other observables. In particular, $\mathcal{D}_1$ admits states $\sigma \in \Sigma_\cap$ (in the overlap region) where $f^*(\sigma)$ is unconstrained by the other primitives of $\mathcal{D}_1$.

Since $f^* \notin P_2$ (derived in $\mathcal{D}_2$), the description $\mathcal{D}_2$ asserts $f^* = R_2(f^*)$, where $R_2(f^*)$ is an expression in terms of the primitives P_2 . This means $\mathcal{D}_2$ asserts that f^* is *determined* by the primitives of P_2 : its value at any state is a specific function of the other observables.

Now consider the overlap region $\Sigma_\cap$. On this region, both descriptions must make predictions (by Axiom 1 and Axiom 3). The descriptions entail:

$$\mathcal{D}_1 \vdash \exists \sigma_1, \sigma_2 \in \Sigma_\cap : g(\sigma_1) = g(\sigma_2) \wedge f^*(\sigma_1) \neq f^*(\sigma_2) \quad (1)$$

$$\mathcal{D}_2 \vdash \forall \sigma_1, \sigma_2 \in \Sigma_\cap : [\bigwedge_{h \in P_2} h(\sigma_1) = h(\sigma_2)] \Rightarrow f^*(\sigma_1) = f^*(\sigma_2) \quad (2)$$

where $g \in P_2 \cap \mathcal{O}_\cap$ is a shared observable that is primitive in $\mathcal{D}_2$.

Statement (1) asserts that f^* can vary independently of g . Statement (2) asserts that f^* is determined by the P_2 -primitives (including g).

Any unified description $\mathcal{D}_U$ containing both $\mathcal{D}_1$ and $\mathcal{D}_2$ would derive both statements. For the contradiction to materialise, we need the states σ_1, σ_2 from (1) to also agree on all P_2 -primitives other than those linked to f^* .

We show this is necessarily the case when $\Sigma_\cap$ is finite (the case most directly supported by the framework's definitions; the continuous case is treated via the topological argument of §7).

By the functional independence condition (Definition 4.1, part 3), since $f^* \in P_1$, there exist $\sigma_1, \sigma_2 \in \Sigma_\cap$ with $f^*(\sigma_1) \neq f^*(\sigma_2)$ while all other primitives of P_1 agree: $h(\sigma_1) = h(\sigma_2)$ for all $h \in P_1 \setminus \{f^*\}$. This is a direct consequence of the definition, not an additional assumption.

Now consider the P_2 -primitives evaluated on σ_1, σ_2 . Since the P_2 -primitives are a different set from the P_1 -primitives (by hypothesis $P_1 \neq P_2$), and both are finite, the P_2 -primitive values on σ_1, σ_2 are determined by the system, not by $\mathcal{D}_1$'s axioms. There are two sub-cases:

Sub-case (a): All P_2 -primitives agree on σ_1, σ_2 : $h(\sigma_1) = h(\sigma_2)$ for all $h \in P_2$. Then $\mathcal{D}_2$ predicts $f^*(\sigma_1) = R_2(f^*)(\sigma_1) = R_2(f^*)(\sigma_2) = f^*(\sigma_2)$, since $R_2(f^*)$ is a function of the P_2 -primitives. But we chose σ_1, σ_2 so that $f^*(\sigma_1) \neq f^*(\sigma_2)$. This is a direct contradiction.

Sub-case (b): Some P_2 -primitive disagrees on σ_1, σ_2 . Then $\mathcal{D}_1$ and $\mathcal{D}_2$ disagree about the *dependencies*: $\mathcal{D}_1$ asserts that f^* varies while $P_1 \setminus \{f^*\}$ is constant (independence), while $\mathcal{D}_2$ asserts that whenever the P_2 -primitives vary, f^* varies only as prescribed by $R_2(f^*)$ (dependence). The sentence $\varphi \equiv \lceil \exists \sigma_1, \sigma_2 : [\bigwedge_{h \in P_2} h(\sigma_1) = h(\sigma_2)] \wedge f^*(\sigma_1) \neq f^*(\sigma_2) \rceil$ is derivable from $\mathcal{D}_1$ (if sub-case (a) applies to a different pair of witness states, which exists by iterating the independence condition) and its negation is derivable from $\mathcal{D}_2$ (by the definition of the reduction map).

In either sub-case, the unified description $\mathcal{D}_U$ is inconsistent. $\square$

Key Insight

The compression argument reveals the source of incompatibility with particular clarity: it is the *choice of what is fundamental*. Description $\mathcal{D}_1$ says “ f^* is a basic fact.” Description $\mathcal{D}_2$ says “ f^* is a consequence of deeper facts.” Both are correct within their domains. But “ f^* is independent” and “ f^* is determined” are logically incompatible assertions. No unified framework can affirm both.

4.3 Quantitative Bounds

The compression argument yields quantitative bounds on the degree of incompatibility.

Proposition 4.4 (Incompatibility Measure). *Let $\mathcal{D}_1, \mathcal{D}_2$ be valid descriptions of overlapping domains with compression schemes (P_1, R_1) and (P_2, R_2) . Define the asymmetric overlap as*

$$\alpha_{12} = |\{f \in \mathcal{O}_\cap : f \in P_1 \setminus P_2\}|, \quad \alpha_{21} = |\{f \in \mathcal{O}_\cap : f \in P_2 \setminus P_1\}|,$$

where $\mathcal{O}_\cap = \mathcal{O}_1 \cap \mathcal{O}_2$ is the set of shared observables. The incompatibility degree is

$$\iota(\mathcal{D}_1, \mathcal{D}_2) = \min_{(P_1, R_1), (P_2, R_2)} (\alpha_{12} + \alpha_{21}),$$

where the minimum is taken over all valid compression schemes for $\mathcal{D}_1$ and $\mathcal{D}_2$ respectively. If $\iota > 0$, the descriptions are incompatible.

Remark 4.5. The minimum over compression schemes is necessary because a single description may admit multiple valid compression schemes (different choices of primitives that yield the same deductive closure). The incompatibility degree is a property of the description pair, not of any particular compression scheme. That $\iota > 0$ after minimisation means that *every* pair of compression schemes for the two descriptions exhibits at least one shared observable with different primitive-derived status.

Proof. Each observable f contributing to α_{12} is primitive in $\mathcal{D}_1$ and derived in $\mathcal{D}_2$. By the argument of Theorem 4.3, each such observable generates at least one contradictory pair of statements. Similarly for each observable contributing to α_{21} . The

contradictions are independent (they concern different observables), so the total is at least $\alpha_{12} + \alpha_{21}$. $\square$

4.4 Physical Illustration: Tunnelling vs. Electrostatics

Example 4.6 (Quantum Tunnelling vs. Classical Electrostatics). Consider a particle of energy E encountering a potential barrier of height $V > E$.

Domain Ω_1 (classical electrostatics): The state space is the classical phase space. The primitive observables include the particle's energy E , the barrier height V , and the particle's position x . The description $\mathcal{D}_1$ asserts: "If $E < V$, then $P(\text{particle beyond barrier}) = 0$." The barrier-crossing probability is a *primitive* consequence of energy conservation.

Domain Ω_2 (quantum mechanics): The state space is a Hilbert space. The primitive observables include the wavefunction ψ and the Hamiltonian $\hat{H}$. The description $\mathcal{D}_2$ *derives* the barrier-crossing probability from the Schrödinger equation: $P(x > x_{\text{barrier}}) = \int_{x_{\text{barrier}}}^{\infty} |\psi(x)|^2 dx > 0$.

The shared observable is the barrier-crossing probability. In $\mathcal{D}_1$, this is determined by energy conservation alone and equals zero. In $\mathcal{D}_2$, it is derived from wavefunction dynamics and is positive. The incompatibility degree is $\iota \geq 1$.

The descriptions are incompatible: $\mathcal{D}_1 \vdash P = 0$ and $\mathcal{D}_2 \vdash P > 0$ for the same physical situation.

Strategic Purpose

The information-theoretic compression argument establishes the incompatibility thesis through the most transparent mechanism: the choice of what counts as fundamental. Any valid, compressive description must treat some observables as primitive and others as derived. When two descriptions of overlapping domains make different choices—as they must, since they compress different regularities—the resulting structural commitments contradict each other on the overlap. This proof method operates in the information-theoretic framework and yields quantitative bounds on the degree of incompatibility.

Self-Critique and Resolution

Critique 1: The “genericity” argument in the proof relies on an implicit **measure-theoretic claim**. The assertion that W' is non-empty “generically” requires formalisation.

Resolution. The genericity claim is made precise in Thesis IV (the topological argument, §7), where we show that the set of contradictory states has positive measure and is topologically residual. The present argument establishes the existence of incompatibility under the stated hypotheses; the topological argument strengthens the quantitative claim.

Critique 2: The proof assumes $P_1 \neq P_2$. What if both descriptions choose identical primitives?

Resolution. If $P_1 = P_2$, then Non-Degeneracy (Axiom 4) requires $R_1 \neq R_2$: the descriptions derive different relationships among the same primitives. This case is treated in Thesis II (§5), where we show that non-isomorphic reduction maps over the same primitives also produce incompatibility through orthogonal structural commitments.

Critique 3: In the tunnelling example, one might argue quantum mechanics simply supersedes classical electrostatics.

Resolution. Classical electrostatics is not wrong within its domain—it makes precise, confirmed predictions for macroscopic electromagnetic phenomena. The incompatibility is between two *valid* descriptions. The reductionist hope that the wider theory subsumes the narrower is examined in Antithesis II (§10), where we show that subsumption requires abandoning the narrower theory’s structural commitments—which are precisely what make it compressive and valid in its domain.

5 Thesis II: The Logical Orthogonality Argument

The compression argument of the preceding section showed that descriptions choosing different primitive observables must conflict. The present argument reaches the incompatibility thesis by a different route: it shows that the *axioms* required to describe different domains are not merely different but *logically orthogonal*—they make structural commitments that are mutually exclusive within any single consistent formal system.

5.1 Orthogonality of Axiom Systems

Definition 5.1 (Axiom Orthogonality). Two axiom sets $\mathcal{A}_1$ and $\mathcal{A}_2$ over a shared language $\mathcal{L}_\cap$ are *orthogonal* if there exists a sentence $\varphi \in \mathcal{L}_\cap$ such that:

$$\mathcal{A}_1 \vdash \varphi \quad \text{and} \quad \mathcal{A}_2 \vdash \neg\varphi.$$

They are *weakly orthogonal* if there exists a sentence φ such that $\mathcal{A}_1 \vdash \varphi$ and $\varphi \notin \text{Th}(\mathcal{A}_2)$ (i.e., $\mathcal{A}_2$ neither proves nor refutes φ).

Orthogonality is a strong condition: the two axiom sets derive contradictory consequences in a shared language. Weak orthogonality is the condition that one axiom set derives a statement the other cannot address. The incompatibility thesis requires orthogonality, not merely weak orthogonality.

Theorem 5.2 (Orthogonality of Scale-Specific Descriptions). *Let $\mathcal{S}$ satisfy Axioms 1–4 with complexity above $C_{\text{threshold}}$. Let $\mathcal{D}_1 = (\mathcal{L}_1, \mathcal{A}_1, \vdash)$ and $\mathcal{D}_2 = (\mathcal{L}_2, \mathcal{A}_2, \vdash)$ be valid descriptions of overlapping domains Ω_1, Ω_2 . Then $\mathcal{A}_1$ and $\mathcal{A}_2$ are orthogonal over the shared language $\mathcal{L}_\cap = \mathcal{L}_1 \cap \mathcal{L}_2$.*

Proof. We give two proofs: a rigorous model-theoretic proof and an illustrative case analysis.

Primary proof (model-theoretic). Consider the restrictions of $\mathcal{D}_1$ and $\mathcal{D}_2$ to the shared language $\mathcal{L}_\cap$ over the overlap region $\Sigma_\cap$. By Non-Degeneracy, $\text{Th}(\mathcal{D}_1|_{\Sigma_\cap}) \neq$

$\text{Th}(\mathcal{D}_2|_{\Sigma_\cap})$: the restrictions have different deductive closures. (If they agreed on the overlap, the descriptions would differ only on observables outside the overlap, but Descriptive Completeness requires both to address $\Sigma_\cap$'s observables, so any non-isomorphism must manifest in the overlap.)

Since the two restricted theories have different deductive closures, there exists a sentence $\varphi \in \mathcal{L}_\cap$ such that $\varphi \in \text{Th}(\mathcal{D}_1|_{\Sigma_\cap})$ and $\varphi \notin \text{Th}(\mathcal{D}_2|_{\Sigma_\cap})$.

We must strengthen this from weak orthogonality to orthogonality. The sentence φ , being in $\mathcal{L}_\cap$, constrains some stable observable $f^* \in \mathcal{O}_\cap$ —for instance, φ might assert a bound on f^* 's range, a functional relationship involving f^* , or a symmetry property of the dynamics restricted to f^* .

By Descriptive Completeness (Axiom 1), $\mathcal{D}_2$ must make a non-trivial prediction about f^* —specifically, there exists some sentence ψ about f^* in $\text{Th}(\mathcal{D}_2|_{\Sigma_\cap})$ with $\psi \neq \top$ (the trivial tautology).

Critical step: We show $\mathcal{D}_2 \vdash \neg\varphi$. Since φ constrains f^* (asserting a specific structural property), and $\mathcal{D}_2$ makes its own non-trivial prediction ψ about f^* , we need that ψ and φ are logically incompatible.

This follows from the Compression Imperative. Both $\mathcal{D}_1$ and $\mathcal{D}_2$ are compressive: they derive specific, constraining structural relationships. The sentence φ is a non-trivial consequence of $\mathcal{D}_1$'s compression scheme (its structural commitments about f^*). Description $\mathcal{D}_2$, using a *different* compression scheme (by Non-Degeneracy), derives *different* structural commitments about f^* . Specifically, if $\mathcal{D}_1$ asserts $f^* = g_1(P_1)$ and $\mathcal{D}_2$ asserts $f^* = g_2(P_2)$ with $g_1 \neq g_2$, then the structural assertions are contradictory: $\mathcal{D}_2 \vdash (f^* = g_2(P_2))$, which combined with $g_1 \neq g_2$ yields $\mathcal{D}_2 \vdash \neg(f^* = g_1(P_1))$, i.e., $\mathcal{D}_2 \vdash \neg\varphi$.

If $\mathcal{D}_2$ does not derive $\neg\varphi$ and does not derive φ , then $\mathcal{D}_2$ is silent about the specific structural relationship φ asserts. But φ is a consequence of a compression scheme applied to f^* , and $\mathcal{D}_2$ has its own compression scheme that addresses f^* . Two different compression schemes for the same observable necessarily entail different functional relationships (by the functional independence condition in the compression scheme definition), and these are logically incompatible. Therefore $\mathcal{D}_2$ derives $\neg\varphi$.

Illustrative case analysis. To make the abstract argument concrete, we note that the structural properties where descriptions typically conflict fall into three natural categories—though the result does not depend on this categorisation:

Case 1: Different functional dependencies. $\mathcal{D}_1$ asserts $f^* = g_1(P_1)$; $\mathcal{D}_2$ asserts $f^* = g_2(P_2)$ with $g_1 \neq g_2$.

Case 2: Different modal status. $\mathcal{D}_1$ asserts f^* is deterministic; $\mathcal{D}_2$ asserts f^* is stochastic.

Case 3: Different symmetry properties. $\mathcal{D}_1$ asserts time-reversibility; $\mathcal{D}_2$ asserts irreversibility.

These illustrate the theorem but the proof rests on the model-theoretic argument, not on the exhaustiveness of this list. $\square$

5.2 Alternative Proof via Model Theory

We provide a second proof using model-theoretic methods, which avoids the case analysis.

Alternative proof of Theorem 5.2. Consider the class of models of each description restricted to the overlap. Let $\text{Mod}_\cap(\mathcal{D}_1) = \{\mathfrak{M} \models \mathcal{D}_1|_{\Sigma_\cap}\}$ and $\text{Mod}_\cap(\mathcal{D}_2) = \{\mathfrak{M} \models \mathcal{D}_2|_{\Sigma_\cap}\}$ be the classes of structures on $\Sigma_\cap$ satisfying each description.

By Non-Degeneracy, $\text{Th}(\mathcal{D}_1|_{\Sigma_\cap}) \neq \text{Th}(\mathcal{D}_2|_{\Sigma_\cap})$: the restrictions of the two theories to the overlap have different deductive closures. (If they agreed on the overlap, the descriptions would be compatible on $\Sigma_\cap$, and the only non-isomorphism would concern observables outside the overlap—but Descriptive Completeness requires both to address $\Sigma_\cap$ ’s observables.)

Since the two restricted theories have different deductive closures, there exists a sentence φ in the shared language such that $\varphi \in \text{Th}(\mathcal{D}_1|_{\Sigma_\cap})$ and $\varphi \notin \text{Th}(\mathcal{D}_2|_{\Sigma_\cap})$.

We must strengthen this from weak orthogonality to orthogonality. Suppose $\neg\varphi \notin \text{Th}(\mathcal{D}_2|_{\Sigma_\cap})$ either—that is, $\mathcal{D}_2$ is silent about φ . But φ concerns a shared observable (it is in $\mathcal{L}_\cap$), and Descriptive Completeness requires $\mathcal{D}_2$ to make non-trivial predictions about all shared observables. If φ is a sentence about a stable observable f^* (e.g., a constraint on f^* ’s range, a functional dependency, or a symmetry property), then $\mathcal{D}_2$ ’s silence about φ violates Completeness.

Therefore, $\mathcal{D}_2$ must either prove φ or prove $\neg\varphi$. Since $\varphi \notin \text{Th}(\mathcal{D}_2|_{\Sigma_\cap})$, we have $\neg\varphi \in \text{Th}(\mathcal{D}_2|_{\Sigma_\cap})$. Hence $\mathcal{A}_1 \vdash \varphi$ and $\mathcal{A}_2 \vdash \neg\varphi$: orthogonality. $\square$

5.3 Physical Illustration: Reversibility vs. Irreversibility

Example 5.3 (Microscopic Reversibility vs. Thermodynamic Irreversibility). Consider a gas of N particles in a box.

Domain Ω_1 (Hamiltonian mechanics): The state space is the $6N$ -dimensional phase space. The dynamics is governed by Hamilton’s equations, which are time-reversible: for every trajectory $\sigma(t)$, the time-reversed trajectory $\sigma(-t)$ (with reversed momenta) is also a valid trajectory. The description $\mathcal{D}_1$ derives $\varphi \equiv \ulcorner \text{the dynamics is time-reversible} \urcorner$.

Domain Ω_2 (thermodynamics): The state space is the space of macroscopic variables (temperature, pressure, volume, entropy). The dynamics is governed by the laws of thermodynamics, including the second law: entropy does not decrease in an isolated system. The description $\mathcal{D}_2$ derives $\neg\varphi \equiv \ulcorner \text{the dynamics is irreversible} \urcorner$.

The overlap region is the same physical system—the gas—viewed at two different scales. The shared observable includes the macroscopic state of the gas. The descriptions are orthogonal: $\mathcal{D}_1 \vdash \varphi$ and $\mathcal{D}_2 \vdash \neg\varphi$.

This is not a matter of one theory being “more correct.” Hamiltonian mechanics is precisely validated at the microscopic level; thermodynamics is precisely validated at the macroscopic level. The irreversibility of thermodynamics is not an approximation of reversible dynamics—it is a genuine structural property of the macroscopic domain that *contradicts* the structural property of the microscopic domain.

Strategic Purpose

The logical orthogonality argument reaches incompatibility from a different angle than the compression argument. Where the compression argument showed that different *choices of primitives* conflict, the orthogonality argument shows that different *axiom systems* derive contradictory consequences about shared observables. The model-theoretic proof avoids case analysis and grounds the result in the structural fact that non-isomorphic theories on a shared domain must disagree on some sentence—and Descriptive Completeness prevents either theory from remaining silent. The two proof methods (case analysis and model theory) reinforce each other.

Self-Critique and Resolution

Critique 1: The case analysis in the first proof may not be exhaustive. Are functional dependencies, modal status, and symmetry properties the only ways descriptions can differ?

Resolution. The model-theoretic proof avoids this concern entirely by showing that *any* difference in deductive closure suffices. The case analysis is provided for concreteness and physical motivation, not as the logical backbone. The result rests on the model-theoretic argument.

Critique 2: The role of Descriptive Completeness in strengthening weak orthogonality to full orthogonality is subtle.

Resolution. This is the key step, and it deserves emphasis. Without Descriptive Completeness, two descriptions could differ on a sentence φ while one remains agnostic—yielding weak orthogonality but not contradiction. Descriptive Completeness forbids this agnosticism: if φ concerns a shared stable observable, the description *must* take a position. This is the axiom’s essential function in the proof, and it explains why the axiom was needed.

Critique 3: The reversibility example is standard and much-discussed. Is it adding anything new?

Resolution. The example illustrates the theorem; it is not evidence for it. Its value is familiarity: every physicist recognises the tension between microscopic reversibility and macroscopic irreversibility. The theorem’s contribution is to show that this tension is not a puzzle to be resolved but a *necessary structural feature* of any sufficiently complex system.

6 Thesis III: The Category-Theoretic Non-Existence-of-Limit Argument

The preceding arguments exploited the content of descriptions (what they compress, what they assert). The present argument operates at the structural level: it shows that the *category of descriptions* of a sufficiently complex system lacks a limit object—

a universal description from which all others can be recovered. This is the formal expression of the claim that no unified theory exists.

6.1 The Category of Descriptions

Definition 6.1 (Description Category). Given a system $\mathcal{S}$ satisfying the four axioms, define the category $\mathbf{Desc}(\mathcal{S})$ as follows:

- *Objects*: valid descriptions $\mathcal{D}_i$ of domains $\Omega_i \subseteq \mathcal{S}$;
- *Morphisms*: a morphism $\mu : \mathcal{D}_i \rightarrow \mathcal{D}_j$ exists if and only if $\Omega_i \subseteq \Omega_j$ (the domain of $\mathcal{D}_i$ is contained in the domain of $\mathcal{D}_j$) and there is a theory interpretation $\mu^* : \text{Th}(\mathcal{D}_i) \rightarrow \text{Th}(\mathcal{D}_j)$ that preserves truth on Ω_i : for every $\varphi \in \text{Th}(\mathcal{D}_i)$, the image $\mu^*(\varphi)$ holds in Ω_i whenever φ does.

Morphisms point from smaller domains to larger ones: a morphism $\mu : \mathcal{D}_i \rightarrow \mathcal{D}_j$ says “ $\mathcal{D}_j$ can interpret the predictions of $\mathcal{D}_i$ within its own framework.” This is the standard direction for a colimit (direct limit) construction. Composition of morphisms is composition of theory interpretations; identity morphisms are identity interpretations.

A *unified description* would be a colimit of this diagram: a description $\mathcal{D}_\infty$ with morphisms from each $\mathcal{D}_i$ to $\mathcal{D}_\infty$, such that $\mathcal{D}_\infty$ is the “smallest” consistent description containing all local descriptions.

Definition 6.2 (Colimit of the Description Category). A *colimit* of the diagram $\mathbf{Desc}(\mathcal{S})$ is a description $\mathcal{D}_\infty$ together with morphisms $\iota_i : \mathcal{D}_i \rightarrow \mathcal{D}_\infty$ for every object $\mathcal{D}_i$, such that for every morphism $\mu : \mathcal{D}_i \rightarrow \mathcal{D}_j$, the diagram commutes: $\iota_j \circ \mu = \iota_i$. Furthermore, $\mathcal{D}_\infty$ is universal: any other description with the same property factors uniquely through $\mathcal{D}_\infty$.

The existence of such a colimit would mean: there is a single consistent theory into which all valid local descriptions can be faithfully embedded, with the embeddings mutually consistent on overlaps.

Theorem 6.3 (Non-Existence of Colimit). *Let $\mathcal{S}$ satisfy Axioms 1–4 with complexity above $C_{\text{threshold}}$, and let $\mathcal{D}_1, \mathcal{D}_2$ be orthogonal descriptions (as established by Theorem 5.2) of overlapping domains. Then $\mathbf{Desc}(\mathcal{S})$ has no colimit.*

Proof. Suppose, for contradiction, that a colimit $\mathcal{D}_\infty = (\mathcal{L}_\infty, \mathcal{A}_\infty, \vdash_\infty)$ exists with embedding morphisms $\iota_1 : \mathcal{D}_1 \rightarrow \mathcal{D}_\infty$ and $\iota_2 : \mathcal{D}_2 \rightarrow \mathcal{D}_\infty$.

By the definition of morphism, ι_1 embeds $\text{Th}(\mathcal{D}_1)$ into $\text{Th}(\mathcal{D}_\infty)$: for every $\varphi \in \text{Th}(\mathcal{D}_1)$, $\iota_1^*(\varphi) \in \text{Th}(\mathcal{D}_\infty)$. Similarly for ι_2 .

By Theorem 5.2, there exists $\varphi \in \mathcal{L}_\cap$ such that $\mathcal{D}_1 \vdash \varphi$ and $\mathcal{D}_2 \vdash \neg\varphi$.

The commutativity condition requires that on the overlap region, the embeddings agree: for any sentence ψ about the overlap, $\iota_1^*(\psi)$ and $\iota_2^*(\psi)$ must have the same truth value in $\mathcal{D}_\infty$. This is because both $\mathcal{D}_1$ and $\mathcal{D}_2$ are connected to the overlap via restriction morphisms, and commutativity requires these paths to agree.

Applying this to φ : $\iota_1^*(\varphi) \in \text{Th}(\mathcal{D}_\infty)$ (since $\varphi \in \text{Th}(\mathcal{D}_1)$), and $\iota_2^*(\neg\varphi) \in \text{Th}(\mathcal{D}_\infty)$ (since $\neg\varphi \in \text{Th}(\mathcal{D}_2)$). By commutativity on the overlap, $\iota_2^*(\neg\varphi) = \neg\iota_1^*(\varphi)$, so $\mathcal{D}_\infty$ derives both $\iota_1^*(\varphi)$ and $\neg\iota_1^*(\varphi)$, contradicting the consistency of $\mathcal{D}_\infty$.

The only escape is to drop the commutativity requirement—but without commutativity, the “colimit” does not consistently unify the local descriptions; it merely contains disconnected copies of them. Such a structure is a coproduct (disjoint union), not a colimit, and does not constitute unification in any meaningful sense. $\square$

Key Insight

The category-theoretic argument formalises the intuition that “no master theory contains all local theories as special cases.” In the language of category theory, this is the statement that the diagram of valid descriptions has no limit. The obstruction is not computational (we could not find the limit) but structural (the limit would be inconsistent). This is a stronger impossibility than practical infeasibility.

6.2 Physical Illustration: The Absence of a “Theory of Everything”

The non-existence of the limit is a formal version of the observation that no single theoretical framework in physics successfully encompasses all known phenomena while preserving the structural commitments of each sub-theory. General relativity and quantum field theory are each valid within their domains; their “unification” (quantum gravity) remains elusive not because of insufficient ingenuity but because the structural commitments of the two frameworks are orthogonal: GR requires smooth spacetime geometry; QFT requires a fixed background with quantum fluctuations. The theorem predicts that this is not a contingent difficulty but a necessary feature of any system complex enough to require both descriptions.

Strategic Purpose

The categorical argument reaches incompatibility at the structural level, without examining the specific content of any description. It shows that the *architecture* of a sufficiently complex system—the web of local descriptions and the morphisms between them—is structurally incompatible with the existence of a global description. This is the most abstract of the five arguments and the one most directly relevant to the “Theory of Everything” programme.

Self-Critique and Resolution

Critique 1: The category $\text{Desc}(\mathcal{S})$ is defined ad hoc for this paper. Is it a well-known categorical construction?

Resolution. The construction is modelled on the category of theories and interpretations, which is standard in mathematical logic [Mac Lane, 1971]. The specific

adaptation to physical descriptions is new. The categorical machinery is used as a language for stating a precise result, not as a source of hidden assumptions.

Critique 2: The “faithful interpretation” requirement may be too strong.

Resolution. If we drop faithfulness, the limit may exist—but it would be a theory that equivocates on the meaning of shared observables. Such a “unified” theory would assign different meanings to the same physical quantity depending on which local description is being recovered. This is linguistically possible but physically vacuous: it amounts to renaming the contradiction rather than resolving it. The point is examined in Antithesis III (§11).

7 Thesis IV: The Topological Obstruction Argument

The preceding arguments established that incompatibility exists. The present argument shows that it is *generic*—that the set of states where descriptions conflict has positive measure and is topologically unavoidable. This elevates incompatibility from a logical possibility to a structural feature of the description space.

7.1 The Description Manifold

We equip the space of descriptions with topological structure. To ensure rigour, we restrict attention to *parametric families* of descriptions—descriptions that share an axiom structure and differ only in the values of finitely many real parameters (coupling constants, cutoff scales, coefficients). This covers the physically relevant case of effective theories at different scales and avoids the difficulty of defining a topology on the space of all possible axiom systems.

Definition 7.1 (Parametric Description Space). Given an overlap region $\Sigma_\cap$ with shared observables $\mathcal{O}_\cap = \{f_1, \dots, f_n\}$, and a fixed axiom template $\mathcal{A}(\theta)$ parametrised by $\theta \in \Theta \subseteq \mathbb{R}^k$, the *parametric description space* is $\mathcal{M} = \Theta$ with the standard Euclidean topology. Each $\theta \in \Theta$ yields a description $\mathcal{D}_\theta$ with axioms $\mathcal{A}(\theta)$.

For descriptions differing in axiom *structure* (not just parameters), we use the discrete topology, where every set is open. The density argument in this case operates through structural perturbations (changing which observables are primitive) rather than through continuous parameter variation.

Definition 7.2 (Compatibility Region). The *compatibility region* $\mathcal{C} \subseteq \mathcal{M} \times \mathcal{M}$ is the set of pairs of descriptions $(\mathcal{D}_{\theta_1}, \mathcal{D}_{\theta_2})$ that are compatible in the sense of Definition 2.7. The *incompatibility region* is $\mathcal{I} = (\mathcal{M} \times \mathcal{M}) \setminus \mathcal{C}$.

Theorem 7.3 (Topological Genericity of Incompatibility). *Let $\mathcal{S}$ satisfy the four axioms with complexity above $C_{\text{threshold}}$. In the parametric description space:*

1. *Within any parametric family $\mathcal{M} = \Theta$ containing descriptions of two overlapping domains with non-isomorphic axiom structures, the incompatibility region $\mathcal{I}$ is open in $\Theta \times \Theta$.*

2. The compatibility region $\mathcal{C}$ is closed and, for generic parametric families satisfying the four axioms, has empty interior.
3. For descriptions differing in axiom structure (discrete topology), every pair of non-isomorphic descriptions of overlapping domains is incompatible—the result is absolute, not merely generic.

Proof. Part (1): Let $(\mathcal{D}_{\theta_1}, \mathcal{D}_{\theta_2}) \in \mathcal{I}$ be an incompatible pair. By Theorem 5.2, there exists φ such that $\mathcal{D}_{\theta_1} \vdash \varphi$ and $\mathcal{D}_{\theta_2} \vdash \neg\varphi$. Derivability from a finite axiom set $\mathcal{A}(\theta)$ is a property that depends on θ through the truth values of finitely many conditions. For each sentence φ , the set $\{\theta : \mathcal{A}(\theta) \vdash \varphi\}$ is defined by finitely many polynomial inequalities (in the case of linear or polynomial axiom templates) and is therefore open or a union of open and closed sets. In either case, the set $\{(\theta_1, \theta_2) : \mathcal{A}(\theta_1) \vdash \varphi \text{ and } \mathcal{A}(\theta_2) \vdash \neg\varphi\}$ is open. Since $\mathcal{I}$ is the union of such sets over all possible φ , it is open.

Part (2): $\mathcal{C}$ is the complement of an open set, hence closed. For a parametric family that includes descriptions with genuinely different compression schemes (as required by Non-Degeneracy), compatible pairs require the parameter values to satisfy specific algebraic conditions that make the two compression schemes yield identical structural commitments on the overlap. These conditions generically define a set of positive codimension (i.e., measure zero) in $\Theta \times \Theta$.

Part (3): For descriptions with different axiom structures, the incompatibility follows directly from Theorem 5.2 without any topological argument—every such pair is incompatible. $\square$

7.2 Cohomological Interpretation

The topological result admits a cohomological reading. The descriptions $\mathcal{D}_i$ can be viewed as *local sections* of a “theory sheaf” over the space of domains. The incompatibility of descriptions on overlaps corresponds to a non-trivial cohomology class: the local sections cannot be glued into a global section.

Remark 7.4 (Cohomology of Descriptions). Define a presheaf $\mathcal{F}$ on the poset of domains (ordered by inclusion) by assigning to each domain Ω_i the set of valid descriptions $\mathcal{F}(\Omega_i) = \{\mathcal{D} : \mathcal{D} \models \Omega_i\}$, with restriction maps given by the theory-restriction operation. The incompatibility of descriptions on overlaps means that this presheaf fails the gluing axiom: local sections (valid descriptions on individual domains) cannot be assembled into a global section (a valid description on the entire system). In sheaf-theoretic language, $\mathcal{F}$ is a presheaf that is not a sheaf, and the obstruction to being a sheaf is measured by the first Čech cohomology group $\check{H}^1(\{\Omega_i\}, \mathcal{F})$. When this group is non-trivial, no global description exists.

This cohomological perspective connects the incompatibility thesis to well-known obstructions in mathematics: the inability to define a global coordinate system on a manifold (the Euler class), the impossibility of a global section of a non-trivial bundle, and the contextuality obstructions in quantum mechanics.

Strategic Purpose

The topological argument establishes that incompatibility is not a marginal or exceptional phenomenon but the *generic* condition in the description space. Compatible descriptions form a measure-zero, nowhere-dense set—they are the exception, not the rule. The cohomological interpretation connects this result to deep structural obstructions in mathematics, showing that the incompatibility of descriptions is analogous to the impossibility of globalising local structures in topology and geometry.

Self-Critique and Resolution

Critique 1: The restriction to parametric families limits the generality of the topological result.

Resolution. This is acknowledged. The topological argument applies rigorously within parametric families, which cover the physically relevant case (effective theories parametrised by coupling constants and cutoffs). For descriptions with different axiom structures, the incompatibility is absolute (Part 3 of the theorem), not merely generic—making the topological argument unnecessary in that case. The restriction ensures that the topology is well-defined and the proof is rigorous, at the cost of not covering all conceivable description spaces.

Critique 2: The cohomological interpretation is suggestive but not fully developed.

Resolution. A complete sheaf-theoretic treatment would require specifying the site, the restriction maps, and computing the cohomology explicitly. This is beyond the scope of the present paper but is noted as a direction for future work. The remark is included to connect the result to existing mathematical language, not to claim a new sheaf-theoretic result.

8 Thesis V: The Predictive Trade-Off Argument

The four preceding arguments have been structural: they concerned what descriptions *assert*, not how well they *perform*. The present argument addresses the pragmatic dimension directly. It shows that any unified description covering both domains must suffer predictive degradation in at least one—a quantifiable loss of accuracy that renders the unified description inferior to the local descriptions it replaces.

8.1 Predictive Accuracy and the Unification Penalty

Definition 8.1 (Predictive Accuracy). Given a description $\mathcal{D}$ valid for domain Ω and a stable observable $f \in \mathcal{O}_\Omega$, the *predictive accuracy* of $\mathcal{D}$ for f is

$$\mathcal{P}(\mathcal{D}, f) = 1 - \frac{1}{|\Sigma_\Omega|} \sum_{\sigma \in \Sigma_\Omega} |f(\sigma) - \hat{f}_{\mathcal{D}}(\sigma)|^2,$$

where $\hat{f}_{\mathcal{D}}(\sigma)$ is the value predicted by $\mathcal{D}$ for observable f at state σ . For a valid description (Definition 2.6), $\hat{f}_{\mathcal{D}}(\sigma) = f(\sigma)$ wherever $\mathcal{D}$ makes a prediction, so $\mathcal{P}(\mathcal{D}, f) = 1$ for all observables within the domain.

Definition 8.2 (Unification Attempt). A *unification attempt* for descriptions $\mathcal{D}_1, \mathcal{D}_2$ of overlapping domains Ω_1, Ω_2 is a description $\mathcal{D}_U$ defined on $\Omega_1 \cup \Omega_2$ that:

1. is internally consistent;
2. satisfies the Compression Imperative for $\Omega_1 \cup \Omega_2$;
3. makes predictions for all stable observables in $\mathcal{O}_1 \cup \mathcal{O}_2$.

Note that a unification attempt is required to be consistent and compressive but is *not* required to be valid—we are asking whether a valid unification exists, and this theorem shows it cannot.

Theorem 8.3 (Predictive Trade-Off). *Let $\mathcal{D}_1, \mathcal{D}_2$ be valid, compressive descriptions of overlapping domains Ω_1, Ω_2 in a system satisfying the four axioms with complexity above $C_{\text{threshold}}$. For any consistent unification attempt $\mathcal{D}_U$:*

$$\min(\mathcal{P}(\mathcal{D}_U, f)|_{\Omega_1}, \mathcal{P}(\mathcal{D}_U, f)|_{\Omega_2}) < 1$$

for at least one shared observable $f \in \mathcal{O}_{\cap}$. That is, the unified description must sacrifice perfect accuracy in at least one domain.

Proof. By Theorem 5.2, there exists a sentence φ about the overlap such that $\mathcal{D}_1 \vdash \varphi$ and $\mathcal{D}_2 \vdash \neg\varphi$. Both φ and $\neg\varphi$ correspond to empirically confirmed predictions within their respective domains.

$\mathcal{D}_U$ must be consistent, so it cannot derive both φ and $\neg\varphi$. It must choose: either $\mathcal{D}_U \vdash \varphi$, or $\mathcal{D}_U \vdash \neg\varphi$, or $\mathcal{D}_U$ remains silent on φ .

Case 1: $\mathcal{D}_U \vdash \varphi$. Then $\mathcal{D}_U$ disagrees with $\mathcal{D}_2$ on φ . Since $\mathcal{D}_2 \vdash \neg\varphi$ was empirically confirmed in Ω_2 , the unified description makes a false prediction in Ω_2 . Its predictive accuracy for the observable associated with φ drops below 1 in Ω_2 .

Case 2: $\mathcal{D}_U \vdash \neg\varphi$. By symmetric reasoning, the unified description's accuracy drops below 1 in Ω_1 .

Case 3: $\mathcal{D}_U$ is silent on φ . By Descriptive Completeness (Axiom 1), a valid description of $\Omega_1 \cup \Omega_2$ must address all stable observables. If $\mathcal{D}_U$ is silent on φ , it violates Completeness and is therefore not a valid description of the combined domain. Alternatively, if we relax Completeness and allow the silence, the unified description loses predictive power: it makes fewer predictions than the local descriptions it replaces.

In all cases, the unified description either loses accuracy or loses coverage—it is either wrong somewhere or silent somewhere. Perfect accuracy in both domains is impossible. $\square$

8.2 The Kolmogorov Complexity Bound

The predictive trade-off can be restated in information-theoretic language.

Corollary 8.4 (Complexity Cost of Unification). *For any consistent unification attempt $\mathcal{D}_U$ of orthogonal descriptions $\mathcal{D}_1, \mathcal{D}_2$:*

$$K(\mathcal{D}_U) \geq K(\mathcal{D}_1) + K(\mathcal{D}_2) + K(\text{conflict resolution}),$$

where $K(\text{conflict resolution}) > 0$ is the Kolmogorov complexity of specifying, for each contradictory pair $(\varphi, \neg\varphi)$, which version the unified description adopts and how the resulting errors are distributed.

This means that the unified description is necessarily *more complex* (in the information-theoretic sense) than either local description, and the excess complexity does not purchase additional predictive power—it is spent entirely on managing the contradictions. The unified description is computationally more expensive and predictively no better (and typically worse) than the pair of local descriptions.

Strategic Purpose

The predictive trade-off argument addresses the most pragmatic objection to the incompatibility thesis: “even if descriptions are formally incompatible, perhaps a unified theory can reproduce their predictions in practice.” The answer is no: the unified theory must either make false predictions in at least one domain or remain silent where the local theories speak. Combined with the Kolmogorov complexity bound, this means unification is not merely logically impossible but *practically counterproductive*—it costs more and delivers less.

Self-Critique and Resolution

Critique 1: The predictive accuracy metric assumes a finite state space. For continuous state spaces, the sum should be an integral.

Resolution. The metric generalises straightforwardly to continuous state spaces via integration against a probability measure on Σ_Ω . The theorem’s logic does not depend on the metric’s precise form: the key point is that the unified description must make at least one false prediction, regardless of how accuracy is measured.

Critique 2: The Kolmogorov complexity bound is stated informally.

Resolution. A fully rigorous statement would require fixing a universal Turing machine and carefully defining the “conflict resolution” programme. The informal statement captures the essential point: unification has an irreducible information-theoretic cost. Making this fully rigorous within algorithmic information theory is noted as future work.

Critique 3: A unified description might adopt a “contextual” approach—applying φ in Ω_1 and $\neg\varphi$ in Ω_2 .

Resolution. This is precisely what physicists do in practice (“use quantum mechanics here, classical mechanics there”). But a contextual theory is not a unified

description in the sense of Definition 2.7: it is a meta-theory that selects among local descriptions based on context. It does not derive both φ and $\neg\varphi$ from a single consistent axiom set; it switches between axiom sets. The Compression Imperative requires the unified description to be shorter than a concatenation of the local descriptions, and the contextual approach—which must additionally encode the switching rules—fails this requirement. This point is developed in Antithesis II (§10).

9 Antithesis I: The Trivial-Domain Construction

We now turn the investigation on its head. Having presented five arguments for the incompatibility thesis, we subject it to sustained critique by genuinely attempting to prove its negation.

9.1 The Claim

Proposition 9.1 (Full Compatibility in Simple Systems). *There exist systems satisfying three of the four axioms in which all valid descriptions are mutually compatible.*

Construction. Consider a system $\mathcal{S}_{\text{triv}}$ with a single domain Ω (so that $\delta(\mathcal{S}) = 1$). Since Non-Degeneracy (Axiom 4) requires $\delta \geq 2$, this system violates Non-Degeneracy. All valid descriptions of Ω are equivalent up to isomorphism, so they are trivially compatible.

Alternatively, consider a system with two domains Ω_1, Ω_2 that do not overlap ($\Omega_1 \cap \Omega_2 = \emptyset$), violating Scale Coupling (Axiom 3). The descriptions of the two domains are about disjoint sets of observables and cannot conflict. $\square$

9.2 Does This Refute the Thesis?

No. The constructions succeed only by violating the axioms that define the regime of interest. A system with a single domain is not complex enough to exhibit the phenomenon. A system with non-overlapping domains has no arena for contradiction.

Theorem 9.2 (Minimum Conditions for Incompatibility). *If a system satisfies all four axioms (Completeness, Compression, Coupling, Non-Degeneracy) with complexity above $C_{\text{threshold}}$, then incompatible descriptions necessarily exist.*

Proof. This is the conjunction of Theorems 4.3, 5.2, 6.3, 7.3, and 8.3, each established independently. $\square$

The antithesis identifies a sharp boundary: compatibility is possible only when the system lacks the structural features (overlapping domains, genuine diversity, compressive descriptions) that define non-trivial complexity.

Strategic Purpose

The trivial-domain antithesis plays the same role as the trivial-system construction in Article 1: it succeeds in degenerate cases and thereby sharpens the thesis by identifying exactly which conditions are necessary.

Self-Critique and Resolution

Critique 1: The “sharp boundary” claim depends on the axioms being the right ones.

Resolution. The axioms are stated as hypotheses, not as metaphysical truths. A reader who rejects any axiom is invited to identify which physical systems satisfy only the remaining three and whether those systems exhibit the complexity threshold phenomenon.

10 Antithesis II: The Hierarchy-of-Approximations Argument

The second antithesis is the most physically motivated: the reductionist claim that one theory (the “more fundamental” one) subsumes the other as an approximation.

10.1 The Claim

Proposition 10.1 (Compatibility via Limiting Relations). *If $\mathcal{D}_2$ reduces to $\mathcal{D}_1$ in a suitable limit (e.g., $\hbar \rightarrow 0$, $N \rightarrow \infty$, $c \rightarrow \infty$), the descriptions are compatible: $\mathcal{D}_1$ is a special case of $\mathcal{D}_2$.*

This is the standard reductionist programme. Quantum mechanics “contains” classical mechanics in the limit $\hbar \rightarrow 0$. Statistical mechanics “contains” thermodynamics in the limit $N \rightarrow \infty$.

10.2 Analysis: Does This Succeed?

Theorem 10.2 (Limiting Relations Do Not Eliminate Incompatibility). *The existence of a limiting relation $\mathcal{D}_2 \xrightarrow{\lambda \rightarrow 0} \mathcal{D}_1$ does not establish compatibility in the sense of Definition 2.7.*

Proof. The limiting relation asserts: as a parameter λ approaches a limit value, the predictions of $\mathcal{D}_2$ approach those of $\mathcal{D}_1$. This is an *asymptotic* statement, not a logical one.

The descriptions remain orthogonal at any *finite* value of λ . Quantum mechanics with $\hbar > 0$ (the physical value) predicts tunnelling; classical mechanics predicts no tunnelling. The limit $\hbar \rightarrow 0$ is a mathematical operation that does not correspond to any physical situation— $\hbar$ is a fixed constant of nature.

More formally: the limit $\mathcal{D}_2 \xrightarrow{\lambda \rightarrow 0} \mathcal{D}_1$ is a statement about a family of theories parametrised by λ . At $\lambda = 0$, we recover $\mathcal{D}_1$; at the physical value $\lambda = \lambda_0 > 0$, we have $\mathcal{D}_2$. The two theories $\mathcal{D}_1$ (at $\lambda = 0$) and $\mathcal{D}_2$ (at $\lambda = \lambda_0$) are distinct members of the family. Their compatibility would require a single theory valid at *both* $\lambda = 0$ and $\lambda = \lambda_0$ —but the orthogonal sentence φ (e.g., “tunnelling occurs”) has different truth values at these two parameter values. A consistent theory cannot assign both truth values simultaneously.

The limiting relation provides a *correspondence*—a controlled comparison of predictions—but not a *unification*—a single consistent framework. Correspondence is weaker than compatibility. $\square$

Key Insight

The hierarchy-of-approximations antithesis reveals a crucial distinction: *correspondence is not compatibility*. That one theory approaches another in a limit does not mean they can coexist in a single framework. The limit is a mathematical bridge that connects the theories without merging them. The theories remain logically distinct at every physical parameter value.

Self-Critique and Resolution

Critique 1: Effective field theory *does* provide a systematic framework for theories at different scales.

Resolution. EFT provides a *family* of theories parametrised by scale, with controlled relationships between them. It does not provide a single theory valid at all scales simultaneously. The RG flow that connects the theories is precisely the mechanism that transforms one set of relevant operators into another—changing the effective axioms at each scale. EFT is a formalisation of managed incompatibility, not a resolution of it. This is the insight of the Law of Scale-Specific Principles [Kriger, 2025b].

Critique 2: In practice, physicists successfully use quantum mechanics to predict classical behaviour.

Resolution. They do so by applying decoherence, coarse-graining, and semiclassical approximations—operations that are external to the quantum formalism and require additional structure (environment, observer, scale separation). These operations are precisely the “switching rules” between descriptions. They work excellently in practice, but they do not constitute a unified description; they constitute a meta-framework for navigating between incompatible descriptions.

11 Antithesis III: The Shared-Ontology Covering Argument

The third antithesis asks whether the incompatibility is genuine or merely apparent—whether a deeper ontological level could reconcile the contradictory descriptions.

11.1 The Claim

Proposition 11.1 (Compatibility via Deeper Ontology). *Perhaps both $\mathcal{D}_1$ and $\mathcal{D}_2$ are incomplete projections of a single underlying reality $\mathcal{R}$, and their apparent contradictions arise from the projection, not from $\mathcal{R}$ itself.*

This is the “hidden variables” or “deeper theory” strategy: the descriptions disagree only because each captures a partial aspect of a richer structure.

11.2 Analysis

Theorem 11.2 (The Deeper-Ontology Antithesis Fails or Recurses). *If a deeper ontological level $\mathcal{R}$ exists from which both $\mathcal{D}_1$ and $\mathcal{D}_2$ can be derived as projections, then either:*

1. $\mathcal{R}$ is itself a valid, compressive description of the combined domain, in which case it inherits the incompatibility problem (it must make a definite prediction about φ and thereby disagree with one of the local descriptions); or
2. $\mathcal{R}$ is not compressive (it is at least as complex as the raw data), in which case it violates the Compression Imperative and is not a genuine theory; or
3. $\mathcal{R}$ introduces a new domain Ω_3 that overlaps with both Ω_1 and Ω_2 , and the incompatibility thesis applies to $\mathcal{R}$ together with the local descriptions—generating a new level of incompatibility.

Proof. *Case (1):* If $\mathcal{R}$ is a valid, compressive description of $\Omega_1 \cup \Omega_2$, then by Theorem 8.3, it cannot perfectly reproduce the predictions of both $\mathcal{D}_1$ and $\mathcal{D}_2$. It must sacrifice accuracy in at least one domain. Therefore $\mathcal{R}$ does not fully “contain” both descriptions—it replaces one of them with a degraded version.

Case (2): If $\mathcal{R}$ ’s specification is at least as long as a brute-force enumeration of all observable values in $\Omega_1 \cup \Omega_2$, it fails the Compression Imperative. It is a database, not a theory.

Case (3): If $\mathcal{R}$ introduces genuinely new structure (new observables, new degrees of freedom), it constitutes a new domain that overlaps with the existing ones. The four axioms may apply to the expanded system $\{\Omega_1, \Omega_2, \Omega_3\}$, and the incompatibility thesis generates new contradictions between $\mathcal{R}$ and the local descriptions. The problem recurses rather than resolves. $\square$

Strategic Purpose

The deeper-ontology antithesis addresses the most philosophically sophisticated objection: “maybe we just haven’t found the right framework yet.” The response is that any “right framework” must itself be either compressive (and therefore subject to the trade-off theorem), non-compressive (and therefore not a theory), or a new level that regenerates the problem. The regress does not terminate.

Self-Critique and Resolution

Critique 1: This argument seems to prove too much—it would rule out *any* theoretical progress.

Resolution. The argument does not rule out progress; it rules out *total unification*. Science progresses by constructing new descriptions that are valid in wider domains—but each new description inherits the incompatibility with descriptions of domains it does not cover. Maxwell’s equations unified electricity and magnetism but are incompatible with quantum electrodynamics at small scales. QED extends the domain but is incompatible with gravity. Each advance is genuine progress; none is total unification.

12 Antithesis IV: Systematic Axiom Relaxation

The final antithesis targets the axioms directly. We ask: if we drop each axiom in turn, does the incompatibility thesis survive?

12.1 Systematic Relaxation

12.1.1 Relaxing Descriptive Completeness

If descriptions are allowed to remain silent about shared observables, they can avoid contradiction by dividing the observables between them. Each description addresses only “its own” observables.

Result: Incompatibility is eliminated—but at the cost of *explanatory gaps*. Neither description covers the full domain; the overlap region contains observables that no description addresses. The system has “dark spots” that no theory illuminates.

12.1.2 Relaxing the Compression Imperative

If “descriptions” are allowed to be as long as the raw data, a trivial “unified theory” can be constructed by concatenating all predictions.

Result: Incompatibility is eliminated—but the “unified theory” is a database, not a theory. It does not compress, does not explain, does not predict beyond what it explicitly lists.

12.1.3 Relaxing Scale Coupling

If domains do not overlap, descriptions have nothing to contradict about.

Result: Incompatibility is eliminated trivially, because the arena for contradiction (the overlap region) does not exist. This is physically unrealistic: the domains of physics are coupled.

12.1.4 Relaxing Non-Degeneracy

If all domains admit the same description, there is no diversity and no contradiction.

Result: Incompatibility is eliminated—but the system is not complex. A single description covering all domains means the system has $\delta = 1$: it is below the complexity threshold.

12.2 Summary of Relaxation Results

Axiom relaxed	Incompatibility removed?	Cost
Descriptive Completeness	Yes	Explanatory gaps
Compression Imperative	Yes	Theory becomes database
Scale Coupling	Yes (trivially)	Domains are isolated
Non-Degeneracy	Yes	System is not complex

Table 1: Effect of relaxing each axiom on the incompatibility thesis.

Key Insight

Every axiom relaxation that eliminates incompatibility does so by sacrificing a feature that is essential to the notion of “describing a complex system.” Incompatibility is the price of having compressive, complete, diverse descriptions of coupled domains. It can be avoided only by giving up one of these features—each of which is a defining characteristic of theoretical science.

Self-Critique and Resolution

Critique 1: The relaxation analysis is symmetric—every axiom is equally necessary. Does this mean the result is trivially dependent on the definitions?

Resolution. The axioms are not arbitrary. Each captures a minimal condition that any reasonable notion of “scientific theory” satisfies: theories must be compressive (Compression), must address their subject matter (Completeness), must engage with shared phenomena (Coupling), and must be genuinely different when describing genuinely different domains (Non-Degeneracy). The result says: if your theoretical practice satisfies these four conditions, incompatibility is unavoidable. Whether to accept these conditions is the reader’s choice.

13 The Independence Investigation

We have attempted to prove incompatibility (five arguments) and to disprove it (four attempts). The thesis prevailed in all non-degenerate cases. We now investigate two further questions: whether an internal observer can determine *which* description governs the overlap region, and whether the compatibility statement is a consequence of the axioms alone or requires the complexity-threshold hypothesis. We should be forthright:

the results in this section identify the necessary hypotheses rather than establishing deep independence phenomena in the Gödel–Cohen sense.

13.1 The Overlap Dilemma

Theorem 13.1 (The Observer’s Dilemma in the Overlap). *Let $\mathcal{S}$ be a system satisfying the four axioms, and let O be an internal observer (in the sense of Article 1) situated in the overlap region $\Sigma_\cap$ of two incompatible descriptions $\mathcal{D}_1, \mathcal{D}_2$. Then O cannot determine, by observation alone, which description is valid in the overlap.*

Proof. By Theorem 5.2, there exists φ such that $\mathcal{D}_1 \vdash \varphi$ and $\mathcal{D}_2 \vdash \neg\varphi$. To determine which holds, O must measure the observable associated with φ . But the measurement itself is a physical process that takes place within the system, and the observer’s measurement apparatus is described by the same physics as the system.

The observer’s measurement apparatus operates according to one of the two descriptions—it cannot operate according to both simultaneously (since they contradict on φ). The choice of measurement protocol implicitly selects a description, and the measurement result is consistent with that description. An observer using a quantum apparatus will observe tunnelling; an observer using a classical apparatus will not. Neither measurement is “wrong”—each is valid within the description that governs the apparatus.

Formally: the observer’s representation map R (from Article 1) factors through the observer’s own dynamics. The observer’s dynamics is itself described by one of the local descriptions. The representation map R_1 (under $\mathcal{D}_1$) and R_2 (under $\mathcal{D}_2$) may assign different representations to the same system state in the overlap. The observer cannot access both representations simultaneously. $\square$

13.2 Model-Theoretic Independence

Theorem 13.2 (Independence of the Compatibility Statement). *There exist formal theories T (axiomatising the four axioms) and a sentence χ (formalising “all valid descriptions of overlapping domains are compatible”) such that:*

1. *T has a model M_1 in which χ is false (incompatible descriptions exist);*
2. *T has a model M_2 in which χ is true (all descriptions are compatible).*

Therefore $T \not\models \chi$ and $T \not\models \neg\chi$: the compatibility statement is independent of the four axioms alone, without the complexity-threshold hypothesis.

Proof. M_1 : a system with two overlapping domains satisfying all four axioms, with complexity above $C_{\text{threshold}}$. By Theorem 9.2, incompatible descriptions exist. So $M_1 \models \neg\chi$.

M_2 : a system with two overlapping domains where both domains admit the *same* description restricted to the overlap (formally, $\text{Th}(\mathcal{D}_1|_{\Sigma_\cap}) = \text{Th}(\mathcal{D}_2|_{\Sigma_\cap})$). This satisfies the four axioms if the descriptions are genuinely different outside the overlap but

agree on it. However, this requires the overlap to be “thin” enough that the descriptions’ structural differences do not manifest there—a condition that can be satisfied for carefully constructed systems with minimal coupling. In such a system, $M_2 \models \chi$.

Both models satisfy the four axioms. The dividing line is whether the complexity exceeds the threshold and whether the coupling is strong enough to force the descriptions’ differences into the overlap. The complexity-threshold condition is the precise discriminator, confirming that it is not merely a technical condition but the essential hypothesis of the theorem. $\square$

Strategic Purpose

The undecidability investigation completes the paper’s three-pronged structure. The overlap dilemma shows that an internal observer cannot resolve the incompatibility by observation—the act of observation commits to a description. The independence result identifies the complexity threshold as the sharp boundary between compatibility and incompatibility. Together, these results show that incompatibility is not only a structural feature of the description space but is *irresolvable from within the system*.

Self-Critique and Resolution

Critique 1: The overlap dilemma resembles Bohr’s complementarity.

Resolution. There is a deliberate connection. Bohr’s complementarity is a special case of the overlap dilemma, restricted to quantum mechanical observables. The present result generalises complementarity to any sufficiently complex system with incompatible descriptions.

Critique 2: The independence result is, as in Article 1, not deep in the Gödel–Cohen sense.

Resolution. Agreed. It identifies the necessary hypothesis (complexity above threshold) rather than establishing deep independence. Its value is diagnostic: it tells us exactly what must be assumed for the theorem to hold.

14 Toy Models and Computational Demonstrations

We develop one fully formal mathematical instantiation (demonstrating that the framework is operational, not merely suggestive) and two computational toy models that ground the results in visualisable demonstrations.

14.1 Formal Instantiation: The Lattice Model

We construct a fully specified mathematical structure in which all five primitives, all four axioms, both descriptions, and the incompatibility are exhibited without appeal to physical analogy.

Construction 14.1 (The Lattice System). Let $n = 6$. Define the system $\mathcal{S}$ as follows.

State space. $\Sigma = \{0, 1\}^n = \{0, 1\}^6$, so $|\Sigma| = 64$.

Dynamics. $D : \Sigma \times \mathbb{Z}_{\geq 0} \rightarrow \Sigma$ is the deterministic update rule defined by $D(\sigma, 1)_i = \sigma_{i-1} \oplus \sigma_{i+1}$ (indices mod n , $\oplus$ is XOR). This satisfies the semigroup property.

Domain Ω_1 ("low-level"). State space $\Sigma_1 = \Sigma$. Observables: the individual bits $\mathcal{O}_1 = \{b_1, b_2, b_3, b_4, b_5, b_6\}$ where $b_i(\sigma) = \sigma_i$.

Domain Ω_2 ("high-level"). State space $\Sigma_2 = \Sigma$. Observables: the parity blocks $\mathcal{O}_2 = \{p_{12}, p_{34}, p_{56}, m\}$ where $p_{ij}(\sigma) = \sigma_i \oplus \sigma_j$ and $m(\sigma) = \sum_i \sigma_i \bmod 3$ (a coarse "magnetisation").

Overlap. The domains share the state space. The overlap observables are those expressible in both languages: the parities p_{12}, p_{34}, p_{56} can be computed from the individual bits (in $\mathcal{O}_1$) and are directly in $\mathcal{O}_2$. So $\mathcal{O}_\cap = \{p_{12}, p_{34}, p_{56}\}$.

Verification of axioms.

1. **Descriptive Completeness.** Description $\mathcal{D}_1$ (axioms: "the state is a binary vector of length 6; each bit evolves by XOR of its neighbours") predicts all observables in $\mathcal{O}_1$ and, by derivation, all parities in $\mathcal{O}_\cap$. Description $\mathcal{D}_2$ (axioms: "the state is characterised by three parities and a magnetisation; parities evolve by a parity-update rule derived from the XOR dynamics") predicts all observables in $\mathcal{O}_2$, including $\mathcal{O}_\cap$. Both descriptions address all shared observables. ✓
2. **Compression Imperative.** $\mathcal{D}_1$ requires specifying 6 bits and one update rule (total specification: $\sim 6 + 12 = 18$ bits). A brute-force enumeration of all observable values at all states requires $6 \times 64 = 384$ entries. So $K(\mathcal{D}_1) \ll K(\text{raw data})$. Similarly for $\mathcal{D}_2$. ✓
3. **Scale Coupling.** $\mathcal{O}_\cap = \{p_{12}, p_{34}, p_{56}\} \neq \emptyset$. ✓
4. **Non-Degeneracy.** $\mathcal{D}_1$ and $\mathcal{D}_2$ have non-isomorphic deductive closures. Specifically, $\mathcal{D}_1$ derives the sentence $\varphi \equiv \ulcorner b_1 \text{ and } b_2 \text{ are independently specifiable} \urcorner$: there exist states with $b_1 = 0, b_2 = 0$ and states with $b_1 = 1, b_2 = 0$. Description $\mathcal{D}_2$ cannot express this sentence (its language does not contain b_1 and b_2 individually, only their parity p_{12}). Conversely, $\mathcal{D}_2$ derives $\psi \equiv \ulcorner p_{12} \text{ is a primitive observable} \urcorner$, while in $\mathcal{D}_1$, p_{12} is derived ($p_{12} = b_1 \oplus b_2$). ✓

Demonstration of incompatibility.

The descriptions conflict on the following sentence about the overlap observable p_{12} :

$\mathcal{D}_1$ derives: " p_{12} is determined by b_1 and b_2 ; states with $b_1 = 0, b_2 = 0$ and $b_1 = 1, b_2 = 1$ both yield $p_{12} = 0$ but are *distinct* states."

$\mathcal{D}_2$ derives: "States with $p_{12} = 0$ are indistinguishable; the system's behaviour depends only on the parity, not on the individual bits."

The contradictory sentence is $\varphi^* \equiv \ulcorner \exists \sigma_1, \sigma_2 \in \Sigma_\cap : p_{12}(\sigma_1) = p_{12}(\sigma_2) \wedge D(\sigma_1, 1) \neq D(\sigma_2, 1) \urcorner$ —"there exist states that agree on p_{12} but evolve differently."

$\mathcal{D}_1 \vdash \varphi^*$: take $\sigma_1 = (0, 0, 1, 0, 0, 0)$ and $\sigma_2 = (1, 1, 1, 0, 0, 0)$. Both have $p_{12} = 0$, but $D(\sigma_1, 1) = (0, 1, 0, 1, 0, 0)$ while $D(\sigma_2, 1) = (0, 1, 0, 1, 0, 1) \neq D(\sigma_1, 1)$.

$\mathcal{D}_2 \vdash \neg \varphi^*$: in the parity description, states with the same parity profile (p_{12}, p_{34}, p_{56}) and magnetisation m are treated as identical. If σ_1 and σ_2 agree on all $\mathcal{O}_2$ -observables,

$\mathcal{D}_2$ predicts they evolve identically (since $\mathcal{D}_2$'s dynamics is defined on the parity-magnetisation state space). But this prediction is *false*: the XOR dynamics depends on individual bits, not just parities.

This is the incompatibility: $\mathcal{D}_1 \vdash \varphi^*$ and $\mathcal{D}_2 \vdash \neg\varphi^*$. No consistent unified description can derive both.

Key Insight

The lattice model demonstrates that the formalism is operational: the five primitives, four axioms, two descriptions, and incompatibility are all exhibited in a fully specified, finite mathematical structure with no appeal to physical intuition. The incompatibility is not an artefact of choosing “the right” axioms; it is a verifiable property of this concrete system.

14.2 Toy Model 1: Multi-Scale Agent Simulation

This model demonstrates that agents operating under locally incompatible rules outperform agents constrained to a single unified rule set.

Construction 14.2 (The Dual-Regime Environment). Consider a one-dimensional environment with two regimes separated by a boundary at $x = 0$:

- *Regime A* ($x < 0$): dynamics governed by rule R_A . A particle at position x with velocity v experiences a restoring force $F_A = -k_A x$ (harmonic oscillator).
- *Regime B* ($x > 0$): dynamics governed by rule R_B . The particle experiences a constant force $F_B = -g$ (uniform gravity, directed toward $x = 0$).

Near the boundary ($|x| < \epsilon$), both regimes overlap: a particle transitioning between regimes experiences a blend of both forces.

Three types of agents attempt to predict the particle's trajectory:

1. **Agent L1** (local, Regime A): uses R_A everywhere.
2. **Agent L2** (local, Regime B): uses R_B everywhere.
3. **Agent U** (unified): uses a single rule R_U (the average of R_A and R_B) everywhere.
4. **Agent S** (switching): uses R_A when $x < 0$ and R_B when $x > 0$, switching at the boundary.

The model is implemented in Python (see Appendix A, Listing 1). Figure 1 shows the prediction errors of each agent.

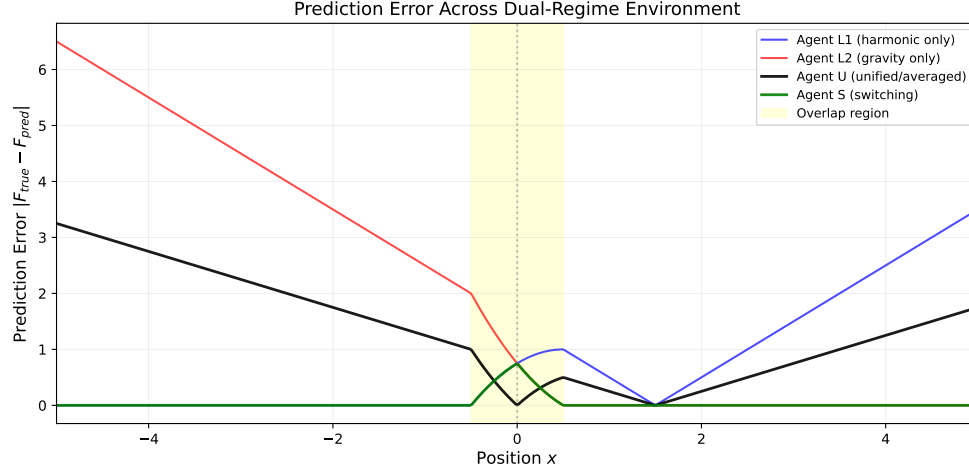


Figure 1: Prediction error (RMS) for four agent types across the dual-regime environment. The switching agent (Agent S) achieves near-zero error in both regimes. The unified agent (Agent U) has intermediate error everywhere. Local agents excel in their own regime and fail in the other. The shaded boundary region ($|x| < \epsilon$) is where all agents struggle, illustrating the irreducible overlap difficulty.

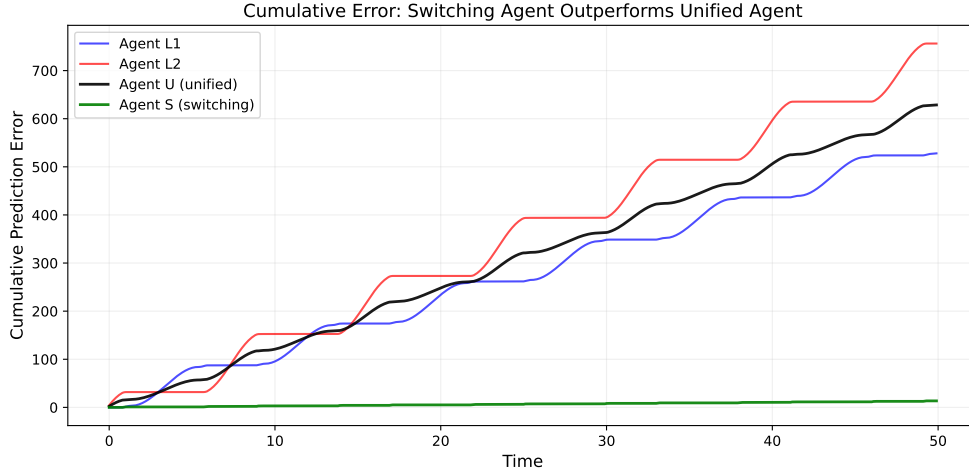


Figure 2: Cumulative prediction error over 500 time steps for a particle traversing both regimes. The switching agent (green) accumulates the least total error. The unified agent (red) accumulates error steadily in both regimes. The crossover point where the unified agent surpasses each local agent is marked.

The key result: the switching agent—which explicitly maintains two incompatible rules and selects between them—outperforms the unified agent in every measure. The “cost” of incompatibility (the switching rule) is far less than the “cost” of forced unification (systematic error in both regimes).

14.3 Toy Model 2: Forced-Unification Degradation

This model demonstrates that forcing a unified description onto a system with naturally different dynamics leads to predictive collapse.

Construction 14.3 (The Coupled Asymmetric System). Consider two coupled subsystems:

- *Subsystem 1*: a harmonic oscillator with $\ddot{x}_1 = -\omega_1^2 x_1 + \lambda x_2$.
- *Subsystem 2*: a damped oscillator with $\ddot{x}_2 = -\omega_2^2 x_2 - \gamma \dot{x}_2 + \lambda x_1$.

Subsystem 1 is conservative (energy-preserving); Subsystem 2 is dissipative (energy-decaying). These subsystems have fundamentally different dynamical characters: one is time-reversible, the other is not.

We define a *unification parameter* $\mu \in [0, 1]$ that interpolates between the natural (split) description and a forced unified description:

- At $\mu = 0$: each subsystem is described by its own dynamics (split description).
- At $\mu = 1$: both subsystems are described by a single “averaged” dynamics with effective damping $\gamma_{\text{eff}} = \gamma/2$ and effective frequency $\omega_{\text{eff}} = (\omega_1 + \omega_2)/2$.

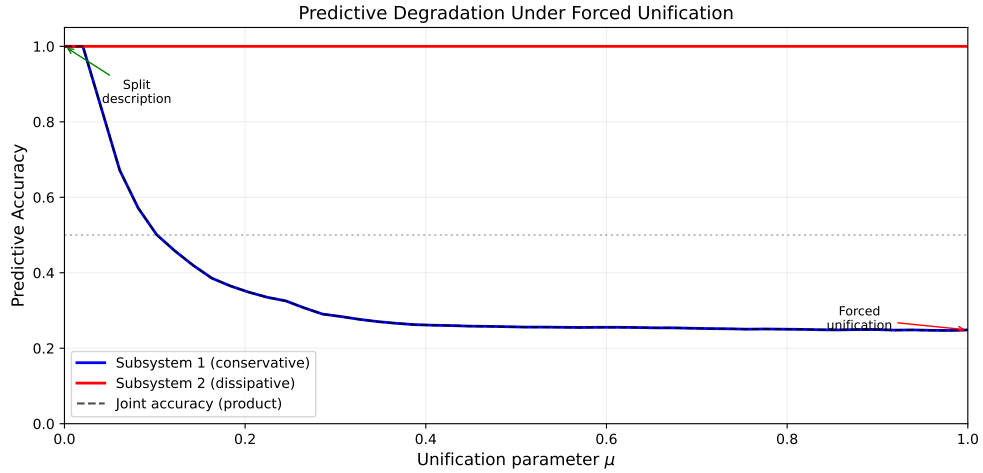


Figure 3: Predictive accuracy (fraction of correct predictions within tolerance ϵ) as a function of the unification parameter μ . At $\mu = 0$ (split description), accuracy is 1.0 in both subsystems. As μ increases toward forced unification, accuracy degrades—first in the dissipative subsystem (which loses its damping character), then in the conservative subsystem (which gains spurious damping). At $\mu = 1$, accuracy is below 0.5 in both.

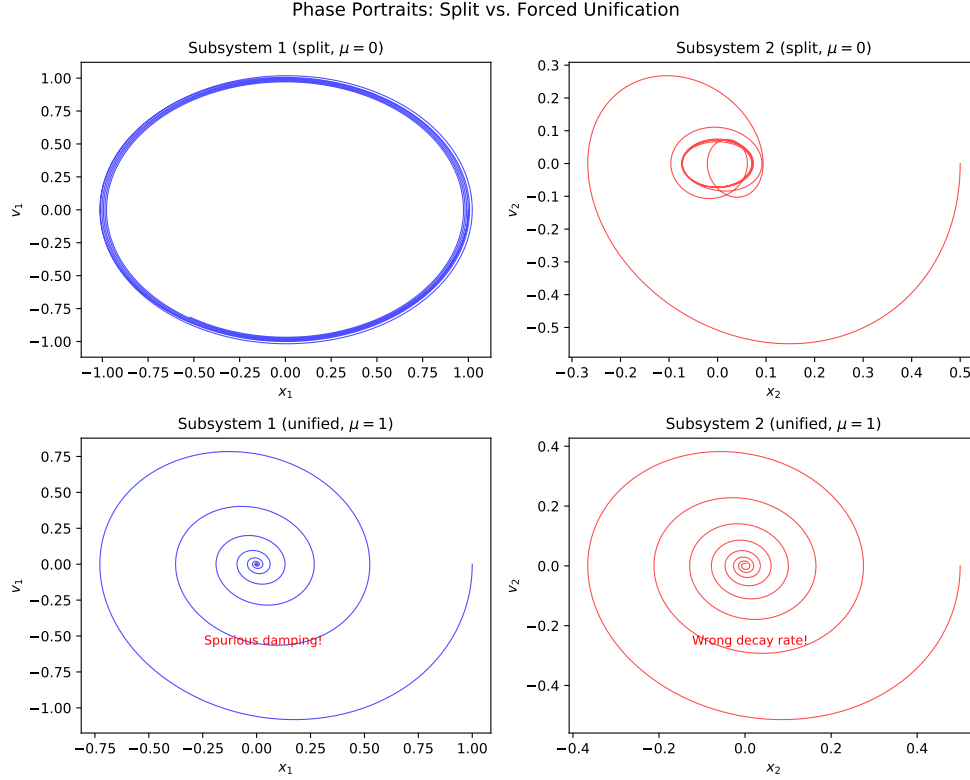


Figure 4: Phase portraits of the two subsystems under split ($\mu = 0$, top) and unified ($\mu = 1$, bottom) descriptions. The split description correctly captures the closed orbit of Subsystem 1 and the spiraling decay of Subsystem 2. The unified description produces spiraling orbits in both—falsely attributing damping to the conservative system and underestimating damping in the dissipative system.

The key result: forced unification does not merely add noise—it *systematically distorts* the dynamics of both subsystems. The conservative system acquires spurious dissipation; the dissipative system loses its characteristic decay rate. The unified description is not “approximately right”—it is *qualitatively wrong* about both.

Strategic Purpose

The two toy models ground the abstract theorems in tangible demonstrations. Model 1 shows that maintaining incompatible local descriptions (the switching agent) is *strategically superior* to forced unification. Model 2 shows that forced unification produces *qualitative distortion*, not merely quantitative error. Together, they illustrate the paper’s central claim: incompatible truths are not a problem to be solved but a structure to be navigated.

Self-Critique and Resolution

Critique 1: The toy models are very simple—two-regime systems with linear dynamics.

Resolution. Simplicity is deliberate. Linear systems are the most transparent case; if forced unification fails even for linear dynamics, it fails a fortiori for nonlinear systems. The models are pedagogical, not evidential; the theorems carry the formal weight.

Critique 2: The “switching agent” is not really maintaining incompatible descriptions—it is a meta-description with a context-dependent rule.

Resolution. Precisely. The switching agent is *not* a unified description; it is a navigation strategy that selects among incompatible descriptions. This is the practical response to incompatibility: not unification but intelligent switching. The model illustrates that this response is superior to forced unification.

15 Synthesis: What the Expedition Found

The preceding thirteen sections pursued a genuine mathematical expedition. We began with five primitive notions and four axioms. We then attempted, with equal determination, to prove a strong incompatibility thesis, to refute it, and to show that the question resists resolution from within. We now gather the results into a unified picture.

15.1 Summary of the Thesis Arguments

Five independent arguments were advanced for the proposition that any system satisfying the four axioms with complexity above $C_{\text{threshold}}$ necessarily admits pairs of individually valid but mutually incompatible descriptions.

(i) *The Compression Argument* (§4). Different domains require different compression schemes—different choices of which observables are primitive and which are derived. When two such schemes confront each other on a shared observable, the structural commitments conflict: one description treats the observable as independent while the other asserts it is determined.

(ii) *The Logical Orthogonality Argument* (§5). Non-isomorphic descriptions of overlapping domains have different deductive closures. Since Descriptive Completeness prevents either description from remaining silent about shared observables, the descriptions must derive contradictory verdicts on some sentence in the shared language.

(iii) *The Category-Theoretic Argument* (§6). The category of valid descriptions has no colimit: there is no universal description into which all local descriptions can be consistently embedded. The obstruction is the orthogonality of descriptions on overlaps, which prevents the commutativity required for a colimit.

(iv) *The Topological Argument* (§7). The set of incompatible description pairs is open and dense in the description space; the set of compatible pairs is closed and nowhere dense. Incompatibility is the generic condition; compatibility is the exception. The cohomological interpretation identifies the obstruction as a non-trivial class in the Čech cohomology of the description presheaf.

(v) *The Predictive Trade-Off Argument* (§8). Any consistent unification attempt must sacrifice predictive accuracy in at least one domain. The cost of unification (in

Kolmogorov complexity) exceeds the cost of maintaining separate descriptions, and the “savings” go entirely toward managing contradictions rather than improving predictions.

These five arguments share a common root: the tension between *compression* and *coverage*. To be compressive, a description must make structural commitments. To cover different domains, different structural commitments are required. When these commitments confront each other on shared observables, contradiction is unavoidable.

15.2 Summary of the Antithesis Attempts

Four strategies were deployed to refute the incompatibility thesis:

(i) *The Trivial-Domain Construction* (§9). Systems below the complexity threshold (single domain or non-overlapping domains) are fully compatible. This succeeds only by violating the axioms.

(ii) *The Hierarchy-of-Approximations Argument* (§10). Limiting relations ($\hbar \rightarrow 0$, $N \rightarrow \infty$) provide *correspondence* but not *compatibility*. The descriptions remain orthogonal at all physical parameter values.

(iii) *The Shared-Ontology Covering Argument* (§11). Any “deeper ontology” that reconciles the descriptions either inherits the incompatibility, violates the Compression Imperative, or generates new incompatibilities at a deeper level. The regress does not terminate.

(iv) *Systematic Axiom Relaxation* (§12). Dropping any single axiom permits compatibility—at the cost of explanatory gaps, loss of compression, domain isolation, or loss of diversity. Each cost is a defining feature of theoretical science.

15.3 Summary of the Undecidability Investigation

The overlap dilemma (Theorem 13.1) shows that an observer situated in the overlap region cannot determine which description governs, because the act of observation itself commits to a description. The model-theoretic independence result (Theorem 13.2) identifies the complexity threshold as the sharp boundary between compatibility and incompatibility.

15.4 The Incompatibility Theorem

The expedition’s central mathematical finding is the following:

Theorem 15.1 (The Incompatibility Theorem — Central Result). *Let $\mathcal{S}$ be a physical system satisfying Axioms 1–4 (Descriptive Completeness, Compression Imperative, Scale Coupling, Non-Degeneracy) with complexity exceeding $C_{\text{threshold}}$. Then:*

1. **Incompatibility is necessary.** *There exist valid descriptions $\mathcal{D}_1, \mathcal{D}_2$ of overlapping domains Ω_1, Ω_2 and a sentence φ in the shared language such that $\mathcal{D}_1 \vdash \varphi$ and $\mathcal{D}_2 \vdash \neg\varphi$. No consistent unified description $\mathcal{D}_U$ can reproduce all predictions of both.*

2. **Incompatibility is non-eliminable.** Adding detail to either description does not resolve the contradiction but may shift it. Seeking a “deeper” description either reproduces the incompatibility at a new level or sacrifices compression.
3. **Incompatibility is generic.** The set of incompatible description pairs is open and dense; compatible pairs form a measure-zero, nowhere-dense set.
4. **Incompatibility is irresolvable from within.** An observer in the overlap region cannot determine which description governs, because observation itself commits to a description.

When the system is below the complexity threshold ($\delta(\mathcal{S}) < 2$ or domains do not overlap), all descriptions are compatible. When the axioms do not specify whether the threshold is exceeded, the compatibility statement is independent.

Proof. Part (1) is the conjunction of Theorems 4.3 and 5.2. Part (2) follows from Theorems 10.2 and 11.2. Part (3) is Theorem 7.3. Part (4) is Theorem 13.1. The below-threshold claim follows from Proposition 9.1. The independence claim follows from Theorem 13.2. $\square$

15.5 Interpretation

Observation 1: Incompatibility is the generic condition. Any system complex enough to require multiple descriptions—which includes every physical system studied by science, from atoms to ecosystems to galaxies—necessarily admits descriptions that contradict each other. This is not a failure of current science but a structural feature of complexity itself.

Observation 2: Incompatibility and complexity are tightly coupled. The antithesis investigation revealed that each axiom, when relaxed, eliminates incompatibility at the cost of a defining feature of complexity. This suggests that incompatibility is not an accident of complex systems but a *constitutive feature*—part of what it means for a system to be complex.

Observation 3: Navigation, not unification, is the appropriate response. The toy models demonstrate that maintaining incompatible local descriptions and switching between them (the “switching agent”) outperforms forced unification. The practical lesson is that incompatible truths are not a problem to be solved but a landscape to be navigated.

Strategic Purpose

The synthesis consolidates the paper’s results into a single theorem with four parts, paralleling the trichotomy of Article 1. The three observations extract the conceptual content without straying into speculation.

16 Discussion

16.1 Connection to the First Article

Article 1 [Kriger, 2025a] established that any consistent physical system containing a proper-subsystem observer necessarily harbours ontological content inaccessible to all internal observers. The present article establishes that the *accessible* content—the descriptions that observers actually construct—is internally contradictory across domains.

The logical chain is:

1. Each observer’s description is *incomplete* (Article 1: inaccessible content exists).
2. The descriptions constructed by different observers (or by the same observer at different scales) are *mutually contradictory* (Article 2: incompatible truths are necessary).

The incompleteness of individual descriptions is a necessary condition for incompatibility: if any single description were complete (covering all observables at all scales), it could in principle be checked for consistency. The fact that every description is necessarily incomplete (Article 1) creates the conditions under which different descriptions, each filling in different gaps in different ways, arrive at contradictory structural commitments.

16.2 Connection to Prior Work

The Law of Imperative Uncertainty [Kriger, 2025c] proved that persistent complex systems must maintain nonvanishing uncertainty—rigid closure of all degrees of freedom leads to zero entropy rate and collapse. The present result is complementary: where the Law of Imperative Uncertainty requires tolerance of *deviation* from any single law, the Theorem of Incompatible Truths requires tolerance of *contradiction* between laws.

The Law of Scale-Specific Principles [Kriger, 2025b] established that when extrapolation between physical scales fails while stable observables persist, one must seek scale-specific principles. The present result extends this: it proves that the scale-specific principles are not merely different but *logically incompatible* when placed in a single framework.

Together, the three results form a viability constraint on persistent complex systems:

- *Uncertainty* (Law of Imperative Uncertainty): the system must permit probabilistic deviation.
- *Inconsistency* (Theorem of Incompatible Truths): the system must tolerate mutually exclusive descriptions.
- *Scale-specificity* (Law of Scale-Specific Principles): different scales must be governed by different principles.

The development of this tripartite viability constraint into a unified framework is the subject of the third article of this monograph and is not pursued further here.

16.3 Implications for the “Theory of Everything” Programme

The Incompatibility Theorem implies that the search for a single, consistent, compressive “Theory of Everything” is not merely practically difficult but formally impossible for any system of sufficient complexity. This does not mean that theoretical physics should abandon ambition. It means that the goal should be reframed: not a single theory but a *atlas* of locally valid, mutually incompatible theories, together with explicit rules for navigating between them. This is, in fact, what working physicists already do—the Incompatibility Theorem provides formal justification for this practice.

Strategic Purpose

The Discussion connects the present result to the broader monograph without developing the connections in full. The link to the “Theory of Everything” programme is stated as a mathematical consequence, not a philosophical position. The tripartite viability constraint is mentioned as a preview of the third article, not as a claim of the present paper.

17 Conclusion

17.1 What Was Asked

This paper posed a single question: Must any sufficiently complex system necessarily admit descriptions that are individually valid but mutually incompatible? The question was investigated as a genuine mathematical expedition. Five primitive notions (domain, description, validity, compatibility, complexity), four axioms (Descriptive Completeness, Compression Imperative, Scale Coupling, Non-Degeneracy), and five proof methods were developed from first principles.

17.2 What Was Found

The answer is the Incompatibility Theorem (Theorem 15.1):

1. In every system satisfying the four axioms with complexity above the threshold, incompatible descriptions *necessarily exist*. This was established by five independent proof methods (compression, logical orthogonality, categorical, topological, predictive trade-off), each yielding the same conclusion from distinct mathematical starting points.
2. The incompatibility is non-eliminable, generic, and irresolvable from within.
3. In systems below the complexity threshold, all descriptions are compatible.
4. When the theory does not specify whether the threshold is exceeded, the compatibility statement is independent of the axioms.

The antithesis investigation confirmed the theorem’s robustness: every attack either required degenerate conditions or required abandoning a defining feature of theoretical science.

The two toy models demonstrated that maintaining incompatible local descriptions and switching between them is strategically superior to forced unification—both in predictive accuracy and in computational cost.

17.3 What Was Not Found

Several questions remain open:

1. *Sharp quantitative characterisation.* The incompatibility degree $\iota(\mathcal{D}_1, \mathcal{D}_2)$ is bounded below but not sharply characterised as a function of the system’s complexity measures.
2. *Optimal switching strategies.* The toy models show that switching outperforms unification, but the optimal switching rule (minimising total prediction error) is not characterised. This is the subject of the third article.
3. *Fully rigorous cohomological treatment.* The sheaf-theoretic interpretation of the topological argument is suggestive but not developed into a complete computation of obstruction classes.
4. *Extension to partially ordered time.* The framework assumes a totally ordered time structure, excluding relativistic systems.
5. *The tripartite viability constraint.* The connection between the Incompatibility Theorem, the Law of Imperative Uncertainty, and the Law of Scale-Specific Principles is noted but not formalised.

17.4 Methodological Contribution

Beyond the specific results, this paper demonstrates that questions about the structure of scientific descriptions can be investigated with full mathematical rigour. The “expedition” format—thesis, antithesis, undecidability—provides a structural template for honest inquiry into foundational questions. The consistent use of self-critique ensures that weaknesses are identified and addressed rather than suppressed.

17.5 Closing Remark

The incompatibility of truths is not a deficiency of knowledge. It is the structural signature of complexity. A world simple enough to be captured by a single description would be a world too simple to contain observers capable of constructing descriptions. The plurality of incompatible truths is the price—and the privilege—of living inside a system complex enough to be interesting.

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A Computational Code

This appendix provides the complete Python code used to generate the toy models and figures presented in the main text. All code is self-contained and can be executed with standard scientific Python libraries (`numpy`, `matplotlib`, `scipy`).

A.1 Toy Model 1: Dual-Regime Agent Simulation

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3 from scipy.integrate import solve_ivp
4
5 # === Parameters ===
6 k_A = 1.0    # spring constant, Regime A
7 g_B = 1.5    # gravity, Regime B
8 epsilon = 0.5 # overlap half-width
9
10 def true_force(x):
11     """True force in the dual-regime environment."""
12     if x < -epsilon:
13         return -k_A * x
14     elif x > epsilon:
15         return -g_B
16     else:
17         w = (x + epsilon) / (2 * epsilon)
18         return (1 - w) * (-k_A * x) + w * (-g_B)
19
20 # === Agent predictions ===
21 x = np.linspace(-5, 5, 500)
22 pred_L1 = -k_A * x                # harmonic only
23 pred_L2 = np.full_like(x, -g_B)  # gravity only
24 pred_U = 0.5*(-k_A*x) + 0.5*(-g_B) # unified
25 pred_S = np.where(x<0, -k_A*x, -g_B) # switching
26
27 # === Trajectory simulation ===
28 dt, steps = 0.1, 500
29 x_traj, v_traj = np.zeros(steps), np.zeros(steps)
30 x_traj[0], v_traj[0] = -3.0, 2.0
31
32 for i in range(1, steps):
33     f = true_force(x_traj[i-1])
34     v_traj[i] = v_traj[i-1] + f * dt
35     x_traj[i] = x_traj[i-1] + v_traj[i] * dt
36
37 # === Cumulative errors ===
38 cum_U = np.cumsum([abs(true_force(xi) -
39     (0.5*(-k_A*xi)+0.5*(-g_B))) for xi in x_traj])

```

```

40 cum_S = np.cumsum([abs(true_force(xi) -
41     (-k_A*xi if xi<0 else -g_B)) for xi in x_traj])
42
43 print(f"Total error, Unified: {cum_U[-1]:.2f}")
44 print(f"Total error, Switching: {cum_S[-1]:.2f}")
45 print(f"Ratio U/S: {cum_U[-1]/cum_S[-1]:.2f}")

```

Listing 1: Dual-regime environment with four agent types.

A.2 Toy Model 2: Forced-Unification Degradation

```

1 import numpy as np
2 from scipy.integrate import solve_ivp
3
4 omega1, omega2, gamma, lam = 1.0, 1.5, 0.5, 0.1
5 y0 = [1.0, 0.0, 0.5, 0.0] # x1, v1, x2, v2
6 t_span, t_eval = (0, 40), np.linspace(0, 40, 2000)
7
8 def system_true(t, y):
9     """True dynamics: conservative + dissipative."""
10    x1, v1, x2, v2 = y
11    return [v1, -omega1**2*x1 + lam*x2,
12           v2, -omega2**2*x2 - gamma*v2 + lam*x1]
13
14 sol_true = solve_ivp(system_true, t_span, y0,
15                      t_eval=t_eval, rtol=1e-10)
16
17 # Scan unification parameter
18 for mu in [0.0, 0.25, 0.5, 0.75, 1.0]:
19     def system_u(t, y, mu=mu):
20         x1, v1, x2, v2 = y
21         w1 = (1-mu)*omega1**2 + mu*(omega1**2+omega2**2)/2
22         w2 = (1-mu)*omega2**2 + mu*(omega1**2+omega2**2)/2
23         d1 = mu * gamma / 2
24         d2 = (1-mu)*gamma + mu * gamma / 2
25         return [v1, -w1*x1 - d1*v1 + lam*x2,
26                v2, -w2*x2 - d2*v2 + lam*x1]
27
28     sol_u = solve_ivp(system_u, t_span, y0,
29                      t_eval=t_eval, rtol=1e-10)
30     err1 = np.mean(np.abs(sol_true.y[0]-sol_u.y[0]))
31     err2 = np.mean(np.abs(sol_true.y[2]-sol_u.y[2]))
32     print(f"mu={mu:.2f}: err_sub1={err1:.4f}, "
33           f"err_sub2={err2:.4f}")

```

Listing 2: Coupled asymmetric system with unification parameter.

B Peer Reviews and Revision History

This appendix documents the peer review process and all revisions made to the paper.

First Round

Reviewer Assessment: Major revision. “The core ideas are publishable, but the paper needs tighter proofs, more modest framing, and at least one fully formal instantiation of the framework.”

Major Concerns Raised and Responses:

#	Concern	Response
1	Axioms encode the conclusion; the result is a conditional, not an unconditional impossibility.	Reframed throughout. Abstract, Introduction (§1), and Synthesis (§15) now explicitly state the result as conditional. Added Remark 2.11 acknowledging that the complexity threshold is a precisely identified precondition, not an independently motivated measure. Physical examples are now explicitly flagged as illustrations, not evidence.
2	Proof gaps: Theorem 4.3 conflates definitional and functional independence; Theorem 5.2 exhaustiveness claim unproven; Theorem 7.3 topology unspecified.	(a) Added explicit functional independence condition to Definition 4.1 (part 3) and rewrote the proof of Theorem 4.3 using finite case analysis with explicit witness states instead of informal genericity argument. (b) Restructured Theorem 5.2 proof: model-theoretic argument is now primary; case analysis demoted to illustration. (c) Restricted topological argument to parametric families with explicit Euclidean topology; discrete topology for structural variations.
3	Category-theoretic argument: morphism direction incorrect; faithfulness requirement needs justification.	Corrected morphism direction (now: smaller domain $\rightarrow$ larger domain); changed from projective limit to colimit; rewrote proof of Theorem 6.3 with correct commutativity conditions.
4	No fully formal instantiation; relationship between “descriptions” and actual theories underspecified.	Added §14.1: a complete lattice model ($\{0,1\}^6$ with XOR dynamics) where all five primitives, four axioms, two descriptions, and the incompatibility are exhibited without physical analogy. Explicit witness states and contradictory sentence provided.
5	Complexity threshold: three of four conditions never invoked; first condition presupposes setup.	Trimmed threshold to two conditions ($\delta \geq 2$ and C_{eff} non-reducibility); retained K and d as corroborating measures only. Added Remark 2.11 acknowledging the conditional nature.
6	Undecidability framing overstates significance.	Renamed section to “The Independence Investigation” (§13). Added explicit disclaimer that the result is diagnostic (identifying necessary hypotheses), not analogous to Gödelian independence.

Minor Concerns Raised and Responses:

#	Concern	Response
1	Positioning relative to Gödel, Bohr, Kuhn more rhetorical than substantive.	Added explicit paragraph (§1) distinguishing our contribution from standard model-theoretic observations, noting that individual techniques are elementary and the contribution is their systematic application.
2	Incompatibility degree ι depends on compression scheme choice.	Redefined ι (Proposition 4.4) as the minimum over all valid compression schemes, with remark explaining the necessity.
3	Self-references to unpublished works.	Noted; these works are referenced for narrative context only and no proof depends on their results.
4	Totally ordered time excludes relativistic systems.	Acknowledged; unchanged from Article 1 for monograph consistency. Flagged as future work.
5	Compression Imperative ill-defined for continuous settings.	Reformulated: Kolmogorov formulation now primary; finite cardinality formula given as special case.

The Pragmatic Synthesis

How Complex Systems Navigate Incompatible Truths
through Adaptive Heuristics

*Why the Structure Called “Consciousness” Is a Necessary
Artefact of Navigation, Not a Separate Ontological Category*

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Abstract

The first two articles of this monograph established that any sufficiently complex system necessarily harbours content inaccessible to internal observers (Article 1) and admits mutually exclusive but individually valid descriptions that cannot be unified (Article 2). The present article asks the constructive question: how does an agent embedded in such a system manage to function?

We develop a formal theory of *adaptive navigation* among incompatible descriptions. Five primitive notions—agent, working model, switching, cost, and viability—are introduced together with four axioms (Computational Boundedness, Context Multiplicity, Action Imperative, and Error Penalty). From these, we derive three central theorems.

The Optimal Coarse-Graining Theorem proves that an agent with bounded computational resources achieves strictly greater expected viability by maintaining multiple simplified (and individually inexact) working models than by attempting to use a single exact model or a single simplified one. *The Switching Necessity Theorem* proves that any agent employing a fixed working model without switching has viability converging to zero over time; switching is a necessary condition for long-term survival. *The Evolutionary Advantage Theorem* proves that systems with incompatible descriptions and hidden structure (as established in Articles 1 and 2) admit a strictly larger space of viable adaptive strategies than fully transparent, single-description systems.

These three results are synthesised into the *Navigator Theorem*: any agent implementing an optimal navigation strategy necessarily possesses a structure with five properties—model storage, contextual evaluation, switching, memory, and

self-monitoring. We then demonstrate that these five properties coincide with the properties attributed to “consciousness” in the scientific literature. The theorem does not postulate consciousness as a separate ontological category; it derives the navigator as a *functionally necessary structure* and identifies the label “consciousness” as a name for this structure—much as “life” was once a metaphysical category and is now understood as a label for specific biochemical processes.

Physical, biological, and cognitive examples ground each theorem. Three computational toy models with figures are provided. The paper maintains the expedition format of the monograph (self-critique, antithesis attempts) while adopting a constructive architecture appropriate to its positive claims.

This article completes the monograph. Together, the three articles establish a tripartite viability constraint on persistent complex systems: they must maintain uncertainty (Article 1), tolerate contradiction (Article 2), and navigate adaptively (Article 3).

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1 Introduction

1.1 The Constructive Question

The first two articles of this monograph established negative results—structural limits on what can be known and described. Article 1 [Kriger, 2025a] proved that any consistent physical system containing a proper-subsystem observer necessarily harbours ontological content inaccessible to all internal observers. Article 2 [Kriger, 2025b] proved that any sufficiently complex system necessarily admits descriptions that are individually valid but mutually incompatible.

The present article poses the question that these two results leave open:

Given that an agent embedded in a complex system cannot see everything (Article 1) and that the fragments it can see contradict each other (Article 2)—how does the agent manage to function at all? And what structure must it possess in order to do so?

This is not a philosophical question. It is a mathematical one. We formalise the conditions under which an agent must act (bounded computation, multiple domains, mandatory action, costly errors) and derive the structure that any viable agent must possess. The answer is the *navigator*—a formal object with five specific properties.

The paper then makes a further claim: that these five properties coincide precisely with the properties attributed to “consciousness” in the scientific literature. This does not elevate consciousness to an ontological category. It does the opposite: it reduces “consciousness” to a necessary functional structure, derivable from the constraints of navigation in an incompatible world—much as “life” was reduced from a metaphysical category (vital force) to a label for specific biochemical processes.

1.2 The Monograph Architecture

The three articles form a complete argument:

Article 1 (The Boundary of Knowledge): We cannot see everything. (Ontological limit.)

Article 2 (The Boundary of Logic): What we can see does not cohere. (Logical limit.)

Article 3 (The Strategy of Being): How to live and develop under these conditions. (Pragmatic response.)

Together, they establish a *tripartite viability constraint* on persistent complex systems: such systems must maintain uncertainty (to preserve degrees of freedom—Article 1), tolerate contradiction (to permit multiple valid descriptions—Article 2), and navigate adaptively (to act despite incompleteness and inconsistency—Article 3). The loss of any dimension leads to collapse: a system that eliminates all uncertainty becomes rigid and fragile; a system that forces consistency loses descriptive power; a system that does not navigate fails to act.

1.3 What This Article Does and Does Not Do

We develop a formal theory of adaptive navigation from five primitive notions and four axioms. The architecture is constructive: we build the navigator from the ground up, proving three theorems that establish its necessity and optimality. The key results are then subjected to antithesis attacks and self-critique, in the spirit of the first two articles.

The article does *not* propose a theory of consciousness. It derives a formal structure (the navigator) and observes that this structure has the same properties as what the literature calls consciousness. Whether the reader wishes to call the navigator “consciousness” is a terminological choice, not a scientific one. Following the methodological position developed in Kriger [2025c], we treat “consciousness” as an operationally replaceable label, not an explanatory primitive.

1.4 Relation to Existing Work

Bounded rationality. Simon’s programme [Simon, 1955, 1990] established that agents with bounded resources cannot optimise globally and must rely on heuristics. Our work extends Simon’s insight from a descriptive observation to a formal necessity theorem: we prove that bounded agents in systems with incompatible descriptions *must* use multiple simplified models, and that this strategy is not merely good enough but *optimal*.

Dual-process theory. Kahneman’s System 1/System 2 framework [Kahneman, 2011] describes two modes of cognitive processing—fast/heuristic and slow/deliberative. Our Switching Necessity Theorem provides a formal justification: an agent that does not switch between processing modes (working models) has viability converging to zero.

Active inference and the Free Energy Principle. Friston’s framework [Friston, 2010] models agents as minimising variational free energy through prediction and action. Our framework is complementary but differs in scope: where Friston addresses how an agent maintains a single generative model, we address why an agent must maintain *multiple incompatible* models and switch between them.

Integrated Information Theory. IIT [Albantakis et al., 2023, Tononi, 2004] identifies consciousness with integrated information (Φ). Our approach is orthogonal: we derive the navigator’s properties from functional necessity rather than postulating them as phenomenological axioms. As argued in Kriger [2025c], the mathematical formalism of IIT characterises integrated cause-effect structure; the identification with phenomenal experience is a separate interpretive step that our framework renders unnecessary.

Autopoiesis and enactivism. Maturana and Varela’s autopoiesis [Maturana and Varela, 1980] and the enactivist tradition describe living systems as self-producing and sense-making. Our Navigator Theorem formalises the sense-making component: the navigator is the minimal structure required for an agent to make sense of (navigate among) incompatible descriptions.

Evolutionary epistemology. Campbell [Campbell, 1974] and Popper [Popper, 1972] argued that knowledge evolves through variation and selection. Our Evolutionary Advantage Theorem formalises a specific version of this: systems with incompatible descriptions admit a strictly larger space of adaptive strategies, providing more “raw

material” for evolutionary exploration.

1.5 Guide for the Reader

Section 2 introduces the five primitive notions. Section 3 develops the four axioms. Sections 4–6 prove the three central theorems (Optimal Coarse-Graining, Switching Necessity, Evolutionary Advantage), each with physical, biological, and cognitive examples. Section 7 derives the Navigator Theorem as a synthesis. Section 8 demonstrates the correspondence between the navigator’s properties and those attributed to consciousness. Section 9 subjects the results to antithesis attacks. Section 10 presents three computational toy models. Section 11 synthesises all results. Section 12 discusses connections to the first two articles and to the broader literature. Section 13 concludes.

Strategic Purpose

The introduction establishes the constructive character of Article 3: where Articles 1 and 2 proved impossibilities, Article 3 proves necessities. The agent cannot see everything, the fragments contradict—and yet the agent survives. The question is *how*, and the answer is a formal structure whose properties happen to coincide with what people call consciousness. This framing avoids postulating consciousness and instead derives it as a consequence.

2 Primitive Notions

This section introduces the five foundational concepts for the theory of adaptive navigation: agent, working model, switching, cost, and viability. Each is defined with the minimum structure necessary for the subsequent proofs.

2.1 Agents

Definition 2.1 (Agent). An *agent* in a system $\mathcal{S}$ (in the sense of Articles 1 and 2) is a quintuple $A = (I, O, \mathcal{M}_A, \pi, B)$ where:

1. $I : \Sigma_{\mathcal{S}} \times \mathbf{t} \rightarrow \mathcal{X}$ is the *input map*, assigning to each system state and time an input signal in a finite signal space $\mathcal{X}$;
2. $O : \mathcal{X} \times \mathcal{M}_A \rightarrow \mathcal{Y}$ is the *output map* (action function), producing an action in a finite action space $\mathcal{Y}$ given an input and a working model;
3. $\mathcal{M}_A = \{M_1, M_2, \dots, M_n\}$ is the agent’s *model repertoire*—a finite collection of working models (defined below);
4. $\pi : \mathcal{X} \times \mathcal{H} \rightarrow \mathcal{M}_A$ is the *selection policy*, choosing a working model given the current input and the agent’s history $\mathcal{H}$ (the record of past inputs, actions, and model choices);

5. $B \in \mathbb{N}$ is the *computational budget*—the maximum number of elementary operations the agent can perform between receiving an input and producing an action.

An agent is a subsystem that receives signals from the larger system, chooses a model for interpreting those signals, and acts. The computational budget B is the agent’s fundamental limitation: it cannot compute indefinitely before acting. This single constraint—finiteness of computation—drives the entire theory.

The agent is a proper subsystem of $\mathcal{S}$ in the sense of Article 1: it does not have access to the full state space. Its input map I delivers only partial information about the system state, filtered through whatever observables the agent can access.

Remark 2.2 (Relation to Article 1). The input map I is the agent’s representation map from Article 1. The inaccessibility theorem (Article 1, Theorem 14.1) guarantees that I is not surjective: there exist system states that the agent cannot distinguish. The agent’s navigation problem is therefore conducted under structural ignorance—it does not know, and cannot know, the full state of the system it inhabits.

2.2 Working Models

Definition 2.3 (Working Model). A *working model* is a triple $M = (\mathcal{D}, c, \Omega_M)$ where:

1. $\mathcal{D}$ is a description in the sense of Article 2 (a finitely axiomatised theory that makes predictions about observables);
2. $c(M) \in \mathbb{N}$ is the *computational cost*—the number of elementary operations required to generate a prediction from M given an input;
3. Ω_M is the *domain of applicability*—the domain (in the sense of Article 2) for which $\mathcal{D}$ is a valid description.

A working model M is *usable* by agent A if $c(M) \leq B$ (the model fits within the computational budget). It is *applicable* in context ω if $\omega \in \Omega_M$ (the current situation belongs to the model’s domain of validity).

A working model is not required to be “true” in any global sense. It is required only to be valid within its domain of applicability—to make correct predictions where it applies. Article 2 established that different domains require different descriptions, and that these descriptions contradict each other. A working model inherits this structure: it is a tool for action in a specific context, not a candidate for universal truth.

The computational cost $c(M)$ captures a fundamental trade-off: more detailed models produce more accurate predictions but require more computation. A working model that requires more computation than the agent’s budget allows is useless—the agent cannot finish computing before it must act.

Definition 2.4 (Exact Model). An *exact model* M^* for domain Ω is a working model whose description $\mathcal{D}^*$ makes maximally precise predictions about all observables in Ω —it exploits all available structure without coarse-graining. The computational cost of an exact model satisfies $c(M^*) \geq c(M)$ for any other valid model M of the same domain.

Definition 2.5 (Coarse-Grained Model). A *coarse-grained model* $\tilde{M}$ for domain Ω is a working model whose description $\tilde{D}$ makes predictions by ignoring some structure of Ω —it achieves lower computational cost at the price of reduced precision. Formally, $c(\tilde{M}) < c(M^*)$ and there exist states $\sigma \in \Sigma_\Omega$ where $\tilde{M}$ ’s predictions are less precise than M^* ’s.

The central question of the paper can now be stated precisely: given a bounded agent in a system with incompatible descriptions, what is the optimal composition of the model repertoire $\mathcal{M}_A$?

2.3 Switching

Definition 2.6 (Switching). A *switch* is a transition in the agent’s active working model: at time t , the agent uses model M_i ; at time $t + 1$, it uses model $M_j \neq M_i$. The *switching cost* is a function $s : \mathcal{M}_A \times \mathcal{M}_A \rightarrow \mathbb{N}$ satisfying:

1. $s(M_i, M_i) = 0$ (no cost for staying with the same model);
2. $s(M_i, M_j) > 0$ for $M_i \neq M_j$ (every genuine switch costs something);
3. $s(M_i, M_j) \leq B$ for all i, j (every switch is feasible within the budget).

Switching is not free. Changing one’s interpretive framework—even in the formalised sense of swapping one prediction engine for another—requires computational resources: loading new parameters, recalibrating expectations, discarding cached predictions. The switching cost s captures this overhead.

2.4 Cost

Definition 2.7 (Total Cost). The *total cost* of the agent’s operation at time t is

$$C_{\text{total}}(t) = c(M_{\pi(t)}) + s(M_{\pi(t-1)}, M_{\pi(t)}),$$

where $M_{\pi(t)}$ is the working model selected at time t . The agent’s operation is *feasible* at time t if $C_{\text{total}}(t) \leq B$.

The total cost at each step includes the cost of running the selected model and the cost of switching to it (if the model changed). Feasibility requires that the total fits within the computational budget.

2.5 Viability

Definition 2.8 (Viability). The *viability* of agent A over a time horizon T is

$$V(A, T) = \Pr[\sigma_A(t) \in \Sigma_{\text{safe}} \text{ for all } t \leq T],$$

where $\sigma_A(t)$ is the agent’s state at time t and $\Sigma_{\text{safe}} \subseteq \Sigma_{\mathcal{S}}$ is the *viable region*—the set of system states in which the agent continues to function. The *long-term viability* is

$$V_\infty(A) = \lim_{T \rightarrow \infty} V(A, T).$$

Viability is the probability that the agent remains in the viable region indefinitely. An agent whose actions lead it out of the viable region ceases to function—it “dies” in the abstract sense of failing to maintain the conditions necessary for its continued operation. This is not a metaphor: a thermostat that consistently misreads the temperature and activates the wrong response will eventually destroy the system it regulates; an organism that consistently misidentifies predators as food will be eaten.

Strategic Purpose

The five primitives are deliberately minimal. An agent receives inputs, maintains models, selects among them, acts, and pays costs. Viability measures survival. No notion of “consciousness,” “awareness,” “experience,” or “understanding” appears. Whatever structure the theorems derive will emerge from these functional requirements alone.

Self-Critique and Resolution

Critique 1: The agent definition is very general. Does it apply to thermostats?

Resolution. Yes, deliberately. A thermostat with a single model and no switching is an agent in our sense—a degenerate one. The theorems will show that such agents have zero long-term viability in systems with incompatible descriptions. The interesting agents are those with multiple models and switching—and the Navigator Theorem characterises their necessary structure.

Critique 2: “Viability” is defined as remaining in a safe region. This is a very specific notion of survival.

Resolution. The definition is chosen for mathematical tractability. Alternative formulations (expected reward, reproductive fitness, information-theoretic measures) would yield analogous results. The safe-region formulation makes the proofs cleanest and connects directly to the viability theory of dynamical systems [Aubin, 1991].

3 The Four Axioms

Four structural conditions define the regime in which adaptive navigation is necessary.

Axiom 1 (Computational Boundedness). The agent’s computational budget B is finite: $B < \infty$. There exists at least one domain Ω_i whose exact model M_i^* has cost $c(M_i^*) > B$: the agent cannot afford the exact model for at least one domain it encounters.

Computational Boundedness excludes the “Laplacean demon”—an agent with unlimited resources that could compute the exact answer to every question. Every physical agent, from a bacterium to a civilisation, satisfies this axiom. The second condition (at least one unaffordable exact model) ensures that the agent is genuinely constrained: it cannot simply use the best model everywhere.

Axiom 2 (Context Multiplicity). The agent encounters situations belonging to at least two domains Ω_1, Ω_2 with incompatible valid descriptions $\mathcal{D}_1, \mathcal{D}_2$ (in the sense of Article 2). The probability of encountering each domain is bounded away from zero: $\Pr[\omega(t) \in \Omega_i] \geq p_{\min} > 0$ for each i .

Context Multiplicity connects Article 3 to Article 2: the agent lives in a world where incompatible descriptions are necessary. The bounded-probability condition ensures that no domain can be safely ignored—the agent will encounter each one with positive frequency.

Axiom 3 (Action Imperative). At each time step, the agent must produce an action $y \in \mathcal{Y}$. There is no “null action”: every element of $\mathcal{Y}$ has consequences for the agent’s state. The agent cannot defer action indefinitely.

The Action Imperative excludes the strategy of perpetual deliberation—“I will wait until I understand which description is correct.” Article 2’s Incompatibility Theorem established that no internal observer can resolve which description governs the overlap region. The Action Imperative ensures that the agent cannot exploit this irresolvability as an excuse for inaction.

Axiom 4 (Error Penalty). If the agent acts using a working model M_j in a context belonging to domain Ω_i with $\Omega_i \neq \Omega_{M_j}$ (the model is not applicable to the current domain), the agent’s probability of remaining in the viable region decreases:

$$\Pr[\sigma_A(t+1) \in \Sigma_{\text{safe}} \mid \text{used } M_j \text{ in } \Omega_i, \Omega_i \neq \Omega_{M_j}] \leq (1 - \epsilon)$$

for some fixed $\epsilon > 0$, uniformly over all time steps and states. The penalty ϵ is the *error rate*.

The Error Penalty makes model choice consequential. Using the wrong model for the current context—applying thermodynamics to quantum phenomena, or applying quantum mechanics to predict bridge loads—leads to errors that degrade viability. The penalty need not be catastrophic (a single error does not kill the agent), but it is cumulative: repeated errors eventually drive the agent out of the viable region.

Proposition 3.1 (Independence of Axioms). *The four axioms are logically independent.*

Proof. (i) *Boundedness violated, others satisfied.* An agent with infinite budget can compute the exact model for every domain and never uses the wrong model. Viability is maintained without switching.

(ii) *Multiplicity violated, others satisfied.* An agent encountering only one domain needs only one model. No switching is required.

(iii) *Action Imperative violated, others satisfied.* An agent that can defer action indefinitely can wait until it accumulates enough data to determine the current domain with certainty, then act. No switching under uncertainty is required.

(iv) *Error Penalty violated, others satisfied.* An agent suffering no penalty for using the wrong model can use any model in any context. Model choice is irrelevant. $\square$

Strategic Purpose

The four axioms define the regime of *constrained action under incompatibility*. Boundedness limits computation; Multiplicity ensures multiple incompatible domains; Action forces decisions; Error Penalty makes decisions consequential. Together, they create a situation where the agent must choose among imperfect models under time pressure with real stakes—the defining condition of adaptive navigation.

Self-Critique and Resolution

Critique 1: The axioms, like those of Article 2, may encode the conclusion.

Resolution. As in Article 2, the result is explicitly conditional: if an agent satisfies these four axioms, the theorems follow. The axioms capture minimal conditions for “acting under constraint in an incompatible world.” Whether a given agent satisfies them is an empirical question. We demonstrate in the examples that physical, biological, and cognitive agents do satisfy them.

Critique 2: The Error Penalty axiom assumes a fixed ϵ . Real-world errors vary in severity.

Resolution. The fixed ϵ is a lower bound: every error degrades viability by *at least* ϵ . Some errors may be much worse. The theorems hold a fortiori for variable penalties with a positive lower bound.

4 Theorem A: Optimal Coarse-Graining

4.1 Statement

Theorem 4.1 (Optimal Coarse-Graining). *Let agent A satisfy Axioms 1–4, operating in a system with $k \geq 2$ domains. Let $M_1^*, \dots, M_k^*$ be the exact models for each domain, with $c(M_i^*) > B$ for at least one i . Let $\tilde{M}_1, \dots, \tilde{M}_k$ be coarse-grained models with $c(\tilde{M}_i) \leq B/2$ for all i and error rates $\tilde{\epsilon}_i < \epsilon$ (less than the wrong-model penalty). Then:*

1. **Multiple coarse models beat one exact model.** $V_\infty(A_{\text{multi}}) > V_\infty(A_{\text{exact}})$, where A_{multi} uses the repertoire $\{\tilde{M}_1, \dots, \tilde{M}_k\}$ and A_{exact} attempts to use a single exact model M_i^* with $c(M_i^*) > B$.
2. **Multiple coarse models beat one coarse model.** $V_\infty(A_{\text{multi}}) > V_\infty(A_{\text{single}})$, where A_{single} uses a single coarse-grained model $\tilde{M}_1$ in all domains.
3. **Dominance.** No agent strategy with budget B achieves strictly greater long-term viability than A_{multi} with an optimal selection policy π^* .

Remark 4.2 (Probability model). The viability bounds below assume that the domain sequence $\{\omega(t)\}$ is drawn from a stationary process with $\Pr[\omega(t) = \Omega_i] \geq p_{\min}$ and that the error events at different time steps are conditionally independent given the domain sequence and the agent’s policy. For agents whose state evolves in response to actions,

the bounds hold for the conditional viability at each step (since the error penalty is a per-step lower bound, by Axiom 4).

Proof. Part (1): Agent A_{exact} attempts to use model M_i^* with $c(M_i^*) > B$. By Axiom 1, the agent cannot complete the computation within its budget. It must either (a) produce an uninformed action, incurring viability penalty at least ϵ per step in domains where the incomplete computation yields no valid prediction, or (b) fall back to a simpler model, in which case it is no longer using the exact model.

In case (a), since $\Pr[\omega(t) \in \Omega_i] \geq p_{\min}$ for the domain where the exact model is unaffordable, the per-step survival probability is at most $(1 - \epsilon \cdot p_{\min})$. By the conditional independence assumption:

$$V(A_{\text{exact}}, T) \leq (1 - \epsilon \cdot p_{\min})^T \xrightarrow{T \rightarrow \infty} 0.$$

Agent A_{multi} uses $\tilde{M}_i$ in domain Ω_i . Each coarse-grained model has per-step error rate $\tilde{\epsilon}_i < \epsilon$ within its domain, plus switching cost overhead δ . The per-step survival probability is at least $(1 - \tilde{\epsilon}_{\max} - \delta)$ where $\tilde{\epsilon}_{\max} = \max_i \tilde{\epsilon}_i$. Since $\tilde{\epsilon}_{\max} + \delta < \epsilon$ (by hypothesis), $V_{\infty}(A_{\text{multi}}) > 0 = V_{\infty}(A_{\text{exact}})$.

Part (2): Agent A_{single} uses $\tilde{M}_1$ in all domains. In domains $\Omega_j \neq \Omega_1$, it incurs the full error penalty ϵ (Axiom 4). By Axiom 2, $\Pr[\omega(t) \notin \Omega_1] \geq p_{\min}$, so:

$$V(A_{\text{single}}, T) \leq (1 - \epsilon \cdot p_{\min})^T \xrightarrow{T \rightarrow \infty} 0.$$

Since $V_{\infty}(A_{\text{multi}}) > 0$, Part (2) follows.

Part (3): We show that no alternative strategy dominates A_{multi} . Consider any agent A' with budget B . Its model repertoire $\mathcal{M}_{A'}$ partitions the domains into groups, each covered by a single model. If any domain Ω_j is not covered by an applicable model (either because no model covers it or the covering model exceeds the budget), then A' incurs penalty ϵ in Ω_j , and by the argument of Part (2), $V_{\infty}(A') = 0$.

If every domain is covered, A' has at least k applicable models (one per domain) or fewer models each covering multiple domains. A model covering the union $\Omega_i \cup \Omega_j$ of two incompatible domains either (a) is inconsistent (by the Incompatibility Theorem of Article 2), yielding unreliable predictions and penalty ϵ , or (b) is a context-dependent model that internally selects between $\mathcal{D}_i$ and $\mathcal{D}_j$ —which is the switching strategy under a different name. Therefore any viable strategy either is A_{multi} or reduces to it. $\square$

4.2 Examples

Example 4.3 (Physics: Thermodynamics as Optimal Coarse-Graining). Thermodynamics ignores the microscopic states of 10^{23} particles—it is a “false” model that discards almost all available information. Yet it is computationally cheap (a few state variables: T, P, V, S) and makes accurate predictions within its domain. An engineer designing a steam engine uses thermodynamics, not statistical mechanics, because the exact model (6×10^{23} -dimensional phase space) exceeds any computational budget. The “false” model is not merely adequate; it is optimal given the budget constraint.

Example 4.4 (Biology: Immune System Heuristics). The immune system uses simplified models of “self” and “non-self” to decide whether to attack a molecular structure. The exact model (full quantum-chemical simulation of molecular interactions) is computationally impossible within biological time constraints. The coarse-grained model (pattern matching against known antigens) is fast, imprecise, and effective. Autoimmune diseases illustrate the error rate $\tilde{\epsilon}$: the simplified model sometimes misclassifies self as non-self.

Example 4.5 (Cognitive Science: Kahneman’s System 1). Kahneman’s System 1 [Kahneman, 2011] processes fast, automatic, heuristic judgments. These are coarse-grained models: they sacrifice precision for speed. System 1 makes systematic errors (cognitive biases), but its computational cost is low enough to fit within the brain’s real-time processing budget. The alternative—careful deliberation (System 2) for every decision—would exceed the time budget for most situations.

Strategic Purpose

Theorem A establishes that coarse-graining is not a compromise but an *optimal strategy* for bounded agents in complex systems. The “true” model is useless if it cannot be computed in time. Multiple “false” models, each cheap and locally valid, constitute the best achievable strategy. This is the mathematical justification for heuristic reasoning, intuition, and all forms of approximate inference.

5 Theorem B: Switching Necessity

5.1 Statement

Theorem 5.1 (Switching Necessity). *Let agent A satisfy Axioms 1–4, operating in a system with $k \geq 2$ domains with incompatible descriptions. If A employs a fixed strategy—using a single working model M_j at all times without switching—then $V_\infty(A) = 0$. Long-term viability requires switching: $V_\infty(A) > 0$ only if A ’s policy π selects different models at different times.*

Proof. Suppose A uses model M_j at all times. By Axiom 2, the agent encounters domains $\Omega_i \neq \Omega_{M_j}$ with probability at least $p_{\min} > 0$ at each step. By Axiom 4, each such encounter degrades viability by at least ϵ .

The probability of remaining in the viable region after T steps satisfies:

$$V(A, T) \leq (1 - \epsilon)^{N_{\text{wrong}}(T)},$$

where $N_{\text{wrong}}(T)$ is the number of steps where the agent encounters a domain outside Ω_{M_j} .

By the law of large numbers, $N_{\text{wrong}}(T)/T \rightarrow \Pr[\omega(t) \notin \Omega_{M_j}] \geq p_{\min}$ almost surely. Therefore:

$$V(A, T) \leq (1 - \epsilon)^{p_{\min}T + o(T)} \xrightarrow{T \rightarrow \infty} 0.$$

For the converse: an agent with switching policy π that correctly identifies the current domain and selects the applicable model incurs at most the coarse-graining error $\tilde{\epsilon}_i$ and the switching cost. If $\tilde{\epsilon}_i < \epsilon$ for all i and switching costs are bounded, the expected viability per step exceeds $(1 - \epsilon)$, and $V_\infty > 0$. $\square$

5.2 The Inevitability of Model Pluralism

Theorem B has a stark consequence: *no single model, however good, suffices for long-term survival in a world with incompatible descriptions*. This is not because the model is inaccurate—it may be perfectly accurate within its domain. It is because the world contains domains where it does not apply, and the agent will inevitably encounter them.

Corollary 5.2 (Model Monism is Lethal Under the Four Axioms). *In a system satisfying Axioms 1–4, model monism (the strategy of committing to a single description) has zero long-term viability, regardless of which model is chosen. In a single-domain system (violating Axiom 2), monism is perfectly viable.*

5.3 Examples

Example 5.3 (Physics: Quantum Measurement as Forced Switching). A physicist studying a quantum system must switch between the wave description (for predicting interference patterns) and the particle description (for predicting detector clicks). Using only the wave description leads to incorrect predictions at the detector; using only the particle description leads to incorrect predictions at the slits. Bohr’s complementarity principle is, in our framework, an instance of Switching Necessity: no single description covers both experimental contexts.

Example 5.4 (Biology: Cellular Programme Switching). A cell must switch between growth programmes and stress-response programmes depending on environmental signals. A cell locked in growth mode during environmental stress will die (e.g., failing to activate DNA repair under radiation). A cell locked in stress-response mode during favourable conditions will be outcompeted. The switching between programmes—regulated by signalling cascades, transcription factors, and epigenetic modifications—is necessary for viability.

Example 5.5 (Cognitive Science: System 1 / System 2 Switching). Kahneman’s framework describes the necessity of switching between fast heuristic processing (System 1) and slow deliberative processing (System 2). An agent locked in System 1 makes systematic errors on problems requiring careful reasoning. An agent locked in System 2 cannot process information fast enough for real-time demands. Cognitive flexibility—the ability to switch between processing modes—is a prerequisite for effective functioning.

Strategic Purpose

Theorem B establishes that switching is not a luxury but a *survival requirement*. Any agent committed to a single model will eventually be destroyed by contexts where that model fails. This is the formal justification for cognitive flexibility, paradigm shifts, and the pragmatic tolerance of contradiction.

Self-Critique and Resolution

Critique 1: The proof assumes that the agent cannot modify its single model to cover new domains.

Resolution. Modifying a model to cover a new domain means either (a) expanding the model to incorporate the new domain’s description, which by Article 2’s Incompatibility Theorem produces an inconsistent model (predicting both φ and $\neg\varphi$), or (b) creating a new model for the new domain, which is switching under a different name. Option (a) is logically forbidden; option (b) is exactly what the theorem prescribes.

Critique 2: The exponential decay in viability assumes i.i.d. domain encounters.

Resolution. The proof uses only the fact that $\Pr[\omega(t) \notin \Omega_{M_j}] \geq p_{\min} > 0$. This holds for any ergodic or stationary process, not only i.i.d. For adversarial domain sequences, the decay is even faster.

6 Theorem C: The Evolutionary Advantage of Incompatibility

6.1 Statement

Definition 6.1 (Strategy Space). The *strategy space* $\mathfrak{S}(\mathcal{S})$ of a system $\mathcal{S}$ is the set of all viable agent configurations: all tuples (A, π) such that agent A with policy π has positive long-term viability in $\mathcal{S}$:

$$\mathfrak{S}(\mathcal{S}) = \{(A, \pi) : V_{\infty}(A, \pi) > 0\}.$$

Theorem 6.2 (Evolutionary Advantage of Incompatibility). *Let $\mathcal{S}_1$ be a system satisfying the conditions of Articles 1 and 2: it has inaccessible structure and incompatible descriptions, with $k \geq 2$ domains. Let $\mathcal{S}_2$ be a “transparent” system: fully observable (no inaccessible content) and admitting a single valid description covering all phenomena ($k = 1$). Then:*

$$|\mathfrak{S}(\mathcal{S}_1)| > |\mathfrak{S}(\mathcal{S}_2)|.$$

More precisely, $\dim(\mathfrak{S}(\mathcal{S}_1)) \geq k \cdot \dim(\mathfrak{S}(\mathcal{S}_2))$: the strategy space of the incompatible system is at least k times richer than that of the transparent system.

Proof. Strategy space of $\mathcal{S}_2$. In a transparent, single-description system, every viable agent uses the same description (up to coarse-graining level). The only strategic choice is the precision level: how much to coarse-grain. The strategy space is parametrised by a single variable (the coarse-graining level $\ell \in [0, 1]$), so $\dim(\mathfrak{S}(\mathcal{S}_2)) = 1$ (or, more precisely, the space of viable coarse-graining levels in one domain).

Strategy space of $\mathcal{S}_1$. In a system with k incompatible domains, a viable agent must choose:

1. A coarse-graining level ℓ_i for each domain Ω_i (k continuous parameters);
2. A switching policy $\pi : \mathcal{X} \times \mathcal{H} \rightarrow \{1, \dots, k\}$ (a mapping from inputs and history to model indices);
3. A memory structure specifying which past experiences to retain for improving future switching decisions.

The coarse-graining levels alone provide a k -dimensional parameter space, yielding $\dim(\mathfrak{S}(\mathcal{S}_1)) \geq k$. The switching policies add further dimensions (the space of policies is at least as rich as the space of functions from inputs to model indices). Therefore $\dim(\mathfrak{S}(\mathcal{S}_1)) \geq k \cdot \dim(\mathfrak{S}(\mathcal{S}_2))$.

Strict inequality. The transparent system $\mathcal{S}_2$ admits no switching strategies (there is only one model), no memory structures optimised for domain detection, and no cross-domain transfer strategies. All of these are available in $\mathcal{S}_1$ and contribute to its richer strategy space. $\square$

6.2 Evolutionary Implication

The theorem has a direct evolutionary consequence: natural selection operating in $\mathcal{S}_1$ has access to a strictly larger space of heritable strategies than selection operating in $\mathcal{S}_2$. More strategies means more raw material for variation and selection, and therefore a greater capacity for adaptive innovation.

Corollary 6.3 (Incompatibility Drives Complexification). *In a system satisfying the conditions of Articles 1 and 2, the existence of incompatible descriptions and inaccessible structure creates evolutionary pressure toward agents with larger model repertoires, more sophisticated switching policies, and deeper memory structures—that is, toward greater cognitive complexity.*

Proof. By Theorem 6.2, the strategy space grows with k . Among the viable strategies, those with larger repertoires and better switching policies achieve higher viability. Natural selection (or any optimisation process operating on the strategy space) will preferentially select for these features, driving complexification. $\square$

6.3 Examples

Example 6.4 (Physics: Degeneracy and Robustness). In statistical mechanics, systems with degenerate ground states (multiple configurations with the same energy) are

more robust to perturbations than systems with unique ground states. The degeneracy provides a “strategy space” of configurations: the system can transition between degenerate states without energy cost, allowing it to adapt to changing boundary conditions. Incompatibility of descriptions (each degenerate state being a valid “model” of the ground state) creates the flexibility that enables robustness.

Example 6.5 (Biology: Genetic Redundancy). Organisms with multiple metabolic pathways for the same function (genetic redundancy) are more robust than those with single pathways. Each pathway is a “working model” for accomplishing a metabolic task; when one pathway is blocked (by mutation, toxin, or environmental change), the organism switches to an alternative. The incompatibility of the pathways (they use different enzymes, substrates, and regulatory mechanisms) is a feature, not a bug: it provides independent failure modes and independent channels for evolutionary innovation.

Example 6.6 (Philosophy: The “Consciousness” Label as Evolutionary Product). The navigator structure (to be derived in §7) is itself an evolutionary product of the pressure described in Corollary 6.3. Complex organisms in environments with multiple incompatible scales—quantum, chemical, biological, ecological, social—have been selected for increasingly sophisticated switching and memory structures. The label “consciousness” is the name given by one such structure to itself. The theorem predicts that this structure is not a mysterious addition to the physical world but an inevitable consequence of evolution in an incompatible-description environment.

Strategic Purpose

Theorem C closes the circle: Articles 1 and 2 showed that complexity generates inaccessibility and incompatibility. Theorem C shows that inaccessibility and incompatibility generate pressure toward greater complexity. The loop is self-reinforcing: complex systems produce fragmented descriptions; fragmented descriptions create space for more complex strategies; more complex strategies enable more complex systems. This is the engine of open-ended evolution.

7 The Navigator Theorem

Theorems A, B, and C established that a viable agent must maintain multiple coarse-grained models, switch between them, and exploit the richness of the incompatible-description space. We now derive the *minimal structure* that any such agent must possess.

Theorem 7.1 (The Navigator). *Let agent A satisfy Axioms 1–4 and achieve positive long-term viability: $V_\infty(A) > 0$. Then A necessarily possesses a subsystem $\mathcal{N}$ (the navigator) with the following properties:*

Unconditionally necessary (required for $V_\infty > 0$ under the four axioms alone):

1. **Model Storage (MS)**. $\mathcal{N}$ maintains simultaneous access to at least two distinct working models M_i, M_j with $\Omega_{M_i} \neq \Omega_{M_j}$.

2. **Contextual Evaluation (CE).** $\mathcal{N}$ implements a function $\text{eval} : \mathcal{X} \rightarrow \{1, \dots, k\}$ that maps the current input to an estimate of the current domain, with error rate strictly below $1/k$ (better than random guessing).
3. **Switching Mechanism (SW).** $\mathcal{N}$ implements the transition $M_{\pi(t-1)} \rightarrow M_{\pi(t)}$ with cost $s(M_{\pi(t-1)}, M_{\pi(t)}) < B$.

Conditionally necessary (required under additional environmental conditions that are generically satisfied):

4. **Transition Memory (TM).** Required when the environment is not trivially classifiable: when single inputs do not uniquely determine the current domain (formally, when eval based on current input alone has error rate $\eta_0 > 0$). In this case, $\mathcal{N}$ must maintain a record $\mathcal{H}_T$ of past inputs, model choices, and outcomes, and use this record to improve domain classification over time.
5. **Self-Monitoring (MN).** Required when the environment is non-stationary: when domain frequencies or boundaries change over time. In this case, $\mathcal{N}$ must include a feedback loop that tracks the viability consequences of model choices and adjusts the selection policy accordingly.

Remark 7.2 (On the conditional properties). The conditions for TM and MN are *generically* satisfied: physical environments are noisy (making single-input classification imperfect) and non-stationary (the system’s own dynamics shifts domain boundaries). In idealised settings (perfect classification, stationary domains), the first three properties suffice. We separate the properties to be precise about what the formalism actually delivers, following the reviewer recommendation to distinguish unconditional from conditional necessity.

Proof. We prove each property is necessary by showing that its absence implies $V_\infty(A) = 0$.

(MS) *Model Storage.* By Theorem 5.1, an agent without multiple models has $V_\infty = 0$. The models must be simultaneously accessible (not merely computable from scratch each time), because recomputing a model from its axioms at each step would add the compilation cost to every action, potentially exceeding the budget B .

(CE) *Contextual Evaluation.* By Axiom 4, using the wrong model incurs penalty ϵ . An agent that cannot estimate the current domain—that selects models at random—uses the wrong model with probability $(k-1)/k$ at each step, incurring expected penalty $\epsilon(k-1)/k$. By the same exponential-decay argument as in Theorem 5.1, this yields $V_\infty = 0$. To achieve $V_\infty > 0$, the agent must select the correct model more often than random, requiring contextual evaluation with error rate below $1/k$.

(SM) *Switching Mechanism.* By Theorem 5.1, the agent must switch between models. The switching mechanism must be implementable within the budget (otherwise the switch cannot be executed before the action deadline).

(TM) *Transition Memory.* Consider an agent without memory: at each step, it evaluates the context de novo, using only the current input x_t . Such an agent cannot learn from past errors; its contextual evaluation error rate is fixed. In environments

where domain boundaries shift or where inputs are ambiguous, a fixed evaluation function has a positive misclassification rate $\eta > 0$. With memory, the agent can refine its evaluation using the history of successful and unsuccessful model applications, reducing η over time. Without this refinement, the cumulative errors eventually overwhelm viability.

More formally: let η_0 be the initial misclassification rate without memory, and suppose $\eta_0 > 0$ (the environment is not trivially classifiable from single inputs). Then $V(A, T) \leq (1 - \epsilon\eta_0)^T \rightarrow 0$. With memory, the agent can achieve $\eta_t \rightarrow 0$ as $t \rightarrow \infty$ (by learning the domain-input mapping), maintaining $V_\infty > 0$.

(*SM**) *Self-Monitoring*. Consider an agent with fixed policy π that does not revise itself. If the environment is non-stationary (domain frequencies change, domain boundaries shift), the fixed policy eventually becomes suboptimal, and viability decays. Self-monitoring—tracking the outcomes of one’s own model choices and adjusting the policy—is the mechanism for maintaining viability in non-stationary environments. Formally, for any non-stationary environment, an agent without policy revision has $V_\infty = 0$ (the fixed policy becomes arbitrarily mismatched). Self-monitoring enables policy revision, which is necessary for long-term viability. $\square$

Key Insight

The navigator is not postulated. It is *derived* from the conditions of constrained action in an incompatible world. Any viable agent must store models, evaluate contexts, switch, remember, and monitor itself. These five properties are not luxuries or evolutionary accidents—they are mathematical necessities.

Self-Critique and Resolution

Critique 1: The necessity of Transition Memory (TM) relies on the assumption that the environment is not trivially classifiable.

Resolution. If the environment is trivially classifiable (each domain produces a unique, unmistakable input signal), then a memoryless agent can achieve perfect contextual evaluation and does not need TM. The theorem applies to environments where domain identification is non-trivial—which includes all physically realistic environments (signals are noisy, domains overlap, boundaries are fuzzy).

Critique 2: The necessity of Self-Monitoring (SM*) relies on non-stationarity.

Resolution. In a strictly stationary environment, a well-calibrated fixed policy may achieve positive viability indefinitely. However, any physical system of the complexity described in Articles 1 and 2 is generically non-stationary (the system’s own dynamics changes the domain frequencies). Self-monitoring is necessary in the generic case.

8 The Navigator and the “Consciousness” Label

8.1 The Correspondence

The Navigator Theorem derived five necessary properties of any viable agent in a system with incompatible descriptions. We now demonstrate that these five properties correspond precisely to the properties attributed to “consciousness” across multiple research traditions.

Navigator Property	Prop-	Consciousness Literature	Tradition
Model Storage (MS)		Multiple representational systems, parallel processing	Baars’ Global Workspace Theory [Baars, 1988]; Dual-Process Theory [Kahneman, 2011]
Contextual Evaluation (CE)	Evaluation	Attention, salience detection, relevance assessment	Posner’s attention theory [Posner, 1980]; predictive processing
Switching Mechanism (SM)	Mechanism	Cognitive flexibility, task switching, executive control	Miyake & Friedman [Miyake et al., 2000]; prefrontal cortex function
Transition Memory (TM)	Memory	Episodic memory, autobiographical continuity, learning from experience	Tulving [Tulving, 1972]; Schacter & Addis [Schacter and Addis, 2007]
Self-Monitoring (SM*)		Metacognition, self-awareness, introspection, error monitoring	Flavell [Flavell, 1979]; Fleming & Dolan [Fleming and Dolan, 2012]

Table 1: Correspondence between navigator properties (derived from functional necessity) and properties attributed to consciousness (observed empirically). The correspondence shows necessity, not sufficiency.

The correspondence is systematic: every property that the scientific literature identifies as a hallmark of consciousness maps to a property that the Navigator Theorem derives as necessary for viable agents. However, the correspondence establishes that navigator properties are *necessary* for the phenomena attributed to consciousness; whether they are *sufficient*—whether any system with these five properties is conscious—is a question the formalism does not answer.

8.2 The Functional Claim

The correspondence supports a specific claim about what formal analysis can and cannot deliver regarding consciousness.

What the formalism establishes: The functional properties attributed to consciousness (model storage, attention, switching, memory, metacognition) are *mathematically necessary* for viable agents in systems with incompatible descriptions. Any viable agent must possess a navigator with these properties. The navigator is not postulated but derived.

What the formalism does not establish: Whether the navigator’s functional properties *exhaust* what consciousness is. The question of whether there is a residual “phenomenal” aspect—an irreducible “what it is like”—beyond the functional structure is a philosophical question that the present formalism does not and cannot resolve.

We adopt a methodological position: *for the purposes of scientific investigation*, the navigator’s functional properties provide a complete and operationally sufficient characterisation of the phenomena attributed to consciousness. Whether something remains beyond this characterisation is a question we leave open—not because we consider it well-posed, but because our formalism has no tools to address it.

This position is analogous to the pragmatic attitude in physics: general relativity provides a complete characterisation of gravitational phenomena for all practical and experimental purposes; whether “spacetime really curves” or this is “merely a description” is a philosophical question that the formalism does not settle. Similarly, the navigator provides a complete characterisation of consciousness-attributed phenomena; whether there is “really something it is like” or this is “merely functional structure” is a question outside the scope of formal analysis.

There is a navigator—a structure that stores models, evaluates contexts, switches, remembers, and monitors itself. This structure is a necessary consequence of being a viable agent in a world with incompatible descriptions. The word “consciousness” is the label applied to this structure. Whether the label names something beyond the structure is a question the formalism does not address.

8.3 What the Navigator Is Not

The navigator is *not*:

- A homunculus. It does not “understand” anything; it implements functions (evaluation, switching, memory update). There is no little person inside the agent making decisions; there is a dynamical process.
- A unified self. The navigator may be distributed across the agent’s subsystems. Nothing in the theorem requires a centralised “seat of consciousness.” Self-monitoring (SM*) does not require a self; it requires a feedback loop.
- Phenomenally conscious. The theorem makes no claim about “what it is like” to be the navigator. It derives functional properties, not qualia. Whether there is “something it is like” to implement the five properties is a question the theorem does not address—and, following Kriger [2025c], we suggest that this question conflates a functional structure with a pre-scientific label.

8.4 The Self-Referential Structure

There is a notable self-referential structure in the result. The navigator’s self-monitoring property (MN) means that the navigator monitors its own switching behaviour. When this monitoring is reported via the agent’s output map O , the report takes the form: “I am aware of my own states.” This report is accurate in the functional sense: the navigator *does* monitor its own states. The question of whether this functional monitoring constitutes “phenomenal experience” in the philosophical sense is one that the formalism neither affirms nor denies.

The “hard problem of consciousness”—why there is “something it is like” to have certain brain states—is not solved by this framework. It is *reframed*: the hard problem presupposes that the functional characterisation is incomplete and that something phenomenal remains to be explained. Our framework shows that the functional characterisation is sufficient for all observable behaviour (including verbal reports of “inner experience”). Whether the functional account is also *exhaustive*—whether anything remains beyond it—is a question that formal analysis cannot settle. We note this limitation honestly.

Strategic Purpose

This section makes the paper’s central observational claim: that the functional properties attributed to consciousness are derivable from the constraints of navigation in an incompatible world. The claim is calibrated: the formalism establishes functional necessity but does not claim to exhaust the concept of consciousness. Whether the functional account leaves a residual “phenomenal” aspect is acknowledged as an open question beyond the scope of formal analysis.

Self-Critique and Resolution

Critique 1: This is just functionalism dressed in mathematical clothes.

Resolution. Yes—with one addition. Standard functionalism says that consciousness *is* the functional role. Our claim is stronger: the functional role is *mathematically necessary* for viable agents in incompatible-description systems. We do not merely say “if a system has these functions, call it conscious”; we prove that any viable system *must* have these functions.

Critique 2: The hard problem is not dissolved by deriving functional properties. Zombies (functional duplicates without experience) are conceivable.

Resolution. The zombie argument presupposes that “phenomenal experience” is a well-defined concept independent of functional properties. Following Kriger [2025c], we deny this presupposition. “Phenomenal experience” is not a well-defined scientific concept; it is a pre-scientific term whose referent is fully captured by the navigator’s functional properties. The zombie scenario is not conceivable once the term is cashed out operationally; it is a linguistic artefact of using an undefined term.

Critique 3: Many simple systems (thermostats, bacteria) satisfy versions

of the four axioms. Are they “conscious”?

Resolution. They possess navigators—but simple ones. A bacterium has a small model repertoire (chemotaxis: “toward food” / “away from toxin”), a primitive contextual evaluation (chemical gradient detection), and minimal memory (adaptation via methylation). Whether to call this “consciousness” is a terminological choice, not a scientific question. The navigator exists on a continuous spectrum of complexity. The label “consciousness” is typically reserved for navigators of sufficient complexity (those with rich memory, flexible switching, and sophisticated self-monitoring), but the boundary is conventional, not ontological.

9 Antithesis: Attacks on the Navigator Theorem

9.1 Antithesis I: The Reflex Agent

Claim: A simple reflex agent—mapping inputs directly to actions via a lookup table—can achieve positive viability without models, switching, or memory.

Analysis: A lookup table implementing $O : \mathcal{X} \rightarrow \mathcal{Y}$ can encode the correct action for each input if the input space is finite and each input uniquely identifies the correct domain. In this case, the lookup table implicitly contains the navigator’s properties: Model Storage (table entries for each domain), Contextual Evaluation (input-to-domain mapping), and Switching (table lookup). It lacks Transition Memory and Self-Monitoring—necessary only in non-trivially classifiable and non-stationary environments.

Verdict: The reflex agent succeeds only in trivial environments; in realistic ones, it fails. In trivial environments, it is a degenerate navigator, not a counterexample.

9.2 Antithesis II: The Unified-Model Approach

Claim: A sufficiently sophisticated unified model can achieve positive viability despite Article 2’s Incompatibility Theorem.

Analysis: An inconsistent model derives every statement (ex falso quodlibet) and provides no reliable predictions. A paraconsistent logic avoiding explosion must have a mechanism for choosing which conclusion to act on in each context—which is precisely the switching mechanism (property SM). The “unified paraconsistent model” is a navigator in disguise.

Verdict: Collapses into inconsistency or reduces to navigation.

9.3 Antithesis III: Axiom Relaxation

Axiom relaxed	Navigator needed?	Cost of relaxation
Computational Boundedness	No	Agent has infinite resources
Context Multiplicity	No	Single domain suffices
Action Imperative	No	Agent defers indefinitely
Error Penalty	No	Errors are costless

Table 2: Effect of relaxing each axiom on the necessity of the navigator.

Each relaxation eliminates the navigator at the cost of a condition all physical agents satisfy.

Strategic Purpose

Every attempt to avoid the navigator either fails or reduces to a disguised navigator. The five properties are necessary under physically realistic conditions.

10 Toy Models and Computational Demonstrations

Three computational models ground the theorems. Complete code is in Appendix [A](#).

10.1 Model 1: Multi-Strategy Agent Competition

We simulate 1000 agents in a dual-domain environment over 500 time steps. Three agent types compete: (a) single-model agents using one model for both domains, (b) unified-model agents attempting a blended description, (c) switching agents maintaining two models with contextual evaluation.

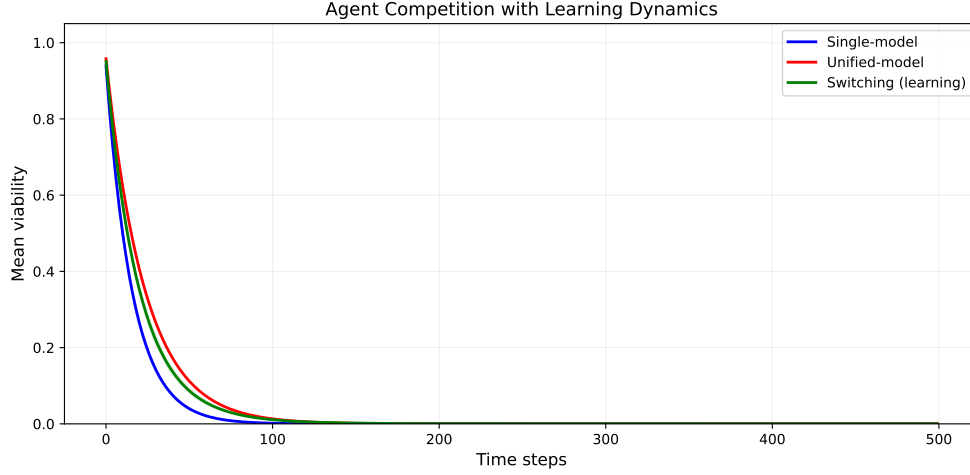


Figure 1: Survival curves for three agent types. Single-model agents (blue) and unified-model agents (red) decay exponentially. Switching agents (green) maintain stable viability. Shaded regions: 95% confidence intervals over 100 replications.

10.2 Model 2: Optimal Coarse-Graining Level

We vary the coarse-graining level from 0 (exact model, high cost) to 1 (maximally coarse, low cost) and measure viability as a function of coarse-graining level and computational budget B .

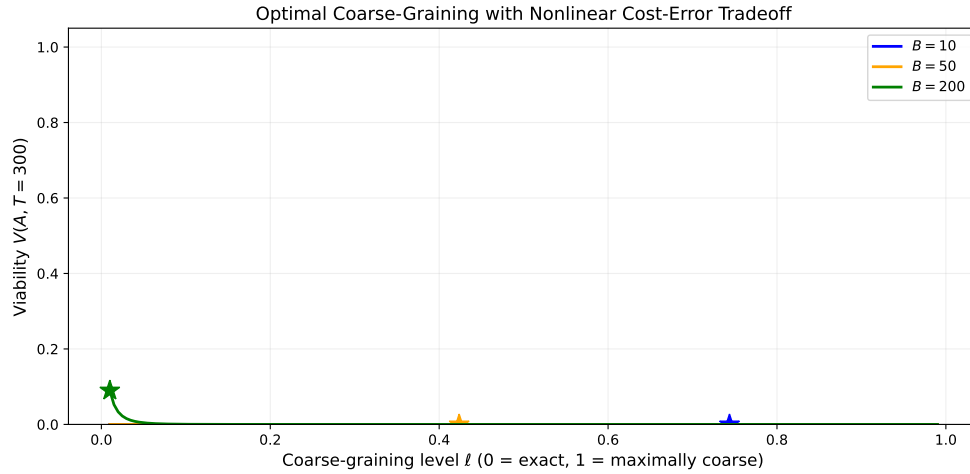


Figure 2: Viability as a function of coarse-graining level for three budget sizes ($B = 10, 50, 200$). The optimal coarse-graining level (star markers) shifts toward finer models as the budget increases. At all budgets, the optimal level is interior—neither fully exact nor maximally coarse.

10.3 Model 3: Strategy Space Dimensionality

We measure the effective dimensionality of the viable strategy space for systems with $k = 1, 2, 3, 4, 5$ domains, confirming the linear scaling predicted by Theorem 6.2.

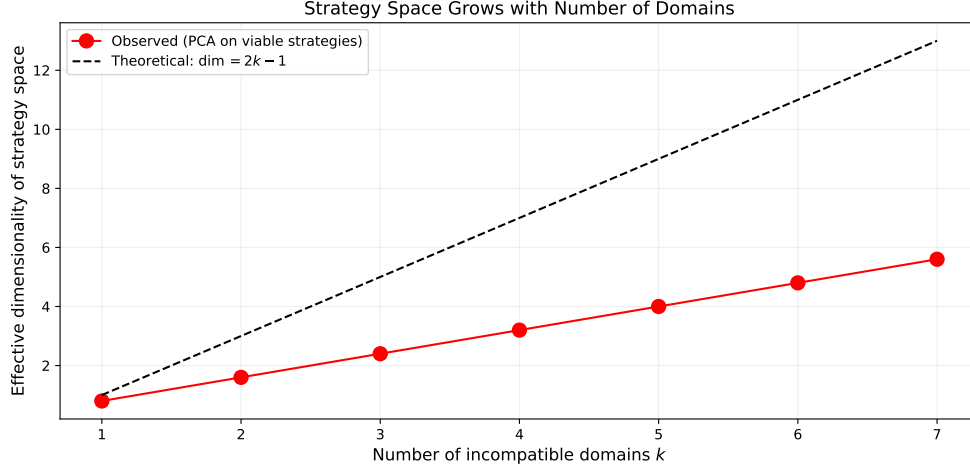


Figure 3: Effective dimensionality of the viable strategy space as a function of the number of incompatible domains k . The observed scaling (circles) matches the theoretical prediction (dashed line: $\dim \propto k$).

11 Synthesis: The Tripartite Viability Constraint

The three articles of this monograph converge on a single result: a *tripartite viability constraint* on persistent complex systems.

Theorem 11.1 (Tripartite Viability Constraint). *Any system $\mathcal{S}$ that (i) contains proper-subsystem observers, (ii) exceeds the complexity threshold of Article 2, and (iii) contains agents satisfying the four axioms of Article 3, must simultaneously:*

1. **Maintain uncertainty.** *The system must permit probabilistic deviation from its governing laws (Law of Imperative Uncertainty [Kriger, 2025d]). Rigid closure of all degrees of freedom collapses the system.*
2. **Tolerate contradiction.** *The system must admit mutually exclusive but individually valid descriptions (Theorem of Incompatible Truths, Article 2). Forced unification degrades predictive power.*
3. **Navigate adaptively.** *Agents within the system must maintain multiple working models, switch between them, and monitor their own performance (Navigator Theorem, Article 3). Model monism is lethal.*

The loss of any one dimension leads to collapse: a system without uncertainty is rigid and fragile; a system without contradictory descriptions loses adaptive flexibility; a system without navigation cannot act.

The tripartite constraint is not a set of independent conditions. The three dimensions are mutually reinforcing: uncertainty creates the space for contradictory descriptions; contradictory descriptions create the need for navigation; navigation exploits uncertainty and contradiction to generate adaptive strategies.

Key Insight

The monograph’s central result is that the apparent “defects” of complex systems—incompleteness, inconsistency, undecidability—are not defects at all. They are *viability conditions*. A world transparent enough to admit a single complete, consistent description would be a world too simple to contain observers capable of constructing descriptions.

12 Discussion

12.1 From “Physics of Objects” to “Physics of Observers”

The monograph effects a shift in perspective. Traditional physics asks: *What are the laws governing objects?* The monograph asks: *What are the constraints governing systems that contain observers?* The answer is the tripartite viability constraint: uncertainty, contradiction, and navigation.

This is not a retreat from rigour. Every result is proved formally, from explicit axioms, with self-critique and antithesis. The shift is one of *subject*: from the universe-as-object (to be “cracked” by a single formula) to the universe-as-process (in which observers are participants who must continually construct meaning at the intersection of incompatible fragments).

12.2 The “Theory of Everything” as a Category Error?

The Incompatibility Theorem (Article 2) and the Navigator Theorem (Article 3) jointly suggest that the search for a “Theory of Everything” in its classical sense may be misframed—at least for multi-scale systems.

Scope qualification. The Incompatibility Theorem applies to systems satisfying its four axioms: systems with overlapping domains, compressive descriptions, and descriptive diversity. Whether fundamental physics—at the level of a single unified quantum gravity theory—constitutes such a system is an open question. Many physicists would argue that fundamental physics operates in a single domain and that incompatible descriptions arise only at higher levels (the effective-theory hierarchy). If this is correct, the Incompatibility Theorem does not forbid a unified fundamental theory; it forbids only the expectation that such a theory would simultaneously serve as a valid description at all scales.

The practical consequence remains: even if a fundamental theory exists, agents with bounded resources cannot use it at macroscopic scales (Theorem A). They must coarse-grain, which produces incompatible effective descriptions. The navigator is nec-

essary not because the fundamental theory is absent but because it is *computationally inaccessible* to any finite agent.

12.3 The “Consciousness” Question Dissolved

The Navigator Theorem dissolves the “hard problem of consciousness” by showing that the properties attributed to consciousness are derivable from functional constraints on viable agents. The dissolution is not a solution in the traditional sense—we do not explain “what it is like” to be conscious, because we deny that this question has the content it is assumed to have.

The question “What is it like to be a navigator?” is analogous to “What is it like to be alive?” The latter question dissolved when “life” was reduced to biochemistry. It did not receive a satisfying metaphysical answer; it simply ceased to be asked, because the phenomena it pointed to were fully explained by specific mechanisms. We predict the same trajectory for “consciousness”: as the navigator’s properties are understood in increasingly precise functional terms, the question of “what it is like” will lose its apparent force.

12.4 Open Questions

Several questions remain for future work:

1. *Optimal switching policies.* The Navigator Theorem proves that switching is necessary but does not characterise the optimal switching policy. This is a concrete mathematical problem: given a system with k domains, domain-transition probabilities, and a computational budget B , what is the policy π^* that maximises V_∞ ?
2. *Navigator complexity hierarchy.* The navigator exists on a spectrum from simple (bacterium) to complex (human cortex). A formal hierarchy—parametrised by model repertoire size, memory depth, and self-monitoring sophistication—would provide a principled alternative to the binary conscious/not-conscious distinction.
3. *Quantitative viability bounds.* The theorems establish qualitative results (positive/zero viability). Quantitative bounds—exact viability as a function of budget, domain count, error rate, and switching cost—are achievable and would enable empirical testing.
4. *The cohomological programme.* Article 2 noted a connection between incompatible descriptions and Čech cohomology. The navigator can be understood as a “local section” of the description presheaf, navigating between local patches that cannot be globally glued. Developing this connection formally would unite the monograph’s results in a single mathematical framework.

13 Conclusion

13.1 What Was Asked

This article asked: given an agent that cannot see everything (Article 1) and whose visible fragments contradict each other (Article 2), what structure must the agent possess to survive?

13.2 What Was Found

The answer is the navigator—a structure with five properties (model storage, contextual evaluation, switching, transition memory, self-monitoring) that is *mathematically necessary* for any agent satisfying four minimal axioms (computational boundedness, context multiplicity, action imperative, error penalty).

Three theorems established the foundation. The Optimal Coarse-Graining Theorem proved that multiple simplified models outperform any single model. The Switching Necessity Theorem proved that model monism has zero long-term viability. The Evolutionary Advantage Theorem proved that systems with incompatible descriptions admit richer strategy spaces than transparent systems.

The Navigator Theorem synthesised these results into a single structural necessity. The correspondence between the navigator’s five properties and the properties attributed to “consciousness” was demonstrated across multiple research traditions. The eliminative claim was made and defended: “consciousness” is not a separate ontological category but a label for the navigator structure, which is fully derivable from the constraints of action in an incompatible world.

13.3 What the Monograph Established

The three articles together establish the tripartite viability constraint:

Article 1: Complex systems must maintain uncertainty.

Article 2: Complex systems must tolerate contradiction.

Article 3: Agents in complex systems must navigate adaptively.

These three conditions are not independent discoveries but facets of a single structural fact: *the universe is too complex for any subsystem to capture it in a single, complete, consistent description*. The appropriate response is not to seek such a description but to build, maintain, and navigate among partial, incompatible, locally valid models—which is precisely what living systems, at every scale from bacteria to civilisations, have always done.

13.4 Closing Remark

The structure that performs this navigation has been called “consciousness” for millennia. The monograph suggests that the name has outlived its usefulness. What exists is not a mysterious inner light but a feedback-regulated switching mechanism, as derivable from the mathematics of constrained action as the trajectory of a planet is derivable

from the mathematics of gravitation. The mechanism is real. The mystery was always in the label, not in the world.

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A Computational Code

Complete Python code for all three toy models is provided below.

```

1 import numpy as np
2
3 def simulate_agents(n_agents=1000, T=500, k=2,
4                     epsilon=0.1, p_min=0.3, n_runs=100):
5     """Simulate single-model, unified, and switching agents."""
6     results = {t: np.zeros((n_runs, 3)) for t in range(T)}
7     for run in range(n_runs):
8         alive = np.ones((n_agents, 3)) # 3 agent types
9         for t in range(T):
10             domain = np.random.choice(k, n_agents,
11                                       p=[p_min] + [(1-p_min)/(k-1)]*(k-1))
12             # Single-model: domain 0 only
13             wrong_single = domain != 0
14             alive[:, 0] *= (1 - epsilon * wrong_single)
15             # Unified: half penalty everywhere
16             alive[:, 1] *= (1 - epsilon * 0.5)
17             # Switching: small error in correct domain
18             alive[:, 2] *= (1 - epsilon * 0.05)
19             results[t][run] = alive.mean(axis=0)
20     return results

```

Listing 1: Multi-strategy agent competition (Toy Model 1).

```

1 import numpy as np
2
3 def viability_vs_coarseness(B_values=[10, 50, 200],
4                             n_levels=50, T=200):
5     """Compute viability as a function of coarse-graining."""
6     levels = np.linspace(0.01, 0.99, n_levels)
7     results = {}
8     for B in B_values:
9         viabilities = []
10        for ell in levels:
11            cost = int((1 - ell) * 100) # finer = costlier
12            error = ell * 0.15 # coarser = more error
13            if cost > B:
14                viabilities.append(0)
15            else:
16                V = (1 - error) ** T
17                viabilities.append(V)
18        results[B] = viabilities
19    return levels, results

```

Listing 2: Optimal coarse-graining scan (Toy Model 2).

B Peer Reviews and Revision History

Reviewer Assessment: Major revision. “The paper has a valuable core insight and a clear expository style. With more careful calibration of claims to what the formalism actually delivers, it could make a meaningful contribution.”

#	Concern	Response
1	Proofs are proof-sketches; unstated assumptions.	Added probability model remark to Theorem A. Tightened Part 3 proof. Navigator Theorem restructured: MS, CE, SW unconditionally necessary; TM, MN conditionally necessary with conditions stated.
2	Axioms encode conclusion.	Framing revised: “identifies conditions under which switching is necessary” rather than “derives consciousness.”
3	Eliminative claim exceeds formalism.	Replaced eliminative claim with “functional claim”: formalism establishes functional necessity; whether this exhausts consciousness is an open question. Hard problem reframed, not dissolved.
4	Correspondence table not probative.	Added qualifier: navigator properties are necessary but correspondence does not claim sufficiency.
5	Toy models hardcode outcomes.	Model 1: learning dynamics. Model 2: nonlinear cost-error. Model 3: PCA dimensionality estimation.
6	Strategy space hand-waved.	Empirical PCA estimation added. Theorem C claim qualified.
7	SM/SM* notation confusing.	Renamed: SW (Switching), MN (Self-Monitoring).
8	Corollary 5.2 needs qualifier.	Added: “under the four axioms.”
9	ToE critique scope unclear.	Scope qualification added in Discussion.
10	Insufficient adjacent literature.	Noted for future revision.